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Volume II
Amendment No. 10**

**AMENDMENT NO. 10
TO THE
PROCEDURES
FOR
AIR NAVIGATION SERVICES**

AIRCRAFT OPERATIONS

**VOLUME II
CONSTRUCTION OF VISUAL AND
INSTRUMENT FLIGHT PROCEDURES**

SEVENTH EDITION — 2020

INTERNATIONAL CIVIL AVIATION ORGANIZATION

Checklist of Amendments
to the PANS-OPS (Doc 8168), Volume II, Seventh Edition

	<i>Date of applicability</i>
Seventh Edition (Approved by the Council on 19 May 2020)	5 November 2020
Amendment No. 10 (Approved by the President of the Council of ICAO on behalf of the Council on 28 March 2024) Replacement pages: vi, viii to x, xiii, xv to xvii xxxi and xxxii I-1-1-4, I-1-2-1 to I-1-2-4, I-2-3-6, I-2-4-1 to I-2-4-6, I-3-3-4, I-3-5-2, I-3-5-3, I-3-6-1, I-4-1-1, I-4-1-2, I-4-3-1, I-4-3-9, I-4-3-10, I-4-4-1, I-4-4-2, I-4-5-1, I-4-5-2, I-4-5-5, I-4-5-8, I-4-5-9, I-4-5-11, I-4-6-1, I-4-9-1, I-4-9-2, I-4-9-6 I-4-10-1, I-4-10-App-1 II-1-1-3 to II-1-1-15, II-1-1-18, II-1-1-21 to II-1-1-24, II-1-1-33, II-1-1-App A-1, II-1-1-App C-1, II-1-1-App D-1 to II-1-1-App D-4, II-1-3-4 to II-1-3-10, II-1-3-14, II-1-3-15, II-1-3-18, II-1-3-22 to II-1-3-25, II-1-3-35 III-1-1-1, III-1-1-2, III-1-2-5, III-1-2-8, III-2-3-1 to III-2-3-5, III-2-6-App B-1 to III-2-6-App B-3, III-3-2-2 to III-3-2-5, III-3-5-2, III-3-5-3, III-3-5-6, III-3-5-9 to III-3-5-12, III-3-5-App-1, III-3-5-App-2, III-3-6-1 to III-3-6-44, III-3-7-1 to III-3-7-3, III-3-7-App A-1 to III-3-7-App A-7, III-3-7-App B-1 to III-3-7-App B-6, III-3-8-1, III-3-8-2, III-5-1-2 to III-5-1-9 IV-1-2 to IV-1-11, IV-1-13, IV-2-4 to IV-2-9, IV-2-14, IV-2-16, IV-2-17, IV-2-19, IV-2-20, IV-3-1 to IV-3-3, IV-3-App A-1, IV-3-App A-2, IV-3-App B-1 to IV-3-App B-4, IV-3-App C-1, IV-3-App C-2	28 November 2024



Transmittal note

Amendment No. 10
to the
Procedures for Air Navigation Services
AIRCRAFT OPERATIONS
(Doc 8168)
Volume II
Construction of Visual and
Instrument Flight Procedures

1. Insert the following replacement pages in the PANS-OPS, Volume II (Seventh Edition) to incorporate Amendment No. 10 which becomes applicable on 28 November 2024.

- | | |
|--|--------------------------------|
| a) Pages vi, viii to x, xiii, xv to xvii | — Table of Contents |
| b) Pages xxxi and xxxii | — Foreword |
| c) Page I-1-1-4 | — Part I, Section 1, Chapter 1 |
| d) Pages I-1-2-1 to I-1-2-4 | — Part I, Section 1, Chapter 2 |
| e) Page I-2-3-6 | — Part I, Section 2, Chapter 3 |
| f) Pages I-2-4-1 to I-2-4-6 | — Part I, Section 2, Chapter 4 |
| g) Page I-3-3-4 | — Part I, Section 3, Chapter 3 |
| h) Pages I-3-5-2 and I-3-5-3 | — Part I, Section 3, Chapter 5 |
| i) Page I-3-6-1 | — Part I, Section 3, Chapter 6 |
| j) Pages I-4-1-1 and I-4-1-2 | — Part I, Section 4, Chapter 1 |
| k) Pages I-4-3-1, I-4-3-9 and I-4-3-10 | — Part I, Section 4, Chapter 3 |
| l) Pages I-4-4-1 and I-4-4-2 | — Part I, Section 4, Chapter 4 |

- m) Pages I-4-5-1, I-4-5-2, I-4-5-5,
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- n) Page I-4-6-1 — Part I, Section 4, Chapter 6
- o) Pages I-4-9-1, I-4-9-2 and I-4-9-6 — Part I, Section 4, Chapter 9
- p) Page I-4-10-1 and — Part I, Section 4, Chapter 10
- q) Page I-4-10-App-1 — Part I, Section 4, Chapter 10, Appendix
- r) Pages II-1-1-3 to II-1-1-15, II-1-1-18,
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- s) Page II-1-1-App A-1 — Part II, Section 1, Chapter 1, Appendix A
- t) Page II-1-1-App C-1 — Part II, Section 1, Chapter 1, Appendix C
- u) Pages II-1-1-App D-1 to II-1-1-App D-4 — Part II, Section 1, Chapter 1, Appendix D
- v) Pages II-1-3-4 to II-1-3-10,
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- w) Pages III-1-1-1 and III-1-1-2 — Part III, Section 1, Chapter 1
- x) Pages III-1-2-5 and III-1-2-8 — Part III, Section 1, Chapter 2
- y) Pages III-2-3-1 to III-2-3-5 — Part III, Section 2, Chapter 3
- z) Pages III-2-6-App B-1 to III-2-6-App B-3 — Part III, Section 2, Chapter 6, Appendix B
- aa) Pages III-3-2-2 to III-3-2-5 — Part III, Section 3, Chapter 2
- bb) Pages III-3-5-2, III-3-5-3, III-3-5-6 and
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- cc) Pages III-3-5-App-1 and III-3-5-App-2 — Part III, Section 3, Chapter 5, Appendix
- dd) Pages III-3-6-1 to III-3-6-44 — Part III, Section 3, Chapter 6
- ee) Pages III-3-7-1 to III-3-7-3 — Part III, Section 3, Chapter 7
- ff) Pages III-3-7-App A-1 to III-3-7-App A-7 — Part III, Section 3, Chapter 7, Appendix A
- gg) Pages III-3-7-App B-1 to III-3-7-App B-6 — Part III, Section 3, Chapter 7, Appendix B
- hh) Pages III-3-8-1 and III-3-8-2 — Part III, Section 3, Chapter 8
- ii) Pages III-5-1-2 to III-5-1-9 — Part III, Section 5, Chapter 1

- jj) Pages IV-1-2 to IV-1-11 and IV-1-13 — Part IV, Chapter 1
 - kk) Pages IV-2-4 to IV-2-9, IV-2-14, IV-2-16, IV-2-17, IV-2-19 and IV-2-20 — Part IV, Chapter 2
 - ll) Pages IV-3-1 to IV-3-3 — Part IV, Chapter 3
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 - nn) Pages IV-3-App B-1 to IV-3-App B-4 — Part IV, Chapter 3, Appendix B
 - oo) Pages IV-3-App C-1 and IV-3-App C-2 — Part IV, Chapter 3, Appendix C
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- a) Page I-4-9-7 — Part I, Section 4, Chapter 9
 - b) Pages II-1-1-App D-5 to II-1-1-App D-8 — Part II, Section 1, Chapter 1, Appendix D
 - c) Page II-1-1-App E-1 — Part II, Section 1, Chapter 1, Appendix E
 - d) Pages III-2-6-App B-4 — Part III, Section 2, Chapter 6, Appendix B
 - e) Pages III-3-6-45 to III-3-6-52 — Part III, Section 3, Chapter 6
 - f) Pages III-3-7-4 to III-3-7-6 — Part III, Section 3, Chapter 7
 - g) Pages III-3-7-App A-8 to III-3-7-App A-13 — Part III, Section 3, Chapter 7, Appendix A
 - h) Pages IV-3-App-1 to IV-3-App-4 — Part IV, Section 3, Appendix
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Note.— Obstacle assessment surfaces for approach with vertical guidance and precision approach (PANS-OPS software) can be found on the ICAO public website (www.icao.int) under “Publications”.

<i>Amendment</i>	<i>Source(s)</i>	<i>Subject(s)</i>	<i>Approved Applicable</i>
8	Thirteenth meeting of the Instrument Flight Procedures Panel (IFPP/13) Third meeting of the Flight Operations Panel (FLTOSP/3) Twelfth meeting of the Aeronautical Information Service (AIS) Aeronautical Information Management (AIM) Study Group (AIS-AIMSG/12) First meeting of the Separation and Airspace Safety Panel (SASP/1)	a) Safety risk assessment for an instrument flight procedure design (IFPD); b) revised criteria for helicopter point-in-space (PinS); c) updated definition of procedure altitude/height; d) revised criteria for naming waypoints on performance-based navigation (PBN) approach procedures; e) revised datum crossing point (DCP) definition; f) revised ground-based augmentation system (GBAS) and GBAS landing system (GLS) and satellite-based augmentation system (SBAS) terminology; g) revised guidance on encoding GBAS final approach segment (FAS) and SBAS FAS data block parameters; h) alignment of localizer performance with vertical guidance (LPV) and localizer performance (LP) criteria; i) introduction of a visual segment surface-obstacle clearance surface (VSS-OCS) around the nominal flight track; j) clarification of intermediate segment protection area limits; k) consequential amendments to references arising from the restructuring of Annex 15 and the introduction of PANS-AIM; and l) new PBN departure procedures for parallel runway operations.	28 August 2018 8 November 2018
9 (Seventh Edition)	Fourteenth meeting of the Instrument Flight Procedures Panel (IFPP/14); fourth meeting of the Flight Operations Panel (FLTOSP/4)	a) helicopter criteria: maintenance of criteria; b) charted altitudes and flight levels; c) GBAS CAT II and III criteria; d) simultaneous operations on parallel and near parallel runways; e) visual segment surface (VSS); f) PBN to xLS criteria (use of RF legs); g) missed approach criteria after xLS procedures; h) PBN approach charting identification; i) consequential amendment related to continuous descent final approach (CDFA).	19 May 2020 5 November 2020 4 November 2021

<i>Amendment</i>	<i>Source(s)</i>	<i>Subject(s)</i>	<i>Approved Applicable</i>
10	The Fifteenth Meeting of the Instrument Flight Procedure Panel (IFPP/5)	Charting navigation specifications and accuracies, instrument flight procedures (IFP) for helicopter PBN operations and altitudes on instrument procedures	28 March 2024 28 November 2024

Direct visual segment (Direct-VS). A visual segment designed as:

- a) a leg in a PinS approach, which may contain a single turn, from the MAPt direct to the heliport or landing location or via a descent point to the heliport or landing location; or
- b) a straight leg from the heliport or landing location to the IDF in a PinS departure.

DME distance. The line of sight distance (slant range) from the source of a DME signal to the receiving antenna.

Elevation. The vertical distance of a point or a level, on or affixed to the surface of the earth, measured from mean sea level.

European GNSS navigation overlay service (EGNOS). A satellite-based augmentation system providing navigation service meeting Annex 10 requirements that provides navigation service in the European Region.

Fictitious heliport point (FHP). The FHP is a point over which the PinS final approach segment path passes at a relative height defined as the fictitious heliport point crossing height (FHPCH). It is defined by the WGS-84 latitude, longitude and ellipsoid height. The FHP replaces the FTP for PinS approaches.

Fictitious threshold point (FTP). The FTP is a point over which the final approach segment path passes at a relative height specified by the reference datum height. It is defined by the WGS-84 latitude, longitude and ellipsoid height. The FTP replaces the LTP when the final approach course is not aligned with the runway extended centre line or when the threshold is displaced from the actual runway threshold. For non-aligned approaches the FTP lies on the intersection of the perpendicular from the FAS to the runway threshold. The FTP elevation is the same as the actual runway threshold elevation.

Final approach and take-off area (FATO). A defined area over which the final phase of the approach manoeuvre to hover or landing is completed and from which the take-off manoeuvre is commenced. Where the FATO is to be used by performance Class 1 helicopters, the defined area includes the rejected take-off area available.

Final approach segment. That segment of an instrument approach procedure in which alignment and descent for landing are accomplished.

Final approach segment (FAS) data block. The set of parameters to identify a single precision approach or APV and define its associated approach path.

Final approach track. The flight track in the final approach segment that is normally aligned with the runway centre line. For offset final approach segments, the final approach track is aligned with the orientation of the FTP and the FPAP.

Flight level (FL). A surface of constant atmospheric pressure which is related to a specific pressure datum, 1 013.2 hectopascals (hPa), and is separated from other such surfaces by specific pressure intervals.

Note 1.— A pressure type altimeter calibrated in accordance with the Standard Atmosphere:

- a) *when set to a QNH altimeter setting, will indicate altitude;*
- b) *when set to a QFE altimeter setting, will indicate height above the QFE reference datum;*
- c) *when set to a pressure of 1 013.2 hPa, may be used to indicate flight levels.*

Note 2.— The terms “height” and “altitude”, used in Note 1 above, indicate altimetric rather than geometric heights and altitudes.

Flight path alignment point (FPAP). The FPAP is a point in the same lateral plane as the LTP or FTP that is used to define the alignment of the final approach segment. For approaches aligned with the runway centre line, the FPAP is located at or beyond the opposite threshold of the runway. The delta length offset from the opposite threshold of the runway defines its location.

Flight procedure designer. A person responsible for flight procedure design who meets the competency requirements as laid down by the State.

Full-scale deflection (FSD). The term used to describe the maximum deviation from centre of either a course deviation indicator (CDI) or a vertical deviation indicator (VDI), such as a glide slope indicator, and that applies to both linear and angular scaling.

GBAS Approach Service Type (GAST). A GAST is defined as the matched set of airborne and ground performance and functional requirements that are intended to be used in concert in order to provide approach guidance with quantifiable performance. Four types of approach service, GAST A, GAST B, GAST C and GAST D are currently defined. GAST A, B and C are intended to support typical APV I, APV II and Category I operations, respectively. GAST D has been introduced to support landing and guided take-off operations in lower visibility conditions including Category III operations. Note that provisions for a separate service type to support Category II operations have not been made as these can be either supported by GAST C or D.

Note.— Further details regarding GBAS approach services can be found in Annex 10, Volume I, Attachment D, 7.1.2.

GBAS landing system (GLS). A system for approach and landing operations utilizing GNSS, augmented by a ground-based augmentation system (GBAS), as the primary navigational reference.

Geoid. The equipotential surface in the gravity field of the Earth, which coincides with the undisturbed mean sea level (MSL) extended continuously through the continents.

Note.— The geoid is irregular in shape because of local gravitational disturbances (wind tides, salinity, current, etc.) and the direction of gravity is perpendicular to the geoid at every point.

Geoid undulation. The distance of the geoid above (positive) or below (negative) the mathematical reference ellipsoid.

Note.— In respect to the World Geodetic System — 1984 (WGS-84) defined ellipsoid, the difference between the WGS-84 ellipsoidal height and orthometric height represents WGS-84 geoid undulation.

Global navigation satellite system (GNSS). A worldwide position and time determination system that includes one or more satellite constellations, aircraft receivers and system integrity monitoring, augmented as necessary to support the required navigation performance for the intended operation.

Note.— GNSS performance standards are found in Annex 10, Volume I, Chapter 3.

GNSS azimuth reference point (GARP). The GARP is defined to be beyond the FPAP along the procedure centre line by a fixed offset of 305 m (1 000 ft). It is used to establish the lateral deviation display limits.

Ground-based augmentation system (GBAS). An augmentation system in which the user receives augmentation information directly from a ground-based transmitter.

Guidance control point. For procedure design purposes, the virtual point of the aircraft that is supposed to follow the lateral and vertical guidance of a computed flight path.

Heading. The direction in which the longitudinal axis of an aircraft is pointed, usually expressed in degrees from North (true, magnetic, compass or grid).

Height. The vertical distance of a level, a point or an object considered as a point, measured from a specified datum.

Chapter 2

ABBREVIATIONS AND ACRONYMS

(used in this document)

AC	Advisory circular
AIP	Aeronautical information publication
AMA	Area minimum altitude
AMSL	Above mean sea level
ANP	Actual navigation performance
AOB	Angle of bank
APV	Approach procedures with vertical guidance
ARP	Aerodrome reference point
ASE	Altimeter system error
ATC	Air traffic control
ATM	Air traffic management
ATS	Air traffic services
ATT	Along-track tolerance
AZM	Azimuth
BV	Buffer value
CA	Course to an altitude
CAT	Category
CCO	Continuous climb operation
CDFA	Continuous descent final approach
CDI	Course deviation indicator
CDO	Continuous descent operation
CF	Course to a fix
C/L	Centre line
COP	Change-over point
CRC	Cyclic redundancy check
CRM	Collision risk model
DA/H	Decision altitude/height
DCP	Datum crossing point
DER	Departure end of the runway
DF	Direction finding
Direct-VS	Direct visual segment
DME	Distance measuring equipment
DP	Descent point
DR	Dead reckoning
DTT	System use accuracy
EDA	Elevation differential area
EUROCAE	European Organization for Civil Aviation Equipment
FA	Course from a fix to an altitude

FACF	Final approach capture fix
FAF	Final approach fix
FAP	Final approach point
FAS	Final approach segment
FAT	Final approach track
FATO	Final approach and take-off area
FHP	Fictitious heliport point
FHPCH	Fictitious heliport point crossing height
FL	Flight level
FM	Course from a fix to manual termination
FMC	Flight management computer
FMS	Flight management system
FPAP	Flight path alignment point
FROP	Final approach roll-out point
FTE	Flight technical error
FTP	Fictitious threshold point
FTT	Flight technical tolerance
GARP	GNSS azimuth reference point
GAST	GBAS approach service type
GBAS	Ground-based augmentation system
GCE	Glide path construction error
GLS	GBAS landing system
GNSS	Global navigation satellite system
GP	Glide path
GPA	Glide path angle
GPWS	Ground proximity warning system
HA	Holding/racetrack to an altitude
HAE	Height above ellipsoid
HAL	Horizontal alarm limit
HCH	Heliport crossing height
HF	Holding/racetrack to a fix
HL	Height loss
HM	Holding/racetrack to a manual termination
HP	Helipoint
HRP	Helipoint reference point
IAC	Instrument Approach Chart
IAF	Initial approach fix
IAP	Instrument approach procedure
IAS	Indicated airspeed
IDF	Initial departure fix
IF	Intermediate approach fix
IFPD	Instrument flight procedure design
IFP	Instrument flight procedure
IFR	Instrument flight rules
ILS	Instrument landing system
IMAL	Integrity monitor alarm
IMC	Instrument meteorological conditions
ISA	International standard atmosphere
KIAS	Knots indicated airspeed
LDAH	Landing distance available — helicopters
LF	Low frequency
LOC	Localizer
LORAN	Long range air navigation system

LP	Localizer performance
LPV	Localizer performance with vertical guidance
LTP	Landing threshold point
MA/H	Minimum altitude/height
MAHF	Missed approach holding fix
Manoeuvring-VS	Manoeuvring visual segment
MAPt	Missed approach point
MATF	Missed approach turning fix
MCA	Minimum crossing altitude
MCH	Minimum crossing height
MDA/H	Minimum descent altitude/height
MEA	Minimum en-route altitude
MLS	Microwave landing system
MM	Middle marker
MOC	Minimum obstacle clearance
MOCA	Minimum obstacle clearance altitude
MSA	Minimum sector altitude
MSD	Minimum stabilization distance
MSL	Mean sea level
NDB	Non-directional beacon
NM	Nautical mile
NPA	Non-precision approach
NSE	Navigational system error
NTZ	No transgression zone
OAS	Obstacle assessment surface
OCA/H	Obstacle clearance altitude/height
OCA/H _{fm}	OCA/H for the final approach and straight missed approach
OCA/H _{ps}	OCA/H for the precision segment
OCS	Obstacle clearance surface
OFZ	Obstacle free zone
OIS	Obstacle identification surface
OJT	On-the-job training
OLS	Obstacle limitation surface
OM	Outer marker
PA	Precision approach
PAOAS	Parallel approach obstacle assessment surfaces
PAPI	Precision approach path indicator
PAR	Precision approach radar
PBN	Performance-based navigation
PDG	Procedure design gradient
PinS	Point-in-space
PRP	Point-in-space reference point
R	Rate of turn
RAIM	Receiver autonomous integrity monitoring
RASS	Remote altimeter setting source
RDH	Reference datum height (for APV and PA)
RF	Constant radius arc to a fix
RNAV	Area navigation
RNP	Required navigation performance
RPDS	Reference path data selector
RSR	En-route surveillance radar
RSS	Root sum square
SA	Safety area

SARPs	Standards and Recommended Practices (ICAO)
SBAS	Satellite-based augmentation system
SD	Standard deviation
SDF	Stepdown fix
SI	International system of units
SID	Standard instrument departure
SIS	Signal in space
SOC	Start of climb
SST	Supersonic transport
ST	System computation tolerance
STAR	Standard instrument arrival
TAA	Terminal arrival altitude
TACAN	UHF tactical air navigation aid
TA/H	Turn at an altitude/height
TAR	Terminal area surveillance radar
TAS	True airspeed
TCH	Threshold crossing height
TF	Track to a fix
THR	Threshold
TMA	Terminal control area
TNA/H	Turn altitude/height
TP	Turning point
TSE	Total system error
TTT	Template tracing technique
V _{mini}	Minimum instrument meteorological conditions indicated airspeed
VA	Heading to an altitude
VAL	Vertical alarm limit
VASIS	Visual approach slope indicator system
VDF	Very high frequency direction-finding station
VEB	Vertical error budget
VHF	Very high frequency
VI	Heading to an intercept
VM	Heading to a manual termination
VOR	Very high frequency omnidirectional radio range
VPA	Vertical path angle
VS	Visual segment
VSDA	Visual segment descent angle
VSDG	Visual segment design gradient
VSS	Visual segment surface
WD	Waypoint distance
WGS	World geodetic system
XTT	Cross-track tolerance
5LNC	Five-letter name code

3.3.5 Secondary areas with additional track guidance

3.3.5.1 A secondary area can be created on the outer side of the turn as soon as the aircraft has track guidance. On the outer edge of the turn this area is based on a 30° line from the nominal track up to the point (P) where the tangent becomes parallel to the nominal track after the turn.

3.3.5.2 The same principle applies for the area on the inner side of the turn, except that the 30° line from the nominal track is up to the point from whichever edge of the area provides the best lateral protection. See Figure I-2-3-6.

Table I-2-3-1. Turn construction parameter

Segment or fix of turn location	Speed (IAS)*	Altitude/height	Wind	Bank angle**	FTT (seconds)			
					c (seconds)		Outbound timing tolerance	Heading tolerance
					Bank establishment time	Pilot reaction time		
Departure	Final missed approach IAS + 10%, see Table I-4-1-1 or I-4-1-2 ³	Turn at altitude/height: Specified altitude/height Turn at turn point: A/D elevation + height based on 10% climb from DER	95% omni- directional wind or 56 km/h (30 kt) for wind spirals	For turn area calculation: 15° For establishment of the average flight path only: 15° until 305 m (1 000 ft) 20° between 305 m (1 000 ft) and 915 m (3 000 ft) 25° above 915 m (3 000 ft)	3	3	N/A	N/A
En-route	CAT A-E: 585 km/h (315 kts) CAT H: 325 km/h (175 kts)	Specified altitude	95% probability wind, or ICAO standard wind ⁴	15°	5	10	N/A	N/A

Chapter 4

QUALITY ASSURANCE

4.1 GENERAL

4.1.1 This chapter provides general provisions on quality assurance for the instrument flight procedure (IFP) process (see 4.3.1). Detailed guidance can be found in the *Quality Assurance Manual for Flight Procedure Design* (Doc 9906). Guidance on regulating the IFP process can be found in the *Manual on the Development of a Regulatory Framework for Instrument Flight Procedure Design Service* (Doc 10068).

4.1.2 The State is responsible to ensure that all published flight procedures in the airspace under the authority of that State can be flown safely by the relevant aircraft. Safety is not only accomplished by application of the technical criteria in PANS-OPS and associated ICAO provisions, but also requires measures that control the quality of the process used. These measures shall ensure the quality and safety of the procedure design product through review, verification, coordination, and validation at appropriate points in the process, so that corrections can be made at the earliest opportunity in the process.

4.1.3 This chapter establishes the high-level requirements for a quality process for procedure design, including inputs, key required elements, recommendations, and expected outputs.

4.1.4 In the interest of safety, and to promote a reasonable degree of standardization, it is desirable, to the greatest degree possible, to implement the criteria in a consistent manner, using processes that will minimize the possibility of errors, identify errors that do occur before they impact safety, and provide for continuous improvement of the IFP process in order to eliminate or reduce future errors. This is especially important in the modern aviation environment, where increasing reliance is placed on computers and the data they process, for navigation and obstacle awareness.

4.1.5 Safety of air navigation is highly dependent on the quality of aeronautical data. Processes for data quality assurance, from data origination through to publication in the State's Aeronautical Information Publication (AIP), is detailed in the *Quality Assurance Manual for Flight Procedure Design* (Doc 9906).

4.2 PROCEDURE DESIGNER QUALIFICATIONS AND TRAINING

4.2.1 Each State shall establish the required competencies for flight procedure design in the State's regulatory framework. Each State shall ensure that flight procedure designers have acquired and maintain this competency level through training and supervised on-the-job training (OJT). This is to ensure that the quality assurance in the procedure design process and its output, including the quality of aeronautical information/data, meets the requirements of Annex 11 — *Air Traffic Services*.

4.2.2 Training for flight procedure design should at least include an initial training and recurrent training at periodic intervals. The State shall establish the appropriate interval for recurrent training in the State's regulatory framework.

4.2.3 Initial training shall ensure that the flight procedure designer is able to demonstrate a basic level of competency that includes at least the following elements:

- a) knowledge of information contained in the PANS-OPS, Volumes I, II and III and other related ICAO provisions relevant for the IFP process; and
- b) skills in the design of procedures.

4.2.4 Recurrent training shall ensure that the flight procedure designer is able to demonstrate a basic level of competency that includes at least the following elements:

- a) knowledge about amendments in ICAO provisions and other provisions pertaining to procedure design; and
- b) maintenance and improvement of knowledge and skills in the design of procedures.

4.2.5 The State shall ensure that flight procedure designers have undergone an adequate, supervised OJT.

4.2.6 Competencies of the flight procedure designer shall be verified at regular intervals according to the State's regulatory framework.

Note.— Guidance material for planning, implementing and evaluating flight procedure designer training is provided in the Quality Assurance Manual for Flight Procedure Design (Doc 9906), Volume 2 — Flight Procedure Designer Training (Development of a Flight Procedure Designer Training Programme).

4.3 IFP PROCESS

4.3.1 Introduction

The IFP process (see Figure I-2-4-1) encompasses the full process from initiation to publication/promulgation of procedures and includes the continuous maintenance of the procedure as well as periodic review. The elements of the process encompass enablers, constraints, output, and post-publication feedback for the procedure under consideration.

Note.— Detailed guidance on every step in the IFP process is contained in the Quality Assurance Manual for Flight Procedure Design (Doc 9906), Volume 1— Flight Procedure Design Quality Assurance System.

4.3.2 Conceptual design

4.3.2.1 The conceptual design shall be coordinated with all relevant stakeholders. As input for the procedure design process the following aspects need to be assessed:

- a) airport, navigation aid, obstacle, and terrain coordinate and elevation data, based on verified surveys and complying with Annexes 11, 14 and 15 requirements;
- b) airspace requirements;
- c) user requirements: needs of air traffic service providers and operators who will use this procedure;
- d) airport infrastructure such as runway classification, lighting, communications, runway markings, and availability of local altimeter setting;
- e) environmental considerations; and
- f) any other potential issue associated with the procedure.

4.3.2.2 The conceptual design process, including all stakeholder feedback, shall be documented. The decision to proceed to the next step in the process shall be made by the owner/sponsor of the procedure in coordination with the regulator as appropriate.

4.3.3 Procedure design

4.3.3.1 Procedures shall be designed according to the State's regulatory framework, taking into account all design inputs. Coordination with all stakeholders should continue throughout the procedure design and also during the validation process to ensure that the procedure meets the needs of stakeholders.

4.3.3.2 As part of the validation process, each new or revised procedure shall be verified by a qualified procedure designer other than the designer who designed the procedure, to ensure compliance with applicable criteria. See 4.3.5.

4.3.3.3 Published procedures shall be subjected to a periodic review to ensure that they continue to comply with changing criteria, and meet user requirements. The individual States shall establish the interval for periodic review of instrument flight procedures according to the State's regulatory framework. The maximum interval for this review shall be five years. (See Annex 11, Appendix 7). Guidance on periodic reviews can be found in the *Quality Assurance Manual for Flight Procedure Design* (Doc 9906), Volume 1, and in the *Manual on the Development of a Regulatory Framework for Instrument Flight Procedure Design Service* (Doc 10068).

4.3.4 Procedure design documentation

4.3.4.1 The documentation provided by the procedure designer shall as a minimum consist of:

- a) documentation required to maintain transparency concerning the details and assumptions used by the procedure designer, which should include any information and data used in the design;
- b) software tools used and versions thereof;
- c) reference documents used and versions thereof;
- d) controlling obstacle for each segment of the procedure;
- e) for modifications or amendments to existing procedures, the reasons for any changes;
- f) for any deviation from existing standards, the reasons for such a deviation and details of mitigations applied to assure continued safe operations;
- g) additional information required to facilitate validation of the procedure;
- h) draft chart;
- i) for PBN procedures, a proposed coding table;
- j) if applicable, the effect of environmental considerations on the design of the procedure;
- k) if applicable, the effect of infrastructure considerations on the design of the procedure; and
- l) if applicable, the effect of airspace structure considerations on the design of the procedure.

4.3.4.2 All documentation should be retained in accordance with the State's regulatory framework to assist in recreating the procedure in the future in the case of incidents and for periodic review and continuous maintenance. The period of retention shall not be less than the operational lifetime of the procedure.

4.3.5 Validation

4.3.5.1 *General.* Validation is the necessary final quality assurance step in the procedure design process, prior to publication. The purpose of validation is the verification of all obstacle and navigation data, assessment of flyability of the procedure, assessment of crew workload and human factors, as well as for confirmation of correct database coding and avionics behaviour for PBN procedures. Validation follows a step-by-step process explained in the *Quality Assurance Manual for Flight Procedure Design* (Doc 9906), Volume 5 – *Validation of Instrument Flight Procedures*.

4.3.5.2 *Independent procedure design review.* The independent review shall be performed by a qualified flight procedure designer who was not involved in the design of the procedure. It is meant to catch errors in the application of criteria and documentation produced during the IFP process, and shall be recorded separately.

4.3.5.3 *Flight operational review of flight procedure design.* The operational review shall be performed by a pilot in accordance with the State's regulatory framework to identify potential issues in the design of the procedure from a flight operational perspective.

4.3.5.4 *Flight validation.* Flight validation of instrument flight procedures shall be carried out in accordance with the State's regulatory framework as part of the initial publication of a procedure or when existing procedures have undergone significant changes. It shall be accomplished by a qualified flight validation pilot approved by the State in accordance with the State's regulatory framework. The objectives of the flight validation of instrument flight procedures are to:

- a) become familiar with the flight procedure and understand the solution chosen by the flight procedure designer;
- b) determine which activities of flight validation are going to be performed (simulator only, aircraft only, or both);
- c) as required, ensure a pre-production navigation database is available, validated and matches with the designed procedure;
- d) provide assurance that adequate obstacle clearance has been provided;
- e) verify that all required infrastructure, such as runway markings, lighting, and communications and navigation sources, are in place and operative;
- f) conduct an assessment of flyability to determine that the procedure can be safely flown;
- g) evaluate human factors such as cockpit workload; and
- h) evaluate the draft charting, required infrastructure, operational factors and meteorological conditions in accordance with the *Manual of All-Weather Operations* (Doc 9365).

4.3.5.5 *Flight validation pilot qualification and training.* The State shall approve flight validation pilots/companies/organizations according to the State's regulatory framework. In order to achieve the safety and quality assurance objectives of the flight validation, each State shall ensure that flight validation pilots have acquired and maintain the required competencies in accordance with the State's regulatory framework.

Note.— Recommended qualifications and training, as well as guidance concerning the skills, knowledge and attitudes to be addressed in the training and evaluation of flight validation pilots, can be found in the Quality Assurance Manual for Flight Procedure Design (Doc 9906), Volume 6 — Flight Validation Pilot Training and Evaluation (Development of a Flight Validation Pilot Training Programme).

4.3.5.6 *Flight procedure designer role in flight validation.* The procedure documentation serving as a basis for conducting a flight validation originates from the procedure designer. The procedure designer shall be prepared to provide briefings to the flight validation crews as required. The procedure designer may participate in the flight validation to assist in its evaluation and obtain direct knowledge of issues related to the procedure design from the flight validation pilot.

4.4 PROCEDURE DESIGN AUTOMATION

4.4.1 Procedure design automation tools have the potential to greatly reduce errors in the procedure design process, as well as to standardize the application of the procedure design criteria. For this reason, flight procedure design service providers should use the available software packages to design instrument flight procedures.

4.4.2 There are numerous software packages available that automate, to varying degrees, the application of procedure design criteria. The advantages are many, including maintaining the integrity of the source data throughout the design phase, reducing human errors, gaining the capability to develop “what-if” scenarios, and standardized application of the criteria. While software developers test their software extensively, there is no absolute guarantee as to the accuracy of any individual application of the criteria.

4.4.3 As States are responsible for the safety of instrument flight procedures, they should ensure that the software packages used in the design of procedures have been validated according to the State’s regulatory framework. The *Quality Assurance Manual for Flight Procedure Design* (Doc 9906), Volume 3 — *Flight Procedure Design Software Validation* provides guidance to assist States in this task.

4.5 DOCUMENTATION FOR REGULATORY APPROVAL

The documents required by the regulator to be submitted for the approval of the designed procedure shall be stated in the State’s regulatory framework. As a minimum these documents shall consist of:

- a) flight procedure design documentation;
- b) independent procedure design review documentation;
- c) documentation of the flight operational review;
- d) flight validation documentation; and
- e) flight inspection documentation if required.

4.6 REGULATORY OVERSIGHT OF THE IFP PROCESS

States shall oversee the IFP process according to the State’s regulatory framework.

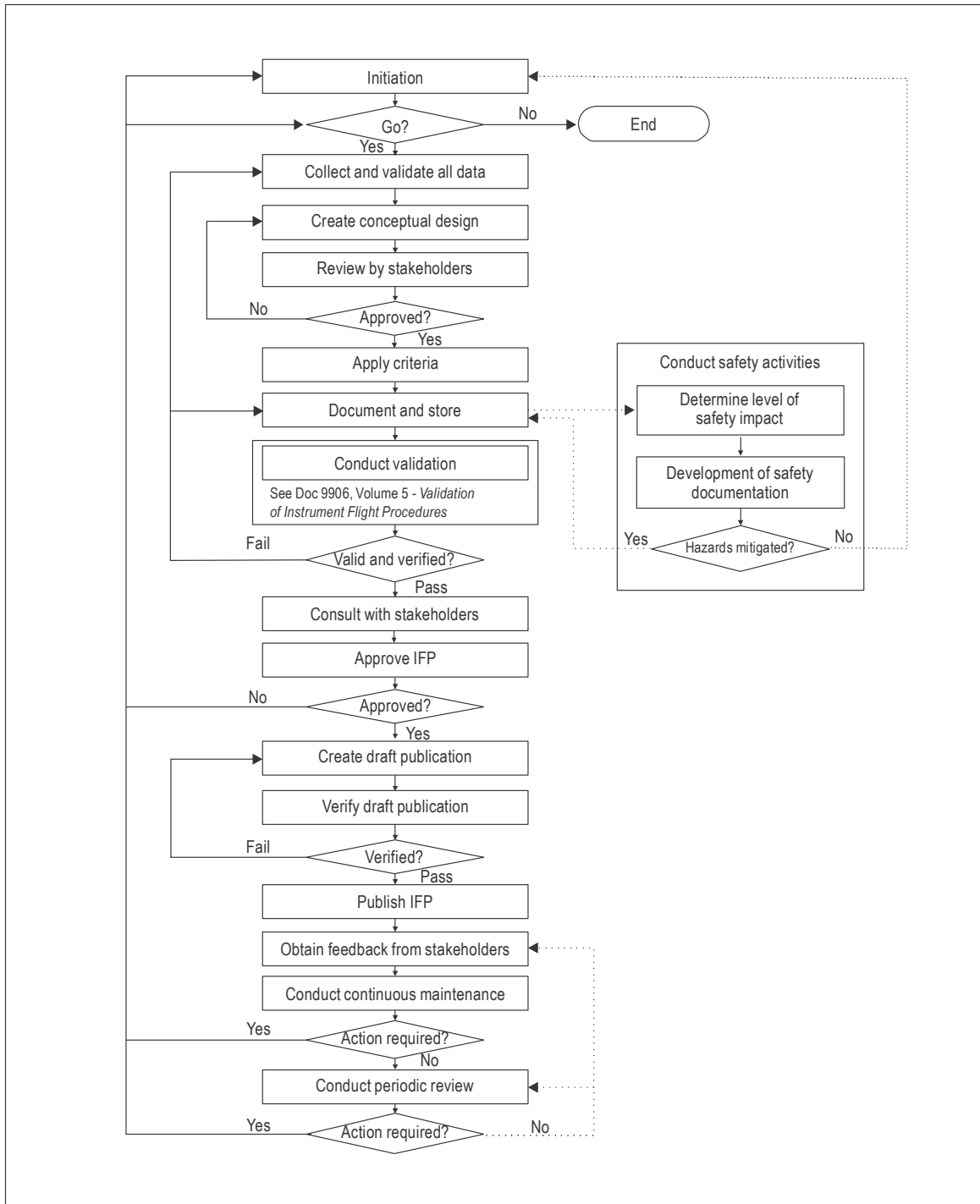


Figure I-2-4-1. Instrument flight procedure process

3.2.5.2 Areas associated with a navigation aid

The areas associated with a navigation aid other than a localizer consist of appropriate portions of the trapezoids specified in Part II, Section 2, Chapters 4 and 6. The general principle of secondary areas is applied.

3.3 TURNING DEPARTURES

3.3.1 General

3.3.1.1 A departure incorporating a turn of more than 15° is a turning departure. Turns may be specified at an altitude/height, or at a fix or at a facility.

3.3.1.2 Straight flight is assumed until reaching a height of at least 120 m (394 ft) (Cat H, 90 m (295 ft)) above the elevation of the DER. No provision is made for turning departures which require a turn below 120 m (394 ft) (Cat H, 90 m (295 ft)) above the elevation of the DER. Where the location and/or height of obstacles makes it impossible to construct turning departures which satisfy the minimum turn height criterion, departure procedures should be developed on a local basis in consultation with the operators concerned.

3.3.1.3 The areas considered in the design of turning departures are defined as:

- a) the turn initiation area; and
- b) the turn area.

The turn initiation area is an area within which the aircraft conducts a straight climb in order to reach the MOC required prior to the beginning of a turn 75 m (246 ft) (CAT H, 65 m (213 ft)). The turn area is the area in which the aircraft is considered to be turning.

3.3.2 Turn initiation area

3.3.2.1 For aeroplanes, the turn initiation area starts at a point 600 m from the start of runway. For helicopters the turn initiation area starts at the beginning of the area available for runway or the start of runway. From the start of the turn initiation area to the DER, the area is 300 m wide (Cat H, 90 m).

3.3.2.2 Where the departure chart prohibits turns prior to the DER the turn initiation area starts at the DER. For helicopters, an earliest limit for a turning departure may be located at an appropriate position along the runway/FATO.

3.3.2.3 The turn initiation area terminates at the TP. The TP may be defined by:

- a) the earliest fix tolerance of the TP fix (turn at designated turn point); or
- b) the position at which the PDG reaches the specified turn altitude/height.

3.3.2.4 The TP may be located no closer to the DER than the distance required at the PDG to reach the higher of 120 m (394 ft) (Cat H, 90 m (295 ft)) or the specified turn altitude/height. The turn initiation area is identical to the area associated with a straight departure with no track guidance as described in 3.2.4, “Straight departure without track guidance.”

(See Figures I-3-3-9 and I-3-3-10.)

3.3.3 Turn area

The turn area is constructed in the same manner as the turning missed approach area (see Section 4, Chapter 6, 6.4, “Turning missed approach”). The inner and outer boundaries of the turn area are constructed as specified in 3.2, “Turn inner boundary construction” and 3.3, “Turn outer boundary construction” in Section 2, Chapter 3, “Turn area construction”.

3.3.4 Turn parameters

The parameters on which turn areas are based are:

- a) altitude:
 - 1) *turn designated at an altitude/height*: turn altitude/height;
 - 2) *turn at a designated turning point*: aerodrome elevation plus 10 per cent of the distance from the DER to the TP (i.e. allowing for a 10 per cent climb);
- b) temperature: ISA + 15°C corresponding to a) above;
- c) *indicated airspeed*: the speed tabulated for “final missed approach” in Section 4, Chapter 1, Tables I-4-1-1 and I-4-1-2 for the applicable aircraft category, increased by 10 per cent to account for increased aircraft mass at departure. However, where operationally required to avoid obstacles, reduced speeds not less than 1.1 times the IAS tabulated for “intermediate missed approach” in Section 4, Chapter 1, Tables I-4-1-1 and I-4-1-2 may be used, provided the procedure is annotated “Departure turn limited to _____ km/h (kt) IAS maximum”. In order to verify the operational effect of a speed limitation, the speed should be compared with the statistical speed as published in the Appendix to Chapter 3.
- d) *true airspeed*: the IAS in c) above adjusted for altitude a) and temperature b);
- e) *wind*: maximum 95 per cent probability wind on an omnidirectional basis, where statistical wind data are available. Where no wind data are available, an omnidirectional 56 km/h (30 kt) wind should be used;
- f) *bank angle*: 15° average achieved;
- g) *fix tolerance*: as appropriate for the type of fix;
- h) *flight technical tolerances*: a distance “c” equivalent to 6 seconds of flight (3 second pilot reaction and 3 second bank establishing time) at the specified speed (see d) above) plus 56km/h (30 kt) tailwind; and
- i) *secondary areas*: secondary areas are applied where track guidance is available.

3.3.5 Turn at a specified altitude/height

3.3.5.1 General

A turn may be prescribed upon reaching a specified altitude/height to accommodate the situation where there is:

- a) an obstacle located in the direction of the straight departure that must be avoided; and/or
- b) an obstacle located abeam the straight departure track that must be overflown after the turn.

Chapter 5

PUBLISHED INFORMATION FOR DEPARTURE PROCEDURES

5.1 GENERAL

The minimum information to be published for a departure procedure is as follows:

- a) all tracks, points, fixes and altitudes/heights (including turn altitudes/heights) required by the procedure;
- b) all navigation facilities, fixes, waypoints, radials and DME distances used to define route segments;
- c) significant obstacles which penetrate the obstacle identification surfaces (OIS);
- d) the position and height of close-in obstacles penetrating the OIS. A note shall be included on the departure chart wherever close-in obstacles exist which were not considered in the determination of the published procedure design gradient (PDG) (see Figure II-3-2-3 in Chapter 2);
- e) the highest obstacle in the departure area, and any significant obstacle outside that area controlling the design of the procedure;
- f) a PDG greater than 3.3 per cent and the altitude/height to which it applies;
- g) the altitude/height or fix at which a gradient in excess of 3.3 per cent (Cat H, 5.0 per cent) ceases to be required (see Figure I-3-5-1);
- h) where an increased procedure design gradient is required by airspace restrictions only, a note stating that condition e.g. “4% climb gradient required due airspace restrictions only”;
- i) altitude/heights to be achieved at significant points in the departure, identified by navigation aids or fixes; and
- j) when close conformance to a track is important (e.g. noise abatement/ATC constraints) a note stating that the average flight path is designed using statistical aircraft performance data (for construction of the average flight path, see the Appendix to Chapter 3).

Note.— Principles governing the identification of standard departure routes are contained in Annex 11, Appendix 3. Specifications for standard instrument departure charts are contained in Annex 4.

5.2 OMNIDIRECTIONAL DEPARTURES

An omnidirectional departure that restricts turn altitudes/heights and/or procedure design gradients to sectors shall be promulgated as follows:

- a) restrictions shall be shown as sectors in which minimum altitudes and minimum turn altitudes/heights are specified, or in which stated procedure design gradients are required;

- b) sectors may be defined in which flight is not permitted;
- c) sectors shall be described by bearings and distance from the centre of the turn area;
- d) sectors shall diverge at least 15° either side of the controlling obstacle; and
- e) when more than one sector is published, the promulgated gradient shall be the highest PDG required in any sector that may be entered. The altitude/height to which the gradient applies must permit the subsequent use of a 3.3 per cent gradient (Cat H, 5.0 per cent) through that sector, a succeeding sector, or to an altitude/height authorized for another phase of flight (i.e. en-route holding or approach). A fix may also be designated to mark the point at which a gradient in excess of 3.3 per cent (Cat H, 5.0 per cent) ceases.

5.3 CHARTED ALTITUDES/FLIGHT LEVELS

Departure procedures may be developed to procedurally separate air traffic. In doing so, the procedure may be accompanied by altitudes/flight levels that are not associated with any obstacle clearance requirements, but are developed to separate arriving and departing air traffic procedurally. These altitudes/flight levels shall be charted as indicated in Table I-3-5-1. The method of charting of altitudes/flight levels to correctly depict the designed procedure may differ between avionics manufacturers.

5.4 OTHER REQUIREMENTS

- a) When departures are limited to a particular category(ies) of aircraft, the procedure shall be clearly annotated.
- b) Where a suitable fix is available, a procedure design gradient requirement may be promulgated by specifying a restriction based on an altitude at a DME distance or an altitude at a significant point (e.g. “reach 5 000 ft by DME 15” or “reach 3 500 ft by VWXYZ”).
- c) A turn may be specified at a fix or an altitude/height, e.g. “at DME 4 turn right, track 170°” or “at 2 500 ft turn left track to VWXYZ”.
- d) When it is necessary, after a turn, to track to intercept a specified radial/bearing, the procedure will specify:
 - 1) the turning point;
 - 2) the track to be made good; and
 - 3) the radial/bearing to be intercepted;(e.g. “at DME 4 turn left, track 340° to intercept BNE R020 (VOR)”; or “at DME 2 turn left, track 340° to intercept 010° track to STN (NDB)”).
- e) Where a PDG in excess of the standard gradient is required to provide obstacle clearance, an alternative procedure using a lower PDG may be published for operations in VMC only.
- f) Gradients to a height of 60 m (200 ft) or less due to close-in obstacles shall not be promulgated. A note shall be published stating that close-in obstacles exist.

- g) Where a suitably located DME exists, or when suitably located RNAV fixes can be established, additional specific height/distance information intended for obstacle avoidance should be published in order to provide a means of monitoring aircraft position relative to critical obstacles.
- h) Where turns prior to the DER are not accommodated, and the procedure design is based upon the turn initiation area commencing at the DER, the departure procedure shall include a note that turns are not permitted prior to the DER.

Table I-3-5-1. Charted procedure altitudes/flight levels

Altitude/Flight level “Window”	<u>17 000</u> <u>FL220</u> <u>10 000</u> <u>10 000</u>
“At or above” altitude/flight level	<u>7 000</u> <u>FL060</u>
“At or below” altitude/flight level	<u>5 000</u> <u>FL050</u>
“At” altitude/flight level	<u>3 000</u> <u>FL030</u>
“Recommended” altitude/flight level	5 000 FL050
“Expected” altitude/flight level	Expect 5 000 Expect FL050

Because of obstacle B, the gradient cannot be reduced to 3.3% (2.5% + 0.8%) (Cat H, 5.0 per cent) just after passing obstacle A. The altitude/height or fix at which a gradient in excess of 3.3% (Cat H, 5.0 per cent) is no longer required is promulgated in the procedure.

Obstacles A and B will be promulgated. Mountain promulgated on Aerodrome Obstacle Chart Type C.

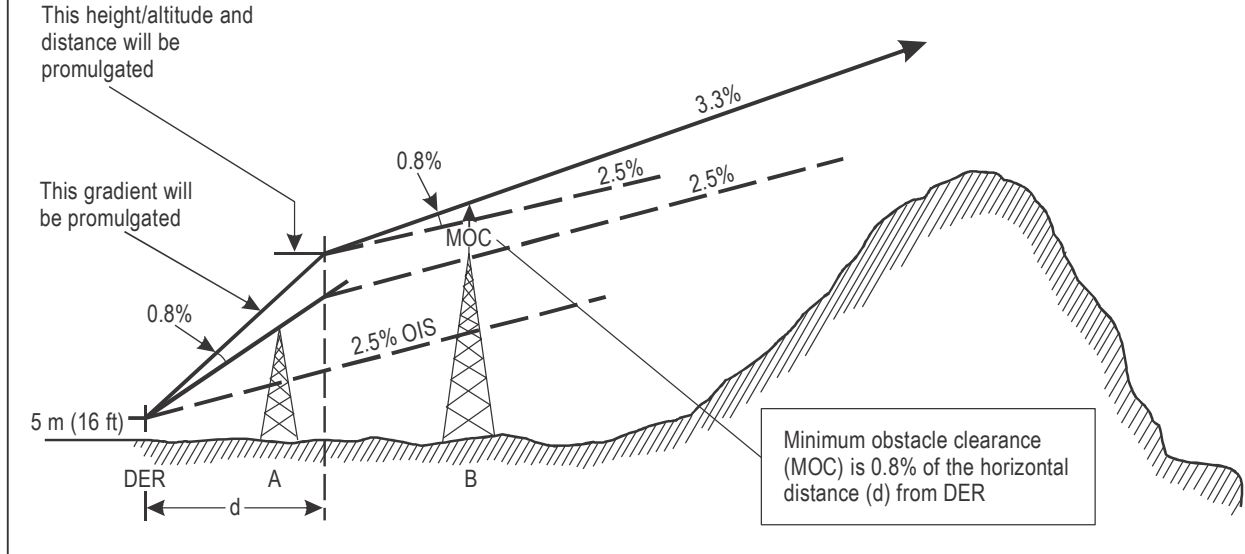


Figure I-3-5-1. Climb gradient reduction in departure

Chapter 6

SIMULTANEOUS OPERATIONS ON PARALLEL OR NEAR-PARALLEL INSTRUMENT RUNWAYS

Note.— Guidance material is contained in the Manual on Simultaneous Operations on Parallel or Near-Parallel Instrument Runways (SOIR) (Doc 9643).

6.1 INSTRUMENT DEPARTURES FROM PARALLEL RUNWAYS

When it is intended to use two instrument departure procedures from parallel runways simultaneously, the nominal departure tracks shall diverge by at least 15 degrees immediately after take-off or at least 10 degrees when both aircraft are flying a PBN SID provided the turn commences no more than 3.7 km (2.0 NM) from the departure end of the runway (see Chapter 3, “Departure routes”).

6.2 SEGREGATED OPERATIONS ON PARALLEL RUNWAYS

When it is intended to use an instrument departure procedure and an instrument approach procedure in the same direction on parallel runways simultaneously, the nominal tracks of the departure procedure and of the missed approach procedure shall diverge by at least 30 degrees as soon as practicable (see Section 4, Chapter 10).

Chapter 1

GENERAL CRITERIA FOR APPROACH/ARRIVAL PROCEDURES

1.1 SCOPE

Section 4 contains criteria common to all types of instrument arrival and approach procedures. Criteria which apply to specific types of facilities, such as ILS, are located in the chapters which deal with these kinds of guidance. Criteria which are specific to their implementation, as well as additions and exceptions to the general criteria, can be found in Part II, “Conventional procedures”, and in Part III, “RNAV procedures and satellite-based procedures”. Criteria for helicopters to runways are found in Parts I, II and III. Criteria for helicopters to heliports are found in Part IV.

Where characteristics of radio facilities are provided in this document, they are intended solely for the construction of procedures, and they do not replace or supplement corresponding material in Annex 10.

1.2 PROCEDURE CONSTRUCTION

An instrument approach procedure may have five separate segments. They are the arrival, initial, intermediate, final and missed approach segments. In addition, an area for circling the aerodrome under visual conditions should be considered. The approach segments begin and end at designated fixes. However, under some circumstances certain segments may begin at specified points where no fixes are available (or necessary). For example, the final approach segment of a precision approach may originate at the point of intersection of the designated intermediate procedure altitude with the nominal glide path; the intermediate segment may begin at the end of the inbound turn.

1.3 FIX NAMES

The fixes are named according to the segment they precede. For example, the intermediate segment begins at the intermediate fix. Where no fix is available, as mentioned above in 1.2, “Procedure construction”, the segments begin and end at specified points (e.g. the point where the glide path intersects the intermediate procedure altitude and the point where the glide path intersects the nominal DA/H). This document discusses the segments in the order in which the pilot would fly them in a complete procedure, that is from arrival through initial and intermediate to a final approach and, if necessary, the missed approach.

1.4 SEGMENT APPLICATION

Only those segments that are required by local conditions need be included in a procedure. In constructing the procedure, the final approach track should be identified first because it is the least flexible and most critical of all the segments. When the final approach has been determined, the other necessary segments should be blended with it to produce an orderly manoeuvring pattern which is responsive to the local traffic flow. See Figure I-4-1-1.

1.5 PROCEDURE ALTITUDE/HEIGHT

1.5.1 The aviation industry has identified that the majority of large aircraft accidents occur lined up with and within 19 km (10 NM) of the landing runway. To support the controlled flight into terrain (CFIT) prevention initiatives, instrument approach charts shall not only provide altitudes/heights to ensure appropriate obstacle clearance, i.e. minimum obstacle clearance altitude (MOCA), but also procedure altitudes/heights. The procedure altitude/height is intended to place the aircraft above the applicable MOCA and to support a stabilized prescribed descent gradient/angle in the final segment.

1.5.2 All non-precision instrument approach procedures shall be developed to include not only the minimum altitudes/heights to ensure obstacle clearance, i.e. MOCA, but also procedure altitudes/heights. Procedure altitudes/heights shall be developed to place the aircraft at altitudes/heights that would normally be flown to intercept and fly the prescribed descent gradient/angle in the final approach segment to a 15 m (50 ft) threshold crossing. In no case shall a procedure altitude/height be less than any OCA/H.

1.6 TRACK GUIDANCE

1.6.1 Track guidance should normally be provided for all phases of flight through the arrival, initial, intermediate, final and missed approach segments. When track guidance is provided, the appropriate segment shall lie within the established coverage of the navigation facility on which the track guidance is based.

1.6.2 When track guidance is not provided the obstacle clearance area shall be expanded as prescribed for dead reckoning (DR) segments in Chapter 3, “Initial approach segment” and in Appendix A to Chapter 3, “Initial approach using dead reckoning (DR)”. Terminal area surveillance radar (TAR), when available, may be used to provide vectors to the final approach (see Part II, Section 2, Chapter 6, “SRE”). En-route surveillance radar (RSR) may be used to provide track guidance through initial approach segments up to and including the intermediate fix. Criteria for the construction of areas for missed approaches without track guidance are provided in Chapter 6, “Missed approach segment”.

Note.— Detailed procedures regarding the use of primary radar in the approach control service are set forth in the PANS-ATM, Doc 4444, Procedures for Air Navigation Services — Air Traffic Management.

1.7 DESCENT GRADIENTS

1.7.1 Throughout the document, optimum and maximum descent gradients are specified. The optimum is the operationally preferred descent gradient. This should only be exceeded where alternative means of satisfying obstacle clearance requirements are impracticable. The maximum gradient shall not be exceeded. (See also Section 4, Chapter 9.)

1.7.2 Optimum and maximum descent gradients are specified depending on the type of procedure and the segment of the approach. At least in the case of the final approach segment for non-precision approach procedures and, preferably, also for other approach segments where appropriate, the descent gradient(s) used in the construction of the procedure shall be published.

Chapter 3

INITIAL APPROACH SEGMENT

3.1 GENERAL

3.1.1 The initial approach segment starts at the initial approach fix (IAF). In the initial approach the aircraft is manoeuvring to enter the intermediate segment. When the intermediate fix (IF) is part of the en-route structure, it may not be necessary to designate an initial approach segment. In this case the instrument approach procedure begins at the intermediate fix and intermediate segment criteria apply. An initial approach may be made along a VOR radial, NDB bearing, specified radar vector or a combination thereof. Where none of these is possible, a DME arc or a specified heading may be used.

3.1.2 Reversal and racetrack procedures as well as holding pattern descents are considered initial segments until the aircraft is established on the intermediate approach track. Where holding is required prior to entering the initial approach segment, the holding fix and initial approach fix should coincide. When this is not possible, the initial approach fix shall be located within the holding pattern on the inbound holding track.

3.1.3 Normally track guidance is required except that dead reckoning tracks may be used for distances not exceeding 19 km (10 NM). Although more than one initial approach may be established for a procedure, the number should be limited to that which is justified by traffic flow or other operational requirements.

3.2 ALTITUDE SELECTION

3.2.1 Minimum obstacle clearance altitudes (MOCAs)

MOCAs in the initial approach segment shall be established in 100-ft or 50-m increments as appropriate. The initial approach segment altitude shall not be below the altitude established for the reversal or racetrack procedure where such a procedure is required. In addition, altitudes specified in the initial approach segment must not be lower than any altitude specified for any portion of the intermediate or final approach segments.

3.2.2 Minimum obstacle clearance altitudes (MOCAs) for different aircraft categories

When different MOCAs are specified for different categories of aircraft, separate procedures shall be published.

3.2.3 Procedure altitude/height

All initial approach segments shall have procedure altitudes/heights established and published. Procedure altitudes/heights shall not be less than the MOCA and shall be developed in coordination with air traffic control requirements. The initial segment procedure altitude/height should be established to allow the aircraft to intercept the final approach segment descent gradient/angle from within the intermediate segment.

3.3 INITIAL APPROACH SEGMENTS (OTHER THAN RADAR VECTORS) UTILIZING STRAIGHT TRACKS AND DME ARCS

3.3.1 Tracks

The angle of intersection between the initial approach track and the intermediate track should not exceed 120°. When the angle exceeds 70°, a radial, bearing, radar vector or DME information providing at least 4 km (2 NM) of lead (Cat H, 1.9 km (1 NM)) shall be identified to assist in leading the turn onto the intermediate track (see Figure I-4-3-1). When the angle exceeds 120°, the use of a racetrack or reversal procedure or dead reckoning (DR) track should be considered. Criteria for such procedures are in 3.4, “Initial approach segment using a racetrack procedure”, 3.5, “Initial approach segment using a reversal procedure” and 3.3.3.3, “Area associated with dead reckoning (DR) track procedures”.

3.3.2 DME arcs

An arc may provide track guidance for all or for a portion of an initial approach. The minimum arc radius shall be 13 km (7 NM) (Cat H, 9.3 km (5 NM)). An arc may join a track at or before the intermediate fix. When joining a track, the angle of intersection of the arc and the track should not exceed 120°. When the angle exceeds 70°, a radial which provides at least 4 km (2 NM) (Cat H, 1.9 km (1 NM)) of lead shall be identified to assist in leading the turn onto the intermediate track.

3.3.3 Area

3.3.3.1 The initial approach segment has no standard length. The length shall be sufficient to permit the altitude change required by the procedure. The width is divided into:

- a) a primary area which extends laterally 4.6 km (2.5 NM) on each side of the track; and
- b) a secondary area which adds an additional 4.6 km (2.5 NM) on each side of the primary area.
(See Figure I-4-3-2.)

3.3.3.2 Area splay

Where, because of an operational requirement, any portion of the initial approach is more than 69 km (37 NM) from the VOR or 52 km (28 NM) from the NDB providing track guidance, the area will start splaying at these distances at an angle of 7.8° for VOR or 10.3° for NDB. Within this splayed area, the width of the primary area shall remain one half of the total width of the area. (See Figure I-4-3-3.) For calculating secondary area width at a given point, see Section 2, Chapter 1, 1.2.1, “Calculating secondary area width at a given point”.

3.6.6 Use of DME or intersecting radial/bearing

If a DME distance or an intersecting radial or bearing is used to limit the outbound leg, the area may be reduced by applying the appropriate adjustments described in Appendix C, in this case the limiting distance or radial/bearing shall allow adequate time for the descent specified. The distance on the outbound track is thereby limited by the timing or by reaching the limiting DME distance or radial/bearing, whichever occurs first.

3.6.7 Secondary areas

Secondary areas shall be added to the outer boundary of all areas calculated using the criteria in 3.6.4, “Area construction”. The width of the secondary area is 4.6 km (2.5 NM).

Note.— See Appendix B, “Reduction of the width of a straight initial approach area after the IAF and interface between straight initial approach area and reversal procedure areas” for possible reduction of secondary area width.

3.7 MAXIMUM DESCENT/NOMINAL OUTBOUND TIMING RELATIONSHIP FOR A REVERSAL OR RACETRACK PROCEDURE

3.7.1 General

Because the actual length of the track will vary, it is not possible to specify a descent gradient for the racetrack or reversal procedures. Instead, the maximum descents which can be specified on the outbound and inbound tracks of the procedure are listed in Table I-4-3-1 as a function of nominal outbound time.

Note.— Where a 45° procedure turn is specified, an additional 1 minute may be added to the nominal outbound time in calculating the maximum descent outbound.

Example: Reversal procedure with 2.5 minutes outbound track (Category A and B aircraft):

- a) maximum descent to be specified on outbound track = 612 m (2 010 ft); and
- b) maximum descent to be specified on inbound track = 500 m (1 638 ft).

3.7.2 Turns

In calculating maximum descents, no descent shall be considered as having taken place during turns.

3.8 OBSTACLE CLEARANCE

The prescribed MOCAs for either the racetrack or the reversal procedure shall not be less than 300 m (984 ft) above all obstacles within the appropriate primary areas. In the secondary area, the minimum obstacle clearance shall be 300 m (984 ft) at the inner edge, reducing linearly to zero at the outer edge. See Part I, Section 2, Chapter 1, 1.3, “Obstacle clearance”.

Table I-4-3-1. Maximum/minimum descent on a reversal or racetrack procedure

		<i>Maximum*</i>	<i>Minimum*</i>
Outbound track	Cat A/B	245 m/min (804 ft/min)	N/A
	Cat C/D/E/H	365 m/min (1 197 ft/min)	N/A
Inbound track	Cat A/B	200 m/min (656 ft/min)	120 m/min (394 ft/min)
	Cat H	230 m/min (755 ft/min)	N/A
	Cat C/D/E	305 m/min (1 000 ft/min)	180 m/min (591 ft/min)

* *Maximum/minimum descent for 1 minute nominal outbound time in m(ft). For maximum descent rates related to a final approach segment, see Chapter 5, 5.3.*

Chapter 4

INTERMEDIATE APPROACH SEGMENT

4.1 GENERAL

4.1.1 The intermediate approach segment blends the initial approach segment into the final approach segment. It is the segment in which aircraft configuration, speed, and positioning adjustments are made for entry into the final approach segment.

4.1.2 There are two types of intermediate approach segments:

- a) one which begins at a designated intermediate approach fix (IF); and
- b) one which begins upon completion of a dead reckoning (DR) track, a reversal or a racetrack procedure.

4.1.3 In both cases, track guidance shall be provided inbound to the final approach fix (FAF) where the intermediate approach segment ends. See Figure I-4-3-2 of Chapter 3 for typical intermediate approach segments.

4.2 ALTITUDE SELECTION

The procedure altitude in the intermediate approach segment shall be established in 100-ft increments or 50-m increments as appropriate.

4.3 INTERMEDIATE APPROACH SEGMENT BASED ON A STRAIGHT TRACK ALIGNMENT

The track to be flown in the intermediate approach segment should normally be the same as the final approach track. Where this is not practicable and the final approach fix in a non-precision procedure is a navigation facility, the intermediate track shall not differ from the final approach track by more than 30° (Cat H, 60°). Where the turn at the FAF is greater than 10° the final approach area should be widened on the outer side of the turn as described in Chapter 6, 6.4.6.3.3, “TP marked by a facility (NDB or VOR)”.

4.3.1 Area

This section deals with the construction of the area of an intermediate approach segment based on a straight track alignment.

4.3.1.1 Length

4.3.1.1.1 The length of the intermediate approach segment shall not be more than 28 km (15 NM) (Cat H, 9.3 km (5.0 NM)), or less than 9.3 km (5.0 NM) (Cat H, 3.7 km (2.0 NM)), (except as provided for in ILS, MLS, RNAV [DME/DME, VOR/DME, GNSS] and radar sections), measured along the track to be flown.

4.3.1.1.2 The optimum length is 19 km (10 NM) (Cat H, 9.3 km (5.0 NM)). A distance greater than 19 km (10 NM) should not be used unless an operational requirement justifies a greater distance. When the angle at which the initial approach track joins the intermediate approach track exceeds 90° (Cat H, 60°), the minimum length of the intermediate approach track is as shown in Table I-4-4-1.

4.3.1.2 Width

In a straight-in approach, the width of the intermediate approach segment tapers from a maximum width of 19 km (10 NM) at the IF to its minimum width at the FAF (or FAP). The segment is divided longitudinally as follows:

- a) a primary area which extends laterally on each side of the track; and
- b) a secondary area on each side of the primary area. (See Figure I-4-3-2 of Chapter 3.)

For calculating secondary area width at a given point, see Section 2, Chapter 1, 1.2.2, “Calculating secondary area width at a given point”.

Note.— See also Appendix B to Chapter 3, “Reduction of the width of a straight initial approach area after the IAF and interface between straight initial approach area and reversal procedure areas” for possible reduction of the width of the initial approach area.

4.3.2 Obstacle clearance

4.3.2.1 A minimum of 150 m (492 ft) of obstacle clearance shall be provided in the primary area of the intermediate approach segment. In the secondary area, 150 m (492 ft) of obstacle clearance shall be provided at the inner edge, reducing to zero at the outer edge. See Figure I-2-1-1 of Section 2, Chapter 1. For calculating obstacle clearance at a given point, see Section 2, Chapter 1, 1.3, “Obstacle clearance”.

4.3.2.2 The MOCAs shall be rounded upwards to the next 50 m or 100 ft as appropriate.

4.3.3 Procedure altitude/height and descent gradient

4.3.3.1 Because the intermediate approach segment is used to prepare the aircraft speed and configuration for entry into the final approach segment, this segment should be flat or at least have a flat section contained within the segment.

4.3.3.2 If a descent is necessary the maximum permissible gradient will be 5.2 per cent (Cat H, 10 per cent or, if the intermediate approach speed is restricted to 165 km/h IAS (90 kt IAS), 13.2 per cent). In this case, a horizontal segment with a minimum length of 2.8 km (1.5 NM) should be provided prior to the final approach for Cat C and D aircraft. For procedures specific to Cat A and B aircraft, this minimum length may be reduced to 1.9 km (1.0 NM). This should allow sufficient distance for aircraft to decelerate and carry out any configuration changes necessary before final approach segment.

Chapter 5

FINAL APPROACH SEGMENT

5.1 GENERAL

5.1.1 States are encouraged to implement approach procedures that have vertical guidance.

5.1.2 In the final approach segment, alignment and descent for landing are carried out. The instrument part of the final approach segment begins at the final approach fix, and ends at the missed approach point (MAPt). Track guidance shall be provided for the instrument phase of the final approach segment. Final approach may be made:

- a) to a runway for a straight-in landing; or
- b) to an aerodrome for a circling approach.

5.1.3 The final approach segment should be aligned with a runway whenever possible. All final approaches with a FAF have an optimum length of 9.3 km (5 NM). The minimum final approach segment length shall not be less than 5.6 km (3.0 NM). This value also applies to the minimum distance from the FAF to the threshold except for non-RNAV procedures constrained by existing installations. Exceptions apply for:

- a) RNAV procedures with large track changes at the FAF for Category D, D_L and E aircraft (see Table I-4-5-1); and
- b) Cat H aircraft (see Table I-4-5-1).

5.2 ALIGNMENT

5.2.1 The final approach and its track guidance should be aligned with a runway whenever possible. An offset final approach increases the complexity of pilot operation. Consequently it should only be designed when siting or obstacle problems permit no other option. An offset final approach track shall not be established as a noise abatement measure. When runway aligned track guidance is not possible it may be offset up to 5 degrees without OCA/H penalty (see 5.4.3.1, “Aligned straight-in approach”). Beyond these limits (or where other requirements cannot be met) a circling approach shall be used.

5.2.2 Straight-in approach

5.2.2.1 This paragraph contains the alignment criteria for non-precision approaches. The alignment criteria for approaches other than non-precision are found in the applicable sections.

5.2.2.2 *Maximum lateral displacement.* The final approach track shall pass a point 1 400 m before the runway threshold within a maximum of 150 m laterally.

5.2.2.3 *Minimum distance.* For offset angles (θ) greater than 5° the distance between the runway threshold and the point at which the final approach track intersects the runway centre line shall not be less than 1 400 m (see Figure I-4-5-1), and the OCA/H shall not be less than that stated in 5.4.3.2.

5.2.2.4 *Maximum offset angle.* For a straight-in approach, the angle formed by the final approach track and the runway centre line shall not exceed:

- a) 30° for procedures restricted to Cat A and B aircraft ; and
- b) 15° for other aircraft categories.

5.2.2.5 *Final approach track angle for helicopters.* For helicopters, the final approach track shall intersect the final approach and take-off area (FATO) axis at an angle not exceeding 30° and at a distance not less than 400 m from the FATO. When the final approach track does not intersect the extended axis of the FATO, the track shall lie within 75 m of it laterally at a point 400 m from the FATO.

5.2.3 Circling approach

The circling approach contains the visual phase of flight after completing an instrument approach, to bring an aircraft into position for landing on a runway that for operational reasons is not suitably located for straight-in approach. In addition, when the final approach track alignment or the descent gradient does not meet the criteria for a straight-in landing, only a circling approach shall be authorized and the track alignment should ideally be made to the centre of the landing area. When necessary, the final approach track may be aligned to pass over some portion of the usable landing surface. In exceptional cases, it may be aligned beyond the aerodrome boundary, but in no case beyond 1.9 km (1.0 NM) from the usable landing surface (see Figure I-4-5-2).

5.3 DESCENT GRADIENT

5.3.1 Gradient/angle limits

5.3.1.1 *Minimum/optimum descent gradient/angle.* The minimum/optimum descent gradient is 5.2 per cent for the final approach segment of a non-precision approach with FAF (3° for a precision approach or approach with vertical guidance). Descent gradients steeper than the optimum should not be used unless all other means to avoid obstacles have been attempted since these steeper descent gradients may result in rates of descent which exceed the recommended limits for some aircraft on final approach.

5.3.1.2 *Maximum descent gradient/angle.* This paragraph provides guidance regarding the maximum descent gradient/angle to be applied for approach procedures. When, because of obstacle clearance reasons, an approach procedure meeting the maximum descent gradient/angle requirement as specified in this paragraph cannot be implemented, then consideration should first be given to more advanced types of approaches that provide vertical guidance and may allow the descent/angle to stay within the limits. If this is not feasible for operational reasons and it is opted for an approach procedure that exceeds the maximum descent gradient/angle then the approach procedure shall be subject to an aeronautical study and requires special approval by the national competent authority. See Appendix B to this chapter for guidance on steep angle non-precision approaches. See Appendix B to Part II, Section 1, Chapter 1 for guidance on steep angle precision approaches.

5.4.1.5 *Aerodrome operating minima*

OCA/H is one of the factors taken into account in establishing operating minima for an aerodrome in accordance with Annex 6. See Figure I-4-5-3 a) to c).

5.4.2 OCA/H for precision approaches and approach procedures with vertical guidance

The determination of OCA/H in precision approaches and approach procedures with vertical guidance is described in Part II, Section 1 and Part III, Section 3, Chapters 4 to 6.

5.4.3 OCA/H for non-precision approach (straight-in)

5.4.3.1 *Aligned straight-in approach*

The OCA/H for a straight-in, non-precision approach where the angle between the track and the extended runway centre line does not exceed 5 degrees shall provide the following minimum obstacle clearance (MOC) over the obstacles in the final approach area:

- a) 75 m (246 ft) with FAF; and
- b) 90 m (295 ft) without FAF.

The OCA/H shall also ensure that missed approach obstacle clearance is provided. See Chapter 6, “Missed approach segment”. A straight-in OCA/H shall not be published where final approach alignment or descent gradient criteria are not met. In this case, only circling OCA/H shall be published.

5.4.3.2 *Non-aligned straight-in approach*

For a final approach with an offset angle great than 5°, OCA/H varies according to the interception angle. The OCH of the procedure shall be equal to or greater than the lower limits shown in Table I-4-5-3. The calculations used to arrive at these values appear in Appendix A to this chapter. For nominal descent gradients above 5.2 per cent, increase by 18 per cent the lower limits shown in the table for each per cent of gradient above 5.2 per cent.

5.4.4 OCA/H for visual manoeuvring (circling)

The OCA/H for visual manoeuvring (circling) shall provide the minimum obstacle clearance (MOC) over the highest obstacle in the visual manoeuvring (circling) area as specified in Table I-4-7-3 of Chapter 7. It shall also be:

- a) above the lower limits (also specified in Table I-4-7-3); and
- b) not less than the OCA/H calculated for the instrument approach procedure which leads to the circling manoeuvre. See Chapter 7, “Visual manoeuvring (circling) area”.

Circling procedures are not provided for helicopters. When a helicopter instrument approach is followed by visual manoeuvring, the OCH shall not be less than 75 m (246 ft). See Chapter 7.

5.4.5 MOC and OCA/H adjustments

5.4.5.1 In certain cases the MOC and/or the OCA/H must be increased. This may involve:

- a) an additional margin that is added to MOC;
- b) a percentage increase in OCA/H; and
- c) applying a lower limit (a minimum value) to OCA/H; as described below.

5.4.5.2 Additional margin applied to MOC

5.4.5.2.1 *Mountainous areas.* See 1.7, “Increased altitudes/heights for mountainous areas” in Section 2, Chapter 1 for guidance on increased MOC in mountainous areas.

5.4.5.2.2 These criteria are applicable to non-precision approach procedures only.

5.4.5.3 Percentage increase in OCA/H

5.4.5.3.1 *Remote altimeter setting.* When the altimeter setting is derived from a source other than the aerodrome, and more than 9 km (5 NM) remote from the threshold, the OCA/H shall be increased at a rate of 0.8 m for each kilometre in excess of 9 km (5 ft for each nautical mile in excess of 5 NM) or a higher value if determined by local authority. In mountainous areas or other areas where reasonably homogenous weather cannot always be expected, a procedure based on a remote altimeter setting source should *not* be provided. In all cases where the source of the altimeter setting is more than 9 km (5 NM) from the threshold, a cautionary note should be inserted on the instrument approach chart identifying the altimeter setting source.

5.4.5.3.2 Remote altimeter setting source (RASS) in mountainous areas

- a) The use of RASS in mountainous areas requires additional calculations to determine the correct OCA/H. The calculation uses the formula

$$\text{OCA/H} = 2.3x + 0.14z \text{ (non SI)}$$

$$\text{OCA/H} = 0.4x + 0.14z \text{ (SI)}$$

where: OCA/H is the RASS increased altitude/height value (m/ft);

x is the distance from the RASS to the landing area (km/NM); and

z is the difference in elevation between the RASS and the landing area (m/ft).

These formulas are used where no intervening terrain adversely influences atmospheric pressure patterns. The use of this criteria is limited to a maximum distance of 138 km (75 NM) laterally or an elevation differential of 1 770 m (6 000 ft) between the RASS and the landing area. An example calculation in nautical miles and feet is illustrated in Figure I-4-5-4.

- b) Where intervening terrain adversely influences atmospheric pressure patterns, the OCA/H shall be evaluated in an Elevation Differential Area (EDA). The EDA is defined as the area within 9 km (5 NM) each side of a line connecting the RASS and the landing area, including a circular area enclosed by a 9 km (5 NM) radius at each

end of the line. In this case, z becomes the terrain elevation difference (m/ft) between the highest and lowest terrain elevation points contained in the EDA. An example of a calculation in nautical miles and feet is illustrated in Figure I-4-5-5.

5.4.5.4 Lower limit (a minimum value) applied to OCA/H

- a) *Forecast altimeter setting.* When the altimeter setting to be used with procedures is a forecast value obtained from the appropriate meteorological office, the OCA/H shall be increased by a value corresponding to the forecasting tolerance for the location as agreed by the meteorological office for the time periods involved. Procedures which require the use of forecast altimeter setting shall be suitably annotated on the approach charts.
- b) *Final approach track intersecting the extended runway centre line between 5° and 30°.* When the final approach track intersects the extended runway centre line between 5° and 30° a lower limit is applied to OCA/H (5.4.3.2, “Non-aligned straight-in approach”).
- c) *Final approach track intersecting the extended runway centre line at more than 30° or descent gradient exceeding 6.5 per cent.* When the final approach track intersects the extended runway centre line at more than 30°, or the descent gradient exceeds 6.5 per cent, the OCA/H for visual manoeuvring (circling) becomes the lower limit and is applied to the approach procedure.
- d) *Visual manoeuvring (circling).* For visual manoeuvring (circling) a lower limit consisting of the OCA/H for the associated instrument approach procedure is applied (see 5.4.4, “OCA/H for visual manoeuvring (circling)”).

5.4.6 Protection for the visual segment of the approach procedure

5.4.6.1 All new straight-in instrument approach procedures published on or after 15 March 2007 shall be protected for obstacles in the visual segment. For this purpose no obstacles, except subject to 5.4.6.4, shall penetrate a Visual Segment Surface (VSS) laterally, defined as follows:

- a) for procedures with localizer or localizer look-alike lateral guidance (LOC only, APV I, and PA approaches) where the final approach track is aligned with the runway centre line, with a base width equal to the inner approach surface as defined in Annex 14, originating 60 m prior to the runway threshold, extending parallel to the extended runway centre line, and terminating at the point where the height of the surface reaches the OCH (see Figure I-4-5-6 a)); and
- b) for all other straight-in instrument approach procedures:
 - 1) a base width equal to the runway strip width originating 60 m prior to the runway threshold, splaying 15 per cent on either side of the extended runway centre line, and terminating at the point where the height of the surface reaches the OCH (see Figure I-4-5-6-b));
 - 2) where the final approach course is offset and intersects the extended runway centre line, the splay on the side closest to the final approach course is increased by the offset angle (see Figure I-4-5-7); and
 - 3) where the final approach course is offset but does not intersect the extended runway centre line, the splay closest to the final approach course is increased by an amount equal to the final approach course offset at 1 400 m from the runway threshold (see Figure I-4-5-8).

5.4.6.2 Vertically, the VSS originates at the runway threshold height and has a slope of 1.12 degrees less than the promulgated approach procedure angle.

5.4.6.3 Straight-in instrument approach procedures published before 15 March 2007 shall be protected in the visual segment by means of the VSS after the periodical review of the procedure, but not later than 15 March 2012.

5.4.6.4 If the VSS is penetrated, the approach procedure shall not be promulgated without an aeronautical study. Mitigation action as a result of such a study may result in the removal or lowering of the obstacle, an increase of the descent gradient/angle, and/or runway threshold displacement. Obstacles with a height less than 15 m above the threshold may be disregarded when assessing the VSS. Temporary moving obstacles such as aircraft holding at the runway hold-point are allowed.

5.4.6.5 Indication that a VSS has been penetrated shall be promulgated in the AIP, Section AD 2.25 Visual segment surface (VSS) penetration.

Note.— Guidance on promulgating VSS penetration is provided in the Aeronautical Information Services Manual (Doc 8126).

5.4.6.6 If no mitigation action as defined in 5.4.6.4 has been deemed operationally acceptable and obstacles remain penetrating the visual segment surface (VSS), none of these obstacles shall require the pilot to destabilize the approach to avoid them.

5.4.6.6.1 For this purpose, no obstacle shall penetrate an obstacle clearance surface (OCS) defined as follows (see Figure I-4-5-9):

Laterally:

- a) for procedures with localizer or localizer look-alike lateral guidance (LOC only, APV 1 and PA approaches), where the final approach track is aligned with the runway centre line:
 - 1) the OCS begins at the THR/LTP;
 - 2) the beginning width is 30 m each side of the runway edge;
 - 3) it extends from the THR up to the point 60 m before the THR at the VSS width and continues with the same width up to the point where the OCH is reached on the promulgated profile (“OCH point”).
- b) for all other straight-in instrument approach procedures:
 - 1) the OCS begins at the THR/LTP and extends up to the point where the OCH is reached on the promulgated profile (“OCH point”);
 - 2) the beginning width is 30 m each side of the runway edge;
 - 3) the semi-width at “OCH point” is equal to $E = 120 \text{ m} + D \cdot \tan(2^\circ)$ where D is the distance between the THR/LTP and the “OCH point”.

Vertically:

- a) The OCS originates at the runway threshold height where RDH is 15 m or smaller and at RDH –15 m above the runway threshold height where RDH is greater than 15 m.

b) the OCS has a slope of θ defined as follows:

- 1) for NPA: θ = promulgated approach procedure angle minus 1° ;
- 2) for APV Baro: θ = minimum cold temperature VPA minus 0.5° ; and
- 3) for APV with geometric vertical guidance: θ = promulgated VPA minus 0.5°

Note 1.— Penetrations caused by airport lighting, airport signage, and their associated equipment, may be disregarded when installed in accordance with ICAO Standards.

Note 2.— These values are not applicable for approaches above 3.5° .

5.4.6.6.2 Where the final approach course is offset and intersects the extended runway centre line, the OCS at the point where the OCH is reached extends perpendicularly to the final approach course on the side of the offset for distance E. On the side closest to the centre line, the area extends perpendicularly to the FAC until intersecting the runway centre line. It then extends perpendicularly to the runway centre line for distance E (see Figure I-4-5-10).

5.4.6.6.3 Where the final approach course is offset but does not intersect the extended runway centre line, the OCS at the point where OCH is reached extends perpendicularly to the final approach course on the side of the offset for distance E (see Figure I-4-5-11).

5.5 PROMULGATION

5.5.1 *Descent gradients/angles for charting.* Descent gradients/angles for charting shall be promulgated to the nearest one-tenth of a per cent/degree. Descent gradients/angles shall originate at a point 15 m (50 ft) above the landing runway threshold. For precision approaches different origination points may apply (see RDH in specific chapters). For non-precision approaches at short runways (Code 1 and 2) the height above threshold can be as low as 12 m (40 ft). Earth curvature is not considered in determining the descent gradient/angle.

5.5.2 *Descent angles for database coding.* Paragraph 5.5.1 applies with the exception that descent angles shall be published to the nearest one-hundredth of a degree.

5.5.3 Both the procedure altitude/height and the minimum altitude for obstacle clearance, i.e. MOCA, shall be published. In no case shall the procedure altitude/height be lower than the MOCA.

5.5.4 The designed stabilized descent path shall clear the stepdown fix MOCA. This can be achieved by increasing the descent gradient by:

- a) increasing the procedure altitude/height at the FAF; or (if a) is not possible)
- b) moving the FAF toward the landing threshold.

5.5.5 *Publication of OCA/H.* An OCA and/or an OCH shall be published for each instrument approach and circling procedure. For non-precision approach procedures, either value shall be expressed in 5-m or 10-ft increments by rounding up as appropriate.

Table I-4-5-1. Minimum length of final approach segment

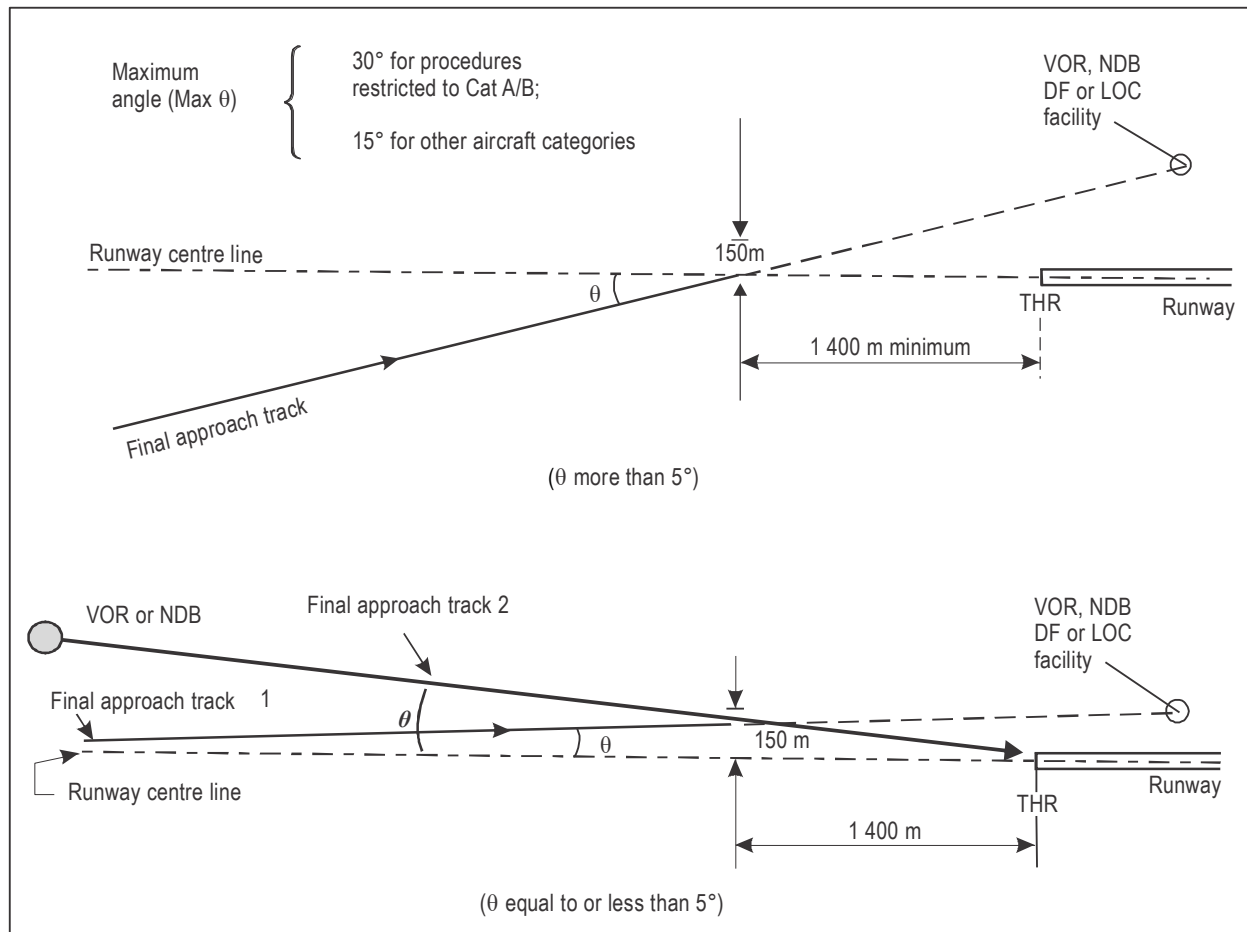
Aircraft category	Magnitude of turn over FAF			
	10° or less	20°	30°	60°
D and D _L	5.6 km (3.0 NM)	5.6 km (3.0 NM)	6.5 km (3.5 NM)	
E	5.6 km (3.0 NM)	6.5 km (3.5 NM)	7.4 km (4.0 NM)	
H	1.9 km (1.0 NM)	2.8 km (1.5 NM)	3.7 km (2.0 NM)	5.6 km (3.0 NM)
The values in this table may be interpolated. If turns of more than 30° (Cat H, 60°) are required, or if the minimum lengths specified in the table are not available for the procedure, straight-in minimums are not authorized and only the circling OCA/H should be published.				

Table I-4-5-2. Rate of descent in the final approach segment of a non-precision procedure with no FAF

Aircraft categories	Rate of descent	
	Maximum	Minimum
Cat A/B	200 m/min (655 ft/min)	120 m/min (394 ft/min)
Cat H	230 m/min (755 ft/min)	N/A
Cat C/D/E	305 m/min (1 000 ft/min)	180 m/min (590 ft/min)

Table I-4-5-3. Lower limit on OCH

Aircraft category	Lower limit on OCH (m (ft))	
	5° < θ ≤ 15°	15° < θ ≤ 30°
A	105 (340)	115 (380)
B	115 (380)	125 (410)
C	125 (410)	
D	130 (430)	
E	145 (480)	

**Figure I-4-5-1. Final straight-in approach alignment**

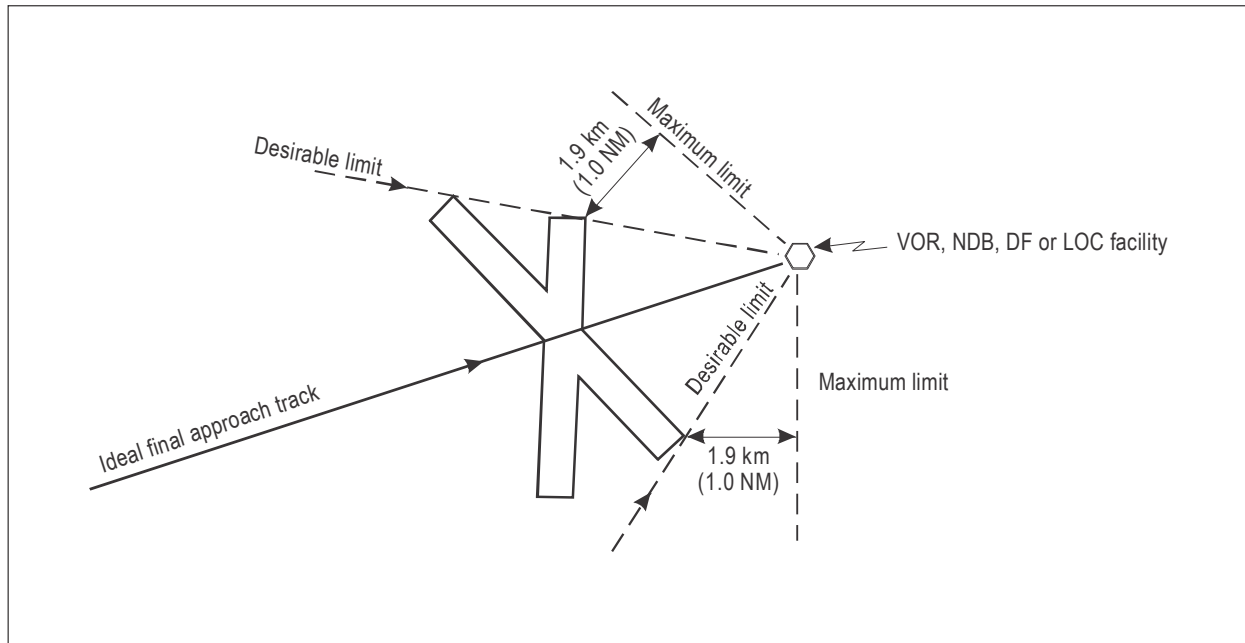


Figure I-4-5-2. Final circling approach alignment

Chapter 6

MISSED APPROACH SEGMENT

6.1 GENERAL

6.1.1 Requirements

6.1.1.1 A missed approach procedure shall be established for each instrument approach and shall specify a point where the procedure begins and a point where it ends. The missed approach procedure is initiated:

- a) at the decision altitude height (DA/H) in precision approach procedures or approach with vertical guidance (APV); or
- b) at the missed approach point (MAPt) in non-precision approach procedures.

6.1.1.2 The missed approach procedure shall terminate at a point or an altitude/height sufficient to permit:

- a) initiation of another approach; or
- b) return to a designated holding pattern; or
- c) resumption of en-route flight.

Note.— This chapter contains general criteria which apply to all types of instrument landing procedures, as well as criteria specific to non-precision procedures. For the details regarding precision approaches and approaches with vertical guidance, see the applicable chapters.

6.1.2 Phases of missed approach segment

In principle the missed approach segment starts at the MAPt and includes the following three phases (see Figure I-4-6-4):

- a) initial phase — begins at the earliest MAPt, and extends until the Start of Climb (SOC);
- b) intermediate phase — extends from the SOC to the point where 50 m (164 ft) (Cat H, 40 m (132 ft)) obstacle clearance is first obtained and can be maintained; and
- c) final phase — extends to the point at which a new approach, holding or return to en-route flight is initiated. Turns may be carried out during this phase.

6.1.3 Types of missed approach

There are two types of missed approach:

- a) straight missed approach (includes turns less than or equal to 15 degrees); and
- b) turning missed approach.

6.1.4 Missed approach area

The area considered for the missed approach shall start at the earliest MAPt tolerance, with a width equal to that of the final approach segment at that point. The subsequent size and shape of the area depends on the missed approach procedure, including the point at which a turn is initiated, if applicable, and the extent of the turn.

6.1.5 Missed approach point (MAPt)

6.1.5.1 *General.* A missed approach begins at the missed approach point (MAPt) and only applies to non-precision approaches. For non-precision approaches, the MAPt shall be defined as follows:

- a) *procedures without a FAF* — by a navigation facility or fix; and
- b) *procedures with a FAF* — the MAPt shall be defined by one of the following three cases:
 - 1) by timing over the distance from the nominal FAF to the nominal MAPt, where the MAPt is not defined by a facility or fix; or
 - 2) by a navigation facility or fix at the MAPt, in which case the procedure must be annotated “timing not authorized for defining the MAPt”; or
 - 3) by both timing over the distance from the nominal FAF to the nominal MAPt, as well as a facility or fix at the missed approach point. In this case a single OCA/H, which shall be the higher of the OCA/H for the specified distance and the OCA/H for the facility or fix, shall be published. However, when an operational advantage can be achieved, both may be published.

6.1.5.2 *Location of MAPt.* The optimum location of the MAPt is the runway threshold. Where necessary, the MAPt may be moved closer to the FAF provided that the OCA/H is not lower than the altitude/height at the MAPt on a nominal 5.2 per cent (3°) descent gradient or the promulgated descent gradient if steeper. An increase in OCA/H may be required to meet this condition.

6.1.5.3 *Determining earliest and latest MAPt for an MAPt determined by distance.* When the MAPt is determined by timing over the distance from the FAF, the total MAPt tolerance (Y) may be determined by taking the values from Table I-4-6-1 and applying them as shown in Figure I-4-6-3. For the refined calculations see the appendix to this chapter.

6.1.6 Calculating start of climb (SOC)

6.1.6.1 There are two methods for calculating SOC. The method used depends on whether:

- a) the MAPt is defined by a navigation facility or fix; or
- b) the MAPt is defined by a specified distance from the FAF.

Chapter 9

CHARTING/AIP

9.1 GENERAL

Material relating to the publication of charts is contained in Annex 4 as follows:

- a) Standard Departure Chart — Instrument (SID) — ICAO, in Annex 4, Chapter 9;
- b) Standard Arrival Chart — Instrument (STAR) — ICAO, in Annex 4, Chapter 10; and
- c) Instrument Approach Chart — ICAO, in Annex 4, Chapter 11.

9.2 CHARTED ALTITUDES/FLIGHT LEVELS

Arrival procedures may be developed to procedurally separate air traffic. In doing so, the procedure may be accompanied with procedure altitudes/flight levels that are developed to separate arriving and departing air traffic. These procedure altitudes/flight levels shall be charted as indicated in Table I-4-9-1.

9.3 ARRIVAL

In some cases it is necessary to designate arrival routes from the en-route structure to the initial approach fix. Only those routes which provide an operational advantage shall be established and published. These should take local air traffic flow into consideration.

9.4 APPROACH

9.4.1 General

9.4.1.1 Optimum and maximum descent gradients and angles are specified depending on the type of procedure and the segment of the approach. The descent gradient(s)/angles used in the construction of the procedure shall be published for the final approach segment. It is preferable that they also be published for the other approach segments, where appropriate.

9.4.1.2 *Fix identification.* The IAF, IF, FAF, MAPt, TP and other essential fixes or points where established are published on the approach chart.

9.4.1.3 *Aeronautical database requirements.* For non-RNAV approaches the following data shall be published in tabular form on the verso of the ICAO Instrument Approach Chart or as a separate, properly referenced sheet (see Annex 4, Chapter 11):

- a) final approach fixes/points and other essential fixes/points comprising the instrument approach procedure identified with their geographical coordinates in degrees, minutes, seconds and tenths of seconds;
- b) instrument approach procedure fix formation bearings to the nearest hundredth of a degree;
- c) instrument approach procedure fix formation distance to the nearest hundredth of a nautical mile; and
- d) for non-precision approaches, the final approach descent angle to the nearest hundredth of a degree.

9.4.2 Initial approach segment

9.4.2.1 Separate procedures shall be published when:

- a) different MOCAs;
- b) different timings; or
- c) different outbound tracks

are specified for different categories of aircraft.

9.4.2.2 Speeds below the minimum value for initial approach in a given aircraft category shall not be specified (see Tables I-4-1-1 and I-4-1-2 of Chapter 1). If procedures are developed which exclude specific aircraft categories due to speed, this must be stated explicitly.

9.4.3 Final approach segment

9.4.3.1 An OCA and/or an OCH shall be published for each instrument approach and circling procedure. For non-precision approach procedures, values shall be expressed in 5-m or 10-ft increments by rounding up as appropriate.

9.4.3.2 A straight-in OCA/H shall not be published where final approach alignment or descent gradient criteria are not met. In this case, only circling OCA/H shall be published.

9.4.3.3 Procedures which require the use of forecast altimeter setting shall be suitably annotated on the approach charts.

9.4.3.4 *Steep angle approach.* For procedures with a final descent gradient/angle greater than the maximum value specified in Chapter 5, “Final Approach Segment”, a warning note shall be published on the instrument approach chart (See Annex 4, 11.10.8.8).

9.4.3.5 *Vertical guidance.* Where distance information is available, to facilitate a continuous descent final approach (CDFA), descent profile advisory information for the final approach should be provided to assist the pilot in maintaining the calculated descent gradient. This information should consist of a table showing altitudes/heights through which the aircraft should be passing at each 2 km or 1 NM as appropriate.

9.5.2.5 *Circling approach.* When on a chart only circling minima are provided, the approach procedure shall be identified by the last navaid providing final approach guidance followed by a single letter, starting with the letter a. When there are two or more approaches at an airport (or a nearby airport), a different letter shall be used. If the IFR portion of the procedure is the same but there are different circling tracks for the same procedure, only one procedure with one title should be promulgated and the different circling procedures indicated in the procedure. The suffix letter shall not be used again for any procedures at that airport, at any other airport serving the same city, or at any other airport in the same State, serving a city with the same name. For example:

VOR-A

VOR-B

NDB-C

9.5.3 Duplicate procedure identification

9.5.3.1 A single letter suffix, starting with the letter z following the radio navigation aid type shall be used if two or more procedures to the same runway cannot be distinguished by the radio navigation aid type only. For example:

VOR Y RWY 20

VOR Z RWY 20

9.5.3.2 The single letter suffix shall be used when:

- a) two or more navigation aids of the same type are used to support different approaches to the same runway;
- b) two or more missed approaches are associated with a common approach, each approach being identified by a single letter suffix;
- c) different approach procedures using the same radio navigation type are provided for different aircraft categories;
- d) two or more arrivals are used to a common approach and are published on different charts, each approach being identified by a single letter suffix. If additional radio navigation aids are required for the arrival they shall be specified on the chart's plan view. For example:

ILS y RWY 20 (“CAB VOR Arrival” shown in the plan view)

ILS z RWY 20 (“DNA VOR Arrival” shown in the plan view)

9.5.3.3 Some avionics systems are not capable of recognizing the duplicate procedure identification as defined in paragraphs 9.5.3.1 and 9.5.3.2. The systems that are not capable will include only one of the duplicate procedures and will not include the duplicate procedure identification letter. States may therefore employ some method of indicating which procedure should be included in those systems, e.g. a note in the AIP, listing of procedures, the consistent use of a defined character for the identification.

9.5.4 Minima boxes

9.5.4.1 The OCA/H for each aircraft category shall be published in the minima box on the chart. Where an OCA/H is predicated on a specific navigation aid (e.g. stepdown fixes), or a specific navigation specification (see Part III, Section 5, Chapter 1, 1.4), this shall be clearly identified. For example:

OCA/(OCH)	CAT A	CAT B	CAT C	CAT D	CAT H
VOR/DME	610 (300)	610 (300)	610 (300)	610 (300)	610 (300)
VOR	660 (350)	660 (350)	660 (350)	660 (350)	660 (350)

or

OCA/(OCH)	CAT A	CAT B	CAT C	CAT D	CAT H
CAT I	210 (170)	210 (170)	220 (180)	230 (190)	210 (170)
LOC/DME	290 (250)	290 (250)	290 (250)	290 (250)	290 (250)

Table I-4-9-1. Charted procedure altitudes/flight levels

Altitude/flight level “Window”	<u>17 000</u> <u>10 000</u>	<u>FL220</u> <u>10 000</u>
“At or above” altitude/flight level	<u>7 000</u>	<u>FL060</u>
“At or below” altitude/flight level	<u>5 000</u>	<u>FL050</u>
“At” altitude/flight level	<u>3 000</u>	<u>FL030</u>
“Recommended” altitude/flight level	5 000	FL050
“Expected” altitude/flight level	Expect 5 000	Expect FL050

Chapter 10

SIMULTANEOUS APPROACHES TO PARALLEL OR NEAR-PARALLEL INSTRUMENT RUNWAYS

Note.— Guidance material is contained in the Manual on Simultaneous Operations on Parallel or Near-Parallel Instrument Runways (SOIR) (Doc 9643).

10.1 GENERAL

10.1.1 Simultaneous parallel approaches may be conducted to parallel runways using any combination of a three-dimensional (3D) instrument approach procedure.

Note.— Detailed procedures regarding the requirements for aircraft guidance during simultaneous parallel approach procedures are set forth in the Procedures for Air Navigation Services — Air Traffic Management (PANS-ATM, Doc 4444).

10.1.2 When it is intended to use approach procedures to parallel runways simultaneously, the following additional criteria shall be applied in the design of both procedures:

- a) when the final approach course or track is intercepted by a published arrival and approach procedure that intercepts the initial approach fix (IAF) or intermediate fix (IF), the minimum altitudes of the intermediate approach segments of the two procedures shall differ by at least 300 m (1 000 ft) unless an RNP AR approach is used in accordance with 10.3 or vectoring is exclusively used to intercept the final approach tracks; and
- b) the nominal tracks of the two missed approach procedures shall diverge by at least 30°. Associated missed approach turns shall be specified as “as soon as practicable”.

10.2 OBSTACLE CLEARANCE

10.2.1 The obstacle clearance criteria for approaches, as specified in the designated chapters, apply for each of the parallel approach procedures.

10.2.2 In order to safeguard early turns that may be required to avoid intruding aircraft from the adjacent runway, independent parallel approach operations shall include an obstacle evaluation of the area adjacent to the final approach segment on the side opposite the other parallel runway. This can be made using a set of separately defined parallel approach obstacle assessment surfaces (PAOAS) included in the appendix to this chapter. Although the method described in the appendix is based on the use of instrument landing system (ILS) approaches, the assessment technique may also be used for evaluation of independent parallel approach operations using other approved approach types.

10.3 ESTABLISHED ON RNP AR APCH

When the ‘established on RNP AR APCH’ procedure is used, the design of independent parallel approach procedures shall include an obstacle evaluation of the area adjacent to the portions of the final approach segment that are aligned with the extended runway centreline (see 10.2.2).

Note.— Detailed provisions regarding the use of the ‘established on RNP AR APCH’ procedure during simultaneous parallel approach operations are set forth in the Procedures for Air Navigation Services — Air Traffic Management (PANS-ATM, Doc 4444).

10.4 PROMULGATION

A note shall be included on the chart indicating the approach procedures that are authorized for simultaneous independent or dependent operations. The note shall include the runway(s) involved and if they are closely spaced.

Note.— Guidance material is provided in the Aeronautical Chart Manual (Doc 8697).

Appendix to Chapter 10

INDEPENDENT PARALLEL APPROACHES TO CLOSELY SPACED PARALLEL RUNWAYS

1. INTRODUCTION

1.1 Independent parallel approaches to closely spaced parallel runways are allowed when the distance between runways is not less than 1 035 m. To guarantee the safety of such operations, an obstacle assessment has to be carried out to protect a lateral break-out manoeuvre, which may need to be executed to avoid collision with a potential blundering aircraft. This will provide obstacle clearance from obstacles in the areas adjacent to the final approach segments.

1.2 The following method provides an example for the assessment of these obstacles and was derived from an existing procedure used by one State. This section includes the considerations made in developing the basis for the assessment.

1.3 It was considered that a difference exists between the current precision approach procedures described in Part II, Section 1, Chapter 1, “Instrument landing system (ILS)” and Part II, Section 1, Chapter 3, “MLS”, and the break-out procedures. For the approach procedures, an extensive data collection could be performed from which statistical probabilities of aircraft distributions could be obtained. In establishing a target level of safety (TLS) of 10^{-7} , obstacle assessment surfaces (OAS) and the collision risk model (CRM) are derived. For the assessment surfaces of the break-out manoeuvre, this type of method was not considered feasible due to the low probability of occurrence of a break-out manoeuvre. From one State’s report, it was learned that the occurrence of a break-out during simultaneous approaches was initially assumed to be in the order of 10^{-4} and 10^{-5} per approach and may even be lower.

1.4 In order to find obstacle clearance criteria for break-out manoeuvres, other methods were considered. One way was to use the existing missed approach criteria. However, these criteria are based on an occurrence of 10^{-2} which may be unduly restrictive at some aerodromes, and missed approaches are not primarily designed for break-out manoeuvres.

2. PARALLEL APPROACH OBSTACLE ASSESSMENT SURFACES (PAOAS)

2.1 The proposed method for the obstacle assessment for simultaneous parallel approaches was based on existing criteria provided by one State (FAA Order 8260.41). An evaluation was carried out by the Obstacle Clearance Panel (OCP). This evaluation was made by means of certification and operational criteria contained in the FAR/JAR 23/25 minimum climb requirements with all engines operating, together with the operational assumptions made by the ICAO Study Group on Simultaneous Operations on Parallel or Near-parallel Instrument Runways (SOIR), which established the minimum runway separation for use with simultaneous independent precision approaches (Cir 207). The evaluation considered that the initial part of the break-out manoeuvre would be executed in the landing configuration, followed by a climb gradient of 8.3 per cent within a height of 120 m (400 ft) above the break-out altitude/height. This evaluation indicated, in general, that the following restrictions to the break-out manoeuvres were necessary:

- a) no break-out manoeuvres below 120 m (400 ft); and
- b) maximum 45° break-out angle.

2.2 Due to the nature of the surfaces, these two parameters are interdependent. During the evaluation, it was found that the lower the break-out was considered, the earlier the assessment surfaces would be penetrated, causing the break-out angle to be reduced, e.g. a minimum break-out height of 60 m (200 ft) would result in a break-out angle of 20°, and a minimum break-out height of 300 m (1 000 ft) would result in a break-out angle of 65°.

2.3 It was considered necessary to restrict the minimum break-out altitude/height. One reason was that break-out manoeuvres at too low heights could be considered unsafe. Moreover, considering the maximum assumed blunder angle of 30° and approach speed of 150 kt, it could be assumed that below a certain height the blundering aircraft could not reach the threatened aircraft before it landed. and therefore it would be of no use to protect for these low heights.

2.4 Information available in respect of flight and simulator tests conducted by one State for these manoeuvres showed that phraseology used by the air traffic services (ATS) was similar to that contained in the PANS-ATM, Chapter 12, on independent parallel approaches. Following the instructions from air traffic services, the pilot actually first arrested the descent and then established climb, crossing the glide path (if below) before turning. This information supported the assumptions used to validate the proposed obstacle assessment criteria.

2.5 The evaluation report further indicated that it was not considered convenient to provide additional obstacle assessment surface (OAS) constant tables in PANS-OPS for these cases for each localizer-threshold distance combination. The proposed surfaces are based on operational rather than statistical considerations. Therefore, it was proposed to use one set of surfaces for all combinations of localizer-threshold distances. These surfaces would guarantee protection for aircraft following the assumed operational scenario.

2.6 A mathematical match was made from the surfaces for an average runway length/localizer distance contained in the Federal Aviation Administration (FAA) Order (8260.41). This approach was considered acceptable for the assessment of rare events for which statistical analysis was not feasible.

3. APPLICATION OF PARALLEL APPROACH OBSTACLE ASSESSMENT SURFACE (PAOAS) CRITERIA

3.1 General

In addition to the application of OAS criteria specified in Part II, Section 1, Chapter 1, 1.4.8, “Obstacle clearance of the precision segment using (OAS) criteria,” parallel approach obstacle assessment surfaces (PAOAS) are defined to safeguard the execution of an immediate climb and turn manoeuvre to the assigned heading and altitude/height. PAOAS criteria are used to demonstrate obstacle clearance, accommodating turns up to 45° from the approach path and a lowest break-out manoeuvre initiation of 120 m (400 ft) above threshold elevation.

3.2 Definition of surfaces

3.2.1 The PAOAS consists mainly of two sloping plane surfaces (denoted P1 and P2) positioned on the side of the runway opposite to the adjacent runway. The geometry of the sloping surfaces is defined, similar to the OAS surfaces (see Part II, Section 1, Chapter 1, 1.4.8.4, “Definition of obstacle assessment surfaces (OAS)”) by a linear equation of the form $z = Ax + By + C$. The constants are related to the minimum effective vertical path angle only. They are independent of the category of ILS/MLS/GLS operations and localizer-threshold/GARP-LTP distance. The constants are given in Table I-4-10-App-1 for ILS, MLS, GLS, and APV I and CAT I procedures based on SBAS, and in Table I-4-10-App-2 for APV procedures based on Baro-VNAV.

1.1.5 Methods of calculating OCA/H

1.1.5.1 *General.* Three methods of calculating OCA/H are presented, which involve progressive increases in the degree of sophistication in the treatment of obstacles. Standard conditions (as specified in 1.1.3) are assumed to exist unless adjustments for non-standard conditions have been made.

1.1.5.2 *First method.* The first method involves a set of surfaces derived from the Annex 14 precision approach obstacle limitation surfaces and a missed approach surface described in 1.4.7.2, “Definition of basic ILS surfaces”. From this point forward, these are termed “basic ILS surfaces”. Where the standard conditions exist as specified in 1.1.3 and where the basic ILS surfaces are free of penetrations (see 1.4.7.1), the OCA/H for Cat I and Cat II is defined by aircraft category margins, and there are no restrictions on Cat III operations. If the basic ILS surfaces are penetrated, then the OCA/H is calculated as described in 1.4.7.3, “Determination of OCA/H with basic ILS surfaces”.

1.1.5.3 *Second method.* The second method involves a set of obstacle assessment surfaces (OAS) above the basic ILS surfaces (see 1.4.8.4, “Definition of obstacle assessment surfaces (OAS). If the OAS are not penetrated — and provided the obstacle density below the OAS is operationally acceptable (see 1.4.8.9, “Effect of obstacle density on OCA/H”) — the OCA/H for Cat I and Cat II is still defined by the aircraft category margins, and Cat III operations remain unrestricted. However, if the OAS are penetrated, then an aircraft category-related margin is added to the height of the highest approach obstacle, or to the adjusted height of the largest missed approach penetration, whichever is greater. This value becomes the OCA/H.

Note.— The OAS surfaces are an approximation of the mathematically derived iso-probability contours. The surfaces are simplified compared to the complex shape of the iso-probability contours. The result is that the OAS method is normally more conservative compared to the CRM method (resulting in a higher OCH). The use of the CRM in addition to the OAS may therefore provide an operational benefit. Refer to the Attachment to Part II for specific areas where these differences may affect the OCH calculation.

1.1.5.4 *Third method.* The third method, using a collision risk model (CRM), is employed either as an alternative to the use of the OAS criteria (second method) or when the obstacle density below the OAS is considered to be excessive. The CRM accepts all objects as an input and assesses, for any specific OCA/H value, both the risk due to individual obstacles and the accumulated risk due to all the obstacles. It is intended to assist operational judgement in the choice of an OCA/H value.

Note.— The CRM does not take into account the characteristics of helicopters. The CRM can be used but the method should be conservative.

1.1.6 References

The following appendices relate to and amplify the material contained in this chapter:

- a) background information relating to the derivation of the OAS material (Attachment to Part II, paragraph 1) and to airborne and ground equipment performance assumed in the derivation of the OAS (Attachment to Part II, paragraph 2);
- b) turning missed approach after precision approach (Appendix A);
- c) determining ILS glide path descents/MLS elevation heights and distances (Appendix C);
- d) PANS-OPS OAS software (ICAO public website (www.icao.int)); and
- e) minimum distance between localizer and glide path interceptions after a constant radius-to-fix (RF) turn to the localizer course (Appendix D).

Examples of OCA/H calculations can be found in the *Instrument Flight Procedures Construction Manual* (Doc 9368).

1.1.7 ILS with glide path inoperative

The ILS with glide path inoperative is a non-precision approach procedure. The principles of Section 2, Chapter 1, “LOC only”, apply.

1.2 INITIAL APPROACH SEGMENT

1.2.1 General

The initial approach segment must ensure that the aircraft is positioned within the operational service volume of the localizer on a heading that will facilitate localizer interception. For this reason, the general criteria which apply to the initial segment (see Part I, Section 4, Chapter 3) are modified in accordance with 1.2.2, “Initial approach segment alignment” and 1.2.3, “Initial approach segment area”. The initial approach segment may be defined by an RNAV or RNP route, using an RNAV or RNP system for track guidance. Only the systems capable of navigation accuracy of 1 NM or lower in this phase of flight can be considered. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for initial approach. The RNAV or RNP route shall terminate at an IF defined by RNAV or RNP located on the LOC course. RNAV/RNP turn construction is applicable for turns within the initial segment and for the turn at the IF on the LOC course (see Figures II-1-1-5 and II-1-1-6). For an RNAV or RNP initial approach segment, criteria in Part III apply. If a course reversal is required with an RNAV or RNP initial approach segment, only a racetrack can be used. The fix and the inbound leg shall be located on the LOC course and the inbound segment defined by the LOC.

1.2.2 Initial approach segment alignment

The angle of interception between the initial approach track and the intermediate track should not exceed 90 degrees. In order to permit the autopilot to couple on to the localizer, an interception angle not exceeding 30 degrees is desirable. When the angle exceeds 70 degrees a radial, bearing, radar vector, or DME information providing at least 4 km (2 NM) (CAT H, 1.9 km (1 NM)) of lead shall be identified to assist the turn onto the intermediate track. When the angle exceeds 90 degrees, the use of a reversal, racetrack, or dead reckoning (DR) track procedure should be considered (see Part I, Section 4, Chapter 3, “Initial Approach Segment” and Part I, Section 4, Appendix A to Chapter 3, “Initial approach using dead reckoning (DR)”).

1.2.3 Initial approach segment area

The area is as described in the general criteria (see Part I, Section 4, Chapter 3, 3.3.3, “Area”). The difference is that the intermediate approach fix (IF) must be located within the service volume of the ILS localizer course signal, (except when an RF turn to the localizer course is used in the intermediate approach segment according to criteria in 1.3.6 below), and normally at a distance not exceeding 46 km (25 NM) from the localizer antenna. When radar is used to provide track guidance to the IF, the area shall be in accordance with 6.2, “Initial approach segment” (Section 2, Chapter 6, “SRE”).

1.3 INTERMEDIATE APPROACH SEGMENT

1.3.1 General

1.3.1.1 The intermediate approach segment for ILS differs from the general criteria in that:

- a) the alignment coincides with the localizer course;
- b) the length may be reduced;
- c) in certain cases the secondary areas may be eliminated; and
- d) if an RF turn to the localizer course is used, specific criteria in 1.3.6 apply.

1.3.1.2 The primary and secondary areas at the final approach point (FAP) are defined in terms of the ILS surfaces. Consequently, the general criteria in Part I, Section 4, Chapter 4, “Intermediate Approach Segment” are applied except as modified or amplified in the paragraphs below with regards to alignment, area length and width, and for obstacle clearance. For RNAV initial approach segments, the criteria in the applicable RNAV chapters apply.

1.3.2 Intermediate approach segment alignment

The intermediate approach segment of an ILS procedure shall be aligned with the localizer course, except when an RF turn to the localizer course is used (see 1.3.6 below).

1.3.3 Intermediate approach segment length

1.3.3.1 The optimum length of the intermediate approach segment is 9 km (5 NM) (Cat H, 3.7 km (2.0 NM)). This segment shall allow interception with the localizer course and with the glide path.

1.3.3.2 Segment length should be sufficient to permit the aircraft to stabilize and establish on the localizer course prior to intercepting the glide path, taking into consideration the angle of interception with the localizer course.

1.3.3.3 Minimum values for distance between localizer and interception of glide path are specified in Table II-1-1-1; however, these minimum values should only be used if usable airspace is restricted. The maximum length of the segment is governed by the requirement that it be located wholly within the service volume of the localizer signal and normally at a distance not exceeding 46 km (25 NM) from the localizer antenna. For RF turns to the localizer course, see 1.3.6 below.

1.3.4 Intermediate approach segment area width

1.3.4.1 The total width at the beginning of the intermediate approach segment is defined by the final total width of the initial approach segment. It tapers uniformly to match the horizontal distance between the OAS X surfaces at the FAP (see 1.4.8.4, “Definition of obstacle assessment surfaces (OAS)”).

1.3.4.2 For obstacle clearance purposes the intermediate approach segment is usually divided into a primary area bounded on each side by a secondary area. However, when a DR track is used in the initial approach segment, the primary area of the intermediate approach segment extends across its full width and secondary areas are not applied.

1.3.4.3 The primary area is determined by joining the primary initial approach area with the final approach surfaces (at the FAP). At the interface with the initial approach segment the width of each secondary area equals half the width of the primary area. The secondary area width decreases to zero at the interface with the final approach surfaces. See Figures II-1-1-1, II-1-1-2 and II-1-1-3. For RF turns to the localizer course, see 1.3.6 below.

1.3.4.4 Where a racetrack or reversal manoeuvre is specified prior to intercepting the localizer course the provisions in Part I, Section 4, Chapter 4, 4.4.4, “Turn not at the facility” apply, the facility being the localizer itself and the FAF being replaced by the FAP. (See Figure II-1-1-4.)

1.3.5 Intermediate approach segment obstacle clearance

The obstacle clearance is the same as defined in Part I, Section 4, Chapter 4, “Intermediate approach segment” except where the procedure permits a straight-in approach in which the aircraft is stabilized on the localizer course prior to crossing the IF. In this case, obstacles in the secondary areas need not be considered for the purpose of obstacle clearance.

1.3.6 RF turn to the localizer course

1.3.6.1 The intermediate approach segment may incorporate a performance-based navigation (PBN) route ending with a radius-to-fix turn to the localizer course. In this case, the RF turn shall terminate at a waypoint located on the localizer course. The position of this waypoint is considered as the point of localizer interception. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for intermediate approach segment.

1.3.6.2 Minimum distance between localizer and glide path interceptions in Table II-1-1-1 apply up to temperatures of ISA +30 degrees and up to glide path interceptions at 3 000 feet above aerodrome elevation. For cases exceeding either of these values, criteria in Appendix D shall be applied. When these temperature and intercept height assumptions are too limiting, Appendix D may be applied. The maximum total length of the intermediate approach segment shall not exceed 18.5 km (10 NM). Additionally, in case of CAT II and III operations, the minimum distance between the end fix of the RF turn and the landing threshold shall be 9.3 km (5 NM).

1.3.6.3 Descent in the intermediate approach segment shall comply with the general criteria in Part I, Section 4, Chapter 4, 4.3.3 “Procedure altitude/height and descent gradient”. If a descent is necessary after the RF turn to the LOC course, then criteria in Appendix D shall be applied, and obstacle clearance shall be provided in the total intermediate approach area according to Part I, Section 4, Chapter 4, 4.3.2 “Obstacle clearance”. The altitude constraint at the end fix of the RF turn shall be an altitude ‘window’. The top of the window shall comply with the calculations in Appendix D or shall be equal to the intercept altitude if the minimum length in 1.3.6.2 is applied. The bottom of the window shall comply with the minimum altitude requirements of the RF turn.

Note.— If the calculations result in the same altitude for the top and the bottom of the window, the constraint at the end fix of the RF turn becomes an ‘at’ altitude.

1.3.6.4 Criteria for the RF turn construction defined in Part III, Section 2, Chapter 2, 2.4 “RF turn method” apply with the following additions (see Figure II-1-1-8):

- a) the boundaries of the RF turn shall be continued one along-track tolerance (ATT) after the end fix of the RF turn assuming the width and fix tolerance values of the navigation specification associated with the RF turn;
- b) the boundary of the succeeding segment is the D”-E” line and the extension of the D-D” line considered as primary only area;
- c) if the bounding arcs (either primary or secondary) on the inner side of the turn intersect the extension of the D-D” line, then the extension becomes the edge of the area after the intersection, otherwise the RF boundaries shall be joined to the succeeding segment from one ATT after the end fix of the turn by a line at 30° to the LOC course; and
- d) if the bounding arcs (either primary or secondary) on the outer side of the turn do not intersect the extension of the D-D” line, then the arcs shall be joined to the extension by a tangent splaying 15° from the LOC course, otherwise the RF boundaries shall be joined to the succeeding segment from one ATT after the end fix of the turn by a line at 30° to the LOC course.

1.4 PRECISION SEGMENT

1.4.1 General

The precision segment is aligned with the localizer course and contains the final descent for landing as well as the initial and intermediate phases of the missed approach segment See Figure II-1-1-8.

1.4.2 Origin

The precision segment starts at the final approach point (FAP), that is, the intersection of the nominal glide path and the procedure altitude specified for the preceding segment. The FAP should not normally be located more than 18.5 km (10.0 NM) before threshold, unless adequate glide path guidance beyond the minimum specified in Annex 10 is provided.

1.4.3 Descent fix

1.4.3.1 A descent fix may be located at the FAP to overcome certain obstacles located before the FAP as an alternative to increasing the glide path (GP) angle. When so located, it becomes the final approach fix. The extension of the precision surfaces into the intermediate segment is then terminated. The descent fix should not normally be located more than 18.5 km (10.0 NM) before threshold, unless adequate GP guidance beyond the minimum specified in Annex 10 is provided. The maximum fix tolerance is ± 0.9 km (± 0.5 NM). Where DME is used to identify the fix, the range shall be stated in tenths of kilometres (nautical miles).

Note.— Guidance material for determining the distance to the descent fix from the threshold is contained in Appendix C.

1.4.3.2 *Obstacle clearance at the descent fix.* When a descent fix is provided, the precision approach surfaces start at the earliest point of the FAF tolerance area (see Figure II-1-1-2). The provisions of Part I, Section 2, Chapter 2, 2.7.4, “Obstacle close to a final approach fix or stepdown fix” which allow obstacles close to the fix to be ignored, apply in the area below the 15 per cent gradient within the precision surfaces (Cat H, 15 per cent gradient or the nominal gradient multiplied by 2.5, whichever is greater). Where a descent fix is not provided at the FAP, no curtailment of the precision surfaces is permitted. If the precision surfaces are extended into the preceding segment, they shall not be extended beyond the early fix tolerance of the IF (see Figure II-1-1-3) or the edge of the intermediate primary area (see Figure II-1-1-7).

1.4.4 Glide path verification check

A fix (outer marker or DME) is necessary so as to permit comparison between the indicated glide path and the aircraft altimeter information. The fix shall not have a fix tolerance exceeding ± 0.9 km (± 0.5 NM). When DME is used to identify the fix, the range shall be stated in tenths of kilometres (nautical miles).

Note.— Guidance material for determining the height crossing the outer marker is contained in Appendix C.

1.4.5 Missed approach

The missed approach point is defined by the intersection of the nominal glide path and the decision altitude/height (DA/H). The DA/H is set at or above the OCA/H, which is determined as specified in 1.4.7 to 1.4.9 and 1.5.

1.4.6 Termination

The precision segment normally terminates at the point where the final phase of the missed approach commences (see Part I, Section 4, Chapter 6, 6.1.2, “Phases of missed approach segment”) or where the missed approach climb surface Z (starting 900 m past threshold) reaches a height of 300 m (984 ft) above threshold, whichever is lower.

1.4.7 Obstacle clearance of the precision segment application of basic ILS surfaces

1.4.7.1 *General.* The area required for the precision segment is bounded overall by the basic ILS surfaces defined in 1.4.7.2, below. In standard conditions there is no restriction on objects beneath these surfaces (see 1.1.3, “Standard conditions”). Objects or portions of objects that extend above these surfaces must be either:

- a) minimum mass and frangible; or
- b) taken into account in the calculation of the OCA/H.

1.4.7.2 *Definition of basic ILS surfaces.* The surfaces to be considered correspond to a subset of Annex 14 obstacle limitation surfaces as specified for precision approach runway code numbers 3 or 4 (see Figure II-1-1-9 b)). These are:

- a) the approach surface continuing to the final approach point (FAP) (first section 2 per cent gradient, second section 2.5 per cent as described in Annex 14);
- b) the runway strip assumed to be horizontal at the elevation of the threshold and has a fixed width of 300 m;
- c) the missed approach surface. This is a sloping surface which:
 - 1) starts at a point 900 m past the threshold (Cat H, a starting point of 700 m past the threshold can be considered if necessary) at threshold elevation;
 - 2) rises at a 2.5 per cent gradient; and
 - 3) splays so as to extend between the transitional surfaces. It extends with constant splay to the level of the inner horizontal surface. Thereafter, it continues at the same gradient but with a 25 per cent splay until the termination of the precision segment; and
- d) the extended transitional surfaces, which continue longitudinally along the sides of the approach and missed approach surfaces and up to a height of 300 m above threshold elevation.

1.4.7.3 *Determination of OCA/H with basic ILS surfaces*

1.4.7.3.1 Where the basic ILS surfaces specified in 1.4.7.2 are not penetrated, the OCA/H for Category I and Category II is defined by the margins specified in Table II-1-1-2, and Category III operations are not restricted. Obstacles may be excluded when they are below the transitional surface defined by Annex 14 for runways with code numbers 3 and 4, regardless of the actual runway code number (i.e., the surfaces for code numbers 3 and 4 are used for the obstacle assessment on runways with code numbers 1 and 2).

1.4.7.3.2 If the basic ILS surfaces listed above are penetrated by objects other than those listed in Table II-1-1-3, the OCA/H may be calculated directly by applying height loss/altimeter margins to obstacles (see 1.4.8.8, “Determination of OCA/H with OAS or basic ILS surfaces”).

1.4.7.3.3 The obstacles in Table II-1-1-3 may only be exempted if the following two criteria are met:

- a) the localizer course sector has the standard width of 210 m (see 1.1.3, “Standard conditions”); and
- b) the Category I decision height is not less than 60 m (200 ft) or the Category II decision height is not less than 30 m (100 ft).

1.4.7.3.4 An object that penetrates any of the basic ILS surfaces and becomes the controlling obstacle, but must be maintained because of its function with regards to air navigation requirements, may be ignored under certain circumstances in calculating the OCA/H, with the following provision. It must be established by the appropriate authority that the portion which penetrates the surface is of minimum mass and frangibly mounted and would not adversely affect the safety of aircraft operations.

1.4.8 Obstacle clearance of the precision segment using obstacle assessment surface (OAS) criteria

1.4.8.1 General

1.4.8.1.1 This section describes the OAS surfaces, the constants which are used to define these surfaces, and the conditions under which adjustments may or must be made. The OAS dimensions are related to:

- a) the ILS geometry (localizer-threshold distance, glide path angle, ILS RDH, localizer sector width);
- b) the category of ILS operation; and
- c) other factors, including aircraft geometry, missed approach climb gradient.

Thus, a table of OCA/H values for each aircraft category may be calculated for Cat I and II ILS operations at the particular airfield.

1.4.8.1.2 Additional material is included to enable appropriate authorities to assess realistic benefits for claims of improved performance and associated conditions. See 1.4.8.7, “Adjustment of OAS constants”.

1.4.8.1.3 Note that the OAS are not intended to replace Annex 14 surfaces as planning surfaces for unrestricted obstacle growth. The obstacle density between the basic ILS surfaces and the OAS must be accounted for (see 1.4.8.9, “Effect of obstacle density on OCA/H”).

1.4.8.2 Frame of reference

Positions of obstacles are related to a conventional x, y, z coordinate system with its origin at threshold. See Figure II-1-1-13. The x-axis is parallel to the precision segment track: positive x is distance before threshold and negative x is distance after threshold. The y-axis is at right angles to the x-axis. Although shown conventionally in Figure II-1-1-13, *in all calculations associated with OAS geometry, the y coordinate is always counted as positive*. The z-axis is vertical, heights above threshold being positive. All dimensions connected with the OAS are specified in metres only. The dimensions should include any adjustments necessary to cater for tolerances in survey data (see Part I, Section 2, Chapter 1, 1.9, “Charting accuracy”).

1.4.8.3 OAS constants — specification

For Category I and II operations the constants A, B and C for each sloping surface are obtained from the PANS-OPS OAS software (ICAO public website (www.icao.int) under “Publications”). The PANS-OPS OAS software gives coefficients for glide path angles between 2.5 and 3.5 degrees in 0.1 degree steps, and for any localizer-threshold distance between 2 000 m and 4 500 m. Extrapolation outside these limits is not permitted. If a localizer threshold distance outside this range is entered, the PANS-OPS OAS software gives the coefficients for 2 000 m or 4 500 m as appropriate, which must be used. For an example of the PANS-OPS OAS software results see Figure II-1-1-15.

1.4.8.4 Definition of obstacle assessment surfaces (OAS)

1.4.8.4.1 The OAS consist of six sloping plane surfaces (denoted by letters W, X, Y, and Z) arranged symmetrically about the precision segment track, together with the horizontal plane which contains the threshold (see Figures II-1-1-11 and II-1-1-12). The geometry of the sloping surfaces is defined by four linear equations of the form $z = Ax + By + C$. In these equations x and y are position coordinates and z is the height of the surface at that position (see Figure II-1-1-10).

1.4.8.4.2 For each surface a set of constants (A , B and C) are obtained from the PANS-OPS OAS software for the operational range of localizer threshold distances and glide path angles. Separate sets of constants are specified for Category I and II. These constants may be modified by the programme (see 1.4.8.7, “Adjustment of OAS constants”).

1.4.8.4.3 The Category I OAS are limited by the length of the precision segment and, except for the W and X surfaces, by a maximum height of 300 m. The Category II OAS are limited by a maximum height of 150 m.

1.4.8.4.4 Where the Annex 14 approach and transitional obstacle limitation surfaces for code numbers 3 and 4 precision approach runways penetrate inside the OAS, the Annex 14 surfaces become the OAS (i.e. the surfaces for code numbers 3 and 4 are used for obstacle assessment on runways with code numbers 1 and 2). The Annex 14 inner approach, inner transitional and balked landing obstacle limitation surfaces protect Category III operations, provided the Category II OCA/H is at or below the top of those surfaces which may be extended up to 60 m if necessary) (see Figure II-1-1-9 a)).

1.4.8.5 Calculation of OAS heights

To calculate the height z of any of the sloping surfaces at a location x' , y' , the appropriate constants should be first obtained from the PANS-OPS OAS software. These values are then substituted in the equation $z = Ax' + By' + C$. If it is not clear which of the OAS surfaces is above the obstacle location this should be repeated for the other sloping surfaces. The OAS height is the highest of the plane heights (zero if all the plane heights are negative).

Note.— The PANS-OPS OAS software also contains an OCH calculator that will show the height of the OAS surface z above any x , y location. It includes all the adjustments specified for ILS geometry, aircraft dimensions, missed approach climb gradient and ILS reference datum height.

1.4.8.6 OAS template construction

1.4.8.6.1 Templates, or plan views of the OAS contours to map scale, are sometimes used to help identify obstacles for detail survey (see Figure II-1-1-14). The OAS data in the PANS-OPS software include the coordinates of the points of intersection:

- a) of the sloping surfaces at threshold level. The intersection coordinates are labeled as C, D and E (Figure II-1-1-12);
- b) at 300 m above threshold level for Cat I; and
- c) at 150 m for Cat II.

1.4.8.7 Adjustment of OAS constants

1.4.8.7.1 *General.* The following paragraphs describe the adjustments that the PANS-OPS OAS software programme makes to the OAS constants. These adjustments are mandatory when the standard conditions are not met (see 1.1.3, “Standard conditions”). Optional adjustments may be made when so specified. For examples of calculations see the Instrument Flight Procedures Construction Manual (Doc 9368).

1.4.8.7.2 *Reasons for adjusting constants.* The constants may be modified to account for the following:

- a) missed approach climb gradient (see 1.4.8.7.7, below);
- b) dimensions of specific aircraft (see 1.4.8.7.3, below);
- c) the height of the ILS reference datum (see 1.4.8.7.4, below);
- d) improved beam holding performance due to use of autopilots certified for Category II operations (see 1.4.8.7.6, below); and
- e) certain Category I localizers having a sector width greater than the nominal 210 m at threshold (see 1.4.8.7.5, below).

1.4.8.7.3 *Specific aircraft dimensions.* An adjustment is mandatory where aircraft dimensions exceed those specified in 1.1.3, “Standard conditions” and is optional for aircraft with smaller dimensions. The PANS-OPS OAS software adjusts the OAS coefficients and template coordinates for the standard dimensions of Category A, B, C, D and D_L aircraft automatically. It will do the same for specific aircraft dimensions in any category. It uses the following correction formula to adjust the coefficient C for the W, W*, X and Y surfaces:

$$\text{W surface: } C_{w\text{corr}} = C_w - (t - 6)$$

$$\text{W* surface: } C_{w^*\text{corr}} = C_{w^*} - (t - 6)$$

$$\text{X surface: } C_{x\text{corr}} = C_x - B_x \cdot P$$

$$\text{Y surface: } C_{y\text{corr}} = C_y - B_y \cdot P$$

$$\text{where: } P = \left[\frac{t}{LB_x} \text{ or } S + \frac{t-3}{B_x}, \text{ whichever is the maximum} \right] - \left[\frac{6}{LB_x} \text{ or } 30 + \frac{3}{B_x}, \text{ whichever is the maximum} \right]$$

and

s = semi-span

t = vertical distance between paths of the GP antenna and the lowest part of the wheels.

1.4.8.7.4 *Height of the ILS reference datum (RDH).* This is based on a reference datum height (RDH) of 15 m. An adjustment to the OAS constants is mandatory for an RDH less than 15 m, and is optional for an RDH greater than 15 m. The PANS-OPS OAS software adjusts the OAS coefficients and template coordinates by correcting the tabulated values of the coefficient C for the W, W*, X and Y surfaces as follows:

$$C_{\text{corr}} = C + (\text{RDH} - 15)$$

where: C_{corr} = corrected value of coefficient C for the appropriate surface

C = tabulated value.

1.4.8.7.5 *Modification for Cat I localizers with course width greater than 210 m at threshold.* Where the ILS localizer sector width at threshold is greater than the nominal value of 210 m, the collision risk model (CRM) method described in 1.4.9 shall be used. Adjustments for sector widths less than 210 m shall not be made, and are inhibited in the PANS-OPS OAS software.

1.4.8.7.6 *Use of autopilot (autocoupled) in Cat II.* The Cat II OAS may be reduced to reflect the improved beam holding of autopilots where these are certificated for the operation by the appropriate authority. This reduction is achieved in the PANS-OPS OAS software by the use of modified A, B and C constants for the X surface, and the introduction of an extra surface (denoted by W*) (see Figure II-1-1-14 c)). The use of these reduced surfaces should not be authorized for non-autocoupled approaches.

1.4.8.7.7 *Missed approach climb gradient.* If equipment is capable of missed approach climb gradients better than the nominal 2.5 per cent, the Y and Z surfaces may be adjusted. This is done by using the desired missed approach climb gradient in the PANS-OPS OAS software. The programme then adjusts the Y and Z surface coefficients.

1.4.8.8 *Determination of OCA/H with OAS or basic ILS surfaces*

1.4.8.8.1 *General.* The OCA/H is determined by accounting for all obstacles which penetrate the basic ILS surfaces defined in 1.4.7.2 and the OAS surfaces applicable to the ILS category of operation being considered. The exemptions listed in 1.4.7.3, “Determination of OCA/H with basic ILS surfaces” for obstacles penetrating the basic ILS surfaces may be applied to obstacles penetrating the OAS, providing the criteria listed in that paragraph are met. The surfaces which apply to each category of operations are:

- a) *ILS Cat I:* ILS Cat I OAS;
- b) *ILS Cat II:* ILS Cat II OAS and those portions of ILS Cat I which lie above the limits of ILS Cat II; and
- c) *ILS Cat III:* Same as ILS Cat II.

1.4.8.8.2 *Calculation of OCA/H values with OAS.* Accountable obstacles, as determined below in 1.4.8.8.2.1, “OCA/H calculation steps” are divided into approach and missed approach obstacles. The standard method of categorization is as follows: approach obstacles are those between the FAP and 900 m after threshold (Category H, 700 m if necessary); missed approach obstacles are those in the remainder of the precision segment (see Figure II-1-1-16). However, in some cases this categorization of obstacles may produce an excessive penalty for certain missed approach obstacles (see Attachment to Part II, 1.9). Alternatively, missed approach obstacles may therefore be defined as those above a plane surface parallel to the plane of the glide path and with origin at –900 m (Category H, –700 m if necessary) (see Figure II-1-1-17), i.e. obstacle height greater than $[(900 + x) \tan \theta]$.

1.4.8.8.2.1 *OCA/H Calculation steps*

- a) Determine the height of the highest approach obstacle.
- b) Convert the heights of all missed approach obstacles (h_{ma}) to the heights of equivalent approach obstacles (h_a) by the formula given below, and determine the highest equivalent approach obstacle.
- c) Determine which of the obstacles identified in steps a) and b) is the highest. This is the controlling obstacle.
- d) Add the appropriate aircraft category related margin (Table II-1-1-2) to the height of the controlling obstacle.

$$h_a = \frac{h_{ma} \cot Z + (x_z + x)}{\cot Z + \cot \theta}$$

where: h_a = height of equivalent approach obstacle

h_{ma} = height of missed approach obstacle

θ	=	angle of glide path (elevation angle)
Z	=	angle of missed approach surface
x	=	range of obstacle relative to threshold (negative after threshold)
x_z	=	distance from threshold to origin of Z surface (900 m (700 m Cat H))

1.4.8.8.3 Adjustment for high airfield elevations and steep glide path angles

1.4.8.8.3.1 *Height loss (HL)/altimeter margins.* The margins in Table II-1-1-2 shall be adjusted as follows:

- for airfield elevation higher than 900 m (2 953 ft), the tabulated allowances shall be increased by 2 per cent of the radio altimeter margin per 300 m (984 ft) airfield elevation; and
- for glide path angles greater than 3.2° in exceptional cases, the allowances shall be increased by 5 per cent of the radio altimeter margin per 0.1° increase in glide path angle between 3.2° and 3.5°.

1.4.8.8.3.1.1 Procedures involving glide paths greater than 3.5° or any angle when the nominal rate of descent (V_{at} for the aircraft type $\times$ the sine of the glide path angle) exceeds 5 m/sec (1 000 ft/min), are non-standard. They require the following:

- increase of height loss margin (which may be aircraft type specific);
- adjustment of the origin of the missed approach surface (applicable for fixed-wing aircraft procedures only);
- adjustment of the slope of the W surface;
- re-survey of obstacles; and
- the application of related operational constraints.

Such procedures are normally restricted to specifically approved operators and aircraft, and are associated with appropriate aircraft and crew restrictions. They are not to be used as a means to introduce noise abatement procedures.

1.4.8.8.3.1.2 Appendix B shows the procedure design changes required and the related operational/certification considerations.

Example: Aircraft Category C — Aerodrome elevation: 1 650 m above MSL; glide path angle 3.5°.

Tabulated allowances: radio altimeter 22 m

(Table II-1-1-2) pressure altimeter 46 m

Correction for aerodrome elevation:

$$22 \times \frac{2}{100} \times \frac{1\,650}{300} = 2.42 \text{ m}$$

Correction for glide path angle:

$$22 \times \frac{5}{100} \times \frac{3.5 - 3.2}{0.1} = 3.30 \text{ m}$$

Total correction 5.72 m rounded up to 6 m.

Corrected radio altimeter margin $22 + 6 = 28$ m.

Corrected pressure altimeter margin $46 + 6 = 52$ m.

1.4.8.8.3.2 *Exceptions and adjustments to values in Table II-1-1-2.* Values in Table II-1-1-2 are calculated to account for aircraft using normal manual overshoot procedures from OCA/H on the nominal approach path. The values in Table II-1-1-2 do not apply to Cat III operations. The values do not consider the lateral displacement of an obstacle nor the probability of an aircraft being so displaced. If consideration of these joint probabilities is required, then the CRM discussed in 1.4.9 shall be used. Values in Table II-1-1-2 may be adjusted for specific aircraft types where adequate flight and theoretical evidence is available, i.e. the height loss value corresponding to a probability of 1×10^{-5} (based on a missed approach rate 10^{-2}).

1.4.8.8.3.3 *Radio altimeter verification.* If the radio altimeter OCA/H is promulgated, operational checks shall have confirmed the repeatability of radio altimeter information.

1.4.8.8.3.4 *Height loss (HL)/altimeter margins for a specific speed at threshold.* If a height loss/altimeter margin is required for a specific V_{at} , the following formulae apply (see also Table II-1-1-4):

Use of radio altimeter:

Margin = $(0.096 V_{at} - 3.2)$ metres where V_{at} in km/h

Margin = $(0.177 V_{at} - 3.2)$ metres where V_{at} in kt

Use of pressure altimeter:

Margin = $(0.068 V_{at} + 28.3)$ metres where V_{at} in km/h

Margin = $(0.125 V_{at} + 28.3)$ metres where V_{at} in kt

where V_{at} is the speed at threshold based on 1.3 times stall speed in the landing configuration at maximum certificated landing mass.

Note.— The equations assume the aerodynamic and dynamic characteristics of the aircraft are directly related to the speed category. Thus, the calculated height loss/altimeter margins may not realistically represent small aircraft with V_{AT} at maximum landing mass exceeding 165 kt.

1.4.8.8.3.5 *Height loss (HL)/altimeter margins for a specific speed at threshold (Helicopters).* For helicopter operations the concept of V_{at} is not applicable. Height loss margins are listed in Table II-1-1-2.

1.4.8.9 *Effect of obstacle density on OCA/H.* To assess the acceptability of obstacle density below the OAS, the CRM described in 1.4.9 may be used. This can provide assistance by comparing aerodrome environments and by assessing risk levels associated with given OCA/H values. It is emphasized that it is not a substitute for operational judgement.

1.4.9 Obstacle clearance of the precision segment — application of collision risk model (CRM)

1.4.9.1 *General.* The CRM is a computer programme that establishes the numerical risk which can be compared to the target level of safety for aircraft operating to a specified OCA/H height. A description of the programme and instructions on its use, including the precise format of both the data required as input and the output results, are given in the *Manual on the Use of the Collision Risk Model (CRM) for ILS Operations* (Doc 9274).

1.4.9.2 *Input.* The CRM requires the following data as input:

- a) *aerodrome details:* name, runway threshold position and runway orientation in threshold elevation above MSL, details of proceeding segment;
- b) *ILS parameters:* category, glide slope angle, localizer-threshold distance, localizer course width, height of ILS reference datum above threshold;
- c) *missed approach parameters:* decision height (obstacle clearance height) and missed approach turn point;
- d) *aircraft parameters:* type, wheel height (antenna to bottom of wheel), and wing semi-span, aircraft category (A, B, C, D or D_L) and missed approach climb gradient; and

Note.— The CRM does not consider Category E aircraft.

- e) *obstacle data:* obstacle boundaries (either as x and y coordinates relative to the runway threshold or as map grid coordinates) and obstacle height (either above threshold elevation or above MSL). For density assessment, all obstacles penetrating the basic ILS surfaces described in 1.4.7.2 must be included.

1.4.9.3 *Output and application.* The output of the programme is:

- a) the overall (total) risk of collision with obstacles for aircraft operating to a specified OCA/H; and
- b) the minimum OCA/H which will provide the target level of safety.

The user, by rerunning the CRM with the appropriate parameters, can assess the effect on the safety of operations of any alteration in the parameters, typically varying the glide path angle or remaining obstacles.

1.5 MISSED APPROACH SEGMENT

1.5.1 General

1.5.1.1 The criteria for the final missed approach are based on those for the general criteria (see Part I, Section 4, Chapter 6). Certain modifications have been made to allow for the different areas and surfaces associated with the precision segment and for the possible variation in OCA/H for that segment with aircraft category. Area construction is according to the navigation system specified for the missed approach.

1.5.1.2 The datum used for calculation of distances and gradients in obstacle clearance calculations is termed “start of climb” (SOC). It is defined by the height and range at which the plane GP’ — a plane parallel with the glide path and with origin at –900 m (Cat H, –700 m) at threshold level — reaches the altitude OCA/H – HL. OCA/H and HL must both relate to the same category of aircraft.

1.5.1.3 If obstacles identified in the final missed approach segment result in an increase in any of the OCA/H calculated for the precision segment, a higher gradient of the missed approach surface (Z) may be specified in addition if this will provide clearance over those obstacles at a specified lower OCA/H (see Part I, Section 4, Chapter 6, 6.2.3.1, “Climb gradient in the final phase”).

1.5.1.4 A missed approach can be based on RNAV or RNP navigation specifications. Only the systems capable of navigation accuracy of 1 NM or better in this phase of flight can be considered. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for missed approach.

1.5.2 Straight missed approach

1.5.2.1 *General.* The precision segment terminates at the point where the Z surface reaches a height 300 m above threshold. The width of the Z surface at that distance defines the initial width of the final missed approach area which splay at an angle of 15 degrees from that point, as shown in Figure II-1-1-18. There are no secondary areas.

1.5.2.2 *Straight missed approach obstacle clearance.* (See Figure II-1-1-19.) Obstacle elevation/height in this final missed approach area shall be less than

$$(OCA/H_{ps} - HL) + d_o \tan Z$$

where:

- a) OCA/H of the precision segment (OCA/H_{ps}) and HL (Table II-1-1-2 value) both relate to the same aircraft category.
- b) d_o is measured from SOC parallel to the straight missed approach track; and
- c) Z is the angle of the missed approach surface with the horizontal plane.

If this requirement cannot be met, a turn shall be prescribed to avoid the obstacle in question. If a turn is not practical, the OCA/H shall be raised.

1.5.2.3 Transition from conventional navigation to an RNAV or RNP missed approach may be defined by an RNAV or RNP fix located on the extended LOC course, or by a turn at an altitude direct to a waypoint (see Figure II-1-1-24). If the RNAV or RNP track is collinear with the LOC course, the area abeam the SOC shall be expanded at a 15-degree angle from the track course until it reaches the applicable width of the RNAV or RNP area constructed on the extended LOC course. In the region between the lateral boundaries of the Z surface and the outer boundaries of the area, the obstacle evaluation shall be based on the Y surface and a mathematical extrapolation of the Y surface where the area is outside the lateral boundaries of the OAS 300 m contour. The Z surface shall continue to splay at the same angle until reaching the width of the RNAV or RNP area. Secondary areas shall apply from the point where the width of Z surface reaches the width of RNAV or RNP primary area (see Figure II-1-1-22). Obstacle clearance up to this point for the extended Y and Z surfaces shall be the same as in the precision segment (see 1.4.8 “Obstacle clearance of the precision segment using obstacle assessment surface (OAS) criteria”), and this shall also apply further out to all portions of the Z surface that are within the RNAV or RNP primary area. The principle of secondary areas shall apply between the edge of the RNAV or RNP primary area and the edge of the total area. If an obstacle penetrates either the extended Y or the Z surface within the secondary area, its elevation/height shall be less than:

$$(OCA/H_{ps} - HL) + d_o \tan Z + M$$

where:

- a) OCA/H of the precision segment (OCA/H_{ps}) and HL (Table II-1-1-2 value) both relate to the same aircraft category;
- b) d_0 is measured from SOC parallel to the straight missed approach track;
- c) Z is the angle of the missed approach surface with the horizontal plane; and
- d) M is zero at the edge of the primary area increasing linearly to 30 m (98 ft) at the edge of the total area.

1.5.2.4 If the penetration in the secondary area is less than “M” as described in 1.5.2.3 d) above, then it may be ignored in OCA/H calculations.

1.5.3 Turning missed approach

1.5.3.1 *General.* Turns may be prescribed at a designated turning point (TP), at a designated altitude/height, or “as soon as practicable”. The criteria used depend on the location of the turn relative to the normal termination of the precision segment (see 1.4.6, “Termination”) and are as follows:

- a) *turn after normal termination of the precision segment.* If a turn is prescribed after normal termination of the precision segment, the general criteria of Part I, Section 4, Chapter 6, 6.4.5, “Turn initiated at a designated altitude/height” and Part I, Section 4, Chapter 6, 6.4.6, “Turn initiated at a designated turning point” apply with the following exceptions:
 - 1) OCA/H is replaced by (OCA/H – HL) as in 1.5.2.2, “Straight missed approach obstacle clearance”; and
 - 2) because SOC is related to OCA/H, it is not possible to obtain obstacle clearance by the means used in non-precision approaches (that is, by independent adjustment of OCA/H or MAPt);
- b) *turn before normal termination of the precision segment.* If a turn is prescribed at a designated altitude/height which is less than 300 m above threshold, or at a designated TP such that the earliest TP is within the normal termination range, the criteria specified in 1.5.3.2 and 1.5.3.3 below shall be applied;
- c) for missed approaches using RNAV or RNP, fly-by or flyover turns should be limited to 90 degrees. RF turns are not permitted as the first RNP leg of the missed approach since there would be no tangent RNP entry track specified. In this case, a TF leg with the first waypoint located on the extended LOC course is required preceding the RF leg;
- d) the earliest location of the first RNAV or RNP fix is at a distance of ATT after the SOC (see Figure II-1-1-25). If the fix designates a fly-by turn, the additional distance $d_1 + 3V/3\ 600$, where V is the TAS shall be added prior to the fix ($d_1 = r \tan A/2$); and
- e) for an RNAV or RNP missed approach, RNAV/RNP turn construction and turning MOC (50 m (164 ft) for turns of more than 15 degrees and 30 m (98 ft) for all other turns) shall apply after the earliest turning point. Secondary areas shall apply between the standard width of the RNAV or RNP primary area and the edge of the total area, except that obstacles located under the Y surface or its extension on the outer side of the turn need not be considered. See Figures II-1-1-22 and II-1-1-23.

Note.— Adjustments to designated TP location or to the designated turn altitude may involve redrawing the associated areas and recalculating the clearances. This can exclude some obstacles or introduce new ones. Thus, when it is necessary to obtain the minimum value of OCA/H – particularly when constraints due to obstacles are very high – it may be necessary to adjust the designated TP or turn altitude by trial and error. (See Appendix A).

If the obstacle elevation/height exceeds this value, the OCA/H must be increased, or the TP moved to obtain the required clearance (see Appendix A).

1.6 PROMULGATION

1.6.1 General

1.6.1.1 The general criteria in Part I, Section 2, Chapter 1, 1.12, “Promulgation” apply as amplified by criteria in Part III, Section 2, Chapter 5 and Section 5, Chapter 1. The instrument approach chart for an ILS approach procedure shall be identified by the title ILS Rwy XX. If CAT II and/or III minima are included on the chart, the title shall read ILS Rwy XX CAT II or ILS Rwy XX CAT II & III, as appropriate. If more than one ILS procedure is published for the same runway, the Duplicate Procedure Title convention shall be applied. An example where the convention is applied is two ILS procedures with different segments, such as different missed approach segments.

1.6.1.2 When a final approach fix is identified at the FAP, a warning shall be appended to the procedure stating that descent on the glidepath below the FAF altitude is not permitted until passing the FAF.

1.6.1.3 When a PBN route with an RF turn to the localizer course is utilized in the intermediate approach segment, then the last item of the PBN procedure description shall be the waypoint at the end of the RF turn with an altitude ‘window’ constraint, or an ‘at’ altitude constraint (see the *Note* to 1.3.6.3).

1.6.2 Promulgation of OCA/H values

1.6.2.1 Promulgation of OCA/H for Cat I and II approach procedures

1.6.2.1.1 The OCA or OCH values, as appropriate, shall be promulgated for those categories of aircraft for which the procedure is designed. The values shall be based on the following standard conditions:

- a) Cat I flown with pressure altimeter;
- b) Cat II flown autocoupled with radio altimeter;
- c) standard aircraft dimensions (see 1.1.3, “Standard conditions”); and
- d) 2.5 per cent missed approach climb gradient.

1.6.2.1.2 Additional values of OCA/H may be agreed upon between operators and the appropriate authority and be promulgated, provided that modifications have been carried out using the guidelines and algorithms defined in 1.4.8.7, “Adjustment of OAS constants”.

1.6.2.1.3 Use of OCA/H values for Category I approach procedures based on radio altimeter height loss margins may be agreed upon between operators and the appropriate authority, and the values promulgated, if the requirement of 1.4.8.8.3.3, “Radio altimeter verification” is met.

1.6.2.2 Promulgation of Category III approach procedures

Category III operations may be permitted subject to the appropriate Category II OCA/H being below the height of the Annex 14 inner horizontal surface. Category III operations may also be permitted with a Category II OCA/H between the height of the inner horizontal surface and 60 m provided the Annex 14 Category II inner approach, inner transitional and balked landing surfaces are extended to protect that OCA/H.

1.6.3 Turn at a designated altitude/height (missed approach)

If the TP is located at the SOC, the chart shall be annotated “turn as soon as practicable to ... (heading or facility)” and shall include sufficient information to identify the position and height of the obstacles dictating the turn requirement.

1.6.4 Turn at a designated TP (missed approach)

Where the procedure requires that a turn be executed at a designated TP, the following information must be published with the procedure:

- a) the TP, when it is designated by a fix; or
- b) the intersecting VOR radial, NDB bearing, or DME distance where there is no track guidance (see Part I, Section 2, Chapter 2, 2.6.5, “Missed approach fixes”).

1.6.5 Procedures involving non-standard glide path angles

Procedures involving glide paths greater than 3.5° or any angle when the nominal rate of descent exceeds 5 m/sec (1 000 ft/min), are non-standard and subject to restrictions (see 1.4.8.8.3.1, “Height loss (HL)/altimeter margins”). They are normally restricted to specifically approved operators and aircraft, and are promulgated with appropriate aircraft and crew restrictions annotated on the approach chart.

1.6.6 Additional gradient for the final missed approach segment

If obstacles identified in the final missed approach segment result in an increase in any of the OCA/H calculated for the precision segment, a higher gradient of the missed approach surface (Z) may be specified in addition if this will provide clearance over those obstacles at a specified lower OCA/H (see Part I, Section 4, Chapter 6, 6.2.3.1, “Climb gradient in the final phase”).

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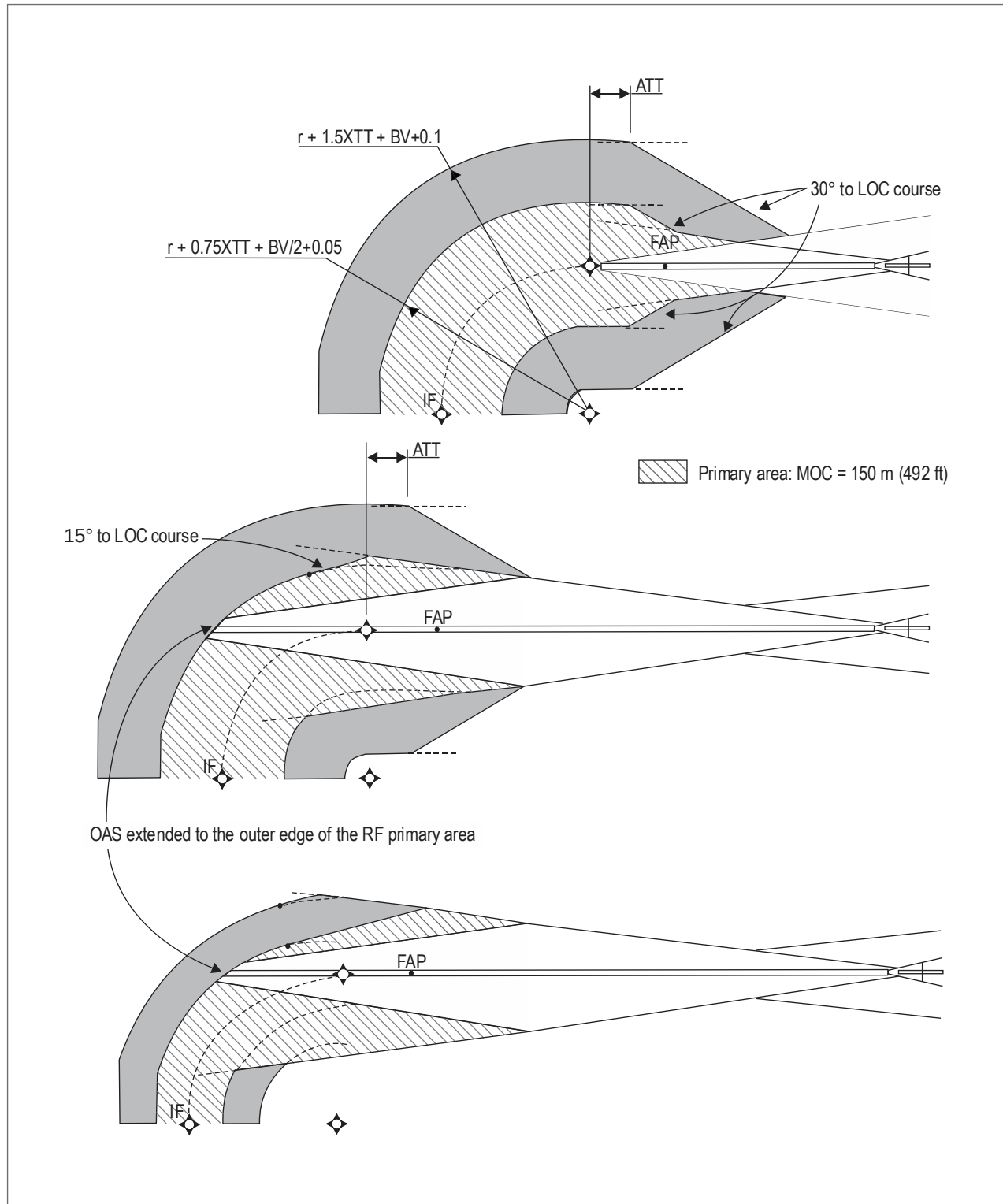


Figure II-1-1-7. Area construction for radius-to-fix turns to the localizer course

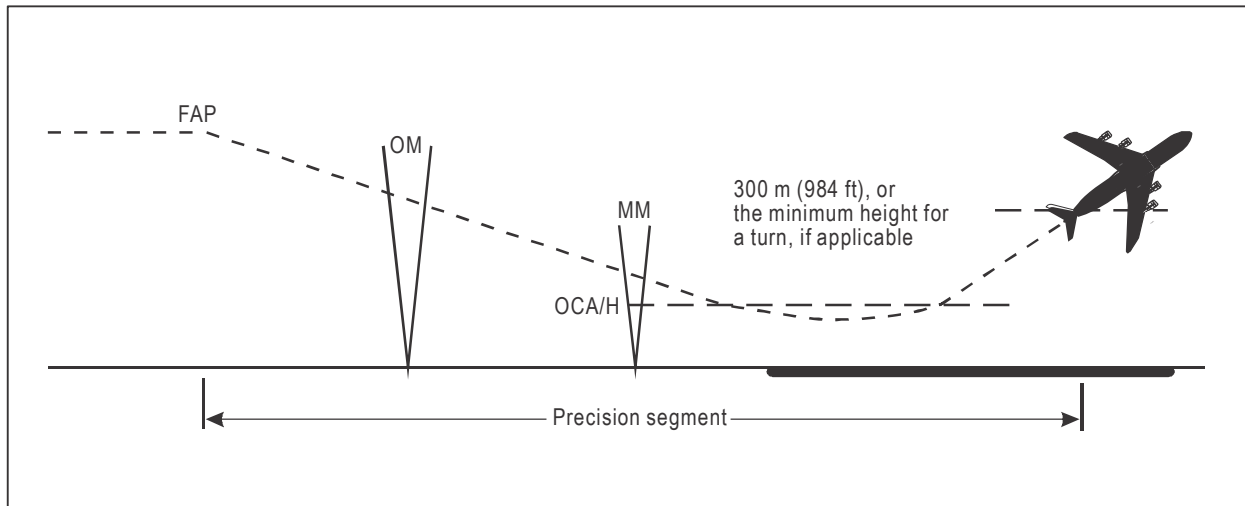


Figure II-1-1-8. Precision segment

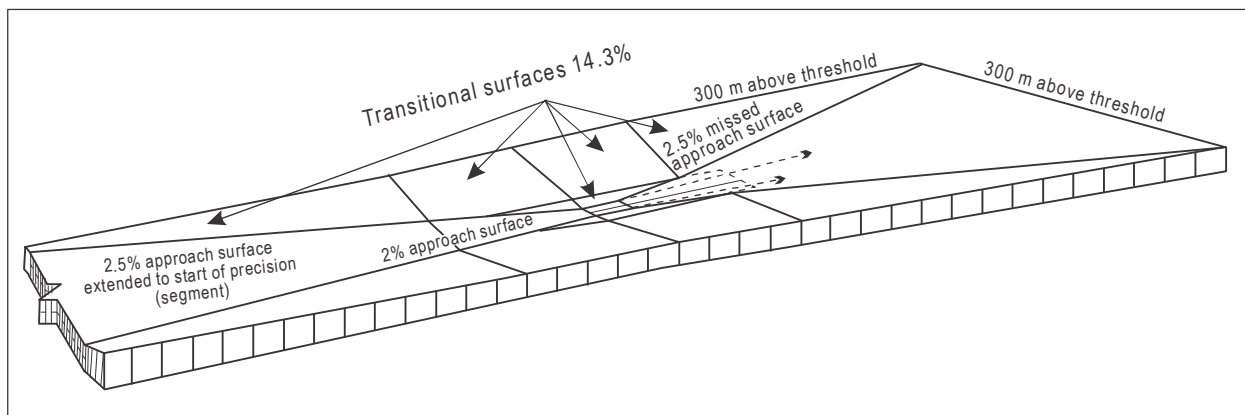


Figure II-1-1-9 a). Annex 14 inner approach, inner transitional and balked landing surfaces as applied for obstacle assessment for Cat II and III ILS operations

Appendix A to Chapter 1

ILS: TURNING MISSED APPROACH ASSOCIATED WITH A PRECISION APPROACH

(see Chapter 1, 1.5.3, “Turning missed approach”)

Note.— This appendix is referenced in Part II, Section 1, Chapter 3 and Part III, Section 3, Chapter 6 and contains material that may amplify the guidance presented on MLS and GLS operations, respectively.

1. INTRODUCTION

1.1 This appendix contains some guidance material about the way to adjust the turn altitude or the TP location in the case of turning missed approach associated with a precision approach, and it gives some simple formulae to use when the OCA/H has to be increased.

1.2 This appendix also describes a method of treating a turn at an altitude from within the precision segment which reduces the penalty some obstacles cause when the more general techniques of Chapter 1, 1.5.3, “Turning missed approach” are used.

2. TURN ALTITUDE/TP LOCATION ADJUSTMENTS

2.1 Turn at an altitude

2.1.1 Assume that a turn at an altitude has been designed to avoid obstacle 01.

2.1.2 *Obstacle straight ahead in the turn area* (see Figure II-1-1-App A-1). If an obstacle in the turn area, located at 02, cannot be overflown with the adequate margin, the options to solve the problem are:

- a) lower the turn altitude in order to exclude 02 from the turn area. In this case, obstacles in the turn initiation area (like 03) might become a problem. This means that it might not be possible to lower the turn altitude as much as needed (since turn altitude must be at least the elevation of obstacle 03 + MOC applicable to turns);
- b) restrict the final missed approach speed. Then the radius of turn will be reduced and the turn area might exclude 02. (Of course, if speed restriction is applied, the published speed shall be kept above the intermediate missed approach speed); and
- c) increase OCA/H. This will raise the turn altitude without moving the latest TP. New OCA/H can be found by using the method described in paragraph 3.

2.1.3 *Obstacle in the turn initiation area.* If an obstacle in the turn initiation area (like 03) is higher than the turn altitude less the margin applicable to turns, then the turn altitude has to be increased. The options are:

- a) increase the turn altitude by moving the latest TP further from the SOC. This is acceptable to the extent that 01 still remains outside the turn area;
- b) if this does not appear to be possible, the final missed approach speed might be restricted, to reduce the radius of turn and keep 01 outside the turn area; and
- c) increase OCA/H without moving the latest TP. New OCA/H can be found by using the formula in paragraph 3.

2.1.4 *Obstacle in the turn area abeam the straight ahead missed approach track.* If an obstacle in the turn area, like 04, cannot be overflown with the appropriate margin, the options a) or b) presented in 2.1.3, “Obstacle in the turn initiation area” above, will be used.

2.2 Turn at a designated TP

2.2.1 *Obstacle straight ahead in the turn area.* If an obstacle straight ahead in the turn area cannot be overflown with the appropriate margin (like 02 in Figure II-1-1-App A-2), the options are:

- a) move the TP closer to the SOC in order to exclude 02 from the turn area. The difficulty in this case, is that it might then be difficult to get the vertical margin applicable to turns at the earliest TP (which will occur at a lower point of the nominal flight path);
- b) if this cannot be solved, the final missed approach speed might be restricted in order to decrease the radius of turn and exclude 02 from the area; and
- c) increase OCA/H by using the method shown in paragraph 3.

2.2.2 *Obstacle abeam the straight ahead missed approach track (and before the earliest TP range).* If an obstacle like 04 (see Figure II-1-1-App A-2) cannot be overflown with the appropriate margin, the options to solve the problem are:

- a) move the TP further from the SOC. This will increase the nominal altitude over the obstacle and could even exclude 04 from the turn area. Of course this is acceptable to the extent obstacle 01 (see Figure II-1-1-App A-2) can be kept outside the area;
- b) if this is not possible, then the final missed approach speed might be restricted; and
- c) increase OCA/H by using the method shown in paragraph 3.

3. CALCULATION OF OCA/H

3.1 Calculation of OCH from obstacle data (h_o , d_o)

Whenever there are obstacles in the turn area it is possible to find the OCH compatible with these obstacles by using the formula:

Appendix C to Chapter 1

DETERMINING ILS GLIDE PATH DESCENT/MLS ELEVATION HEIGHTS AND DISTANCES

Note.— This appendix is referenced in Part II, Section 1, Chapter 3 and Part III, Section 3, Chapter 6 and contains material that may amplify the guidance presented on GLS operations.

1. ILS/MLS glide path heights (H) and horizontal distances (D) from the threshold are calculated by solving a right-angle triangle:

SI units

$$H = h + 1\,000 D \tan \theta \text{ and } D = 0.001(H - h) \cot \theta$$

where: H = height in metres

h = reference datum height in metres

D = distance from the threshold in kilometres

θ = glide path angle in degrees

Non-SI units

$$H = h + 6\,076 D \tan \theta \text{ and } D = 0.0001646(H - h) \cot \theta$$

where: H = height in feet

h = reference datum height in feet

D = distance from the threshold in nautical miles

θ = glide path angle in degrees

2. The influence of the curvature of the earth's surface should be considered in order to check that the heights and distances to the threshold determined in this manner meet the Annex 10 and PANS-OPS requirements. To perform such a check, Tables II-1-1-App C-1 and II-1-1-App C-2 may be used. For intermediate distances, heights and glide path angles, the linear interpolation method is used.

For reference datum heights (h) other than 15 m (49 ft):

a) the values obtained from Table II-1-1-App C-1 should be corrected by adding ΔH where:

SI units: $\Delta H = h - 15$ (Table II-1-1-App C-1a))

and

Non-SI units: $\Delta H = h - 49$ (Table II-1-1-App C-1b))

b) the values obtained from Table II-1-1-App C-2 should be corrected by adding ΔD where:

SI units: $\Delta D = 0.00092(15 - h) \cot \theta$ (Table II-1-1-App C-2a))

and

Non-SI units: $\Delta D = 0.0001514(49 - h) \cot \theta$ (Table II-1-1-App C-2b))

The following formulae may be used for intermediate distances, heights and glide path angles as well as for values which are greater than the maximum values indicated in Tables II-1-1-App C-1 and II-1-1-App C-2:

SI units: $H = h + 1\,000\,D \tan \theta + 0.0785\,D^2$

and

Non-SI units: $H = h + 6\,076\,D \tan \theta + 0.8833\,D^2$

3. Heights are rounded up to the nearest multiple of 5 m (10 ft), and distances are rounded to the nearest tenth of a kilometre (nautical mile).

Note 1.— When heights are rounded up to the nearest multiple of 5 m (10 ft), the check referred to in paragraph 2 will not result in significant differences from the conventional geometric right-angle triangle calculation for threshold distances of less than 8 km or 4 NM. This also applies when distances are rounded to the nearest tenth of a kilometre (NM) for heights less than 500 m or 2 100 ft.

Note 2.— To determine glide path heights at the outer marker fix or other fix, unrounded height values are used.

Appendix D to Chapter 1

MINIMUM DISTANCE BETWEEN LOCALIZER AND GLIDE PATH INTERCEPTIONS AFTER AN RF TURN TO THE LOCALIZER/FINAL APPROACH COURSE

Note.— This appendix is referenced in Part II, Section 1, Chapter 3 and Part III, Section 3, Chapter 6 and contains material that may amplify the guidance presented on MLS and GLS operations, respectively.

1. The computation starts from the height of the final approach point (FAP) above the threshold elevation, h_{FAP} :

$$h_{FAP} = alt_{FAP} - LTP_{elev} \quad (1)$$

where: alt_{FAP} = the altitude of the final approach point
 LTP_{elev} = the elevation of the landing threshold point

2. The distance of the FAP from the threshold is given as follows (see Figure II-1-1-App F-1):

$$dist_{FAP} = \frac{r \cdot \pi}{180} \cdot \left(\cos \left[\cos(GPA) \cdot \frac{r + RDH + LTP_{elev}}{r + h_{FAP} + LTP_{elev}} \right] - GPA \right) \quad (2)$$

where: GPA = glide path angle
RDH = the reference datum height
r = the radius of the earth (6 371 000 m)

Note.— The computation can also start with selecting a distance of the FAP from the threshold, $dist_{FAP}$, after which h_{FAP} could be determined using the following rewritten form of equation (2).

$$h_{FAP} = (r + RDH + LTP_{elev}) \frac{\cos(GPA)}{\cos \left[dist_{FAP} \cdot \frac{180}{r \cdot \pi} + GPA \right]} - r - LTP_{elev} \quad (3)$$

3. Assume a capture segment with length L and vertical barometric angle θ (see the green line in Figure II-1-1-App F-2), which starts at the end fix of the RF turn located on the localizer/final approach course (the final approach capture fix – FACP) and ends at the FAP. Note that both the capture altitude at the FAP and the altitude at the FACP must still be promulgated at 100 ft discrete intervals. This segment should be designed flat. Should a descent be necessary, then the FACP altitude for promulgation shall be rounded down based on the result of the calculations of this Appendix.

4. The procedure height at the FACP is:

$$h_{FACP} = h_{FAP} + L \cdot \tan(\theta) \quad (4)$$

5. The height of the theoretical glide path at the FACP is calculated using a rewritten form of equation (3):

$$h_{GP,FACF} = (r + RDH + LTP_{elev}) \frac{\cos(GPA)}{\cos \left[(dist_{FAP} + L) \cdot \frac{180}{r \cdot \pi} + GPA \right]} - r - LTP_{elev} \quad (5)$$

6. The significant factors having an effect on the vertical capture of the geometric final approach vertical guidance after navigating vertically based on barometric inputs are as follows (see Figure II-1-1-App F-2):

- a) the real height of the aircraft at the FACF can be affected by an assumed maximum ISA deviation (ΔISA) causing a height error (Δh_T) and an Altimeter System Error (ASE); and
- b) the real glide path can be affected by an alignment error that will be referred here as glide path construction error (GCE).

7. The length of the capture segment shall ensure that these errors will not result in a vertical capture from above. Other vertical errors are negligible and vertical flight technical tolerance is not taken into account, because it is directly known to the crew, and based on its maximum allowable value with reference to the other factors, such a vertical capture –that might happen from above with the crew’s intervention – is acceptable.

8. Δh_T is computed using the temperature compensation formula from Part III, Section 3, Appendix A to Chapter 4:

$$\Delta h_T = \frac{\Delta ISA}{\lambda} \cdot \ln \left[1 + \lambda \cdot \frac{h_{FAP} + L \cdot \tan(\theta)}{288,15 + \lambda \cdot LTP_{elev}} \right] \quad (6)$$

where λ = the temperature lapse rate (= -0.00198 °/ft)
 ΔISA = the temperature deviation from ISA.

9. The altimeter system error (ASE) is computed based on the formula in the ICAO *Performance-based Navigation (PBN) Manual* (Doc 9613), Volume II, Attachment A, 4.5.1:

$$ASE = -8.8 \cdot 10^{-8} (LTP_{elev} + h_{FAP} + L \cdot \tan\theta)^2 + 6.5 \cdot 10^{-3} (LTP_{elev} + h_{FAP} + L \cdot \tan\theta) + 50 \quad (7)$$

Note.— L is expressed in feet in formula (7).

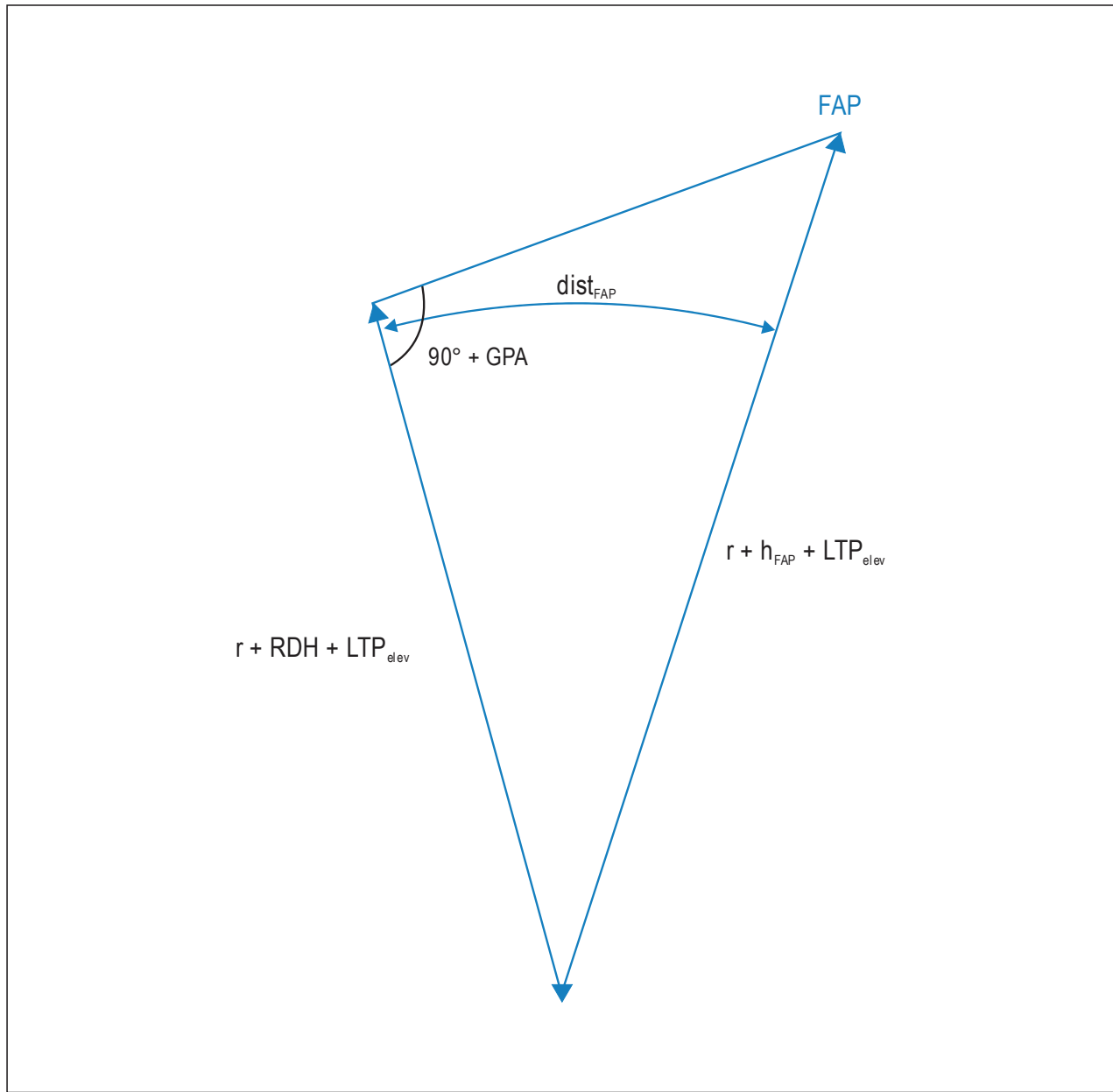
10. The GCE caused by glide path misalignment is quantified according to ICAO Annex 10 — *Aeronautical Telecommunications*, Volume I — *Radio Navigation Aids*, Chapter 3 as follows:

$$GCE = (dist_{FAP} + L) \cdot [\tan(GPA) - \tan(GPA - \alpha \cdot GPA)] \quad (8)$$

where: α = 0.075 in case of ILS Category I, ILS Category II and MLS
 α = 0.04 in case of ILS Category III
 α = 0 in case of GLS and LPV (the path definition error is negligible)

Note.— According to Annex 10, Volume I, 3.11.4.9.6.1, about MLS degradation allowance with distance, the MLS glide path degradation is considered to be within the ILS Category I and Category II limits.

11. To intercept the glide path from below, L and θ should be designed such that the difference between the height of the theoretical glide path at the FACF ($h_{GP,FACF}$) and the procedure height at the FACF (h_{FACF}) is greater than or equal to the sum of the error components. As the three error components Δh_T , ASE and the GCE can be considered independent, their root sum square may be used. This leads to the following formula giving the most optimal values for θ and L (highest θ for a given L or lowest L for a given θ):



$$h_{GP,FACF} - h_{FACF} \geq \sqrt{\Delta h_T^2 + ASE^2 + GCE^2} \quad (9)$$

12. An iterative method has to be used, choosing an initial value of θ and L and then adjusting the value of θ or L until equation (9) is satisfied.

Figure II-1-1-App D-1 Relation of distance and height of FAP in relation to threshold considering earth's curvature

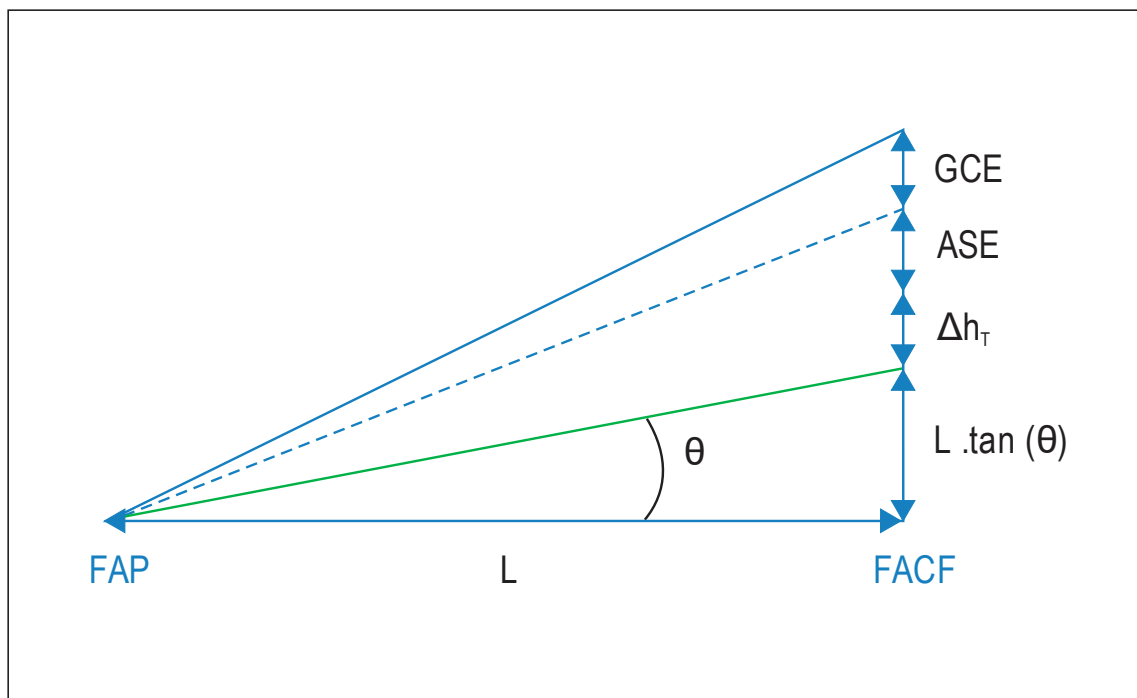


Figure II-1-1-App D-2 Errors affecting the vertical capture

below 60 m, Category III operations can be accommodated provided the inner approach, inner transitional and balked landing surfaces are extended to the Cat II OCA/H level (see Figure II-1-3-10 a)).

When azimuth antenna to threshold distance is less than 2 000 m, the obstacle assessment surface (OAS) tables for a 2 000 m localizer to threshold are to be used. When using the ILS CRM or the OAS table, the ATT distances and displacement sensitivity characteristics shown in 3.1.3 d) above are to be used.

3.1.4 Obstacle clearance altitude/height (OCA/H)

3.1.4.1 The MLS criteria enable an OCA/H to be calculated for each category of aircraft. See Part I, Section 4, Chapter 1, 1.8, “Categories of aircraft”. Where statistical calculations were involved, the OCA/H values were designed against an overall safety target of 1×10^{-7} (1 in 10 million) per approach for risk of collision with obstacles.

3.1.4.2 The OCA/H ensures clearance of obstacles from the start of the final approach to the end of the intermediate missed approach segment.

Note.— This OCA/H is only one of the factors to be taken into account in determining decision height as defined in Annex 6.

3.1.4.3 Additional material is included to allow operational benefit to be calculated for the improved missed approach climb performance in Cat I, II and III.

3.1.4.4 Benefit may also be calculated for aircraft with dimensions smaller than the standard size assumed in the basic calculations and adjustments must be made for larger aircraft. An OCA/H is not associated with Cat III operations. These are supported by the obstacle limitation surfaces defined in Annex 14, in association with overlapping protection from the Cat II criteria.

3.1.5 Methods of calculating OCA/H

3.1.5.1 *General.* Three methods of calculating OCA/H are presented, which involve progressive increases in the degree of sophistication in the treatment of obstacles. Standard conditions (as specified in 3.1.3) are assumed to exist unless adjustments for non-standard conditions have been made.

3.1.5.2 *First method.* The first method involves a set of surfaces derived from the Annex 14 precision approach obstacle limitation surfaces and a missed approach surface described in 3.4.7.2, “Definition of basic ILS surfaces”. From this point forward, these are termed “basic ILS surfaces”. Where the standard conditions exist as specified in 3.1.3 and where the basic ILS surfaces are free of penetrations (see 3.4.7.1), the OCA/H for Cat I and Cat II is defined by aircraft category margins, and there are no restrictions on Cat III operations. If the basic ILS surfaces are penetrated, then the OCA/H is calculated as described in 3.4.7.3, “Determination of OCA/H with basic ILS surfaces”.

3.1.5.3 *Second method.* The second method involves a set of obstacle assessment surfaces (OAS) above the basic ILS surfaces (see 3.4.8.3, “Definition of obstacle assessment surfaces (OAS)”). If the OAS are not penetrated, — and provided the obstacle density below the OAS is operationally acceptable (see 3.4.8.9, “Effect of obstacle density on OCA/H”) — the OCA/H for Cat I and Cat II is still defined by the aircraft category margins, and Cat III operations remain unrestricted. However, if the OAS are penetrated, then an aircraft category-related margin is added to the height of the highest approach obstacle, or to the adjusted height of the largest missed approach penetration, whichever is greater. This value becomes the OCA/H.

3.1.5.4 *Third method.* The third method, using the ILS collision risk model (CRM), is employed either as an alternative to the use of the OAS criteria (second method) or when the obstacle density below the OAS is considered to be excessive. The ILS CRM accepts all objects as an input and assesses, for any specific OCA/H value, both the risk due

to individual obstacles and the accumulated risk due to all the obstacles. In this way it assists operational judgement in the choice of an OCA/H value which will ensure that the hazard due to obstacles, both individually and collectively, can be contained within the overall safety target.

3.1.6 References

The following appendices relate to and amplify the material contained in this chapter:

- a) background information relating to the derivation of the OAS material (Attachment to Part II, paragraph 1) and to airborne and ground equipment performance assumed in the derivation of the OAS (Attachment to Part II, paragraph 2);
- b) turning missed approach after precision approach (Appendix A to Chapter 1);
- c) determining ILS glide path descents/MLS elevation heights and distances (Appendix C to Chapter 1);
- d) PANS-OPS OAS software (ICAO public website (www.icao.int)); and
- e) minimum distance between localizer and glide path interceptions after an RF turn to the localizer course (Appendix D to Chapter 1).

Examples of OCA/H calculation for ILS can be found in the *Instrument Flight Procedures Construction Manual* (Doc 9368).

3.1.7 MLS with glide path inoperative

The MLS with glide path inoperative is a non-precision approach procedure. The principles of Section 2, Chapter 2, “MLS azimuth only”, apply.

3.2 INITIAL APPROACH SEGMENT

3.2.1 General

The initial approach segment for MLS must ensure that the aircraft is positioned within the operational service volume of the azimuth on a track that will facilitate azimuth interception. Consequently, the general criteria applicable to the initial segment (see Part I, Section 4, Chapter 3) are modified in accordance with 3.2.2, “Initial approach segment alignment” and 3.2.3, “Initial approach segment area”, below. The initial approach segment may be defined by an RNAV or RNP route, using an RNAV or RNP system for track guidance. Only the systems capable of navigation accuracy of 1 NM or better in this phase of flight can be considered. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for initial approach. The RNAV or RNP route shall terminate at an IF defined by RNAV or RNP located on the azimuth course. RNAV/RNP turn construction is applicable for turns within the initial segment and for the turn at the IF on the azimuth course (see Figures II-1-3-6 and II-1-3-7). For RNAV and RNP initial approach segments, criteria in Part III apply. If a course reversal is required with an RNAV or RNP initial approach segment, only a racetrack can be used. The fix and the inbound leg shall be located on the azimuth course and the inbound segment defined by the azimuth.

3.2.2 Initial approach segment alignment

The angle of interception between the initial approach track and the intermediate track should not exceed 90°. In order to permit the autopilot to couple on to the azimuth, an interception angle not exceeding 30° is desirable. When the angle exceeds 70°, a radial, bearing, radar vector, or DME information providing at least 4 km (2 NM) of lead shall be identified to assist the turn onto the intermediate track. When the angle exceeds 90°, the use of a reversal, racetrack, or dead reckoning (DR) track procedure should be considered (see Part I, Section 4, Chapter 3, “Initial approach segment” and Part I, Section 4, Appendix A to Chapter 3, “Initial approach using dead reckoning (DR)”).

3.2.3 Initial approach segment area

The area is as described in the general criteria (see Part I, Section 4, Chapter 3, 3.3.3, “Area”). The difference is that the intermediate approach fix (IF) must be located within the service volume of the MLS azimuth signal, (except when an RF turn to the azimuth course is used in the intermediate approach segment according to criteria in 3.3.6 below), and normally at a distance not exceeding 41.7 km (22.5 NM) from the azimuth antenna. When radar is used to provide track guidance to the IF, the area shall be in accordance with 6.2, “Initial approach segment” (Section 2, Chapter 6, “SRE”).

3.3 INTERMEDIATE APPROACH SEGMENT

3.3.1 General

3.3.1.1 The intermediate approach segment for MLS differs from the general criteria in that:

- a) the alignment coincides with the MLS azimuth specified for final approach track;
- b) the length may be reduced;
- c) in certain cases the secondary areas may be eliminated; and
- d) if an RF turn to the azimuth course is used, specific criteria in 3.3.6 apply.

3.3.1.2 The primary and secondary areas at the FAP are defined in terms of the ILS surfaces. Consequently, the general criteria in Part I, Section 4, Chapter 4, “Intermediate approach segment” are applied except as modified or amplified in the paragraphs below with regards to alignment, area length and width, and for obstacle clearance. For RNAV initial approach segments, the criteria in the applicable RNAV chapters apply.

3.3.2 Intermediate approach segment alignment

The intermediate approach segment of an MLS procedure shall be aligned with the MLS azimuth specified for the final approach track, except when an RF turn to the azimuth course is used (see 3.3.6 below).

3.3.3 Intermediate approach segment length

3.3.3.1 The optimum length of the intermediate approach segment is 9 km (5 NM). This segment shall allow interception with the final approach track and with the glide path (MLS elevation angle).

3.3.3.2 Segment length should be sufficient to permit the aircraft to stabilize and establish its course on the final approach track prior to intercepting the glide path (MLS elevation angle), taking into consideration the angle of interception of the final approach track.

3.3.3.3 Minimum values for distance between interception of final approach track and interception of glide path are specified in Table II-1-3-1; however, these minimum values should only be used if usable airspace is restricted. The maximum length of the segment is governed by the requirement that it be located wholly within the operational coverage region of the approach azimuth, and normally at a distance not exceeding 37 km (20 NM) from the runway threshold. For RF turns to the azimuth course, see 3.3.6 below.

3.3.4 Intermediate approach segment area width

3.3.4.1 The total width at the beginning of the intermediate approach segment is defined by the final total width of the initial approach segment. It tapers uniformly to match the horizontal distance between the OAS X surfaces at the FAP (see 3.4.8.3, “Definition of obstacle assessment surfaces (OAS)”).

3.3.4.2 For obstacle clearance purposes the intermediate approach segment is usually divided into a primary area bounded on each side by a secondary area. However, when a DR track is used in the initial approach segment, the primary area of the intermediate approach segment extends across its full width and secondary areas are not applied.

3.3.4.3 The primary area is determined by joining the primary initial approach area with the final approach surfaces (at the FAP). At the interface with the initial approach segment the width of each secondary area equals half the width of the primary area. The secondary area width decreases to zero at the interface with the final approach surfaces. See Figures II-1-3-2, II-1-3-3 and II-1-3-4. For RF turns to the azimuth course, see 3.3.6 below.

3.3.4.4 Where a racetrack or reversal manoeuvre is specified prior to intercepting the final approach track, the provisions in Part I, Section 4, Chapter 4, 4.4.4, “Turn not at the facility” apply, the facility being the MLS azimuth itself and the FAF being replaced by the FAP (see Figure II-1-3-5).

3.3.5 Intermediate approach segment obstacle clearance

The obstacle clearance is the same as defined in Part I, Section 4, Chapter 4, “Intermediate approach segment” except where the procedure permits a straight-in approach in which the aircraft is stabilized on the final approach track prior to crossing the IF. In this case, obstacles in the secondary areas need not be considered for the purpose of obstacle clearance.

3.3.6 RF turn to the azimuth course

3.3.6.1 The intermediate approach segment may incorporate a performance-based navigation (PBN) route ending with a radius-to-fix (RF) turn to the azimuth course. In this case, the constant RF turn shall terminate at a waypoint located on the azimuth course. The position of this waypoint is considered as the point of azimuth interception. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for intermediate approach segment.

3.3.6.2 Minimum distance between azimuth and glide path (elevation angle) interceptions in Table II-1-3-1 apply up to temperatures of ISA +30 degrees and up to glide path interceptions at 3000 feet above aerodrome elevation. For cases exceeding either of these values, criteria in Appendix D to Chapter 1 shall be applied. When these temperature and intercept height assumptions are too limiting, Appendix D to Chapter 1 may be applied. The maximum total length of the intermediate approach segment shall not exceed 18.5 km (10 NM). Additionally, in case of Category II and III operations, the minimum distance between the end fix of the RF turn and the landing threshold shall be 9.3 km (5 NM).

3.3.6.3 Descent in the intermediate approach segment shall comply with the general criteria in Part I, Section 4, Chapter 4, 4.3.3 “Procedure altitude/height and descent gradient”. If a descent is necessary after the RF turn to the azimuth course, then criteria in Appendix D to Chapter 1 shall be applied, and obstacle clearance shall be provided in the total intermediate approach area according to Part I, Section 4, Chapter 4, 4.3.2 “Obstacle clearance”. The altitude constraint at the end fix of the RF turn shall be an altitude ‘window’. The top of the window shall comply with the calculations in Appendix D to Chapter 1 or shall be equal to the intercept altitude if the minimum length in 1.3.6.2 is applied. The bottom of the window shall comply with the minimum altitude requirements of the RF turn.

Note.— If the calculations result in the same altitude for the top and the bottom of the window, the constraint at the end fix of the RF turn becomes an ‘at’ altitude.

3.3.6.4 Criteria for the RF turn construction defined in Part III, Section 2, Chapter 2, 2.4 “RF turn method” apply with the following additions (see Figure II-1-3-9):

- a) the boundaries of the RF turn shall be continued one along-track tolerance (ATT) after the end fix of the RF turn assuming the width and fix tolerance values of the navigation specification associated with the RF turn;
- b) the boundary of the succeeding segment is the D”-E” line and the extension of the D-D” line considered as primary only area;
- c) if the bounding arcs (either primary or secondary) on the inner side of the turn intersect the extension of the D-D” line, then the extension becomes the edge of the area after the intersection, otherwise the RF boundaries shall be joined to the succeeding segment from one ATT after the end fix of the turn by a line at 30° to the azimuth course; and
- d) if the bounding arcs (either primary or secondary) on the outer side of the turn do not intersect the extension of the D-D” line, then the arcs shall be joined to the extension by a tangent splaying 15° from the azimuth course, otherwise the RF boundaries shall be joined to the succeeding segment from one ATT after the end fix of the turn by a line at 30° to the azimuth course.

3.4 PRECISION SEGMENT

3.4.1 General

The precision segment for MLS is aligned with the specified MLS azimuth and contains the final descent for landing as well as the initial and intermediate phases of the missed approach segment. Criteria are generally the same as for ILS, except as amended below. See Figure II-1-3-9.

3.4.2 Origin

The precision segment starts at the final approach point (FAP), that is, the intersection of the glide path (elevation angle) and the procedure altitude specified for the preceding segment. The FAP should not normally be located more than 18.5 km (10.0 NM) before threshold. This distance may be extended for operational requirements provided that:

- a) adequate guidance is available; and
- b) obstacle clearance requirements are not compromised (extension of the W and X surfaces of the OAS).

3.4.3 Descent fix

3.4.3.1 A descent fix may be located at the FAP to overcome certain obstacles located before the FAP as an alternative to increasing the glide path (GP) angle. When so located, it becomes the final approach fix, linking the MOC in the preceding segment smoothly with the precision surfaces. The descent fix should not normally be located more than 18.5 km (10.0 NM) before threshold, unless adequate glide path guidance beyond the minimum specified in Annex 10 is provided. The maximum fix tolerance is ± 0.9 km (± 0.5 NM). The range shall be stated in tenths of kilometres (nautical miles).

Note.— Guidance material for determining the distance to the descent fix from the threshold is contained in Chapter 1, Appendix C.

3.4.3.2 *Obstacle clearance at the descent fix.* When a descent fix is provided, the precision approach surfaces start at the earliest point of the FAF tolerance area (see Figure II-1-3-3). The provisions of Part I, Section 2, Chapter 2, 2.7.4, “Obstacle close to a final approach fix or stepdown fix” which allow obstacles close to the fix to be ignored, apply in the area below the 15 per cent gradient within the precision surfaces (Cat H, 15 per cent gradient or the nominal gradient multiplied by 2.5, whichever is greater). Where a descent fix is not provided at the FAP, no curtailment of the precision surfaces is permitted (see Figure II-1-3-4). If the precision surfaces are extended into the preceding segment, they shall not be extended beyond the early fix tolerance of the IF (see Figure II-1-3-4) or the edge of the intermediate primary area (see Figure II-1-3-9).

3.4.4 Glide path verification check

A DME fix is necessary so as to permit comparison between the indicated glide path (elevation angle) and the aircraft altimeter information. The fix shall not have a fix tolerance exceeding ± 0.9 km (± 0.5 NM). The range shall be stated in tenths of kilometres (nautical miles).

Note.— Guidance material for determining the height crossing the DME fix is contained in Chapter 1, Appendix C.

3.4.5 Missed approach

The missed approach point is defined by the intersection of the nominal glide path and the decision altitude/height (DA/H). The DA/H is set at or above the OCA/H, which is determined as specified in 3.4.7 to 3.4.9 and 3.5.

3.4.6 Termination

The precision segment normally terminates at the point where the final phase of the missed approach commences (see Part I, Section 4, Chapter 6, 6.1.2, “Phases of missed approach segment”) or where the missed approach climb surface Z (starting 900 m past threshold) reaches a height of 300 m (984 ft) above threshold, whichever is lower.

3.4.7 Obstacle clearance in the precision segment application of basic ILS surfaces

3.4.7.1 *General.* The area required for the precision segment is bounded overall by the basic ILS surfaces defined in 3.4.7.2, below. In standard conditions there is no restriction on objects beneath these surfaces (see 3.1.3, “Standard conditions”). Objects or portions of objects that extend above these surfaces must be either:

- a) minimum mass and frangible; or
- b) taken into account in the calculation of the OCA/H.

3.4.7.2 *Definition of basic ILS surfaces.* The surfaces to be considered correspond to a subset of Annex 14 obstacle limitation surfaces as specified for precision approach runway code numbers 3 or 4 (see Figure II-1-3-10 b)). These are:

- a) the approach surface, continuing to the final approach point (FAP) (first section 2 per cent gradient, second section 2.5 per cent as described in Annex 14);
- b) the runway strip assumed to be horizontal at the elevation of the threshold and has a fixed width of 300 m;
- c) the missed approach surface. This is a sloping surface which:
 - 1) starts at a point 900 m past threshold at threshold elevation;
 - 2) rises at a 2.5 per cent gradient; and
 - 3) splays so as to extend between the transitional surfaces.

It extends with constant splay to the level of the inner horizontal surface. Thereafter, it continues at the same gradient but with a 25 per cent splay until the termination of the precision segment; and

- d) the extended transitional surfaces, which continue longitudinally along the sides of the approach and missed approach surfaces and up to a height of 300 m above threshold elevation.

3.4.7.3 Determination of OCA/H with basic ILS surfaces

3.4.7.3.1 Where the basic ILS surfaces specified in 3.4.7.2 are not penetrated, the OCA/H for Category I and Category II is defined by the margins specified in Table II-1-3-2, and Category III operations are not restricted. Obstacles may be excluded when they are below the transitional surface defined by Annex 14 for runways with code numbers 3 and 4, regardless of the actual runway code number (i.e., the surfaces for code numbers 3 and 4 are used for the obstacle assessment on runways with code numbers 1 and 2).

3.4.7.3.2 If the basic ILS surfaces listed above are penetrated by objects other than those listed in Table II-1-3-3 the OCA/H may be calculated directly by applying height loss/altimeter margins to obstacles (see 3.4.8.8.2, “Calculation of OCA/H values with OAS”).

3.4.7.3.3 The obstacles in Table II-1-3-3 may only be exempted if the following two criteria are met:

- a) the nominal course has the standard width of 210 m (see 3.1.3, “Standard conditions”); and
- b) the MLS Category I decision height is not less than 60 m (200 ft) or the MLS Category II decision height is not less than 30 m (100 ft).

3.4.7.3.4 An object which penetrates any of the basic ILS surfaces and becomes the controlling obstacle, but which must be maintained because of its function with regards to air navigation requirements, may be ignored under certain circumstances in calculating the OCA/H with the following provision. It must be established by the appropriate authority that the portion which penetrates the surface is of minimum mass and frangibly mounted and would not adversely affect the safety of aircraft operations.

- b) the height of the MLS approach reference datum above the nominal value (see 3.4.8.7.4, below);
- c) Category I azimuths having a sector width greater than 210 m at threshold (see 3.4.8.7.5, below);
- d) use of flight director (manually flown) in Cat II (see 3.4.8.7.6, below); and
- e) missed approach climb gradient (see 3.4.8.7.7, below).

3.4.8.7.3 *Specific aircraft dimensions.* An adjustment is mandatory where aircraft dimensions exceed those specified in 3.1.3, “Standard Conditions” and is optional for aircraft with smaller dimensions. The PANS-OPS OAS software adjust the OAS coefficients and template coordinates for the standard dimensions of Category A, B, C, D and D_L aircraft automatically. It will do the same for specific aircraft dimensions in any category. It uses the following correction formula to adjust the coefficient C for the W, X and Y surfaces:

$$\text{W surface: } C_{w\text{corr}} = C_w - (t - 6)$$

$$\text{X surface: } C_{x\text{corr}} = C_x - B_x \cdot P$$

$$\text{Y surface: } C_{y\text{corr}} = C_y - B_y \cdot P$$

where:

$$P = \left[\frac{t}{B_x} \text{ or } S + \frac{t - 3}{B_x}, \text{ whichever is the maximum} \right] - \left[\frac{6}{B_x} \text{ or } 30 + \frac{3}{B_x}, \text{ whichever is the maximum} \right]$$

and: s = semi-span

t = vertical distance between paths of the GP antenna and the lowest part of the wheels.

3.4.8.7.4 *Height of the MLS approach reference datum.* The constants are based on an MLS approach reference datum height (RDH) of 15 m. An adjustment to the OAS constants is mandatory for an RDH less than 15 m, and is optional for an RDH greater than 15 m. The PANS-OPS OAS software adjusts the OAS coefficients and template coordinates by correcting the tabulated values of the coefficient C for the W, X and Y surfaces as follows:

$$C_{\text{corr}} = C + (\text{RDH} - 15)$$

where: C_{corr} = corrected value of coefficient C for the appropriate surface

C = tabulated value.

3.4.8.7.5 *Modification for Cat I azimuths with nominal course width greater than 210 m at threshold.* Where the MLS azimuth sector width at threshold is greater than the nominal value of 210 m, the ILS collision risk model (CRM) method described in 3.4.9 shall be used. Adjustments for sector widths less than 210 m shall not be made and are inhibited in the PANS-OPS OAS software.

3.4.8.7.6 *Use of flight director (manually flown) in Cat II.* The Cat I OAS shall be used.

3.4.8.7.7 *Missed approach gradient.* If equipment is capable of missed approach climb gradients better than the nominal 2.5 per cent, the Y and Z surfaces may be adjusted. This adjustment is achieved by selecting the desired missed approach climb gradient in the PANS-OPS OAS software. The programme then adjusts the Y and Z surface coefficients.

3.4.8.8 *Determination of OCA/H with OAS or basic ILS surfaces*

3.4.8.8.1 *General.* The OCA/H is determined by accounting for all obstacles which penetrate the basic ILS surfaces defined in 3.4.7.2 and the OAS surfaces applicable to the ILS category of operation being considered. The exemptions listed in 3.4.7.3, “Determination of OCA/H with basic ILS surfaces” for obstacles penetrating the basic ILS surfaces may be applied to obstacles penetrating the OAS, providing the criteria listed in that paragraph are met. The surfaces which apply to each MLS category of operations are:

- a) *MLS Cat I:* ILS Cat I OAS;
- b) *MLS Cat II flight director:* ILS Cat I OAS;
- c) *MLS Cat II autocoupled:* ILS Cat II OAS, using flight director and those portions of ILS Cat I which lie above the limits of ILS Cat II; and
- d) *MLS Cat III autocoupled:* Same as MLS Cat II autocoupled.

3.4.8.8.2 *Calculation of OCA/H values with OAS.* Accountable obstacles, as determined below in 3.4.8.8.2.1, “OCA/H calculation steps” are divided into approach and missed approach obstacles. The standard method of categorization is as follows: approach obstacles are those between the FAP and 900 m after threshold; missed approach obstacles are those in the remainder of the precision segment (see Figure II-1-3-17). However, in some cases this categorization of obstacles may produce an excessive penalty for certain missed approach obstacles (see Attachment to Part II, 1.9). Alternatively, missed approach obstacles may therefore be defined as those above a plane surface parallel to the plane of the glide path (elevation angle) and with origin at –900 m (see Figure II-1-3-18), i.e. obstacle height greater than $[(900 + x) \tan \theta]$.

3.4.8.8.2.1 *OCA/H calculation steps*

- a) Determine the height of the highest approach obstacle.
- b) Convert the heights of all missed approach obstacles (h_{ma}) to the heights of equivalent approach obstacles (h_a) by the formula given below, and determine the highest equivalent approach obstacle.
- c) Determine which of the obstacles identified in steps a) and b) is the highest. This is the controlling obstacle.
- d) Add the appropriate aircraft category related margin (Table II-1-3-2) to the height of the controlling obstacle.

$$h_a = \frac{h_{ma} \cot Z + (x_z + x)}{\cot Z + \cot \theta}$$

where: h_a = height of equivalent approach obstacle

h_{ma} = height of missed approach obstacle

θ = angle of glide path (elevation angle)

Z = angle of missed approach surface

x = range of obstacle relative to threshold (negative after threshold)

x_z = distance from threshold to origin of Z surface (900 m (700 m Cat H))

3.4.8.8.3 *Adjustment for high airfield elevations and steep glide path angles*

3.4.8.8.3.1 *Height loss (HL)/altimeter margins.* The margins in Table II-1-3-2 shall be adjusted as follows:

- a) for airfield elevation higher than 900 m (2 953 ft), the tabulated allowances shall be increased by 2 per cent of the radio altimeter margin per 300 m (984 ft) airfield elevation; and
- b) for glide path (elevation angles) greater than 3.2° in exceptional cases, the allowances shall be increased by 5 per cent of the radio altimeter margin per 0.1° increase in glide path (elevation angle) between 3.2° and 3.5°.

3.4.8.8.3.1.1 Procedures involving glide paths greater than 3.5° or any angle when the nominal rate of descent (V_{at} for the aircraft type $\times$ the sine of the glide path angle) exceeds 5 m/sec (1 000 ft/min), are non-standard. They require the following:

- a) increase of height loss margin (which may be aircraft type specific);
- b) adjustment of the origin of the missed approach surface (applicable for fixed-wing aircraft procedures only);
- c) adjustment of the slope of the W surface;
- d) re-survey of obstacles; and
- e) the application of related operational constraints.

Such procedures are normally restricted to specifically approved operators and aircraft, and are associated with appropriate aircraft and crew restrictions. They are not to be used as a means to introduce noise abatement procedures.

3.4.8.8.3.1.2 Appendix B to Chapter 1 shows the procedure design changes required and the related operational/certification considerations.

Example: Aircraft Category C — Aerodrome elevation: 1 650 m above MSL; glide path angle 3.5°.

Tabulated allowances: radio altimeter 22 m

(Table II-1-3-2) pressure altimeter 46 m

Correction for aerodrome elevation:

$$22 \times \frac{2}{100} \times \frac{1\,650}{300} = 2.42 \text{ m}$$

Correction for glide path angle:

$$22 \times \frac{5}{100} \times \frac{3.5 - 3.2}{0.1} = 3.30 \text{ m}$$

Total correction 5.72 m rounded up to 6 m.

Corrected radio altimeter margin $22 + 6 = 28$ m.

Corrected pressure altimeter margin $46 + 6 = 52$ m.

3.4.8.8.3.2 *Exceptions and adjustments to values in Table II-1-3-2.* Values in Table II-1-3-2 are calculated to account for an aircraft using normal manual overshoot procedures from OCA/H on the nominal approach path. The values

in Table II-1-3-2 do not apply to Cat III operations. The values do not consider the lateral displacement of an obstacle nor the probability of an aircraft being so displaced. If consideration of these joint probabilities is required, then the ILS CRM discussed in 3.4.9 shall be used. Values in Table II-1-3-2 may be adjusted for specific aircraft types where adequate flight and theoretical evidence is available, i.e. the height loss value corresponding to a probability of 1×10^{-5} (based on a missed approach rate of 10^{-2}).

3.4.8.8.3.3 Radio altimeter verification. If the radio altimeter OCA/H is promulgated, operational checks shall have confirmed the repeatability of radio altimeter information.

3.4.8.8.3.4 Height loss (HL)/altimeter margins for a specific speed at threshold. If a height loss/altimeter margin is required for a specific V_{at} , the following formulae apply (see also Table II-1-3-4):

Use of radio altimeter:

Margin = $(0.096 V_{at} - 3.2)$ metres where V_{at} in km/h

Margin = $(0.177 V_{at} - 3.2)$ metres where V_{at} in kt

Use of pressure altimeter:

Margin = $(0.068 V_{at} + 28.3)$ metres where V_{at} in km/h

Margin = $(0.125 V_{at} + 28.3)$ metres where V_{at} in kt

where V_{at} is the speed at threshold based on 1.3 times stall speed in the landing configuration at maximum certificated landing mass.

Note.— The equations assume the aerodynamic and dynamic characteristics of the aircraft are directly related to the speed category. Thus, the calculated height loss/altimeter margins may not realistically represent small aircraft with V_{at} at maximum landing mass exceeding 165 kt.

3.4.8.9 Effect of obstacle density on OCA/H. To assess the acceptability of obstacle density below the OAS, the ILS CRM described in 3.4.9 may be used. This can provide assistance by comparing aerodrome environments and by assessing risk levels associated with given OCA/H values. It is emphasized that it is not a substitute for operational judgement.

3.4.9 Obstacle clearance in the precision segment — application of the ILS collision risk model (CRM) to MLS operations

3.4.9.1 General. The ILS CRM is a computer program that establishes the numerical risk which can be compared to the target level of safety for aircraft operating to a specified OCA/H height. A description of the programme and instructions on its use, including the precise format of both the data required as input and the output results, are given in the *Manual on the Use of the Collision Risk Model (CRM) for ILS Operations* (Doc 9274).

3.4.9.2 Input. When applied to MLS operations, the ILS CRM requires the following data as input:

- a) *aerodrome details:* name, runway threshold position and runway orientation in map grid coordinates (optional), threshold elevation above MSL;
- b) *MLS parameters:* category (the appropriate ILS category as defined in 3.4.8.8.1), glide path (elevation angle), azimuth-threshold distance, azimuth nominal course width, height of MLS reference datum above threshold;

- c) *missed approach parameters*: decision height (obstacle clearance height) and missed approach turn point;
- d) *aircraft parameters*: type, wheel height (antenna to bottom of wheel), and wing semi-span, aircraft category (A, B, C, D or D_L) missed approach climb gradient; and

Note.— The CRM does not consider Category E aircraft.

- e) *obstacle data*: obstacle boundaries (either as x and y coordinates relative to the runway threshold or as map grid coordinates) and obstacle height (either above threshold elevation or above MSL). For density assessment, all obstacles penetrating the basic ILS surfaces described in 3.4.7.2 must be included.

3.4.9.3 *Output and application.* The output of the programme is:

- a) the overall (total) risk of collision with obstacles for aircraft operating to a specified OCA/H; and
- b) the minimum OCA/H which will provide the target level of safety.

These options are detailed in Doc 9274, *Manual on the Use of the Collision Risk Model (CRM) for ILS Operations*. The user, by rerunning the ILS CRM with the appropriate parameters, can assess the effect on the safety of operations of any alteration in the parameters, typically varying the glide path, elevation angle or remaining obstacles.

3.5 MISSED APPROACH SEGMENT

3.5.1 General

3.5.1.1 The criteria for the final missed approach are based on those for the general criteria (see Part I, Section 4, Chapter 6). Certain modifications have been made to allow for the different areas and surfaces associated with the MLS precision segment and for the possible variation in OCA/H for that segment with aircraft category. Area construction is according to the navigation system specified for the missed approach.

3.5.1.2 The datum used for calculation of distances and gradients in obstacle clearance calculations is termed “start of climb” (SOC). It is defined by the height and range at which the plane GP’ — a plane parallel with the glide path (elevation angle) and with origin at –900 m at threshold level — reaches an altitude OCA/H – HL (OCA/H and HL must both relate to the same category of aircraft).

3.5.1.3 Where obstacles identified in the final missed approach segment result in an increase in any of the OCA/H calculated for the precision segment, a higher gradient of the missed approach surface (Z) may be specified in addition if this will provide clearance over those obstacles at a specified lower OCA/H (see Part I, Section 4, Chapter 6, 6.2.2.1, “Climb gradient in the intermediate phase”).

3.5.1.4 Missed approach segment using RNAV or RNP systems for track guidance can be utilized. Only the systems capable of navigation accuracy of 1 NM or better in this phase of flight can be considered. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for missed approach.

3.5.2 Straight missed approach

3.5.2.1 *General.* The precision segment terminates at the point where the Z surface reaches a height 300 m above threshold. The width of the Z surface at that distance defines the initial width of the final missed approach area which splays at an angle of 15 degrees from that point, as shown in Figure II-1-3-19. There are no secondary areas.

3.5.2.2 *Straight missed approach obstacle clearance.* (See Figure II-1-3-20.) Obstacle elevation/height in this final missed approach area shall be less than

$$(OCA/H_{ps} - HL) + d_o \tan Z$$

where:

- a) OCA/H of the precision segment (OCA/H_{ps}) and HL (Table II-1-3-2 value) both relate to the same aircraft category;
- b) d_o is measured from SOC parallel to the straight missed approach track; and
- c) Z is the angle of the missed approach surface with the horizontal plane.

If this requirement cannot be met, a turn shall be prescribed to avoid the obstacle in question. If a turn is not practical, the OCA/H shall be raised.

3.5.2.3 Transition from conventional navigation to an RNAV or RNP missed approach may be defined by an RNAV or RNP fix located on the extended azimuth course, or by a turn at an altitude direct to a waypoint (see Figure II-1-3-25). If the RNAV or RNP track is collinear with the azimuth course, the area abeam the SOC shall be expanded at a 15-degree angle from the track course until it reaches the applicable width of the RNAV or RNP area constructed on the extended azimuth course. In the region between the lateral boundaries of the Z surface and the outer boundaries of the area, the obstacle evaluation shall be based on the Y surface and a mathematical extrapolation of the Y surface where the area is outside the lateral boundaries of the OAS 300 m contour. The Z surface shall continue to splay at the same angle until reaching the width of the RNAV or RNP area. Secondary areas shall apply from the point where the width of the Z surface reaches the width of the RNAV or RNP primary area (see Figure II-1-3-23). Obstacle clearance up to this point for the extended Y and Z surfaces shall be the same as in the precision segment (see 1.4.8 “Obstacle clearance of the precision segment using obstacle assessment surface (OAS) criteria”), and this shall also apply further out to all portions of the Z surface that are within the RNAV or RNP primary area. The principle of secondary areas shall apply between the edge of the RNAV or RNP primary area and the edge of the total area. If an obstacle penetrates either the extended Y or the Z surface within the secondary area, its elevation/height shall be less than:

$$(OCA/H_{ps} - HL) + d_o \tan Z + M$$

where:

- a) OCA/H of the precision segment (OCA/H_{ps}) and HL (Table II-1-3-2 value) both relate to the same aircraft category;
- b) d_o is measured from SOC parallel to the straight missed approach track;
- c) Z is the angle of the missed approach surface with the horizontal plane; and
- d) M is zero at the edge of the primary area increasing linearly to 30 m (98 ft) at the edge of the total area.

3.5.2.4 If the penetration in the secondary area is less than “M” as described in 3.5.2.3 d) above, then it may be ignored in OCA/H calculations.

3.5.3.2.4.2 *Safeguarding of early turns.* Where the published procedure does not specify a fix to limit turns for aircraft executing a missed approach from above the designated turn altitude/height, an additional check of obstacles shall be made. The general criteria of Part I, Section 4, Chapter 6, 6.4.5.6, “Safeguarding of early turns” and general principles of Part I, Section 4, Chapter 6, Figure I-4-6-14 apply with the following modifications:

- a) the limit of the final approach area is replaced by the line DD” of the OAS surfaces and its extension;
- b) the FAF is replaced by the FAP;
- c) the earliest MAPt is replaced by the line D”D” (earliest limit of the turn initiation area); and
- d) if the criterion cannot be met, then the procedure must prohibit turns before a point equivalent to the MAPt and a note must be added on the profile view of the approach chart.

3.5.3.3 *Turn at a designated TP with earliest TP before normal termination of precision segment*

3.5.3.3.1 Where a turn is specified at a designated TP, and the earliest TP is before the normal termination range of the precision segment, the precision segment terminates at the earliest TP. This allows the calculation of OCA/H_{ps} and $(OCA/H_{ps} - HL)$; SOC is then determined.

3.5.3.3.2 *Turn area.* The turn area is constructed as specified in Part I, Section 4, Chapter 6, 6.4.6.3, “Construction of the turn area” except that it is based on the width of the 300 m OAS Y surface contours at the earliest and latest TP (see Figure II-1-3-22).

3.5.3.3.3 *Obstacle clearance.* Obstacle elevation/height shall be less than:

$$(OCA/H_{ps} - HL) + d_o \tan Z - MOC$$

where:

d_o = d_z + shortest distance from obstacle to line K-K,

d_z = horizontal distance from SOC to the earliest TP,

and MOC is:

50 m (164 ft) for turns more than 15° and

30 m (98 ft) for turns 15° or less.

If the obstacle elevation/height exceeds this value, the OCA/H must be increased, or the TP moved to obtain the required clearance (see Appendix A to Chapter 1).

3.6 PROMULGATION

3.6.1 General

3.6.1.1 The general criteria in Part I, Section 2, Chapter 1, 1.12, “Promulgation” apply as amplified by criteria in Part III, Section 5, Chapter 1, 1.3.4 for chart notes. The instrument approach chart for an MLS approach procedure shall be identified by the title MLS Rwy XX. If CAT II and/or III minima are included on the chart, the title shall read MLS Rwy XX CAT II or MLS Rwy XX CAT II & III, as appropriate. If more than one MLS procedure is published for the same runway, the Duplicate Procedure Title convention shall be applied. An example where the convention is applied is two MLS procedures with different segments, such as different missed approach segments.

3.6.1.2 When a final approach fix is identified at the FAP, a warning shall be appended to the procedure stating that descent on the glidepath below the FAF altitude is not permitted until passing the FAF.

3.6.1.3 When a PBN route with an RF turn to the azimuth course is utilized in the intermediate approach segment, then the last item of the PBN procedure description shall be the waypoint at the end of the RF turn with an altitude ‘window’ constraint, or an ‘at’ altitude constraint (see the *Note* to 3.3.6.3).

3.6.2 Promulgation of OCA/H values

3.6.2.1 Promulgation of OCA/H for MLS Cat I and II approach procedures

3.6.2.1.1 The OCA or OCH values, as appropriate, shall be promulgated for those categories of aircraft for which the procedure is designed. The values shall be based on the following standard conditions:

- a) Cat I flown with pressure altimeter;
- b) Cat II flown autocoupled with radio altimeter;
- c) Cat II flown with radio altimeter and flight director;
- d) standard aircraft dimensions (see 3.1.3, “Standard conditions”); and
- e) 2.5 per cent missed approach climb gradient.

3.6.2.1.2 Additional values of OCA/H may be agreed upon between operators and the appropriate authority and promulgated, on the basis of evidence supporting the modifications defined in 3.4.8.7, “Adjustment of OAS constants”.

3.6.2.1.3 Use of OCA/H values for MLS Category I approach procedures based on radio altimeter height loss margins may be agreed upon between operators and the appropriate authority, and the values promulgated, if the requirement of 3.4.8.3.3, “Radio altimeter verification” is met.

3.6.2.2 Promulgation of MLS Category III approach procedures

Category III operations may be permitted subject to the appropriate Category II OCA/H being below the height of the Annex 14 inner horizontal surface. Category III operations may also be permitted with a Category II OCA/H between the height of the inner horizontal surface and 60 m provided the Annex 14 Category II inner approach, inner transitional and balked landing surfaces are extended to protect that OCA/H (see Figure II-1-3-10 a)).

3.6.3 Degrees magnetic

The final approach track for the MLS procedure is always specified and promulgated in degrees magnetic.

3.6.4 Turn at a designated altitude/height (missed approach)

If the TP is located at the SOC, the chart shall be annotated “turn as soon as practicable to ... (heading or facility)” and shall include sufficient information to identify the position and height of the obstacles dictating the turn requirement.

3.6.5 Turn at a designated TP (missed approach)

Where the procedure requires that a turn be executed at a designated TP, the following information must be published with the procedure:

- a) the TP, when it is designated by a fix; or
- b) the intersecting VOR radial, NDB bearing, or DME distance where there is no track guidance (see Part I, Section 2, Chapter 2, 2.6.5, “Missed approach fixes”).

3.6.6 Procedures involving non-standard glide path angles

Procedures involving glide paths greater than 3.5° or any angle when the nominal rate of descent exceeds 5 m/sec (1 000 ft/min), are non-standard and subject to restrictions (see 3.4.8.8.3.1, “Height loss (HL)/altimeter margins”). They are normally restricted to specifically approved operators and aircraft, and are promulgated with appropriate aircraft and crew restrictions annotated on the approach chart.

3.6.7 Additional gradient for the final missed approach segment

If obstacles identified in the final missed approach segment result in an increase in any of the OCA/H calculated for the precision segment, an additional steeper gradient may also be specified for the gradient of the missed approach surface (Z) for the purposes of lowering the OCA/H (see Part I, Section 4, Chapter 6, 6.2.3.1, “Climb gradient in the final phase”).

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Table II-1-3-1. Minimum length of intermediate segment

<i>Intercept angle with the final approach track (degree)</i>	<i>Minimum distance between the interception of the final approach track and the interception of the glide path</i>	
	<i>Cat A/B</i>	<i>Cat C/D/E</i>
≤ 0-15 or RF turn to the final approach track	2.8 km (1.5 NM)	2.8 km (1.5 NM)
16 – 30	3.7 km (2.0 NM)	3.7 km (2.0 NM)
31 – 60	3.7 km (2.0 NM)	4.6 km (2.5 NM)
61 – 90 or within a racetrack or reversal procedure	3.7 km (2.0 NM)	5.6 km (3.0 NM)

Table II-1-3-2. Height loss/altimeter margin

<i>Aircraft category (Vat)</i>	<i>Margin using radio altimeter</i>		<i>Margin using pressure altimeter</i>	
	<i>Metres</i>	<i>Feet</i>	<i>Metres</i>	<i>Feet</i>
A — 169 km/h (90 kt)	13	42	40	130
B — 223 km/h (120 kt)	18	59	43	142
C — 260 km/h (140 kt)	22	71	46	150
D — 306 km/h (165 kt)	26	85	49	161

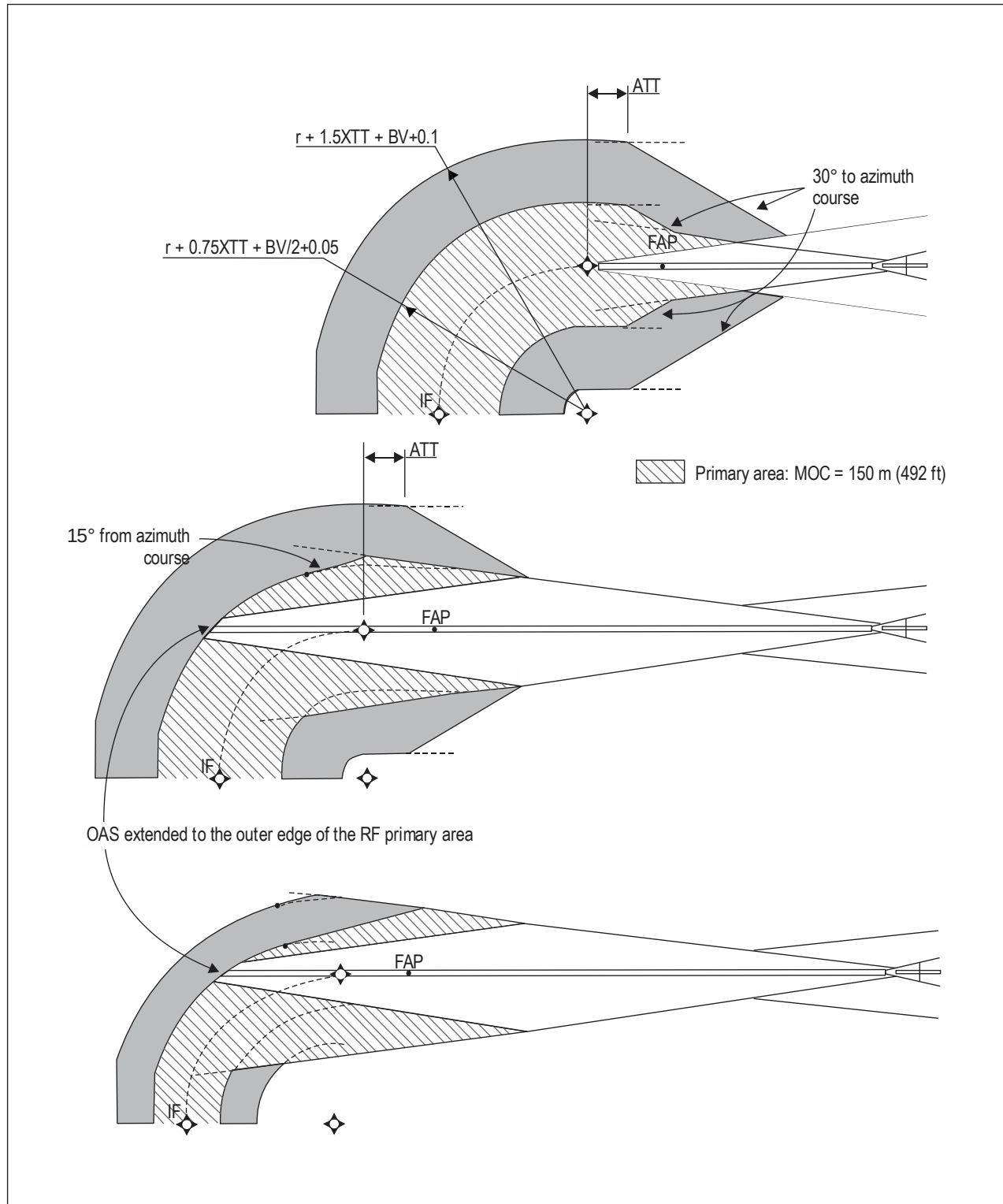


Figure II-1-3-8. Area construction for radius-to-fix turns to the azimuth course

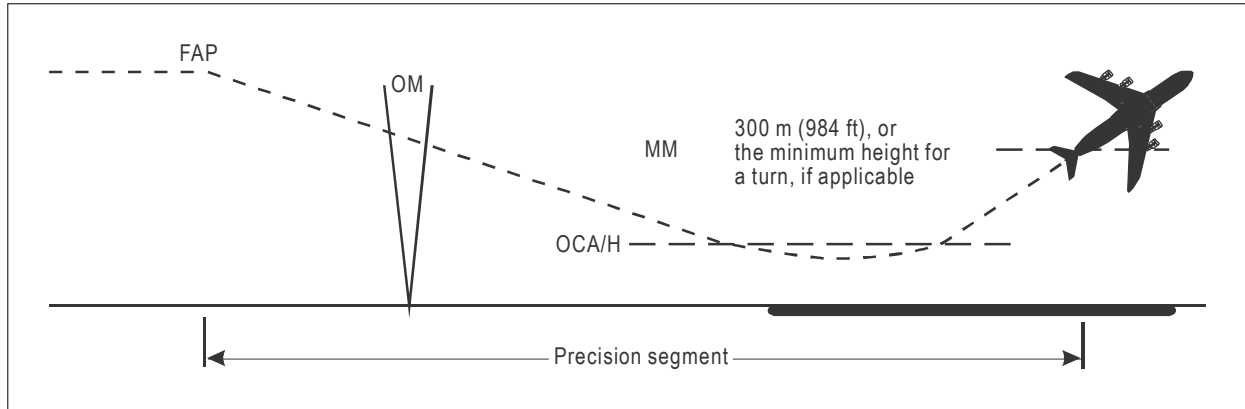


Figure II-1-3-9. Precision segment

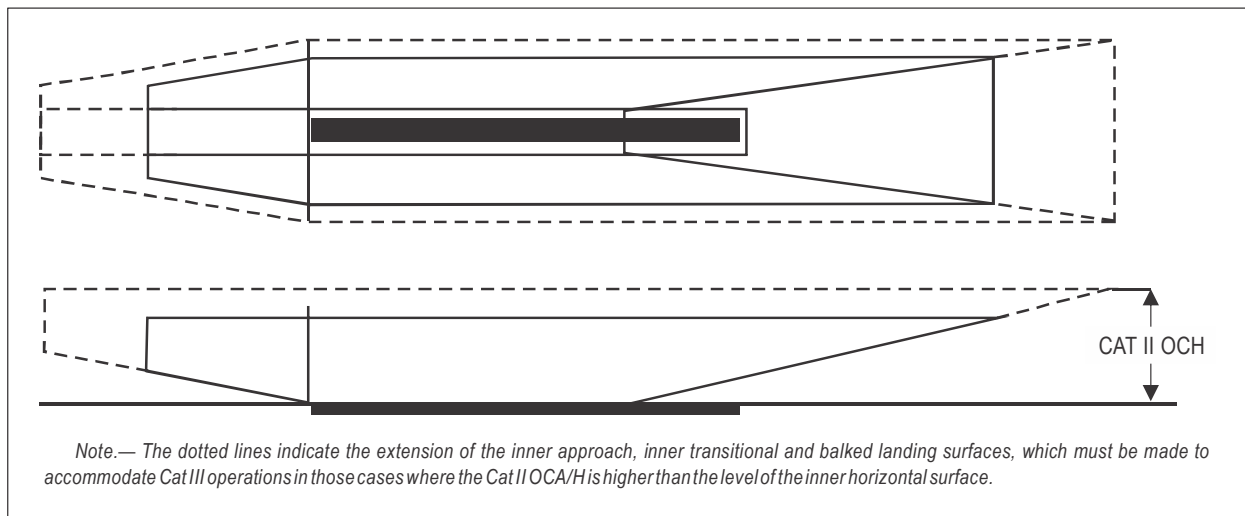


Figure II-1-3-10 a). Annex 14 inner approach, inner transitional and balked landing surfaces as applied for obstacle assessment for Cat II and III ILS operations

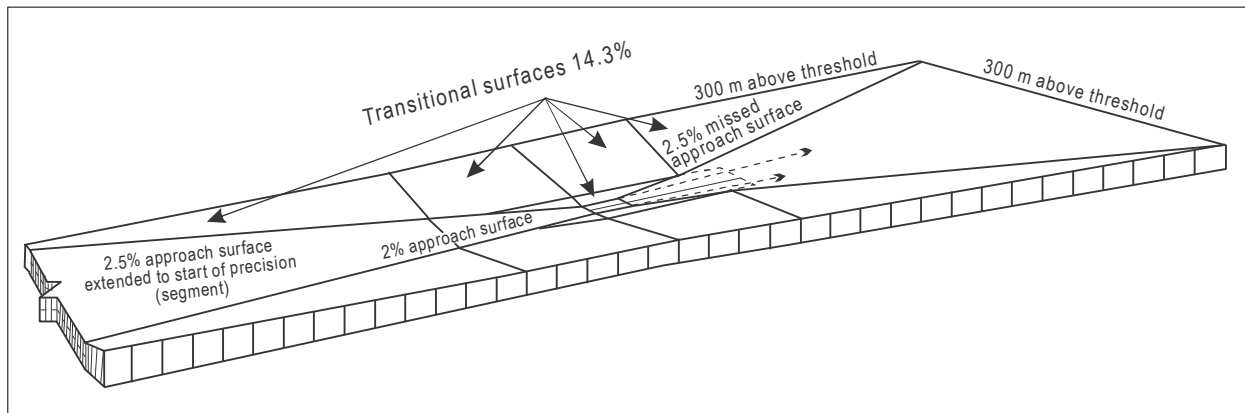


Figure II-1-3-10 b). Illustration of basic ILS surfaces as described in 3.4.7.2

Chapter 1

RNAV CONCEPTS

1.1 GENERAL

1.1.1 The chapters in this section provide the components (XTT, ATT and area semi-width) which are required for the construction of instrument flight procedures, detailed in Sections 2 and 3. It should be noted that this does not apply to GLS procedures and to procedures based on SBAS, as the new error components for such procedures are considered equivalent to the ILS approach, which are angular in nature.

1.1.2 Performance-based navigation (PBN) is defined as a type of area navigation (RNAV) in which the navigation performance requirements are prescribed in navigation specifications. A navigation specification is defined as a set of aircraft and aircrew requirements needed to support PBN operations within a defined airspace.

1.1.3 There are two types of navigation specifications:

- a) *RNAV specification*. A navigation specification designation that does not include requirements for on-board performance monitoring and alerting.
- b) *RNP specification*. A navigation specification designation that includes requirements for on-board performance monitoring and alerting.

1.1.4 The ICAO *Performance-based Navigation (PBN) Manual* (Doc 9613) provides a detailed explanation of the PBN concept and guidance on how to implement PBN applications as well as navigation specifications for the following applications (see also Table III-1-1-1):

- a) *RNAV 10*: used to support RNAV operations in the en-route phase of flight to support longitudinal distance-based separation minima in oceanic or remote area airspace. Procedure design criteria have not been developed.
- b) *RNAV 5*: used to support RNAV operations in the en-route phase of flight for continental airspace. Obstacle clearance criteria are detailed in Part III, Section 1, Chapters 2, 3 and 4, and Section 3, Chapters 7 and 8.
- c) *RNAV 1 and 2*: used to support RNAV operations in the en-route phase of flight, on SIDs, on STARs and on approaches outside the final approach or precision segment. Obstacle clearance criteria are detailed in Part III, Section 1, Chapters 2 and 3, and Section 3, Chapters 1, 2, 3, 7 and 8.

Note.— The criteria are also applicable to regional/national navigation specifications published prior to RNAV 1, such as P-RNAV.

- d) *RNP 4*: used to support RNAV operations in the en-route phase of flight to support longitudinal distance-based separation minima in oceanic or remote area airspace. Obstacle clearance criteria are detailed in Part III, Section 1, Chapter 2, and Section 3, Chapter 8.

- e) *RNP 2*: used to support RNP operations in the en-route phase of flight in oceanic, remote area and continental airspace. Obstacle clearance criteria are detailed in Part III, Section 1, Chapter 2, and Section 3, Chapter 8.
- f) *RNP 1*: used to support RNP operations on SIDs, on STARs and on approaches up to the FAF/FAP outside the final approach or precision segment with no, or limited, ATS surveillance and with low to medium density traffic. Obstacle clearance criteria are detailed in Part III, Section 1, Chapter 2, and Section 3, Chapters 1, 2, 7 and 8.
- g) *Advanced RNP (ARNP)*: used to support RNP operations in en-route continental airspace and on SIDs, STARs and approach procedures. ARNP requirements include the following values for navigation accuracy: 0.3 NM in final approach; 1 or 2 NM in en-route continental; and navigational accuracy of 1 NM in SIDs, STARs, initial/intermediate approaches and missed approaches. Optional requirements include applications for oceanic/remote airspace and an incremental option which allows any navigational accuracy value between 1.0 NM and 0.3 NM, in 0.1NM increments, in all terminal flight phases outside final approach. Obstacle clearance criteria are detailed in Part III, Section 1, Chapter 2, and Section 3, Chapters 1, 2, 3, 4, 7 and 8.
- h) *RNP 0.3*: used to support helicopter RNP operations in all phases of flight except final approach. Obstacle clearance criteria are detailed in Part III, Section 1, Chapter 2 and Part IV.
- i) *RNP APCH*: used to support RNP approach operations to LNAV, LNAV/VNAV, LP and LPV minima. Obstacle clearance criteria are detailed in Part III, Section 1, Chapter 2, and Section 3, Chapters 2, 3, 4, 5 and 7.
- j) *RNP AR APCH*: used to support RNP approach operations, including the final approach segment, which consist of straight and/or fixed radius segments, with navigation accuracies equal to or less than 0.3 NM on final and 1 NM on the other approach segments.

Note 1.— The criteria to develop RNP AR approach procedures are detailed in the Required Navigation Performance Authorization Required (RNP AR) Procedure Design Manual (Doc 9905).

Note 2.— The navigation specifications, together with associated appendices and attachments, in the ICAO Performance-based Navigation (PBN) Manual (Doc 9613) also contain details of specific navigation functions that may be optional or mandatory, depending upon the navigation specification:

- a) *All navigation specifications intended for terminal and approach procedures require the ability to execute leg transitions and maintain tracks consistent with specific ARINC 424 path terminators. In particular, the ability to fly constant radius turns in terminal airspace, using the RF path terminator, is addressed in the RNP AR APCH navigation specification and in Doc 9613, Volume II, Part C, Appendix 1. The RF path terminator functionality is required in the ARNP navigation specification and is an optional function in the RNP 1, RNP 0.3 and RNP APCH navigation specifications. Its application in RNP AR APCH operations is addressed in Doc 9905.*
- b) *The ability to fly constant radius turns in the en-route phase of flight is addressed in Doc 9613, Volume II, Part C, Appendix 2. Fixed radius transitions (FRT) are associated with individual waypoints on an en-route structure and may be used in ARNP and RNP 2 applications.*

Note.— Guidance on how to employ FRT is still being developed and once it is mature, design criteria for FRT will be included in PANS-OPS.

- c) *The use of barometric altitude and RNAV information in the definition of vertical flight paths and vertical guidance in relation to a path, known as Baro-VNAV, is addressed in Doc 9613, Volume II, Attachment A.*

Table III-1-2-5. XTT, ATT and area semi-width for RNP-1 in arrival and departure phases of flight (m)

<i>STAR/SID/missed approach (>56 km ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<56 km ARP)</i>			<i>SID/missed approach (<28 km ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W
1 852	1 482	6 482	1 852	1 482	4 630	1 852	1 482	3 704

Table III-1-2-6. XTT, ATT and area semi-width for RNP-1 (aeroplane) in arrival and departure phases of flight (NM)

<i>STAR/SID/missed approach (>30 NM ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<30 NM ARP)</i>			<i>SID/missed approach (<15 NM ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W
1.00	0.80	3.50	1.00	0.80	2.50	1.00	0.80	2.00

Table III-1-2-7. XTT, ATT and area semi-width for RNP-1 (CAT H) in arrival and departure phases of flight (m)

<i>STAR/SID/missed approach (>56 km ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<56 km ARP)</i>			<i>SID/missed approach (<28 km ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W
1 852	1 482	4 630	1 852	1 482	4 074	1 852	1 482	3 426

Table III-1-2-8. XTT, ATT and area semi-width for RNP 1 (CAT H) in arrival and departure phases of flight (NM)

<i>STAR/SID/missed approach (>30 NM ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<30 NM ARP)</i>			<i>SID/missed approach (<15 NM ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ A/W
1.00	0.80	2.50	1.00	0.80	2.20	1.00	0.80	1.85

Table III-1-2-9. XTT, ATT and area semi-width for Advanced RNP in all phases of flight (aeroplane) (m)

RNP	En-route (Continental & Remote)			STAR/SID (>56 km ARP)			STAR/SID/IF/IAF/ Missed Approach (<56 km ARP)			FAF			MAPt			Missed Approach/ SID (<28 km ARP)		
	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W
2	3704	2964	9260	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1	1852	1482	6482	1852	1482	6482	1852	1482	4630	—	—	—	—	—	—	1852	1482	3704
0.3	—	—	—	—	—	—	—	—	—	556	444	2685	556	444	1759	—	—	—

Table III-1-2-10. XTT, ATT and area semi-width for Advanced RNP in all phases of flight (Aeroplane) (NM)

RNP	En-route (Continental & Remote)			STAR/SID (>30 NM ARP)			STAR/SID (<30 NM ARP) IAF/IF/Missed Approach			FAF			MAPt			Missed Approach/ SID (<15 NM ARP)		
	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W
2	2	1.6	5.0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1	1	0.8	3.5	1	0.8	3.5	1	0.8	2.5	—	—	—	—	—	—	1	0.8	2
0.3	—	—	—	—	—	—	—	—	—	0.3	0.24	1.45	0.3	0.24	0.95	—	—	—

Table III-1-2-11. XTT, ATT and area semi-width for RNP 0.3 (m) (CAT H)

En-route/STAR/SID (>56 km ARP)			STAR/SID/IF/IAF/Missed Approach (≤56 km ARP)			SID/Missed Approach (<28 km ARP)		
XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W
556	444	2 685	556	444	2 130	556	444	1 482

Table III-1-2-12. XTT, ATT and area semi-width for RNP 0.3 (NM) (CAT h)

En-route/STAR/SID (>30 NM ARP)			STAR/SID/IF/IAF/Missed Approach (≤30 NM ARP)			SID/Missed Approach (<15 NM ARP)		
XTT	ATT	½ A/W	XTT	ATT	½ A/W	XTT	ATT	½ A/W
0.30	0.24	1.45	0.30	0.24	1.15	0.30	0.24	0.80

Table III-1-2-13. XTT, ATT and area semi-width for RNP APCH (aeroplane) in initial/intermediate/final approach and missed approach phases of flight (m)

<i>IF/IAF/missed approach (<56 km ARP)</i>			<i>FAF</i>			<i>MAPt/Initial Straight Missed Approach (LP/LPV only)</i>			<i>Missed approach (<28 km ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>
1 852	1 482	4 630	556	444	2 685	556	444	1 759	1 852	1 482	3 704

Table III-1-2-14. XTT and ATT, area semi-width for RNP APCH (aeroplane) in initial/intermediate/final approach and missed approach phases of flight (NM)

<i>IF/IAF/missed approach (<30 NM ARP)</i>			<i>FAF</i>			<i>MAPt/Initial Straight Missed Approach (LP/LPV only)</i>			<i>Missed approach (<15 NM ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>
1.00	0.80	2.50	0.30	0.24	1.45	0.30	0.24	0.95	1.00	0.80	2.00

Table III-1-2-15. XTT, ATT and area semi-width for RNP APCH (CAT H) in initial/intermediate/final approach and missed approach phases of flight (m)

<i>IF/IAF/missed approach (<56 km ARP)</i>			<i>FAF</i>			<i>MAPt/Initial Straight Missed Approach (LP/LPV only)</i>			<i>Missed approach (<28 km ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>
1 852	1 482	4 074	556	444	2 130	556	444	1 482	1 852	1 482	3 426

Table III-1-2-16. XTT, ATT and area semi-width for RNP APCH (CAT H) in en-route, arrival, initial/intermediate/final approach and missed approach phases of flight (NM)

<i>IF/IAF/missed approach (<30 NM ARP)</i>			<i>FAF</i>			<i>MAPt/Initial Straight Missed Approach (LP/LPV only)</i>			<i>Missed approach (<15 NM ARP)</i>		
<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>	<i>XTT</i>	<i>ATT</i>	$\frac{1}{2}$ <i>A/W</i>
1.00	0.80	2.20	0.30	0.24	1.15	0.30	0.24	0.80	1.00	0.80	1.85

Table III-1-2-17. XTT, ATT and area semi-width for RNAV 1 and RNAV 2 (Cat A to E) in en-route, arrival, initial/intermediate approach and departure phases of flight (m)

<i>En-route/STAR/SID/missed approach (>56 km ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<56 km ARP)</i>			<i>SID/missed approach (<28 km ARP)</i>		
<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>
3 704	2 963	9 260	1 852	1 482	4 630	1 852	1 482	3 704

Table III-1-2-18. XTT, ATT and area semi-width for RNAV 1 and RNAV 2 (Cat A to E) in en-route, arrival, initial/intermediate approach and departure phases of flight (NM)

<i>En-route/STAR/SID/missed approach (>30 NM ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<30 NM ARP)</i>			<i>SID/missed approach (<15 NM ARP)</i>		
<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>
2.00	1.60	5.00	1.00	0.80	2.50	1.00	0.80	2.00

Table III-1-2-19. XTT, ATT and area semi-width for RNAV 1 and RNAV 2 (Cat H) in en-route, arrival, initial/intermediate approach and departure phases of flight (m)

<i>En-route/STAR/SID/missed approach (>56 km ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<56 km ARP)</i>			<i>SID/missed approach (<28 km ARP)</i>		
<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>
3 700	2 960	7 400	1 850	1 480	4 070	1 850	1 480	3 420

Table III-1-2-20. XTT, ATT and area semi-width for RNAV 1 and RNAV 2 (Cat H) in en-route, arrival, initial/intermediate approach and departure phases of flight (NM)

<i>En-route/STAR/SID/missed approach (>30 NM ARP)</i>			<i>STAR/IF/IAF/SID/missed approach (<30 NM ARP)</i>			<i>SID/missed approach (<15 NM ARP)</i>		
<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>	<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>
2.00	1.60	4.00	1.00	0.80	2.20	1.00	0.80	1.85

Table III-1-2-21. XTT, ATT and area semi-width for RNAV 5 in the en-route phase of flight (m)

<i>En-route/STAR/SID (>56 km ARP)</i>		
<i>XTT</i>	<i>ATT</i>	<i>½ A/W</i>
4 650	3 720	10 690

Chapter 3

PBN T- OR Y-BAR PROCEDURE CONSTRUCTION

3.1 GENERAL CONCEPT

3.1.1 *Introduction.* An approach procedure based on the PBN navigation specification RNP APCH incorporating a T- or Y-bar arrangement is based in principle on a runway aligned final segment preceded by an intermediate segment and up to three initial segments arranged either side of and along the final approach track to form a T or a Y (see Figure III-2-3-1 and Figure III-2-3-2). For final approach tracks not aligned with the runway centre line, the general criteria in Part I apply.

3.1.2 *Capture region.* A T- or Y-bar arrangement permits direct entry to the procedure from any direction, provided entry is made from within the capture region associated with an IAF. A capture region is defined in terms of an included angle at the IAF (see Figure III-2-3-1 and Figure III-2-3-2).

3.1.3 The lateral initial segments are based on course differences of 70° to 90° from the intermediate segment track. This arrangement ensures that entry from within a capture region requires a change of course at the IAF not greater than 110°.

3.1.4 The central initial segment may commence at the IF.

3.1.5 Where one or both offset IAFs are not provided, a direct entry will not be available from all directions. In such cases, a holding pattern should be provided at the IAF to enable entry to the procedure via a holding pattern entry procedure.

3.1.6 Terminal Arrival Altitudes (TAAs) may be provided to facilitate descent and entry to the procedure. (See Chapter 4.)

3.1.7 The IAF, IF and FAF shall be fly-by waypoints. The missed approach segment starts with a flyover waypoint (MAPt) and ends at a missed approach holding fix (MAHF). For turning missed approaches, a missed approach turning fix (MATF) may also be established to define the turn point.

3.1.8 Area widths are determined in accordance with the tolerances applicable to the PBN navigation specification associated with the procedure.

3.2 INITIAL APPROACH SEGMENT

3.2.1 *Alignment.* Offset IAFs shall be located such that a course change of 70° to 90° is required at the IF. The capture region for tracks inbound to the offset IAF shall extend 180° about the IAFs, providing a direct entry when the course change at the IF is 70° or more. The central IAF should normally be aligned with the intermediate segment. Its capture region shall be 70° to 90° either side of the initial segment track, the angle being identical to the course change at the IF for the corresponding offset IAF. (See Figure III-2-3-1 and Figure III-2-3-2). For turns greater than 110° at the IAFs, a holding/racetrack procedure with Sector 1 or 2 entries should be used (see Figure III-2-3-3).

3.2.2 *Length.* The initial approach segment length shall be according to general criteria in Part III, Section 3, Chapter 2.

3.2.3 *Descent gradient.* The initial approach segment descent gradient shall be according to general criteria in Part III, Section 3, Chapter 2.

3.2.4 *Calculation of track distance (TRD).* The TRD between two fly-by waypoints is defined as the segment length reduced by the stabilization distance at both turns ($r \tan \theta/2$) and increased by the distance flown in the turn from abeam the waypoint to the tangent point ($2\pi r \times 0.5\theta/360$).

$$\text{TRD} = \text{segment length} - r(\tan \theta_1/2 + \tan \theta_2/2) + \pi r (\theta_1 + \theta_2)/360$$

where:

- θ_1 = turn angle (degrees) at the beginning of the segment
- θ_2 = turn angle (degrees) at the end of the segment
- r = turn radius at 25° bank angle

Example for a first 110° turn and a second 70° turn:

$$\text{TRD} = \text{segment length} - 0.56r$$

3.2.5 *Procedure entry altitude.* The procedure is entered at the 46 km (25 NM) minimum sector altitude, terminal arrival altitude or via a designated standard instrument arrival (STAR). Where the initial approach waypoint forms part of an air route, the procedure shall be entered not lower than at the minimum en-route altitude applicable to the route segment.

3.2.6 *Reversal procedures.* When all three initial segment legs are implemented there is no need for reversal procedures. Should one of the legs not be implemented, a racetrack pattern should be established at either or both of the existing IAFs. In the event that the central IAF leg is one of the remaining legs, its capture region shall be adjusted to accommodate normal sector entries into a reversal procedure (see Figure III-2-3-3).

3.2.7 *Holding.* A holding pattern may be provided at any IAF and should be aligned with the initial segment track.

3.3 INTERMEDIATE APPROACH SEGMENT

3.3.1 *Alignment.* The intermediate approach segment alignment shall be according to the general criteria in Part III, Section 3, Chapter 2.

3.3.2 *Length.* The intermediate segment length shall be according to the general criteria in Part III, Section 3, Chapter 2.

3.3.3 *Descent gradient.* The general criteria at Part I, Section 4, Chapter 4, 4.3.3, “Procedure altitude/height and descent gradient” apply. Where a descent is required, the descent gradient shall be calculated for the shortest possible track distance for the fastest category of aircraft, and not the segment length. (For calculation of TRD see 3.2.4).

3.3.4 Where a track change occurs at the FAF, the reduction in track distance may be ignored as the difference is negligible. (Maximum angle of turn is 30°.)

3.4 FINAL APPROACH SEGMENT

3.4.1 *Alignment.* The optimum alignment of the final approach segment is the runway centre line. If this is not possible, the general criteria apply.

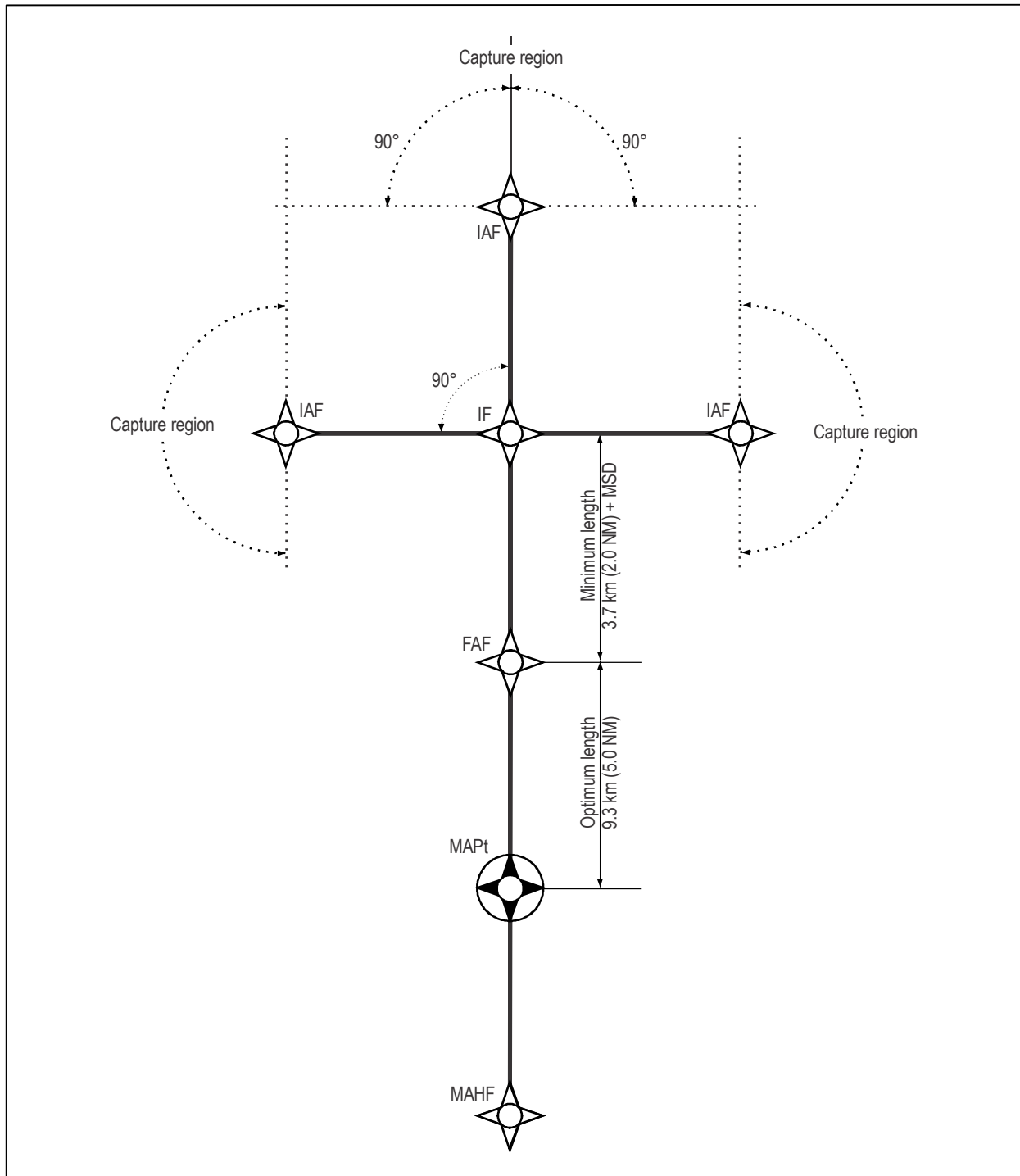
3.4.2 *Length.* The general criteria in Part III, Section 3, Chapter 3 apply.

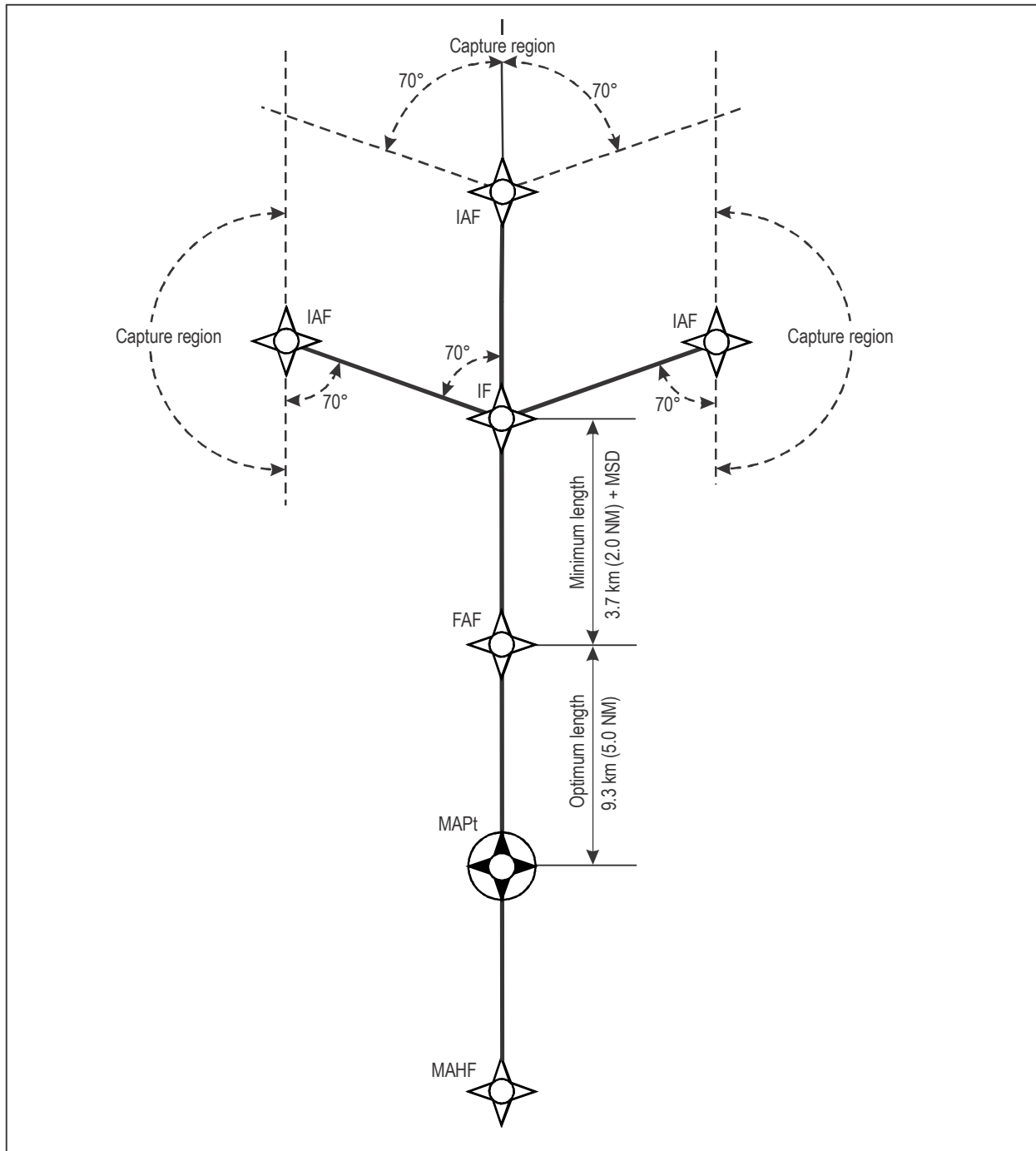
3.4.3 *Descent gradient.* The general criteria in Part I, Section 4, Chapter 5, 5.3, “Descent gradient” apply.

3.5 MISSED APPROACH SEGMENT

3.5.1 *Straight missed approach.* The general criteria in Part III, Section 3, Chapter 3, 3.3, “Initial and Intermediate Missed Approach Segment” apply.

3.5.2 *Turning missed approach.* The general criteria in Part III, Section 3, Chapter 2, 2.5, “Turning missed approach” apply.

**Figure III-2-3-1. T-bar general arrangement**

**Figure III-2-3-2. Y-bar general arrangement**

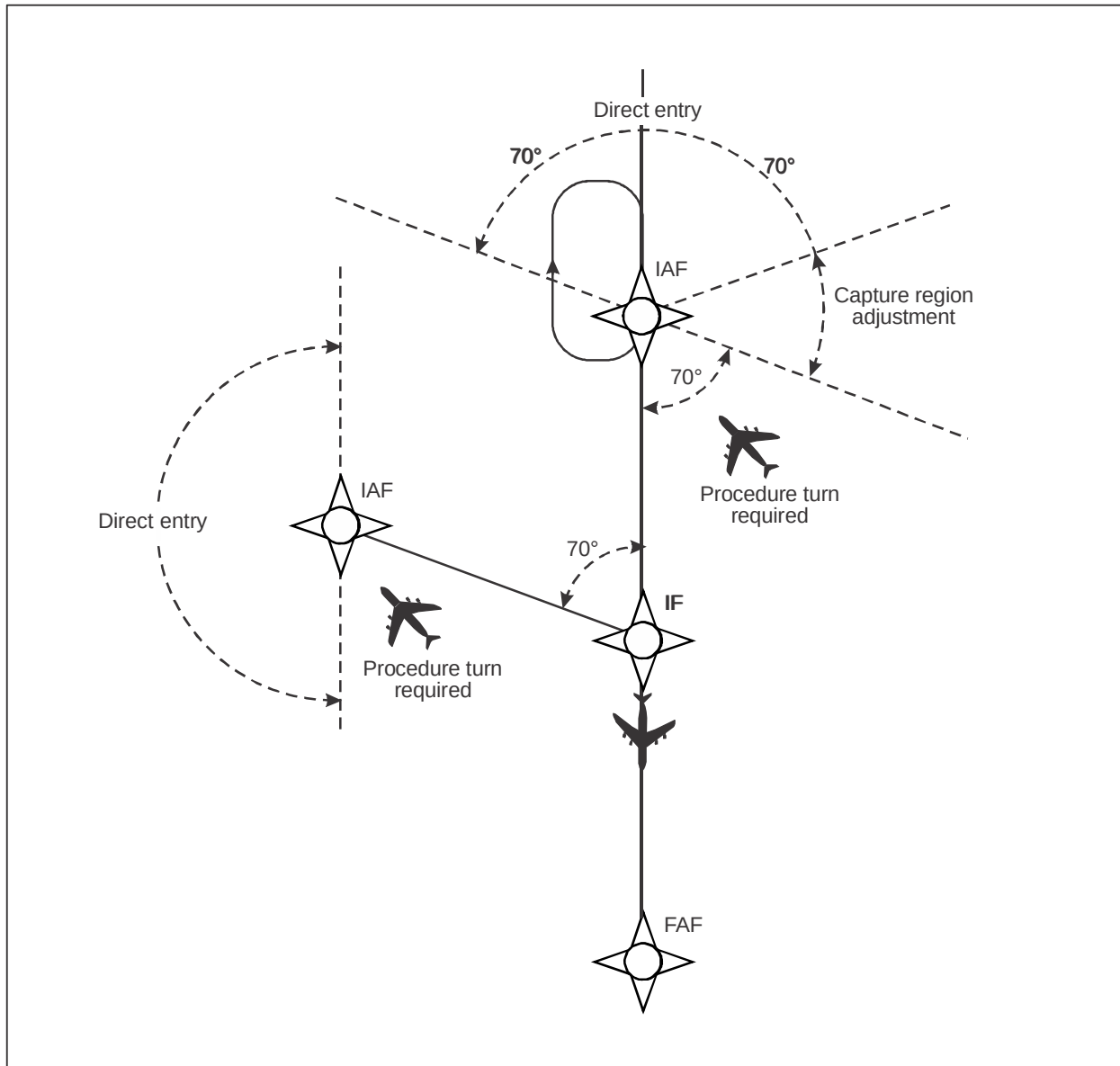


Figure III-2-3-3. Reversal procedures where offset initial not provided

Appendix B to Chapter 6

INFORMATION TO BE PROVIDED BY THE PROCEDURE DESIGNER CONCERNING THE GBAS FAS DATA BLOCK

1. GENERAL

1.1 Some elements of the FAS data block are not the responsibility of the procedure designer. The procedure designer should provide alphanumeric input to an appropriate software tool that generates the binary string describing the FAS data block as well as calculating the cyclic redundancy check (CRC) remainder. The standardized alphanumeric input of the elements into the FAS data block tool is described in this appendix.

1.2 All data used in the construction of the FAS data block requires the use of a high integrity quality control process. The FAS data block content must be protected by the quality control process. The software tools used in this quality control process must ensure the procedure designer's intent is what is provided to the end user.

2. STRUCTURE AND CONTENT OF THE GBAS FAS DATA BLOCK

2.1 *Structure.* There are twenty fields including the CRC remainder field. The first nineteen data fields are protected by the CRC. Additional information in the form of range of values and resolution for entry into the data fields is provided in Annex10, Volume I, Appendix B, Table B-66.

Note.— More material on the encoding of the GBAS FAS data block is found in Annex 10, Volume I, Appendix B, 3.6.4.5 and Table B-66 and Attachment D, 6.6 and 7.11.

2.2 *Geographic relationship of FAS data block elements.* The geographic relationship of various FAS data block elements for a GLS procedure that is not offset is found in Figure III-3-6-1.

2.3 *Content.* The FAS data fields shall contain the parameters that define a single precision approach. FAS data parameters are defined as follows:

Operation type: The operation type is always a straight-in approach procedure. Offset procedures are regarded as straight-in approach procedures.

Values:

0 = straight-in approach

Values 1-15 currently not used (spare)

SBAS service provider ID: Indicates the SBAS service provider associated with a particular FAS data block. Although GBAS does not use information in this field, for precision approaches based on GBAS this field is coded as 14.

Approach performance designator: Indicates GBAS Approach Service Type (GAST) to be used for the operation.

Values:

- 0 = GAST A or B (no criteria exist for this approach performance designator)
- 1 = GAST C
- 2 = GAST C and D
- 3 - 7 = Reserved for future use

Route indicator: The one-letter identifier used to differentiate between multiple GLS approach procedures to the same runway end. The route indicator field reflects the duplicate procedure identifier in the chart identification when it is present (see Part I, Section 4, Chapter 9, 9.5.3 for guidance on duplicate procedure identification).

Reference path data selector (RPDS) and reference path identifier (RPI): The entries for these fields are determined in consultation with system engineers and spectrum management personnel.

FAS data point set: (LTP/FTP latitude, longitude, LTP/FTP height, FPAP latitude, longitude, approach TCH). The latitude and longitude of the LTP/FTP and FPAP are defined as WGS-84 coordinates and entered to the resolution of 5/10 000 of an arc second.

LTP/FTP latitude: The WGS-84 latitude of the LTP/FTP.

Example: DDMMSS.0005N where:

DD = degrees

MM = minutes

SS = seconds

N = Northern Hemisphere

Note.— A software tool may accept data replacing the trailing alpha character used to determine hemisphere with “+” to denote Northern Hemisphere and “-” to denote Southern Hemisphere.

LTP/FTP longitude: The WGS-84 longitude of the LTP/FTP.

Example: DDDMMSS.0005W where:

DDD = degrees

MM = minutes

SS = seconds

W = Western Hemisphere

Note.— A software tool may accept data replacing the trailing alpha character used to determine hemisphere with “+” to denote Eastern Hemisphere and “-” to denote Western Hemisphere.

LTP/FTP height: The height of the LTP/FTP in relationship to the WGS-84 ellipsoid.

ΔFPAP latitude: The difference in the latitude of the FPAP from the LTP/FTP in arc seconds.

ΔFPAP longitude: The difference in the longitude of the FPAP from the LTP/FTP in arc seconds.

Note 1.— The procedure designer is expected to provide FPAP latitude/longitude (WGS-84) and the software tool used for FAS data block encoding will derive the ΔFPAP latitude and longitude based on this information and LTP/FTP latitude.

Note 2.— The FPAP is a point at the same height as the LTP/FTP that is used to define the alignment of the approach. The point of origin of angular deviations in the lateral direction is called GBAS Azimuth Reference Point (GARP) and is defined to be 305 m (1 000 ft) beyond the FPAP along the FAS path.

Course width: The lateral displacement, in metres, from the path defined by the FAS at the LTP/FTP at which full-scale deflection of the course deviation indicator is attained.

Δ length offset: The distance, in metres, from the stop end of the runway to the FPAP. When the stop end of the runway cannot be identified, such as with offset approaches or when the FPAP is located prior to the stop end of the runway, the field data entry is 2 040 m.

Chapter 2

ARRIVAL AND APPROACH PROCEDURES

2.1 GENERAL

2.1.1 Application

2.1.1.1 This chapter describes the arrival, initial and intermediate approach and final missed approach criteria for PBN procedures (see Table III 1 1-1). The criteria for the final approach, initial and intermediate missed approach are specific to the approach classification (NPA, APV and precision) and are dealt with in separate chapters.

Note.— The criteria for RNP AR APCH procedures are detailed in the Required Navigation Performance Authorization Required (RNP AR) Procedure Design Manual (Doc 9905).

2.1.1.2 The general criteria of Part I and Part III, Sections 1 and 2, as amplified or modified by the criteria in this chapter, apply to PBN arrival and approach procedures.

2.1.2 Secondary areas

The general criteria for secondary areas apply (see Part I, Section 2, Chapter 1, 1.2 and 1.3).

2.1.3 Minimum segment length

Minimum segment length distances are listed in the tables in Section 2, Chapter 1.

2.1.4 Area widths

2.1.4.1 For the calculations of area widths and the underlying tolerances involved in these calculations, see the paragraph entitled “XTT, ATT and area semi-width” in Section 1 for the appropriate sensor. These are:

- a) DME/DME, Section 1, Chapter 3, 3.6; and
- b) GNSS, Section 1, Chapter 2, 2.2.

2.1.4.2 The total area width results from joining the various area widths at the relevant fixes, with the exception of the interfaces between the intermediate segment and the final approach segment, and between the en-route segment and the initial approach segment where the methodology detailed in Part III, Section 1, Chapter 1, 1.5 shall be applied.

2.1.5 Y- or T-bar design concept for RNAV procedures

For a detailed description of non-precision approach procedures based on the Y- or T-bar concept, refer to Section 2, Chapter 3, “RNAV T- or Y-bar procedure construction”.

2.2 ARRIVAL ROUTES

2.2.1 General

Arrival obstacle clearance criteria shall apply up to the initial or intermediate approach fix (see Part I, Section 4, Chapter 2). See Figures III-3-2-1 and III-3-2-2.

2.2.2 Minimum sector altitude/terminal arrival altitude

For terminal arrival altitude see Section 2, Chapter 4, “Terminal arrival altitude (TAA)”. Where TAAs are not provided, a minimum sector altitude shall be published. The provisions of Part I, Section 4, Chapter 8, “Minimum sector altitudes (MSA)” apply. The sector is centred on the latitude and longitude of the aerodrome reference point.

2.3 INITIAL APPROACH SEGMENT

2.3.1 Straight segments

2.3.1.1 Initial approach alignment

2.3.1.1.1 For non-precision approach procedures, the angle of interception between an initial approach track and another initial track shall not exceed 120°. Track changes greater than 90° at a waypoint should be avoided. Turns greater than 90° should be accommodated by more than one waypoint or by use of an RF leg.

2.3.1.1.2 The angle of interception between an initial approach track and the intermediate track shall not exceed 90°.

2.3.1.2 *Initial approach area length.* The optimum length is dictated by the necessary descent to be achieved with an optimum descent gradient. The minimum length is dictated by the maximum allowed descent gradient as well as the required minimum distance between waypoints for turn stabilization and anticipation.

2.3.1.3 *Initial approach area width.* The general criteria in Part I, Section 4, Chapter 3, 3.3.3, “Area”, apply as modified in this chapter. The total area width results from joining the various area widths at the relevant fixes. The principle of secondary areas applies. For area widths, see 2.1.4, “Area widths”.

2.3.2 Turn protection

For turn protection at a fly-by, flyover or fixed radius turn, see Section 2, Chapter 2, “Turn protection and obstacle assessment.”

(See also examples in Figure III-3-2-3.)

2.3.3 Reversal procedures

RNAV procedures should be designed to avoid the need for reversal procedures. However, when a procedure requires a track reversal, a racetrack pattern shall be established.

2.4 INTERMEDIATE APPROACH SEGMENT

2.4.1 Intermediate approach alignment

2.4.1.1 For non-precision approach procedures based on basic GNSS, the intermediate approach segment should be aligned with the final approach segment whenever possible. If a turn at the FAF is required, the angle shall not exceed 30° (Cat H, 60°) on a fly-by turn. If the intermediate segment contains an RF leg, the criteria in paragraph 2.4.1.4 apply. Fly-over turns are not permitted.

2.4.1.2 For approach procedures with barometric vertical guidance, the intermediate segment should be aligned with the final approach segment. If a fly-by turn at the FAF is required, the angle shall not exceed 15° (CAT H 30°). Fly-over turns are not permitted. If the intermediate segment contains an RF leg, the criteria in paragraph 2.4.1.4 apply.

2.4.1.3 For approach procedures based on SBAS or GBAS, the intermediate segment shall be aligned with the final approach segment. Fly-by and fly-over turns at the FAF/FAP are not permitted. If the intermediate segment contains an RF turn to the final approach course, then it shall end before the FAP, at a fix aligned with the final approach course; and criteria in Part III, Section 3, Chapter 5, 5.3 “Intermediate approach segment” and in Part III, Section 3, Chapter 6, 6.3 “Intermediate approach segment” shall be applied for SBAS and GBAS, respectively.

2.4.1.4 If the intermediate segment contains an RF segment that ends at the FAF, the following criteria apply:

- a) the track angle change of the RF leg shall not exceed 45°; and
- b) the minimum radius shall be 4.72 km (2.55 NM).

2.4.2 Intermediate approach length

2.4.2.1 The intermediate segment with a fly-by turn at the IF consists of two components:

- a) a turning component followed by
- b) a straight component immediately before the final approach waypoint.

2.4.2.2 The length of the straight component is variable but will not be less than 3.70 km (2.00 NM). This allows the aircraft to be stabilized prior to overflying the final approach waypoint. The length of the turning component is the minimum stabilization distance for the turn angle at the IF and can be determined from Section 2, Chapter 1, Table III-2-1-3 or III-2-1-9.

2.4.2.3 When an RF is used in the intermediate segment, the total length of the intermediate segment, including curved and straight legs, shall not be less than 3.70 km (2 NM), and more than 18.5 km (10 NM).

2.4.2.4 For GLS procedures specific criteria apply (see Chapter 6).

2.4.3 Intermediate approach area width

The total area width is derived from the area widths at the IF and the FAF:

- a) for LNAV approaches, see III-3-3-2-3 and Figure III-3-3-2;
- b) for SBAS approaches, see III-3-5-3.3

The principle of secondary areas applies.

2.4.4 Protection of turns at the FAF

See Section 2, Chapter 2, “Turn protection and obstacle assessment.”

2.5 TURNING MISSED APPROACH

The general criteria in Part I, Section 4, Chapter 6, 6.4.2, “General”, 6.4.3, “Turn parameters”, 6.4.6.4, “Obstacle clearance in the turn area” and 6.4.7, “Turn specified at the MAPt” apply. For turn protection and obstacle assessment see Section 2, Chapter 2, “Turn protection and obstacle assessment.” For missed approach procedures with GNSS receivers which do not provide continuous track guidance after the MAPt, only a DF leg should be used for the design of the first segment of the missed approach.

2.6 END OF THE MISSED APPROACH SEGMENT

A waypoint defining the end of the missed approach segment shall be located at or after the point where the aircraft, climbing at the minimum prescribed gradient for each segment, reaches the minimum altitude for en route or holding, whichever is appropriate.

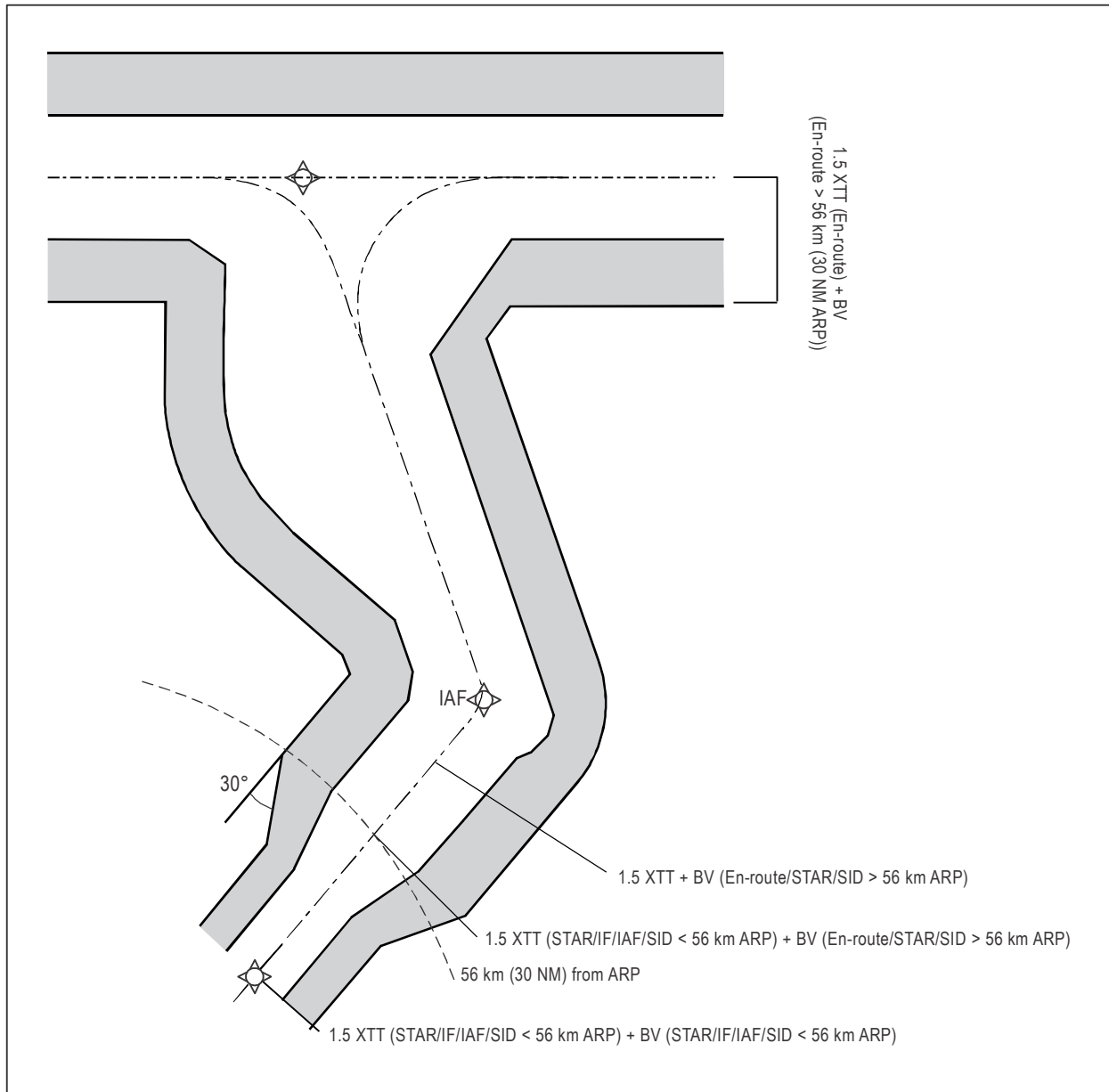


Figure III-3-2-1. Arrival — IAF greater than or equal to 56 km (30 NM) ARP

Chapter 5

SBAS NON-PRECISION APPROACH, APV I AND PRECISION APPROACH CATEGORY I CRITERIA

5.1 INTRODUCTION

5.1.1 Procedure construction

This chapter describes the SBAS criteria for the NPA, APV I and PA Category I procedure segment, which are specific to the performance of SBAS systems. Throughout this Chapter SBAS OAS refers to both SBAS APV I OAS and SBAS CAT I OAS. The APV I or CAT I segment includes the final approach, and the initial and intermediate phases of the missed approach segment. The other phases of flight are generic in character and are presented in Part III, Section 3, Chapter 1 and Chapter 2.

Note.— SBAS Category I may be constructed using the ILS Category I CRM and/or the ILS Category I OAS. GNSS requirements for SBAS category I procedures can be found in Annex 10, Volume I, Attachment D, paragraphs 3.3.6 to 3.3.13.

5.1.2 Standard conditions

The following list contains the standard assumptions on which procedures are developed. Provisions are made for adjustments where appropriate. Adjustments are mandatory when conditions differ adversely from standard conditions and are optional when so specified.

- a) maximum aircraft dimensions are assumed to be the following:

<i>Aircraft category</i>	<i>Wingspan</i>	<i>Vertical distance between the flight paths of the wheels and the centre of navigation (m)</i>
H	30	3
A, B	60	6
C, D	65	7
D _L	80	8

Note 1.— OCA/H for D_L aircraft are published when necessary.

Note 2.— The dimensions shown are those which encompass current aircraft types. They are chosen to facilitate OCA/H calculations and promulgation of aircraft category related minima. It is assumed that these dimensions are not intended to be used for other purposes than the OCA/H calculations in other ICAO documents.

Note 3.— Current Category E aircraft are not normally civil transport aircraft and their dimensions are not necessarily related to V_{at} at maximum landing mass. For this reason, they should be treated separately on an individual basis.

- b) missed approach climb gradient 2.5 per cent;
- c) glide path angle:
 - 1) minimum/optimum: 3.0°
 - 2) maximum: 3.5°
- d) reference datum height 15 m (50 ft); and
- e) all obstacle heights are referenced to threshold elevation.

5.1.3 Obstacle clearance altitude/height (OCA/H)

The SBAS criteria in this chapter enable an OCA/H to be calculated for each category of aircraft.

The OCA/H ensures clearance of obstacles from the start of the final approach to the end of the intermediate missed approach segment.

Note.— This OCA/H is only one of the factors to be taken into account in determining decision height as defined in Annex 6.

5.2 INITIAL APPROACH SEGMENT

See Chapter 2, 2.3.1.1, “Initial approach alignment”.

5.3 INTERMEDIATE APPROACH SEGMENT

5.3.1 *General.* In the intermediate approach segment any sensor can be used; however, the transition to the SBAS navigation shall be made 3.7 km (2.0 NM) prior to the FAF.

5.3.2 *Alignment.* The intermediate approach segment of an SBAS procedure shall be aligned with the final approach segment. For constant radius-to-fix (RF) turns to the final approach course, see 5.3.5 below.

5.3.3 *Intermediate approach segment length.* The segment length should be sufficient to permit the aircraft to stabilize and establish on the final approach course prior to intercepting the glide path, taking into consideration the angle of interception with the final approach course. Minimum values for distance between final approach course and glide path interceptions are specified in Table III-3-6-1. For RF turns to the final approach course, see 5.3.5 below.

5.3.4 *Area width.* From 3.7 km (2.0 NM) to the FAF the area tapers uniformly to match the horizontal distance between the SBAS APV I or Category I OAS X surfaces at the FAF. The secondary area width decreases to zero at the interface with the final approach surfaces (see Figures III-3-5-1 a), III-3-5-1 c) and III-3-5-1 d)). For RF turns to the final approach course, see 5.3.5 below.

Note.— According to the length of the final approach segment, the SBAS APV I or Category I OAS X surface width at the final approach fix can be less than 3.52 km (1.90 NM). In this case, to provide protection to an aircraft that initiates an early missed approach, a 3.52 km (1.90 NM) value (for helicopters 2.96 km (1.60 NM) is considered for the area width of the intermediate approach segment at the final approach fix (see Figure III-3-5-1 b)).

5.3.5 RF turn to the final approach course

5.3.5.1 The intermediate approach segment may incorporate a PBN route ending with a radius-to-fix turn to the final approach course. In this case, the RF turn shall terminate at a waypoint aligned with the final approach course. The position of this waypoint is considered as the point of final approach course interception. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for intermediate approach segment.

5.3.5.2 Minimum distance between final approach course and glide path interceptions in Table III-3-6-1 apply up to temperatures of ISA +30 degrees and up to glide path interceptions at 3 000 ft above aerodrome elevation. For cases exceeding either of these values, criteria in Part II, Section 1, Chapter 1, Appendix D shall be applied. When these temperature and intercept height assumptions are too limiting, Part II, Section 1, Chapter 1, Appendix D may be applied. The maximum total length of the intermediate approach segment shall not exceed 18.520 km (10 NM).

5.3.5.3 Descent in the intermediate segment shall comply with the general criteria in Part I, Section 4, Chapter 4, 4.3.3 “Procedure altitude/height and descent gradient”. If a descent is necessary after the RF turn to the final approach course, then criteria in Part II, Section 1, Chapter 1, Appendix D shall be applied, and obstacle clearance shall be provided in the total intermediate approach area according to Part I, Section 4, Chapter 4, 4.3.2 “Obstacle clearance”. The altitude constraint at the end fix of the RF turn shall be an altitude ‘window’. The top of the window shall comply with the calculations in Part II, Section 1, Chapter 1, Appendix D or shall be equal to the intercept altitude if the minimum length in 5.3.5.2 is applied. The bottom of the window shall comply with the minimum altitude requirements of the RF turn.

Note.— If the calculations result in the same altitude for the top and the bottom of the window, the constraint at the end fix of the RF turn becomes an ‘at’ altitude.

5.3.5.4 For RF turn construction and blending with the final approach segment OAS, criteria in Part II, Section 1, Chapter 1, 1.3.6.4 apply.

5.4 APV I OR CAT I SEGMENT

5.4.1 *General.* The APV I or CAT I segment shall be aligned with the runway centre line and contain the final approach, the initial and the intermediate missed approach segments.

5.4.2 *Origin.* The APV I or CAT I segment starts at the final approach point (the intersection of the nominal vertical path and the minimum altitude specified for the preceding segment). For navigation database coding purposes, the waypoint located at the FAP shall not be considered as a descent fix. The SBAS OAS surfaces extend into the intermediate approach segment but not beyond this segment (see Figure III-3-5-2).

Note.— Guidance material for determining the distance to the FAP from the threshold is contained in Part II, Section 1, Chapter 1, Appendix C.

5.4.3 *Missed approach point.* The missed approach point is defined by the intersection of the nominal vertical path and the decision altitude/height (DA/H). The DA/H is set at or above the OCA/H, which is determined as specified in 5.4.5 and 5.5.

5.4.4 *Termination.* The APV I or CAT I segment terminates at the point where the final phase of the missed approach commences or where the missed approach climb surface Z reaches a semi-width of 1.76 km (0.95 NM) (for helicopters 1.48 km (0.8 NM)), whichever occurs first.

5.4.5 Obstacle clearance of the SBAS APV I or CAT I segment

5.4.5.1 *General.* The method of calculating OCA/H involves a set of obstacle assessment surfaces (SBAS APV I OAS or CAT I OAS). If the SBAS APV I OAS are not penetrated, the OCA/H is still defined by the aircraft category margins. However, if the SBAS APV I OAS are penetrated, the aircraft category margin is added to the highest approach obstacle, or the adjusted height of the largest missed approach penetration, whichever is greater. The value becomes the OCA/H.

5.4.5.2 The SBAS OAS dimensions are related to the approach geometry (GARP/THR distance, GP, RDH) and the SBAS procedure type (APV I or CAT I). The obstacles penetrating the SBAS OAS are divided into two classes, approach obstacles and missed approach obstacles. The height of the highest approach obstacle or the adjusted missed approach surface penetration (see 5.4.5.9.2) is determined and added to an aircraft category related margin to obtain the appropriate OCA/H. Thus, a table of OCA/H values for each aircraft category may be promulgated for SBAS operations at the particular aerodrome.

Note.— At this stage, the SBAS APV I OAS method is the only one applicable to calculate the OCA/H of the APV I segment. A CRM for these operations is currently under development. Use of the ILS Category I CRM is permitted to calculate the SBAS CAT I OCA/H.

5.4.5.3 *Definition of surfaces.* The SBAS APV I OAS consists of up to seven sloping plane surfaces (denoted by letters W, W', X, Y, and Z) disposed symmetrically about the APV I or CAT I segment track and the horizontal plane containing the threshold (see Figure III-3-5-2). The SBAS CAT I OAS contains the following sloping surfaces: W, X, Y and Z, which are equal to the ILS CAT I OAS surfaces. The geometry of the sloping surfaces is precisely defined by four simple linear equations of the form $z = Ax + By + C$. In these equations, x and y are position coordinates and z is the height of the surface at that position. For each surface the constants A, B and C are obtained from the PANS-OPS OAS software (see <http://www.icao.int/safety/AirNavigation/OPS/Pages/PANS-OPS-OAS-Software.aspx>) for the operational range of GARP/THR distances and GP. Separate sets of constants are provided for APV I or CAT I. The SBAS CAT I OAS uses the ILS CAT I OAS constants. The constants may be modified by the programme to account for the following:

- a) missed approach climb gradient;
- b) dimensions of specific aircraft; and
- c) the height of the RDH above the nominal value.

The SBAS OAS Y and Z surfaces are laterally limited by a corridor of 1.76 km (0.95 NM) (for helicopters 1.48 km (0.8 NM)) area semi-width (see Figure III-3-5-2).

5.4.5.4 *Frame of reference.* Positions of obstacles are related to a conventional x, y, z coordinate system with its origin at the threshold (see Figure III-3-5-3). The x axis is parallel to the precision segment track, positive x being measured before threshold and negative x being measured after threshold. The y-axis is at right angles to the x-axis. Although shown conventionally in Figure III-3-5-3, in all calculations associated with SBAS OAS geometry, the y coordinate is always counted as positive. All dimensions connected with the SBAS OAS are specified in metres only. The z-axis is vertical, heights above threshold being positive.

5.4.5.5 *SBAS OAS constants — specification.* For SBAS procedures the constants A, B and C for each sloping surface are obtained from the PANS-OPS OAS software. The PANS-OPS OAS software gives coefficients for GP angles between 2.5 and 3.5 degrees in 0.1-degree steps, and for any GARP-threshold distance between 2 000 m and 4 500 m. For an example of the PANS-OPS OAS software results see Figure III-3-5-4.

5.4.5.6 *Calculation of SBAS APV I OAS heights.* To calculate the height of z of any of the sloping surfaces at a location x', y', the appropriate constants should be first obtained from the PANS-OPS OAS software. These values are then substituted in the equation $z = Ax' + By' + C$. If it is not apparent which SBAS APV I OAS is above the obstacle location, this should be repeated for the other sloping surfaces. The SBAS APV I OAS height is the highest of the X, Y, Z plane heights and the height of the lowest W-W' plane heights (zero if all plane heights are negative). The SBAS CAT I OAS heights are calculated in the same way using the ILS CAT I OAS constants.

For APV I OAS, W and W' planes intersect, and the accountable W-W' plane is always the lower of these two planes, i.e. height of OAS = max [ZX; ZY; ZZ; min ZW; ZW']

Where ZX, ZY, ZZ, ZW and ZW' are the heights of the X, Y, Z, W and W' planes.

Note.— The PANS-OPS software also contains an OCH calculator that will show the height of the SBAS OAS surface Z above any X, Y location. It includes all the adjustments specified for the APV I or CAT I approach geometry, aircraft dimensions, missed approach climb gradient and RDH.

5.4.5.7 *SBAS OAS template construction.* Templates, or plan views of the SBAS OAS contours to map scale, are sometimes used to assist identification of obstacles for detail survey (see Figure III-3-5-5). The SBAS OAS data in the PANS-OPS software includes the coordinates of the points of intersection of the sloping surfaces at threshold level and at 1.9 km (1.0 NM) laterally from the final approach track (see Figure III-3-5-5). The intersection coordinates at threshold level are labelled as C, D and E.

5.4.5.8 *Adjustment of SBAS OAS constants.* Adjustments of SBAS OAS constants for specific aircraft dimensions and RDH different from 15 m are applied as in Part II, Section 1, Chapter 1, 1.4.8.7, “Adjustment of OAS constants”.

5.4.5.9 *Determination of OCA/H*

5.4.5.9.1 *General.* The OCA/H is determined by accounting for all obstacles which penetrate the SBAS OAS surfaces applicable to the operation performance level being considered. The surfaces which apply to each operation type are:

Type A, 3D operation: SBAS APV I OAS.

Type B, 3D operation: SBAS CAT I OAS.

5.4.5.9.2 *Determination of approach and missed approach obstacles.* The accountable obstacles, as determined in 5.4.5.9.1, are divided into approach and missed approach obstacles. The simplest method of partition is by range:

approach obstacles are those between the FAP and range X_E after threshold, and missed approach obstacles are those in the remainder of the APV I or Category I segment (see Figure III-3-5-6). However, in some cases, it may produce an excessive penalty for certain missed approach obstacles. Alternatively, missed approach obstacles may therefore be defined as those above a plane surface parallel to the plane of the GP and with origin at X_E (see Figure III-3-5-7), i.e. obstacle height greater than $[(X - X_E) \tan \theta]$.

5.4.5.9.3 *Calculation of OCA/H.* After the approach and missed approach obstacles have been identified by one of the above described methods, the OCA/H is determined as follows:

- a) determine the height of the highest approach obstacle;
- b) reduce the heights of all missed approach obstacles to the heights of equivalent approach obstacles by the formula given below; and
- c) determine OCA/H by adding the appropriate Table II-1-1-2, “Height loss altimeter margin” aircraft category related margin to the height of the highest approach obstacle (real or equivalent).

$$h_a = \frac{h_{ma} * \cot Z + (X - X_E)}{[\cot Z + \cot \theta]}$$

where:

h_a	=	height of equivalent approach obstacle
h_{ma}	=	height of missed approach obstacle
θ	=	GPA
Z	=	angle of missed approach surface
X	=	range of obstacles relative to threshold (negative after threshold)
X_E	=	$-[900 + (38/\tan \theta)]$ for APV I
For CAT H, X_E	=	-1 000

Note 1.— h_{ma} , X and X_E are expressed in metres (m).

Note 2.— For SBAS Category I operations, OCA/H calculations may use the ILS Category I OCA/H calculation.

5.4.5.9.4 Adjustment for high airfield elevations and steep glide path angles

5.4.5.9.4.1 *Height loss (HL)/altimeter margins.* The margins in Table II-1-1-2 shall be adjusted as follows:

- a) for airfield elevations higher than 900 m (2 953 ft), the tabulated allowances shall be increased by 2 per cent of the radio altimeter margin per 300 m (984 ft) airfield elevation; and
- b) for glide path angles greater than 3.2°, in exceptional cases, the allowances shall be increased by 5 per cent of the radio altimeter margin per 0.1° increase in glide path angle between 3.2° and 3.5°.

5.4.5.9.4.2 *Steep glide path angle.* Procedures involving glide paths greater than 3.5° or any angle when the nominal rate of descent (V_{at} for the aircraft type multiplied by the sine of the glide path angle) exceeds 5 m/s (1 000 ft/min) are non-standard for fixed-wing aircraft. They require the following:

- a) increase of height loss margin (which may be aircraft-type specific);
- b) adjustment of the origin of the missed approach surface (applicable for fixed-wing aircraft procedures only);

5.5.3.3.2 *Areas.* The turn area is constructed as specified in Part I, Section 4, Chapter 6, except that it is based on the width of the SBAS OAS Y surface contours at the earliest and latest TP (see Figure III-3-5-11).

5.5.3.3.3 *Obstacle clearance.* Obstacle elevation/height shall be less than:

$$(OCA/H_{APV} - HL) + d_o \tan Z - MOC$$

where:

d_o = shortest distance from obstacle to line K-K',

and MOC is:

50 m (164 ft) (Cat H, 40 m (132 ft)) for turns more than 15° and 30 m (98 ft) for turns 15° or less.

5.6 SBAS APPROACH WITH OFFSET FINAL APPROACH TRACK ALIGNMENT

5.6.1 Use of SBAS approach with offset alignment

5.6.1.1 In certain cases it may not be physically practicable to align the final approach segment with the runway centre line because of obstacle problems. An offset final approach track shall not be established as a noise abatement measure. The final approach track shall intersect the runway extended centre line:

- a) at an angle not exceeding 5°; and
- b) at a point where the nominal glide path reaches a height called intercept height of at least 55 m (180 ft) above threshold.

5.6.1.2 The procedure shall be annotated: “final approach track offset ... degrees” (tenth of degrees).

5.6.1.3 The general arrangement is shown in Figure III-3-6-24.

5.6.2 Obstacle clearance criteria

The provisions contained in 5.1 to 5.5 apply except that:

- a) all the obstacle clearance surfaces and calculations are based on a fictitious runway aligned with the final approach track. This fictitious runway has the same length and the same landing threshold elevation as the real one. The FTP and the course width at the FTP are analogous to the LTP for an aligned procedure. The DCP is located 15 m (50 ft) above the FTP; and
- b) the OCA/H for this procedure shall be at least: intercept altitude/height + 20 m (66 ft).

5.7 SBAS NPA

5.7.1 *Final approach segment.* The final approach segment primary area is formed by using the outer lateral boundaries of the X surfaces beginning at threshold and extending until the FAF. The secondary area extends laterally up to a total area semi-width of 0.95 NM (0.8 NM for helicopters).

5.7.1.1 *Final approach segment semi-width surfaces.* The semi-width of the final approach surfaces shall be determined using the following formulae:

$$Y_{LTP} = [-0.0031 (GARP - LTP) + 182.83] \text{ metres and}$$

$$\theta_x = [-0.0006 (GARP - LTP) + 9.4367] \text{ degrees}$$

where:

Y_{LTP} is the semi-width of the final approach surface at the LTP/FTP.

θ_x is the angle of splay outward from the LTP/FTP of the final approach surface.

$W/2$ is computed as: $Y_{LTP} +$ the distance from the LTP/FTP multiplied by $\tan \theta_x$.

5.7.2 *Intermediate segment.* The total area width is as described in Chapter 2, 2.4.3, “Intermediate approach area width”. From 3.7 km (2.0 NM) to the FAF, the area tapers uniformly to match the lateral boundaries of the X surface at the FAF. The secondary area width decreases to zero at the FAF when DD'' is larger than 1 759 m (0.95 NM); and to 1 759 m (0.95 NM) when DD'' line is smaller than 1 759 m (0.95 NM). (See Figures III-3-5-12 and III-3-5-13). If the intermediate segment contains an RF turn to the final approach course, then it shall end at a fix aligned with the final approach course, at a minimum distance of 1 852 m (1.0 NM) before the FAF.

5.7.3 *Missed approach segment.* The missed approach area shall start at the early ATT of the MAPt, with a splay of 15 degrees each side of the outer boundary of the final segment. Secondary areas shall be applied when the expanded semi-width reaches the appropriate dimension for the RNP or RNAV navigation accuracies applied for missed approach guidance.

5.7.3.1 The obstacle evaluations and establishment of the OCA/H shall be carried out in the same manner as the LNAV criteria.

5.8 PROMULGATION

5.8.1 The general criteria in Part I, Section 4, Chapter 9, 9.5, “Procedure naming for arrival and approach charts” apply. The instrument approach chart for an SBAS approach procedure shall be identified by the title RNAV_(GNSS) or RNP Rwy XX in accordance with Part III, Section 5, Chapter 1.

5.8.2 When an RF turn to the final approach course is utilized in the intermediate approach segment, then the waypoint at the end of the RF turn shall be promulgated with an altitude ‘window’ constraint, or an ‘at’ altitude constraint (see the Note to 5.3.5.3).

5.8.3 *Promulgation of OCA/H for SBAS approach procedures.* The OCA/H values shall be promulgated for those categories of aircraft for which the procedure is designed. The values shall be based on the following standard conditions:

- a) approach flown with barometric altimeter;

- b) standard aircraft dimensions (see 6.1.3); and
- c) 2.5 per cent missed approach climb gradient.

Additional values of OCA/H may be agreed between operators and appropriate authority and promulgated, on the basis of evidence supporting the modifications defined in 5.4.5.7.

5.8.4 *Minima box.* All SBAS APV I and CAT I OCA/Hs are promulgated as LPV lines of minima. All NPA SBAS OCA/Hs shall be promulgated as LP (localizer performance) lines of minima. LPV and LP lines of minima shall not be published on the same chart.

5.8.5 *Additional gradient for the final missed approach segment.* If obstacles identified in the final missed approach segment result in an increase in any of the OCA/H calculated for the precision segment, an additional steeper gradient may also be specified for the gradient of the missed approach surface (Z) for the purpose of lowering the OCA/H (see Part I, Section 4, Chapter 6, 6.2, “Climb gradient and MOC”).

Note.— For further requirements in PBN charting, see Part III, Section 5, Chapter 1.

5.8.6 *Final approach segment data block (FAS DB).* The FAS DB is specified in Part III, Section 3, Chapter 6. It shall be promulgated in a textual format on the verso of the approach chart or a separate sheet, and shall contain at least the data indicated in Table III-3-5-1.

Table III-3-5-1. Example of textual description of data required for the final approach segment data block

<i>Data Content</i>	<i>Example Data</i>
Operation Type	0
SBAS Provider Identifier	1
Airport Identifier	LFLC
Runway Number	26
Runway Letter	
Approach Performance Designator	0
Route Indicator	Z
Reference Path Data Selector	0
Reference Path Identifier	E26A
LTP/FTP Latitude	454718.3185N
LTP/FTP Longitude	0031114.4545E
LTP/FTP Height above ellipsoid	372.3
FPAP Latitude	454705.1260N
FPAP Longitude	0030900.4790E
Approach Threshold Crossing Height (TCH)	15
Approach TCH Units Selector	1
Glidepath Angle (GPA)	3
Course Width at threshold	105
Length Offset	48
Horizontal Alert Limit (HAL)	40
Vertical Alert Limit (VAL)	0
Final Approach Segment CRC	AB8761C6

5.8.7 SBAS FAS DB Information to be promulgated. The following information shall be promulgated for procedures based on SBAS:

- a) *Channel number*. A globally unique channel number in the range 40 000 to 99 999.
- b) *Reference path identifier (RPI)*. The RPI as defined in Part III, Section 2, Chapter 6, Appendix B, paragraph 3.5, with the procedure information. This is required for charting and data base encoding.
- c) *SBAS service provider*. Since for SBAS vertically guided procedures information from different SBAS cannot be mixed, the leading alpha character in the Reference Path Identifier (RPI) field may be used to identify the service provider.

Examples:

W17A for WAAS
E22A for EGNOS.

Appendix to Chapter 5

STEEP GLIDE PATH ANGLE APPROACHES UP TO 6.3 DEGREES (11 per cent)

1. GENERAL

1.1 For fixed-wing aircraft, glide path angles above the standard values (see Chapter 5, 5.1) should be used in approach procedure design only for obstacle clearance purposes and must not be used as a means to introduce noise abatement procedures. Such procedures are non-standard and require special approval.

1.2 For helicopter procedures, glide path angles above 5.7° (10 per cent) or 7.5° (13.2 per cent), when authorized and provided the speed is restricted to a maximum of 130 km/h IAS (70 kt IAS), are non-standard and require special approval.

1.3 The use of the SBAS APV I or Category I criteria in this appendix is limited to procedures with a glide path angle smaller than or equal to 6.3° (11 per cent).

Note.— For point-in-space (PinS) RNP APCH approach procedures with glide path angle above 6.3° (11 per cent) up to 9° (15.8 per cent), refer to Part IV, Appendix C to Chapter 3.

2. PROCEDURE DESIGN

2.1 Obstacle clearance criteria

The following obstacle clearance criteria should be adjusted for the specific glide path angle:

- a) the W and W' surfaces of the SBAS APV I OAS;
- b) origin of the Z surface of the SBAS APV I OAS (applicable for fixed-wing aircraft procedures only); and
- c) height loss/altimeter margin (see paragraph 3).

2.2 Determination of the SBAS APV I OAS coefficients

- *W surface*: Coefficient A_W is determined by the formula

$$A_W = 0.0239 + 0.0092 (\theta - 2.5)$$

in which θ is the glide path angle in degrees.

$$\text{Coefficient } C_W = -6.45$$

- *W' surface*: Coefficients $A_{W'}$ and $C_{W'}$ are determined by the formula

$$A_{W'} = \tan(0.75\theta)$$

$$C_W = -50 + RDH \tan(0.75\theta)/\tan(\theta) \text{ for APV I}$$

where

RDH = reference datum height (m)

θ = glide path angle

- *X and Y surfaces*: The X and Y surface coefficients for 3.5° glide path at the appropriate GARP/threshold distance are used for all glide path angles greater than 3.5°.
- *Z surface*: The coefficient C_Z for the Z surface is determined by the formula

$$C_Z = -A_Z X_E$$

where A_Z is the A coefficient for the selected missed approach gradient; and X_E is the new coordinate of the Z surface origin:

$$X_E = -[900 + (38/\tan\theta) + 50(\theta - 3.5^\circ)/0.1^\circ] \text{ for APV I}$$

For helicopter procedures, the following coordinate of the Z surface origin is used for all glide path angles:

$$X_E = -1\,000 \text{ for APV I.}$$

2.3 Determination of the height of equivalent approach obstacle

Use the formula:

$$h_a = [h_{ma} \cot Z + (x - X_E)]/(\cot Z + \cot \theta)$$

where

h_a = height of equivalent approach obstacle

h_{ma} = height of missed approach obstacle

θ = glide path angle

Z = angle of missed approach surface

X_E = new coordinate of the Z surface origin

x = range of obstacle relative to threshold (negative after threshold).

2.4 Re-survey of obstacles

As the configuration of the SBAS APV I OAS is changed, a re-survey of obstacles may be required.

2.5 Promulgation

If the glide path angle exceeds the standard values, a special note shall be included on the instrument approach chart stating that appropriate aircraft and crew authorizations are required to use such a procedure (see Annex 4, 11.10.8.8).

Chapter 6

PRECISION APPROACH PROCEDURES — GLS

6.1 INTRODUCTION

6.1.1 Application

The GLS criteria in this chapter are based on ILS criteria and are related to the ground and airborne equipment performance and integrity required to meet the operational objectives described in Annex 10. An illustration of the specific definitions used in this chapter is given in Figure III-3-6-1.

Note 1.— While specific GLS criteria are in preparation, the criteria contained in this chapter are based on an ILS equivalency method.

Note 2.— GBAS is the ground-based GNSS augmentation system. The ground facility includes GBAS reference receivers, a processor which prepares messages for broadcast and a VHF data broadcast (VDB) system. The VDB is automatically tuned with the selection of the channel number in the range of 20 000 to 39 999.

Note 3.— The sensor minimum operational performance standards are found in RTCA DO-253C or equivalent documents.

Note 4.— For each runway end supported by the system, the minimum CAT I service volume is described in Annex 10, Volume I, Chapter 3, 3.7.3.5.3 and Attachment D, Chapter 7, 7.3 and Figure D-4.

6.1.2 Procedure construction

The procedure from en route to the GLS final approach segment and in the final missed approach phase conforms with the general criteria. The differences are found in the physical requirements for the GLS precision segment which contains the final approach segment as well as the initial and intermediate phases of the missed approach segment. These requirements are related to the performance of the GBAS system.

6.1.3 Standard conditions

6.1.3.1 The following list contains the standard assumptions on which procedures are developed. Provisions are made for adjustments where appropriate. Adjustments are mandatory when conditions differ adversely from standard conditions and are optional when so specified (see 6.4.8.7, “Adjustment of OAS constants”).

- a) Maximum aircraft dimensions are assumed to be the following:

<i>Aircraft category</i>	<i>Wing span (m)</i>	<i>Vertical distance between the flight paths of the wheels and the guidance control point (m)</i>
H	30	3
A, B	60	6
C, D	65	7
D _L	80	8

Note 1.— OCA/H for Cat D_L aircraft is published when necessary.

Note 2.— The dimensions shown are those which encompass current aircraft types. They are chosen to facilitate OCA/H calculations and promulgation of aircraft category related minima. It is assumed that these dimensions are not intended to be used for other purposes than the OCA/H calculations in other ICAO documents. The use of OAS surfaces to calculate OCA/H may result in significant differences between aircraft categories because of small differences in size. For this reason, it is always preferable to use the Collision Risk Model (6.4.9) which will allow for more realistic assessment for both height and position of obstacles.

Note 3.— Current Category E aircraft are not normally civil transport aircraft and their dimensions are not necessarily related to V_{at} at maximum landing mass. For this reason, they should be treated separately on an individual basis.

- b) Missed approach climb gradient: 2.5 per cent.
- c) GLS course width: 210 m (± 105 m each side of the final approach track) at threshold.
- d) Glide path angle:
 - 1) Minimum 2.5°;
 - 2) optimum: 3.0°; and
 - 3) maximum: 3.5° (3.2° for Category II/III operations).
- e) GLS reference datum height: 15 m (50 ft).
- f) All obstacle heights are referenced to threshold elevation. A declaration by the procedure designer shall be made for the value of undulation (N) at each runway threshold.
- g) The delta length offset is zero (see also 6.1.3.2).
- h) Category II flown with flight director.
- i) For Category II and Category III operations, the Annex 14 inner approach, inner transitional and balked landing surfaces have not been penetrated. Where the Category II OCA/H is higher than the level of the inner horizontal

surface, but below 60 m, Category III operations can be accommodated, provided the inner approach, inner transitional and balked landing surfaces are extended to the Category II OCA/H level to accommodate Category III operations (see Figure II-1-1-8a).

6.1.3.2 Final approach segment (FAS) data. The final approach segment is defined by data prepared by the procedure designer. The accuracy of the path is therefore totally dependent on the accuracy and integrity of the original data on the runway and calculations carried out by the designer. The total description of the path, including the glide-path, lateral guidance sector width, alignment and all other parameters describing the path are originated by the designer and are not affected by the location of ground facilities. The path parameters are designed using geodetic and geometric calculations and the parameters are formatted into a FAS data block in electronic media as described in the appendix to this chapter. Data are then added to provide a cyclic redundancy check (CRC), and the complete block is transferred to users to insure the integrity of the data throughout the process leading to inclusion of the path data in the GBAS system for transmission to user airborne systems. A complete description of the FAS data block is included in Doc 9368, *Instrument Flight Procedures Construction Manual*, Attachment C.5, along with an example of the process and product.

6.1.3.3 In addition to the constraints specified for GLS FAS Data (PANS-OPS, Volume II, Part III, Section 2, Chapter 6, Appendix B), the following constraints apply for procedures using the GAST D service type:

- a) an FTP is not permissible – the LTP shall be located at the desired threshold location;
- b) the FPAP shall be located on the extended runway centre line;
- c) the authentication protocol shall be applied; and
- d) approach performance designator shall be set in accordance with the desired performance.

Note.— Additional constraints for several elements of the FAS data block in case of GBAS for GAST D procedures are contained in Annex 10, Volume 1 Appendix B, 3.6.7.2.1.3, 3.6.7.2.4 and 3.6.7.2.4.5).

6.1.4 Obstacle clearance altitude/height (OCA/H)

6.1.4.1 The GLS criteria enable an OCA/H to be calculated for each category of aircraft. See Part I, Section 4, Chapter 1, 1.8, “Categories of aircraft”. Where statistical calculations were involved, the OCA/H values were designed against an overall safety target for risk of collision with obstacles of 1×10^{-7} , i.e. 1 in 10 million per approach. The OCA/H ensures clearance of obstacles from the start of the final approach to the end of the intermediate missed approach segment.

Note.— This OCA/H is only one of the factors to be taken into account in determining decision height as defined in Annex 6.

6.1.4.2 Additional material is included to allow operational benefit to be calculated for the improved beam holding performance of autopilots meeting national certification standards (as opposed to flight directors) in Category II, and for improved missed approach climb performance in Category I, II and III.

6.1.4.3 Benefit may also be calculated for aircraft with dimensions other than the standard size assumed in the basic calculations. An OCA/H is not associated with Category III operations. These are supported by the obstacle limitation surfaces defined in Annex 14, in association with overlapping protection from the Category II criteria.

6.1.5 Methods of calculating OCA/H

6.1.5.1 General. Three methods of calculating OCA/H are presented, which in turn involve progressive increases in the degree of sophistication in the treatment of obstacles. Standard conditions (as specified in 6.1.3) are assumed to exist unless adjustments for non-standard conditions have been made.

6.1.5.2 *First method.* The first method involves a set of surfaces derived from the Annex 14 precision approach obstacle limitation surfaces and a missed approach surface described in 6.4.7.2, “Definition of basic ILS surfaces” and from this point forward termed “Basic ILS surfaces”. Where the standard conditions exist as specified in 6.1.3 and where the basic ILS surfaces are free of penetrations (see 6.4.7.1, “General”) the OCA/H for Cat I is defined by aircraft category margins. If the basic ILS surfaces are penetrated, then the OCA/H is calculated as described in 6.4.7.3, “Determination of OCA/H with basic ILS surfaces”.

6.1.5.3 *Second method.* The second method involves a set of obstacle assessment surfaces (OAS) above the basic ILS surfaces (see 6.4.8.3, “Definition of obstacle assessment surfaces (OAS)”). If the OAS are not penetrated, and provided the obstacle density below the OAS is operationally acceptable (see 6.4.8.9, “Effect of obstacle density on OCA/H”), the OCA/H for Cat I is still defined by the aircraft category margins. However, if the OAS are penetrated, then the aircraft category related margin is added to the height of the highest approach obstacle, or to the adjusted height of the largest missed approach penetration, whichever is greater. This value becomes the OCA/H.

6.1.5.4 *Third method.* The third method, using a collision risk model (CRM), is employed either as an alternative to the use of the OAS criteria (second method) or when the obstacle density below the OAS is considered to be excessive. The CRM accepts all objects as an input and assesses, for any specific OCA/H value, both the risk due to individual obstacles and the accumulated risk due to all the obstacles. It is intended to assist operational judgement in the choice of an OCA/H value.

Note 1.— While specific GBAS distributions for the existing CRM are being developed, use should be made of the current ILS CRM.

Note 2.— The CRM does not take into account the characteristics of helicopters. The CRM can be used but the method should be conservative.

6.1.6 References

The following relate to and amplify the material contained in this chapter:

- a) background information relating to the derivation of the OAS material (Attachment to Part II, paragraph 1) and to airborne and ground equipment performance assumed in the derivation of the OAS (paragraph 2);
- b) turning missed approach after precision approach (Part II, Section 1, Chapter 1, Appendix A);
- c) determining ILS glide path descents/MLS elevation heights and distances (Part II, Section 1, Chapter 1, Appendix C);
- d) PANS-OPS OAS software (ICAO public website (www.icao.int)); and
- e) minimum distance between localizer and glide path interceptions after a constant radius-to-fix (RF) turn to the localizer course (Part II, Section 1, Chapter 1, Appendix D).

Examples of OCA/H calculations can be found in the *Instrument Flight Procedures Construction Manual* (Doc 9368).

6.1.7 GLS with glide path inoperative

The GLS with glide path inoperative is a non-precision approach procedure. The principles of Chapter 3, “Non-precision approach procedures” apply.

6.2 INITIAL APPROACH SEGMENT

6.2.1 General

The initial approach segment for a GLS procedure must ensure that the aircraft is positioned within the operational service volume of the GBAS on a track or heading that will facilitate the final approach course interception. For this reason, the general criteria, which apply to the initial segment (see Chapter 2), are modified in accordance with 6.2.2, “Alignment” and 6.2.3, “Area”. The initial approach segment may be defined by an RNAV or RNP route, using an RNAV or RNP system for track guidance. Only the systems capable of navigation accuracy of 1 852 m (1 NM) or better in this phase of flight can be considered. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for initial approach. The RNAV or RNP route shall terminate at an IF defined by RNAV or RNP located on the final approach course (except when an RF turn to the final approach course is used in the intermediate approach segment according to criteria in 6.3.6 below).. RNAV/RNP turn construction is applicable for turns within the initial segment and for the turn at the IF on the final approach course (see Figures III-3-6-5 and III-3-6-6). For RNAV and RNP initial approach segments, criteria in Part III apply. If a course reversal is required with an RNAV or RNP initial approach segment, only a racetrack can be used. The fix and the inbound leg shall be located on the final approach course and the inbound segment defined by GBAS.

6.2.2 Initial approach segment alignment

The angle of interception between the initial approach track and the intermediate track should not exceed 90°. In order to permit the auto pilot to couple on to the final approach course, an interception angle not exceeding 30° is desirable. When the angle exceeds 70° a radial, bearing, radar vector, DME information providing at least 4 km (2 NM) (CAT H, 1.9 km (1 NM)) of lead shall be identified to assist the turn onto the intermediate track. When the angle exceeds 90°, the use of a reversal, racetrack, or dead reckoning (DR) track procedure (see Part I, Section 4, Chapter 3, Appendix A, “Initial approach using dead reckoning (DR)”) should be considered.

6.2.3 Initial approach segment area

The area is as described in the general criteria (see Part I) The only exception to these criteria is that the intermediate approach fix (IF), must be located within the service volume of the GBAS (except when an RF turn to the final approach course is used in the intermediate approach segment according to criteria in 6.3.6 below),, and normally at a distance not exceeding 37 km (20 NM) from the landing threshold point (LTP). When radar is used to provide track guidance to the IF, the area shall be in accordance with Part II, Section 2, Chapter 6, 6.2, “Initial approach segment”.

6.3 INTERMEDIATE APPROACH SEGMENT

6.3.1 General

6.3.1.1 The intermediate approach segment for a GLS procedure differs from the general criteria in that:

- a) the alignment coincides with the final approach course;
- b) the length may be reduced;
- c) in certain cases the secondary areas may be eliminated; and
- d) if an RF turn to the final approach course is used, specific criteria in 6.3.6 apply.

6.3.1.2 The primary and secondary areas at the FAP are defined in terms of the ILS surfaces. Consequently, the criteria in Chapter 5 are applied except as noted for alignment, area length, width and obstacle clearance in 6.3.2 through 6.3.6 below. For RNAV intermediate approach segments, the criteria in the applicable RNAV chapters apply.

6.3.2 Intermediate approach segment alignment

The intermediate approach segment of a GLS procedure shall be aligned with the final approach segment, except when an RF turn to the final approach course is used (see 6.3.6 below).

6.3.3 Intermediate approach segment length

6.3.3.1 The optimum length of the intermediate approach segment is 9 km (5 NM) (Cat H, 3.7 km (2.0 NM)). This segment shall allow interception with the final approach course and with the glide path.

6.3.3.2 The segment length should be sufficient to permit the aircraft to stabilize and establish on the final approach course prior to intercepting the glide path, taking into consideration the angle of interception with the final approach course.

6.3.3.3 Minimum values for distance between final approach and interception of the glide path are specified in Table III-3-6-1; however, these minimum values should only be used if usable airspace is restricted. The maximum length of the segment is governed by the requirement that it be located wholly within the service volume of the GBAS, and normally at a distance not exceeding 37 km (20 NM) from the landing threshold point (LTP). For RF turns to the final approach course, see 6.3.6 below.

6.3.4 Intermediate approach segment area width

6.3.4.1 The total width at the beginning of the intermediate approach segment is defined by the total width of the initial approach segment and tapers uniformly to match the horizontal distance between the OAS X surfaces at the FAP (see 6.4.8.3, “Definition of OAS”).

6.3.4.2 For obstacle clearance purposes the intermediate approach segment is divided into a primary area bounded on each side by a secondary area. However, when a DR track is used in the initial approach segment, the primary area of the intermediate segment extends across the full width and secondary areas are not applied.

6.3.4.3 The primary area is determined by joining the primary initial approach area with the final approach surfaces (at the FAP). At the interface with the initial approach segment the width of each secondary area equals half the width of the primary area. The secondary area width decreases to zero at the interface with the final approach surfaces. See Figure III-3-6-2.

6.3.4.4 Where a racetrack or reversal manoeuvre is specified prior to intercepting the final approach course, the provisions in Part I, Section 4, Chapter 4, 4.4.4, “Turn not at the facility” apply, the facility being the GARP itself and the FAF being replaced by the FAP. (See Figure III-3-6-3).

6.3.5 Intermediate approach segment obstacle clearance

The obstacle clearance is the same as defined in Part I, Section 4, Chapter 4, except where the procedure permits a straight-in approach in which the aircraft is stabilized on the final approach course prior to crossing the IF. In this case, obstacles in the secondary areas need not be considered for the purpose of obstacle clearance.

6.3.6 RF turn to the final approach course

6.3.6.1 The intermediate approach segment may incorporate a PBN route ending with a radius-to-fix turn to the final approach course. In this case, the RF turn shall terminate at a waypoint located on the final approach course. The position of this waypoint is considered as the point of localizer interception. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for intermediate approach segment.

6.3.6.2 Minimum distance between final approach course and glide path interceptions in Table III-3-6-1 apply up to temperatures of ISA +30 degrees and up to glide path interceptions at 3 000 ft above aerodrome elevation. For cases exceeding either of these values, criteria in Part II, Section 1, Chapter 1, Appendix D shall be applied. When these temperature and intercept height assumptions are too limiting, Part II, Section 1, Chapter 1, Appendix D may be applied. The maximum total length of the intermediate approach segment shall not exceed 18.5 km (10 NM). Additionally, in case of Category II and III operations, the minimum distance between the end fix of the RF turn and the landing threshold shall be 9.3 km (5 NM).

6.3.6.3 Descent in the intermediate approach segment shall comply with the general criteria in Part I, Section 4, Chapter 4, 4.3.3 “Procedure altitude/height and descent gradient”. If a descent is necessary after the RF turn to the final approach course, then criteria in Part II, Section 1, Chapter 1, Appendix D shall be applied, and obstacle clearance shall be provided in the total intermediate approach area according to Part I, Section 4, Chapter 4, 4.3.2 “Obstacle clearance”. The altitude constraint at the end fix of the RF turn shall be an altitude ‘window’. The top of the window shall comply with the calculations in Part II, Section 1, Chapter 1, Appendix D or shall be equal to the intercept altitude if the minimum length in 6.3.6.2 is applied. The bottom of the window shall comply with the minimum altitude requirements of the RF turn.

Note.— If the calculations result in the same altitude for the top and the bottom of the window, the constraint at the end fix of the RF turn becomes an ‘at’ altitude.

6.3.6.4 Criteria for the RF turn construction defined in Part III, Section 2, Chapter 2, 2.4 “RF turn method” apply with the following additions (see Figure III-3-6-4):

- a) the boundaries of the RF turn shall be continued one along-track tolerance (ATT) after the end fix of the RF turn assuming the width and fix tolerance values of the navigation specification associated with the RF turn;
- b) the boundary of the succeeding segment is the D”-E” line and the extension of the D-D” line considered as primary only area;
- c) if the bounding arcs (either primary or secondary) on the inner side of the turn intersect the extension of D-D” line, then the extension becomes the edge of the area after the intersection, otherwise the RF boundaries shall be joined to the succeeding segment from one ATT after the end fix of the turn by a line at 30° to the final approach course; and
- d) if the bounding arcs (either primary or secondary) on the outer side of the turn do not intersect the extension of the D-D” line, then the arcs shall be joined to the extension by a tangent splaying 15° from the azimuth course, otherwise the RF boundaries shall be joined to the succeeding segment from one ATT after the end fix of the turn by a line at 30° to the final approach course.

6.4 PRECISION SEGMENT

6.4.1 General

The precision segment for a GLS procedure is aligned with the final approach course and contains the final descent for landing, the initial and the intermediate missed approach. See Figure III-3-6-7.

6.4.2 Origin

The precision segment starts at the final approach point, that is the intersection of the nominal glide path and the minimum altitude specified for the preceding segment. The FAP should not normally be located more than 18.5 km (10.0 NM) before threshold, unless adequate glide path guidance beyond the minimum specified in Annex 10 is provided.

6.4.3 Glide path verification check

A fix at the FAP is necessary so as to permit comparison between the indicated glide path and the aircraft altimeter information.

6.4.4 Descent fix

A descent fix shall be located to start the final approach segment and it becomes the final approach point linking the MOC in the preceding segment smoothly with the precision surfaces. The descent fix should not normally be located more than 18.5 km (10.0 NM) before threshold, unless adequate glide path guidance beyond the minimum specified in Annex 10 is provided. The tolerance of the descent fix does not need to be considered due to accuracy.

Note.— Guidance material for determining the distance to the descent fix from the landing threshold is contained in Part II, Section 1, Chapter 1, Appendix C.

6.4.4.1 The provisions of Part I, Section 2, Chapter 2, 2.7.4 which allow obstacles close to the fix to be ignored, apply in the area below the 15 per cent gradient within the precision surfaces.

6.4.5 Missed approach

The missed approach shall be initiated no lower than the intersection of the nominal glide path with the decision altitude/height (DA/H). The DA/H is set at or above the OCA/H, which is determined as specified in 6.4.7 to 6.4.9 and 6.5.

6.4.6 Termination

The precision segment normally terminates at the point where the final phase of the missed approach commences (see Part I, Section 4, Chapter 6, 6.2.3, “Final phase”) or where the missed approach climb surface Z starting 900 m past threshold reaches a height of 300 m (1 000 ft) above threshold, whichever is lower.

6.4.7 Obstacle clearance of the precision segment using basic ILS surfaces for GLS operations

6.4.7.1 *General.* The area required for the precision segment is bounded overall by the basic ILS surfaces defined in 6.4.7.2. In standard conditions there is no restriction on objects beneath these surfaces (see 6.1.3, “Standard Conditions”). Objects or portions of objects that extend above these surfaces must be either:

- a) minimum mass and frangible; or
- b) taken into account in the calculation of the OCA/H.

6.4.7.2 *Definition of basic ILS surfaces.* The surfaces to be considered correspond to a subset of Annex 14 obstacle limitation surfaces specified for precision approach runway code numbers 3 or 4. These are (see Figure III-3-6-8):

- a) the approach surface continuing to the final approach point (first section 2 per cent gradient, second section 2.5 per cent gradient as described in Annex 14);
- b) the runway strip assumed to be horizontal at the elevation of the threshold and has a fixed width of 300 m;
- c) the missed approach surface. This is a sloping surface which:
 - 1) starts at a point 900 m past the threshold (Cat H, a starting point of 700 m past the threshold can be considered if necessary) at threshold elevation;
 - 2) rises at a 2.5 per cent gradient; and
 - 3) splays so as to extend between the transitional surfaces. It extends with constant splay to the level of the inner horizontal surface, and thereafter, continues at the same gradient but with a 25 per cent splay until the termination of the precision segment; and
- d) the extended transitional surfaces, which continue longitudinally along the sides of the approach and missed approach surfaces and to a height of 300 m above threshold elevation.

6.4.7.3 Determination of OCA/H with basic ILS surfaces.

6.4.7.3.1 Where the basic ILS surfaces specified in 6.4.7.2 are not penetrated, the OCA/H for Category I and Category II is defined by the margins specified in Table III-3-6-3 and Category III operations are not restricted. Obstacles may be excluded when they are below the transitional surface defined by Annex 14 for runways with code numbers 3 and 4, regardless of the actual runway code number (i.e. the surfaces for code numbers 3 and 4 are used for the obstacle assessment on runways with code numbers 1 and 2).

6.4.7.3.2 If the basic ILS surfaces listed above are penetrated by objects other than those tabulated in Table III-3-6-2, the OCA/H may be calculated directly by applying height loss/altimeter margins to obstacles (see 6.4.8.8). The obstacles in Table III-3-6-2 may only be exempted if the GLS course width meets the standard condition (see 6.1.3).

6.4.7.3.3 An object which penetrates any of the basic ILS surfaces and becomes the controlling obstacle, but which must be maintained because of its function with regard to air navigation requirements, may be ignored under certain circumstances in calculating the OCA/H, with the following provision. It must be established by the appropriate authority that the portion which penetrates the surface is of minimum mass and frangibly mounted and would not adversely affect the safety of aircraft operations.

6.4.8 Obstacle clearance of the precision segment using obstacle assessment surfaces (OAS) criteria for GLS operations

6.4.8.1 General

6.4.8.1.1 This section describes the OAS surfaces, the constants which are used to define these surfaces, and the conditions under which adjustments may be made. The OAS dimensions are related to the GLS procedure geometry (GARP – LTP distance, glide path angle), the category of operation and other factors, including aircraft geometry and missed approach climb gradient. A table of OCA/H values for each aircraft category may be promulgated for GLS Cat I and II operations at the particular airfield.

6.4.8.1.2 Additional material is included to enable appropriate authorities to assess realistic benefits for claims of improved performance and associated conditions (see 6.4.8.7, “Adjustment of OAS constants”).

6.4.8.1.3 Note that the OAS are not intended to replace Annex 14 surfaces as planning surfaces for unrestricted obstacle growth. The obstacle density between the basic ILS surfaces and the OAS must be accounted for (see 6.4.8.9, “Effect of obstacle density on OCA/H”).

6.4.8.2 Frame of reference

Positions of obstacles are related to a conventional x, y, z coordinate system with its origin at threshold. See Figure III-3-6-12. The x-axis is parallel to the precision segment track, positive x coordinates measured before landing threshold and negative x coordinates measured after landing threshold. The y-axis is at right angles to the x-axis. Although shown conventionally in Figure III-3-6-12, *in all calculations associated with OAS geometry, the y-coordinate is always counted as positive*. The z-axis is vertical, heights above threshold being positive. All dimensions connected with the OAS are specified in metres only. The dimensions should include any adjustments necessary to cater for tolerances in survey data (see Part I, Section 2, Chapter 1, 1.8).

6.4.8.3 Definition of obstacle assessment surfaces (OAS)

6.4.8.3.1 The OAS consist of six sloping plane surfaces (denoted by letters W, X, Y, and Z) arranged symmetrically about the precision segment track, together with the horizontal plane which contains the threshold (see Figures III-3-6-10 and III-3-6-11). The geometry of the sloping surfaces is precisely defined by four simple linear equations of the form $z = Ax + By + C$. In these equations x and y are position coordinates and z is the height of the surface at that position (see Figure III-3-6-9).

6.4.8.3.2 For each surface a set of constants A, B and C are obtained from PANS-OPS OAS software for the operational range of GARP-threshold distances and glide path angles. These constants may be modified as specified in 6.4.8.7, “Adjustment of OAS constants”.

6.4.8.3.3 The Category I OAS are limited by the length of the precision segment and, except for the W and X surfaces, by a maximum height of 300 m. The Category II OAS are limited by a maximum height of 150 m.

6.4.8.3.4 Where the Annex 14 approach and transitional obstacle limitation surfaces for code numbers 3 and 4 precision approach runways penetrate inside the OAS, the Annex 14 surfaces become the OAS (i.e. the surfaces for code numbers 3 and 4 are used for obstacle assessment on runways with code numbers 1 and 2). The Annex 14 inner approach, inner transitional and balked landing obstacle limitation surfaces protect Category III operations provided the Category II OCA/H is at or below the top of those surfaces, which may be extended up to 60 m if necessary (see Figure III-3-6-8).

6.4.8.4 OAS constants – specification

For Category I operations the constants A, B and C for each sloping surface are obtained from the PANS-OPS OAS software. The PANS-OPS OAS software gives coefficients for glidepath angles between 2.5 and 3.5 degrees in 0.1 degree steps, and for any GARP-LTP distance between 2 000 m and 4 500 m. Extrapolation outside these limits is not permitted. If a GARP-LTP distance outside this range is entered, the PANS-OPS software gives the coefficients for 2 000 m or 4 500 m as appropriate, which must be used. For an example of the PANS-OPS OAS software results see Figure III-3-6-14.

6.4.8.5 Calculation of OAS heights

To calculate the height z of any of the sloping surfaces at a location x' , y' , the appropriate constants should be first obtained from the PANS-OPS OAS software. These values are then substituted in the equation $z = Ax' + By' + C$. If it is not clear which of the OAS surfaces is above the obstacle location, this should be repeated for the other sloping surfaces. The OAS height is the highest of the plane heights (zero if all the plane heights are negative).

Note.— The PANS-OPS OAS software also contains an OCH calculator that will show the height of OAS surface Z above any X, Y location. It includes all the adjustments specified for ILS geometry, aircraft dimensions, missed approach climb gradient and RDH.

6.4.8.6 OAS template construction

Templates, or plan views of the OAS contours to map scale, are sometimes used to help identify obstacles for detail survey (see Figure III-3-6-11). The OAS data on the PANS-OPS OAS software includes the coordinates of the points of intersection of the sloping surfaces at threshold level and at 300 m above threshold level for Cat I (see Figure III-3-6-12). They are at 150 m for Cat II. The intersection coordinates at threshold level are labelled as C, D and E.

6.4.8.7 Adjustment of OAS constants

6.4.8.7.1 General. The following paragraphs describe the adjustments which may be made to the OAS constants. These adjustments are mandatory when the standard conditions are not met (See 6.1.3, “Standard conditions”). Optional adjustments may be made when so specified. For examples of calculations see the *Instrument Flight Procedures Construction Manual* (Doc 9368).

6.4.8.7.2 Reasons for adjusting constants. The constants may be modified by the PANS-OPS OAS software to account for the following:

- a) dimensions of specific aircraft;
- b) the height of the DCP;
- c) course width greater than 210 m at threshold (+/- 105 m each side of the final approach track);
- d) missed approach climb gradient; and
- e) improved beam holding performance due to use of autopilots certified for Category II operations (see 6.4.8.7.6, below).

6.4.8.7.3 Specific aircraft dimensions. An adjustment is mandatory where aircraft wing span exceed those specified in 6.1.3, “Standard conditions” and is optional for aircraft with smaller wing span. The PANS-OPS OAS software adjusts the OAS coefficients and template coordinates for the standard dimensions of category A, B, C, D, and D_L aircraft

automatically. It will do the same for specific aircraft dimensions in any category. It uses the following correction formula to adjust the coefficient C for the X and Y surfaces:

$$\text{X surface: } C_{x\text{corr}} = C_x - B_x \times P$$

$$\text{Y surface: } C_{y\text{corr}} = C_y - B_y \times P$$

where:

$$P = [t/B_x \text{ or } s + (t-3)/B_x], \text{ whichever is the maximum} - [6/B_x \text{ or } 30 + 3/B_x, \text{ whichever is the maximum}]; \text{ and}$$

s = semi-span

t = vertical distance between paths of the guidance control point and the lowest part of the wheels as listed in 6.1.3 a).

6.4.8.7.4 *Height of the datum crossing point (RDH).* The constants are based on a reference datum height (RDH) of 15 m. An adjustment to the OAS constants is mandatory for an RDH less than 15 m, and is optional for an RDH greater than 15 m. The PANS-OPS OAS software adjusts the OAS coefficients and template co-ordinates by correcting the tabulated values of the coefficient C for the W, W*, X and Y surfaces as follows:

$$C_{\text{corr}} = C + (\text{RDH} - 15)$$

where: C_{corr} = corrected value of coefficient C for the appropriate surface

C = tabulated value.

6.4.8.7.5 *Course width greater than 210 m at threshold.* Where the course width at threshold is greater than the nominal value of 210 m (+/-105 m each side of the final approach track), the collision risk model (CRM) method described in 6.4.9 shall be used. Adjustments for sector widths less than 210 m shall not be made, and are inhibited on the PANS-OPS OAS software.

6.4.8.7.6 *Use of autopilot (autocoupled) in Category II.* The Category II OAS may be reduced to reflect the improved beam holding of autopilots where these are certificated for the operation by the appropriate authority. This reduction is achieved in the PANS-OPS OAS software by the use of modified A, B and C constants for the X surface, and the introduction of an extra surface (denoted by W*) (see Figure II-1-1-11 c)). The use of these reduced surfaces should not be authorized for non-autocoupled approaches.

6.4.8.7.7 *Missed approach gradient.* If missed approach climb gradients better than the nominal 2.5 per cent can be achieved, the Y and Z surfaces may be adjusted. This is done by selecting the desired missed approach climb gradient in the PANS-OPS OAS software. The programme then adjusts the Y and Z surface constants.

6.4.8.8 *Determination of OCA/H with OAS*

6.4.8.8.1 *General.* The OCA/H is determined by accounting for all obstacles which penetrate the basic ILS surfaces defined in 6.4.7.2 and the OAS applicable to the Category of operation being considered. The exemptions listed in 6.4.7.3, "Determination of OCA/H with basic ILS surfaces" for obstacles penetrating the basic ILS surfaces may be applied to obstacles penetrating the OAS, providing the criteria listed in that paragraph are met. For Category I operations, ILS Cat I OAS apply; for Category II and Category III operations, the Cat II OAS apply, as well as those portions of Cat I OAS which lie outside the limits of Cat II, higher than 150 m above the threshold.

6.4.8.8.2 *Calculation of OCA/H values with OAS.* Accountable obstacles, as determined below in 6.4.8.8.2.1, “OCA/H calculation steps”, are divided into approach and missed approach obstacles. The standard method of categorization is as follows: approach obstacles are those between the FAP and 900 m after threshold (Category H, 700 m if necessary); missed approach obstacles are those in the remainder of the precision segment (see Figure III-3-6-15). However, in some cases this categorization may produce an excessive penalty for certain missed approach obstacles (see Attachment to Part II, 1.9). Alternatively, missed approach obstacles may therefore be defined as those above a plane surface parallel to the plane of the glide path and with origin at -900 m (Category H, -700 m if necessary) (see Figure III-3-6-16), i.e. obstacle height greater than $(900 + x) \tan \theta$.

6.4.8.8.2.1 *OCA/H calculation steps*

- a) Determine the height of the highest approach obstacle.
- b) Convert the heights of all missed approach obstacles (h_{ma}) to the heights of equivalent approach obstacles (h_a) by the formula given below, and determine the highest equivalent approach obstacle.
- c) Determine which of the obstacles identified in steps a) and b) is the highest. This will give the controlling obstacle.
- d) Add the appropriate aircraft category related margin (Table III-3-6-3) to the height of the highest controlling obstacle.

$$h_a = \frac{h_{ma} \cot Z + (-x_z + x)}{\cot Z + \cot \theta}$$

where: h_a = height of equivalent approach obstacle

h_{ma} = height of missed approach obstacle

θ = glide path angle

Z = angle of missed approach surface

x = range of obstacle relative to landing threshold point (negative after LTP)

x_z = distance from threshold to origin of Z surface (-900 m, -700 m for Cat H)

6.4.8.8.3 *Adjustments for high airfield elevations and steep glide path angles.*

6.4.8.8.3.1 The margins shall be adjusted as follows:

- a) for airfield elevation higher than 900 m (2 953 ft), the allowances shall be increased by 2 per cent of the radio altimeter margin per 300 m (1 000 ft) airfield elevation; and
- b) for glide path angles greater than 3.2° in exceptional cases, the allowances shall be increased by the 5 per cent of the radio altimeter margin per 0.1° increase in glide path angle between 3.2° and 3.5°.

6.4.8.8.3.1.1 Procedures involving glide paths greater than 3.5° or any angle when the nominal rate of descent (V_{at} for the aircraft type $\times$ the sine of the glide path angle) exceeds 5 m/sec (1 000 ft/min), are non-standard. They require the following:

- a) increase of height loss margin (which may be aircraft type specific);

- b) adjustment of the origin of the missed approach surface (applicable for fixed-wing aircraft procedures only);
- c) adjustment of the slope of the W surface;
- d) re-survey of obstacles; and
- e) the application of related operational constraints.

Such procedures are normally restricted to specifically approved operators and aircraft, and are associated with appropriate aircraft and crew restrictions. They are not to be used as a means to introduce noise abatement procedures.

6.4.8.8.3.1.2 Part II, Section 1, Chapter 1, Appendix B shows the procedure design changes required and the related operational/certification considerations.

Example: Aircraft Category C — Aerodrome elevation:

1 650 m above MSL; glide path angle 3.5°

Tabulated allowances: radio altimeter 22 m

(Table III-3-6-3) pressure altimeter 46 m

Correction for aerodrome elevation:

$$22 \times 2/100 \times 1\,650/300 = 2.42 \text{ m}$$

Correction for glide path angle:

$$22 \times 5/100 \times (3.5-3.2)/0.1 = 3.30 \text{ m}$$

Total correction 5.72 m rounded up to 6 m

Corrected radio altimeter margin $22 + 6 = 28 \text{ m}$

Corrected pressure altimeter margin $46 + 6 = 52 \text{ m}$

6.4.8.8.3.2 *Exceptions and adjustments to values in Table III-3-6-3.* Values in Table III-3-6-3 are calculated to account for aircraft using normal manual overshoot procedures from OCA/H on the nominal approach path. The values in Table III-3-6-3 do not apply to Category III operations. The values do not consider the lateral displacement of an obstacle nor the probability of an aircraft being so displaced. If consideration of these joint probabilities is required, then the CRM discussed in 6.4.9 shall be used. Values in Table III-3-6-3 may be adjusted for specific aircraft types where adequate flight and theoretical evidence is available, i.e. the height loss value corresponding to a probability of 1×10^{-5} (based on a missed approach rate 10^{-2}).

6.4.8.8.3.3 *Radio altimeter verification.* If the radio altimeter OCA/H are promulgated, operational checks shall have confirmed the repeatability of radio altimeter information.

6.4.8.8.3.4 *Height loss (HL)/altimeter margins for a specific speed at threshold.* If a height loss/altimeter margin is required for a specific V_{at} , the following formulae apply (see also Table III-3-6-4):

Use of radio altimeter:

$$\text{Margin} = (0.096 V_{at} - 3.2) \text{ metres where } V_{at} \text{ in km/h}$$

Margin = $(0.177 V_{at} - 3.2)$ metres where V_{at} in kt

Use of pressure altimeter:

Margin = $(0.068 V_{at} + 28.3)$ metres where V_{at} in km/h

Margin = $(0.125 V_{at} + 28.3)$ metres where V_{at} in kt

where V_{at} is the speed at threshold based on 1.3 times stall speed in the landing configuration at maximum certificated landing mass.

Note.— The equations assume the aerodynamic and dynamic characteristics of the aircraft are directly related to the speed category. Thus, the calculated height loss/altimeter margins may not realistically represent small aircraft with V_{at} at maximum landing mass exceeding 165 kt.

6.4.8.8.3.5 *Height loss (HL)/altimeter margins for a specific speed at threshold (Helicopters).* For helicopter operations the concept of V_{at} is not applicable. Height loss margins are listed in Table III-3-6-3.

6.4.8.9 *Effect of obstacle density on OCA/H.* To assess the acceptability of obstacle density below the OAS, the CRM described in 6.4.9 may be used. This can provide assistance by comparing aerodrome environments and assessing risk levels associated with given OCA/H values. It is emphasized that it is not a substitute for operational judgement.

6.4.9 Obstacle clearance of the precision segment — application of collision risk model (CRM) for GLS operations

Note.— A specific GLS implementation of the CRM is in preparation.

6.4.9.1 *General.* The ILS CRM is a computer programme that establishes the numerical risk which can be compared to the target level of safety for aircraft operating to a specified OCA/H height. This ILS CRM can be used for GLS operations while the specific GLS CRM is in preparation. A description of the ILS CRM programme and instructions on its use, including the precise format of both the data required as input and the output results, are given in the *Manual on the Use of the Collision Risk Model (CRM) for ILS Operations* (Doc 9274).

6.4.9.2 *Input.* The CRM requires the following data as input:

- a) *Aerodrome details:* name, runway threshold position and runway orientation, threshold elevation above MSL, details of preceding segment;
- b) *GLS parameters:* category, glide path angle, GARP – LTP distance, course width and height of DCP;
- c) *Missed approach parameters:* decision height/altitude (obstacle clearance height) and missed approach turn point;
- d) *Aircraft parameters:* type, wheel height (antenna to bottom of wheel), and wing semi-span, aircraft category (A, B, C, D or D_L) and missed approach climb gradient; and

Note.— The CRM does not consider Category E aircraft.

- e) *Obstacle data:* obstacle boundaries (either as x and y coordinates relative to the runway threshold or as map grid coordinates) and obstacle height (either above threshold elevation or above MSL). For density assessment, all obstacles penetrating the basic ILS surfaces described in 6.4.7.2 must be included.

6.4.9.3 *Output and application.* The output of the programme is the overall (total) risk of collision with obstacles to the aircraft of operating to the specified OCA/H and through the missed approach. Other information may also be produced using various output options.

6.4.9.3.1 For example, the risks associated with individual obstacles may be given, and these risks can be ordered, either in terms of obstacle range, or more usefully in terms of risk magnitude, so that the user may see at a glance which obstacles are the major contributors to the total risk.

6.4.9.3.2 The user, by rerunning the CRM with the appropriate parameters, can assess the effect on the safety of operations of any alteration in the parameters, typically varying the glide path angle, or increasing/reducing the OCA/H. The computed risk is compared with a pre-specified acceptable level of risk (not worse than 1×10^{-7} per approach) which meets the overall safety target.

6.4.9.4 *Determination of OCA/H.* The determination of OCA/H is a process in which the CRM is successively rerun with changing values of OCA/H until the computed risk meets the target level of safety (i.e. better than 1×10^{-7} per approach).

6.5 MISSED APPROACH AFTER THE PRECISION SEGMENT (FINAL MISSED APPROACH)

6.5.1 General

The criteria for the final missed approach are based on those for the general criteria (see Chapter 7). Certain modifications have been made to allow for the different areas and surfaces associated with the GLS precision segment and the possible variation in OCA/H for that segment with aircraft category.

6.5.1.1 The datum used for calculation of distances and gradients in obstacle clearance calculations is termed “start of climb” (SOC). It is defined by the height and range at which the plane GP’ (a plane parallel with the glide path and with origin at –900 m (Cat H, –700 m) at threshold level) reaches an altitude OCA/H – HL. Area construction is according to the navigation system specified for the missed approach (where OCA/H and HL both relate to the same category of aircraft).

6.5.1.2 If obstacles identified in the final missed approach segment result in an increase in any of the OCA/H calculated for the precision segment, an additional steeper gradient may also be specified for the missed approach surface (Z) for the purpose of lowering the OCA/H (see Part I, Section 4, Chapter 6, 6.2.2.2, “Climb gradient in the intermediate phase”).

6.5.1.3 Missed approach using an RNAV or RNP system for track guidance can be utilized. Only the systems capable of navigation accuracy of 1 NM or better in this phase of flight can be considered. Refer to Part III, Section 1, Chapter 1, Table III-1-1-1 for the navigation specifications that can be used for missed approach.

6.5.2 Straight missed approach

6.5.2.1 *General.* The precision segment terminates at the range where the Z surface reaches a height 300 m above threshold LTP. The width of the Z surface at that range defines the initial width of the final missed approach area which is developed as shown in Figure III-3-6-17. There are no secondary areas.

6.5.2.2 *Straight missed approach obstacle clearance.* (See Figure III-3-6-18.) Obstacle elevation/height in this final missed approach area shall be less than

$$(OCA/H_{ps} - HL) + d_o \tan Z$$

where:

OCA/H for precision segment (OCA/H_{ps}) and HL (Table III-3-6-3) both relate to the same aircraft category;

d_o is measured from SOC parallel to the straight missed approach track; and

Z is the angle of the missed approach surface with the horizontal plane.

If this requirement cannot be met, a turn shall be prescribed to avoid the obstacle in question. If a turn is not practical, the OCA/H shall be raised.

6.5.2.3 Transition from conventional navigation to an RNAV or RNP missed approach may be defined by an RNAV or RNP fix located on the extended final approach course, or by a turn at an altitude direct to a waypoint (see Figure III-3-6-24). If the RNAV or RNP track is collinear with the extended final approach course, the area abeam the SOC shall be expanded at a 15-degree angle from the track course until it reaches the applicable width of the RNAV or RNP area constructed on the extended final approach course. In the region between the lateral boundaries of the Z surface and the outer boundaries of the area, the obstacle evaluation shall be based on the Y surface and a mathematical extrapolation of the Y surface where the area is outside the lateral boundaries of the OAS 300 m contour. The Z surface shall continue to splay at the same angle until reaching the width of the RNAV or RNP area. Secondary areas shall apply from the point where the width of the Z surface reaches the width of the RNAV or RNP primary area (see Figure III-3-6-22). Obstacle clearance up to this point for the extended Y and Z surfaces shall be the same as in the precision segment (see 6.4.8 “Obstacle clearance of the precision segment using obstacle assessment surface (OAS) criteria”), and this shall also apply further out to all portions of the Z surface that are within the RNAV or RNP primary area. The principle of secondary areas shall apply between the edge of the RNAV or RNP primary area and the edge of the total area. If an obstacle penetrates either the extended Y or the Z surface within the secondary area, its elevation/height shall be less than:

$$(OCA/H_{ps} - HL) + d_o \tan Z + M$$

where:

- a) OCA/H for precision segment (OCA/H_{ps}) and HL (Table III-3-6-3) both relate to the same aircraft category;
- b) d_o is measured from SOC parallel to the straight missed approach track;
- c) Z is the angle of the missed approach surface with the horizontal plane; and
- d) M is zero at the edge of the primary area increasing linearly to 30 m (98 ft) at the edge of the total area.

6.5.2.4 If the penetration in the secondary area is less than “M” as described in 6.5.2.3 d) above, then it may be ignored in OCA/H calculations.

6.5.3 Turning missed approach

6.5.3.1 *General.* Turns may be prescribed at a designated TP, at a designated altitude/height, or “as soon as practicable”. The criteria used depend on the location of the turn relative to the normal termination of the precision segment and are as follows:

- a) *turn after normal termination of the precision segment.* If a turn is prescribed after the normal termination range of the precision segment, the criteria of Part I, Section 4, Chapter 6, 6.4.5, “Turn initiated at a designated altitude/height” apply with the following exceptions:

- 1) OCA/H is replaced by (OCA/H – HL) as in 6.5.2.2, “Straight missed approach obstacle clearance”; and
- 2) Because SOC is related to OCA/H, it is not possible to obtain obstacle clearance by the means used in non-precision approaches by independent adjustment of OCA/H or MAPt;
- b) *turn before normal termination of the precision segment.* If a turn is prescribed at a designated altitude/height less than 300 m above threshold or at a designated TP such that the earliest TP is within the normal termination range, the criteria specified in 6.5.3.2 and 6.5.3.3 below shall be applied;
- c) for missed approaches using RNAV or RNP, fly-by or flyover turns should be limited to 90 degrees. RF turns are not permitted as the first RNP leg of the missed approach since there would be no tangent RNP entry track specified. In this case, a TF leg with the first waypoint located on the extended LOC course is required, preceding the RF leg;
- d) the earliest location of the first RNAV or RNP fix is at a distance of ATT after the SOC (see Figure III-3-6-25). If the fix designates a fly-by turn, the additional distance $d_1 + 3V/3600$ where V is the TAS must be added prior to the fix ($d_1 = r \tan A/2$); and
- e) for an RNAV or RNP missed approach, standard turn construction and turning MOC (50 m (164 ft) for turns of more than 15 degrees and 30 m (98 ft) for all other turns) shall apply after the earliest turning point. Secondary areas shall apply between the standard width of the RNAV or RNP primary area and the edge of the total area, except that obstacles located under the Y surface on the outer side of the turn need not be considered. See Figures III-3-6-22 and III-3-6-23.

Note.— Adjustments to designated TP location or to the designated turn altitude may involve redrawing the associated areas and recalculating the clearances. This can exclude some obstacles or introduce new ones. Thus, to obtain the minimum value of OCA/H it may be necessary to adjust the designated TP or turn altitude by trial and error. (See Part II, Section 1, Chapter 1, Appendix A.)

6.5.3.2 Turn at a designated altitude/height less than 300 m above threshold.

6.5.3.2.1 The general criteria apply (see Part I, Section 4, Chapter 6, 6.4.5, “Turn initiated at a designated altitude/height”) as amplified or modified by the contents of this section. Construction of the turn initiation area and the subsequent turn are illustrated in Figure III-3-6-19.

6.5.3.2.2 *Turn altitude/height.* The precision segment terminates at the TP. This allows the calculation of OCA/H_{ps} and $(OCA/H_{ps} - HL)$. SOC is then determined, and turn altitude/height (TNA/H) is computed from the following relationship:

$$TNA/H = OCA/H_{ps} - HL + d_z \tan Z$$

where: d_z = is the horizontal distance from SOC to the TP

OCA/H_{ps} = OCA/H calculated for the precision segment

If the TP is located at the SOC, the chart shall be annotated “turn as soon as practicable to... (heading or facility)” and include sufficient information to identify the position and height of the obstacles dictating the turn requirement.

6.5.3.2.3 Areas

6.5.3.2.3.1 *Turn initiation area.* (See Figure III-3-6-19). The turn initiation area is bounded by the 300 m Category I Y surface contour, and it terminates at the range of the TP.

Note.— The earliest TP is considered to be at the beginning of the 300 m Category I Y surface contour (point D”) unless a fix is specified to limit early turns (see 6.5.3.2.6, “Safeguarding of early turns”).

6.5.3.2.3.2 *Turn area.* The turn area is constructed as specified in the general criteria (Part I, Section 4, Chapter 6, 6.4.3, “Turn parameters”).

6.5.3.2.4 *Obstacle clearance*

- a) *Obstacle clearance in the turn initiation area.* Obstacle elevation/height in the turn initiation area shall be less than:

- 1) turn altitude/height –50 m (164 ft) (Cat H, 40 m (132 ft)) for turns more than 15°; and
- 2) turn altitude/height –30 m (98 ft) for turns 15° or less except that obstacles located under the Y surface on the outer side of the turn need not be considered when calculating turn altitude/height.

- b) *Obstacle clearance in the turn area.* Obstacle elevation/height in the turn area and subsequently shall be less than:

$$\text{turn altitude/height} + d_o \tan Z - \text{MOC}$$

where d_o is measured from the obstacle to the nearest point on the turn initiation area boundary and MOC is:

- 1) 50 m (164 ft) (Cat H, 40 m (132 ft)) for turns more than 15°; and
- 2) 30 m (98 ft) for turns 15° or less,

reducing linearly to zero at the outer edge of the secondary areas, if any.

6.5.3.2.5 *Turn altitude/height adjustments.* If the criteria specified in 6.5.3.2.3 a) and/or b) above cannot be met, the turn altitude/height shall be adjusted. This can be done in two ways:

- a) *adjust turn altitude/height without changing OCA/H:* this means that the TP will be moved and the areas redrawn accordingly; and
- b) *raise turn altitude/height by increasing OCA/H:* this results in a higher turn altitude over the same TP. The turn areas remain unchanged.

6.5.3.2.6 *Safeguarding of early turns.* Where the published procedure does not specify a fix to limit turns for aircraft executing a missed approach from above the designated turn altitude/height, an additional check of obstacles shall be made (see Part I, Section 4, Chapter 6, 6.4.5.6, “Safeguarding of early turns”).

6.5.3.3 *Turn at a designated TP with earliest TP before normal termination of precision segment*

6.5.3.3.1 Where a turn is specified at a designated TP, and the earliest TP is before the normal termination range of the precision segment, the precision segment terminates at the earliest TP. This allows the calculation of OCA/H_{ps} and $(\text{OCA}/H_{ps} - \text{HL})$; SOC is then determined.

6.5.3.3.2 Where the procedure requires that a turn be executed at a designated TP, the following information must be published with the procedure:

- a) the TP, when it is designated by a fix; or

- b) the intersecting VOR radial NDB bearing DME distance where there is no track guidance (see Part I, Section 2, Chapter 2, 2.6.4, “Missed approach fixes”).

6.5.3.3.3 *Turn area.* The turn area is constructed as specified in Part I, Section 4, Chapter 6, 6.4.6.3, except that it is based on the width of the 300 m OAS Y surface contours at the earliest and latest TP (see Figure III-3-6-20).

6.5.3.3.4 *Obstacle clearance.* Obstacle elevation/height shall be less than:

$$(OCA/H_{ps} - HL) + d_o \tan Z - MOC$$

where: d_o = d_z + shortest distance from obstacle to line K-K,

d_z = horizontal distance from SOC to the earliest TP,

and MOC is:

50 m (164 ft) (Cat H, 40 m (132 ft)) for turns more than 15° and

30 m (98 ft) for turns 15° or less.

If the obstacle elevation/height exceeds this value, the OCA/H must be increased, or the TP moved to obtain the required clearance (see Part II, Section 1, Chapter 1, Appendix A).

6.6 GLS CAT I WITH OFFSET AZIMUTH FINAL APPROACH TRACK ALIGNMENT

6.6.1 Use of GLS Cat I with offset azimuth final approach track alignment

The final approach track shall intersect the runway extended centre line:

- a) at an angle not exceeding 5°; and
- b) at a point where the nominal glide path reaches a height called intercept height of at least 55 m (180 ft) above threshold elevation. The procedure shall be annotated: “final approach track offset... degrees” (tenth of degrees).

The general arrangement is shown in Figure III-3-6-21.

6.6.2 Obstacle clearance criteria

The provisions contained in 6.1 to 6.6 apply except that:

- a) all the obstacle clearance surfaces and calculations are based on a fictitious runway aligned with the final approach track. This fictitious runway has the same length and the same landing threshold elevation as the real one. The FTP is analogous to the LTP for aligned procedures. The GLS course width at the FTP is the same as at the LTP. The DCP is located 15 m (50 ft) above the FTP; and
- b) the OCA/H for this procedure shall be at least: intercept altitude/height + 20 m (66 ft).

6.7 PROMULGATION

6.7.1 General

6.7.1.1 The general criteria in Part I, Section 4, Chapter 9 apply as amplified by criteria in Part III, Section 5, Chapter 1, 1.3.4 for chart notes. The instrument approach chart for a GLS approach procedure shall be identified by the title GLS Rwy XX. If more than one GLS approach is published for the same runway, the Duplicate Procedure Title convention shall be applied.

6.7.1.2 When a PBN route with an RF turn to the final approach course is utilized in the intermediate approach segment, then the last item of the PBN procedure description shall be the waypoint at the end of the RF turn with an altitude ‘window’ constraint, or an ‘at’ altitude constraint (see the Note to 6.3.6.3).

6.7.2 Promulgation of OCA/H values

Promulgation of OCA/H for GLS Cat I approach procedures. The OCA or OCH values, as appropriate, shall be promulgated for those categories of aircraft for which the procedure is designed. The values shall be based on the following standard conditions:

- a) Cat I flown with pressure altimeter;
- b) standard aircraft dimensions (see 6.1.3); and
- c) 2.5 per cent missed approach climb gradient.

Additional values of OCA/H may be agreed between operators and the appropriate authority and promulgated, on the basis of evidence supporting the modifications defined in 6.4.8.7.

6.7.3 Minima box

A table of OCA/H values for each aircraft category may be promulgated for Cat I operations at the particular airfield.

6.7.4 Procedures involving non-standard glide path angles

Procedures involving glide paths greater than 3.5 degrees or any angle when the nominal rate of descent exceeds 5 m/s (1 000 ft/min), are non-standard and subject to restrictions (see 6.4.8.3.1). They are normally restricted to specifically approved operators and aircraft, and are promulgated with appropriate aircraft and crew restrictions annotated on the approach chart.

6.7.5 Additional gradient for the final missed approach segment

If obstacles identified in the final missed approach segment result in an increase in any of the OCA/H calculated for the precision segment, an additional steeper gradient may also be specified for the gradient of the missed approach surface (Z) for the purpose of lowering the OCA/H (see Part I, Section 4, Chapter 6, 6.2.2.2, “Climb gradient in the intermediate phase”).

6.7.6 Turns

6.7.6.1 *Turn at a designated altitude/height.* If the turn point is located at the SOC, the chart shall be annotated “turn as practicable to... (heading or facility)” and shall include sufficient information to identify the position and height of the obstacles dictating the turn requirement.

6.7.6.2 *Turn at a designated TP.* Where the procedure requires that a turn be executed at a designated TP, the following information must be published with the procedure:

- a) the TP, when it is designated by a fix; or
- b) the intersecting VOR radial, NDB bearing, or DME distance where there is no track guidance (see Part I, Section 2, Chapter 2, 2.6.4, “Missed approach fixes”).

6.7.7 Reference path identifier

Reference path identifier (RPI). The RPI, as defined in Part III, Section 2, Chapter 6, Appendix B, paragraph 3.5, shall be promulgated with the procedure information.

6.7.8 Channel number

GBAS channel number. The channel number for the procedure as defined in Annex 10, Volume I, Attachment D, paragraph 7.7, shall be promulgated with the procedure information.

6.7.9 GBAS reference point detail

GBAS reference point information. The following information about the GBAS reference point shall be promulgated with the procedure information: latitude/longitude of the GBAS reference point receivers, identifier, service volume radius, MSL elevation, and ellipsoid height.

Table III-3-6-1. Minimum distance between final approach and glide path interceptions

<i>Intercept angle with final approach (degrees)</i>	<i>Cat A/B/H</i>	<i>Cat C/D/D_I/E</i>
RF turn to the final approach course	1.8 km (1.0 NM)	1.8 km (1.0 NM)
< 15	2.8 km (1.5 NM)	2.8 km (1.5 NM)
16 — 30	3.7 km (2.0 NM)	3.7 km (2.0 NM)
31 — 60	3.7 km (2.0 NM)	4.6 km (2.5 NM)
61 — 90 or within a racetrack or reversal procedure	3.7 km (2.0 NM)	5.6 km (3.0 NM)

Table III-3-6-2. Objects which may be ignored in OCA/H calculations

	<i>Maximum height above landing threshold</i>	<i>Minimum lateral distance from runway centre line</i>
Landing system antenna	17 m (55 ft)	120 m
Aircraft taxiing	22 m (72 ft)	150 m
A/C in holding bay or in taxi holding position at a range between threshold and 250 m (Cat I only)	15 m (50 ft)	75 m

Table III-3-6-3. Height loss/altimeter margin

Aircraft category (V_{at})	<i>Margin using radio altimeter</i>		<i>Margin using pressure altimeter</i>	
	Metres	Feet	Metres	Feet
A - 169 km/h (90 kt)	13	42	40	130
B - 223 km/h (120 kt)	18	59	43	142
C - 260 km/h (140 kt)	22	71	46	150
D/D _L - 306 km/h (165 kt)	26	85	49	161
H - 167 km/h (90 kt)	8	25	35	115

Note 1 — Cat H speed is the maximum final approach speed, not V_{at}

Note 2 — For Category E aircraft refer directly to the equations given in 6.4.8.8.3.4

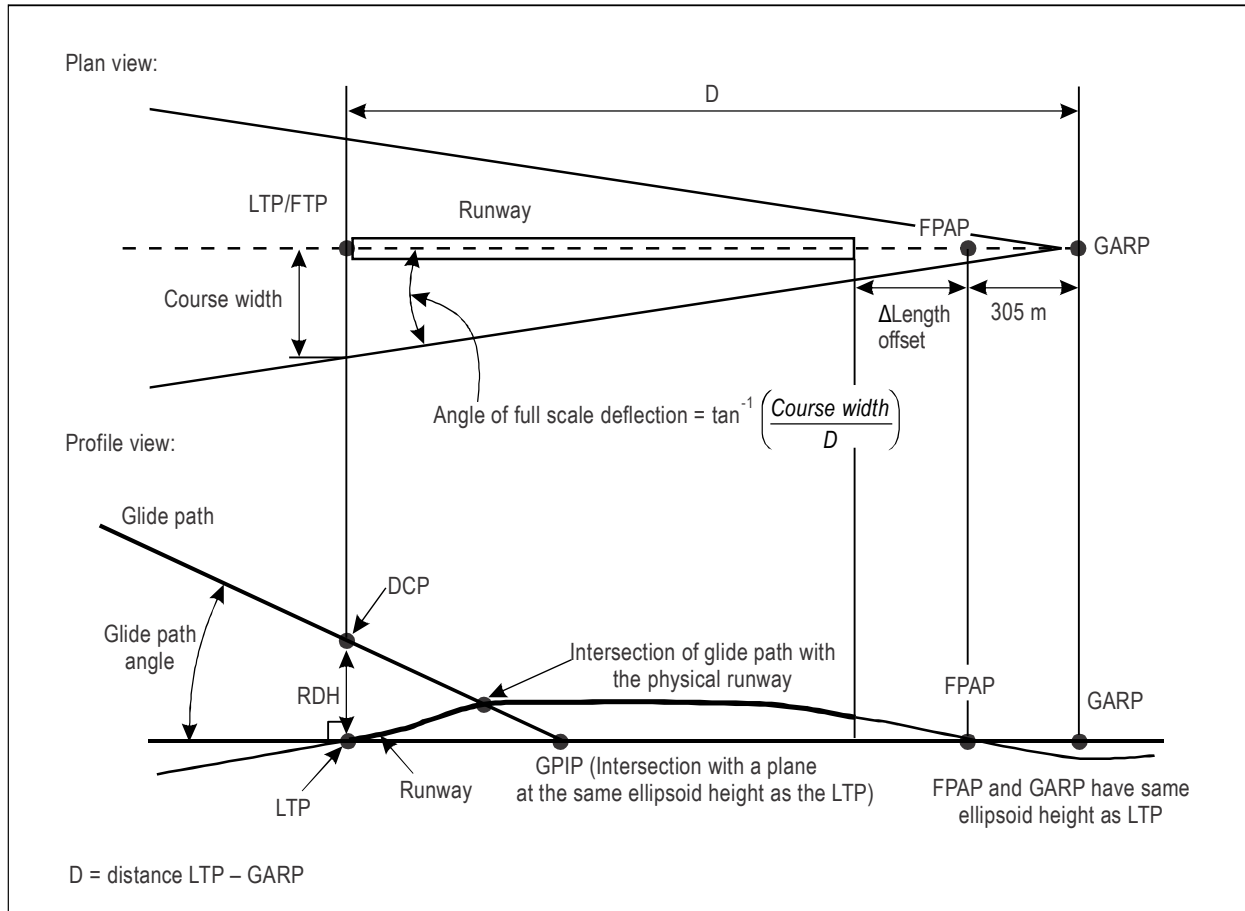


Figure III-3-6-1. Illustration of definitions

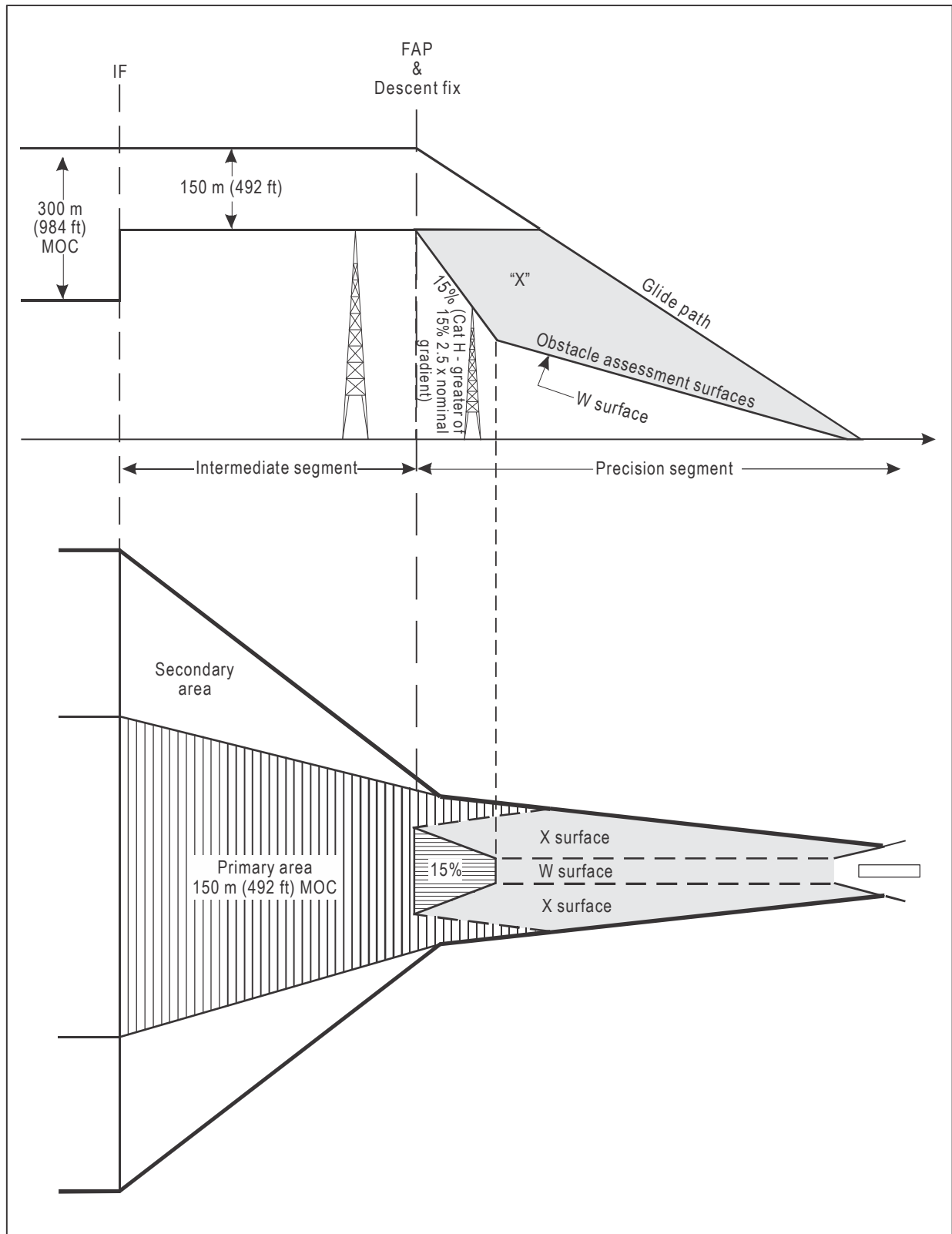


Figure III-3-6-2. Final approach point defined by descent fix

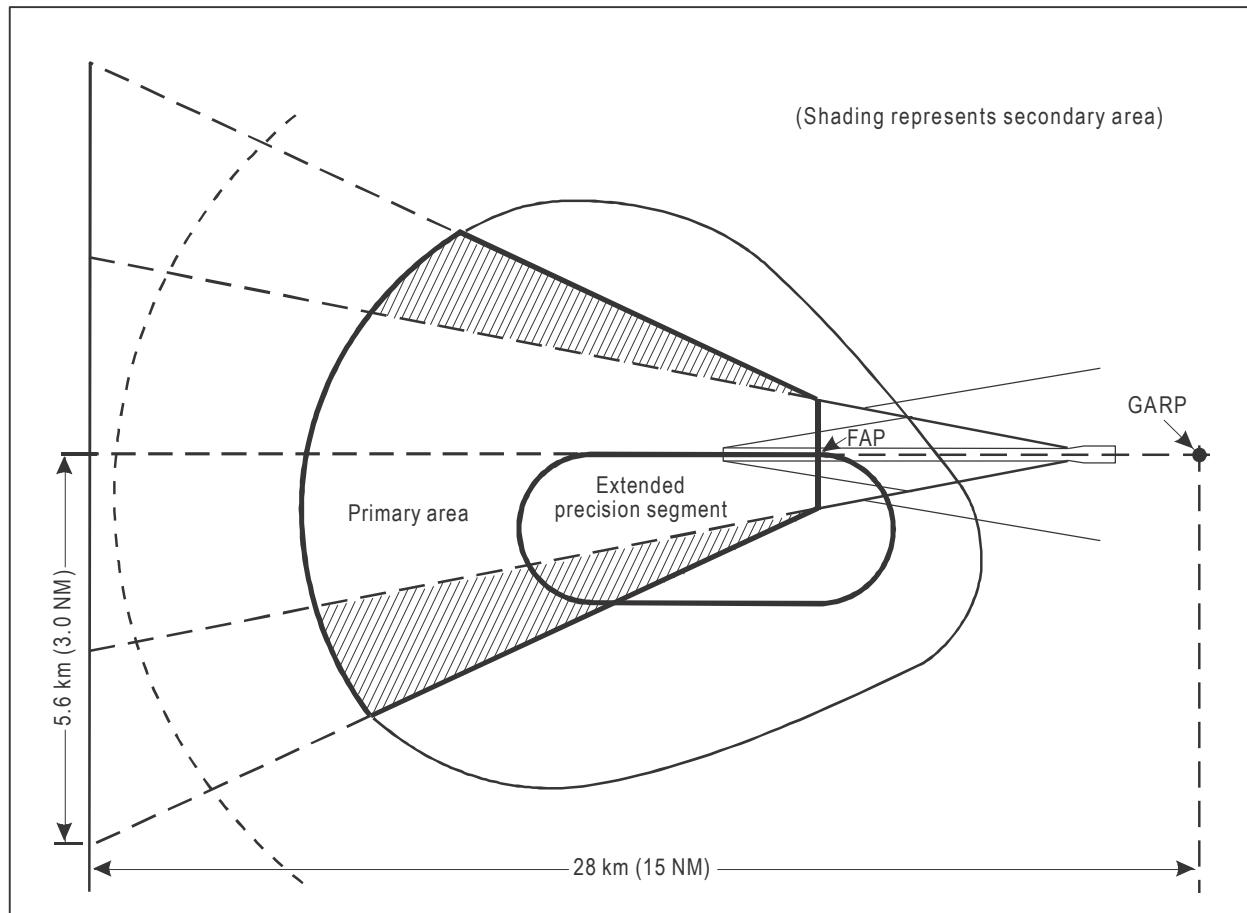


Figure III-3-6-3. Intermediate approach area. GLS approach using reversal or racetrack procedure

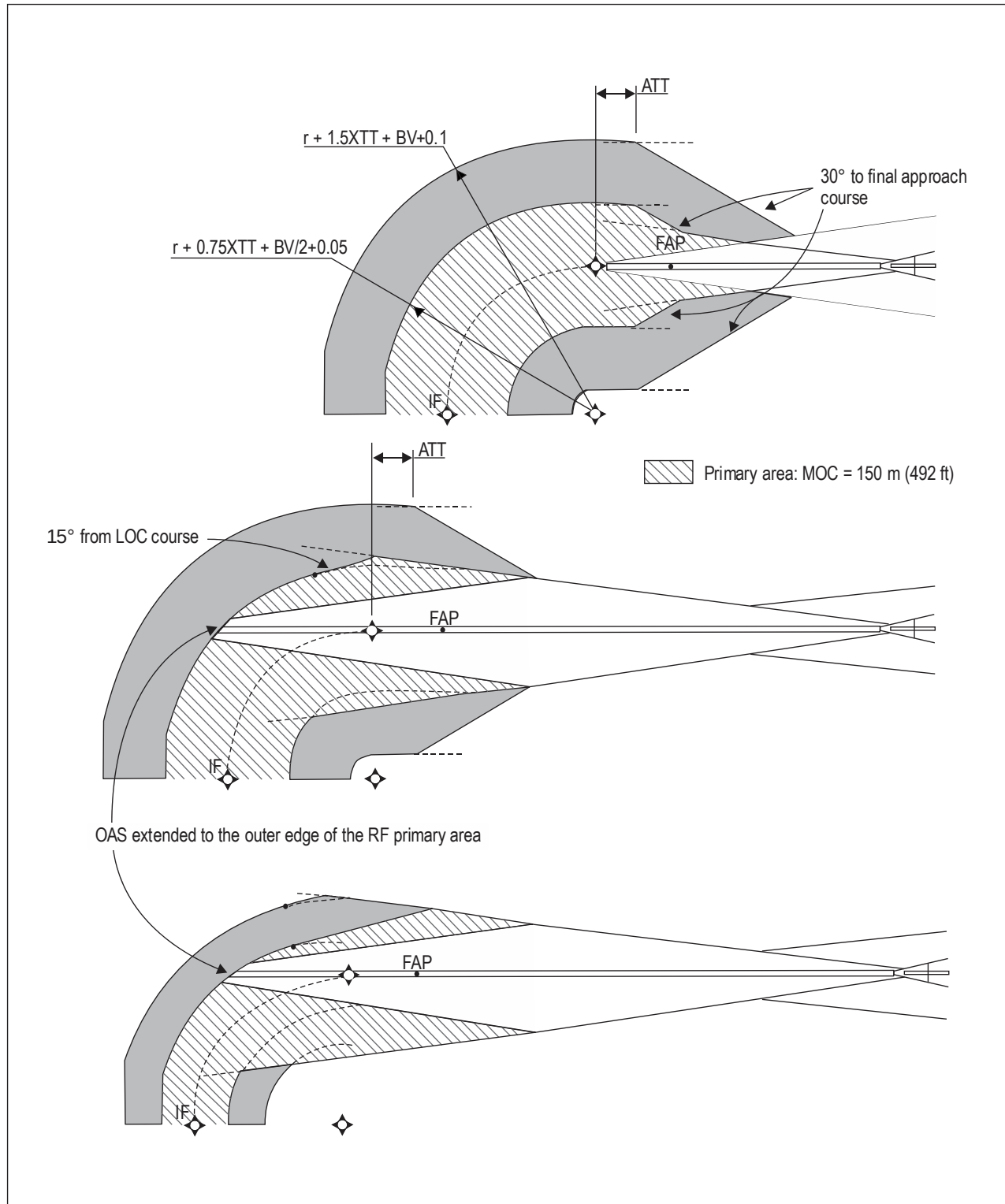


Figure III-3-6-4. Area construction for radius-to-fix turns to the final approach course

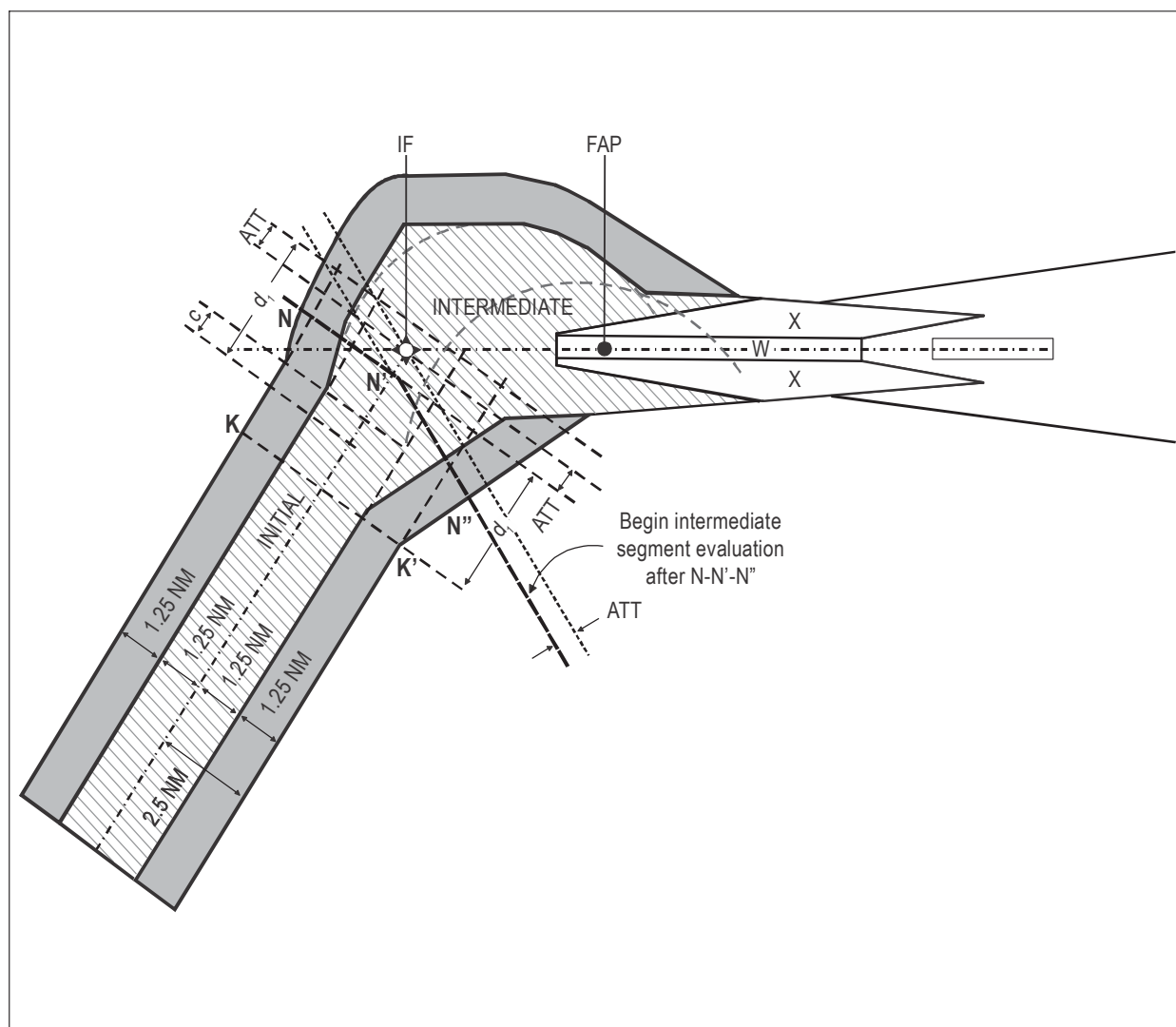


Figure III-3-6-5. Example: RNAV 1 or RNP 1 to GBAS transition, 60° intercept

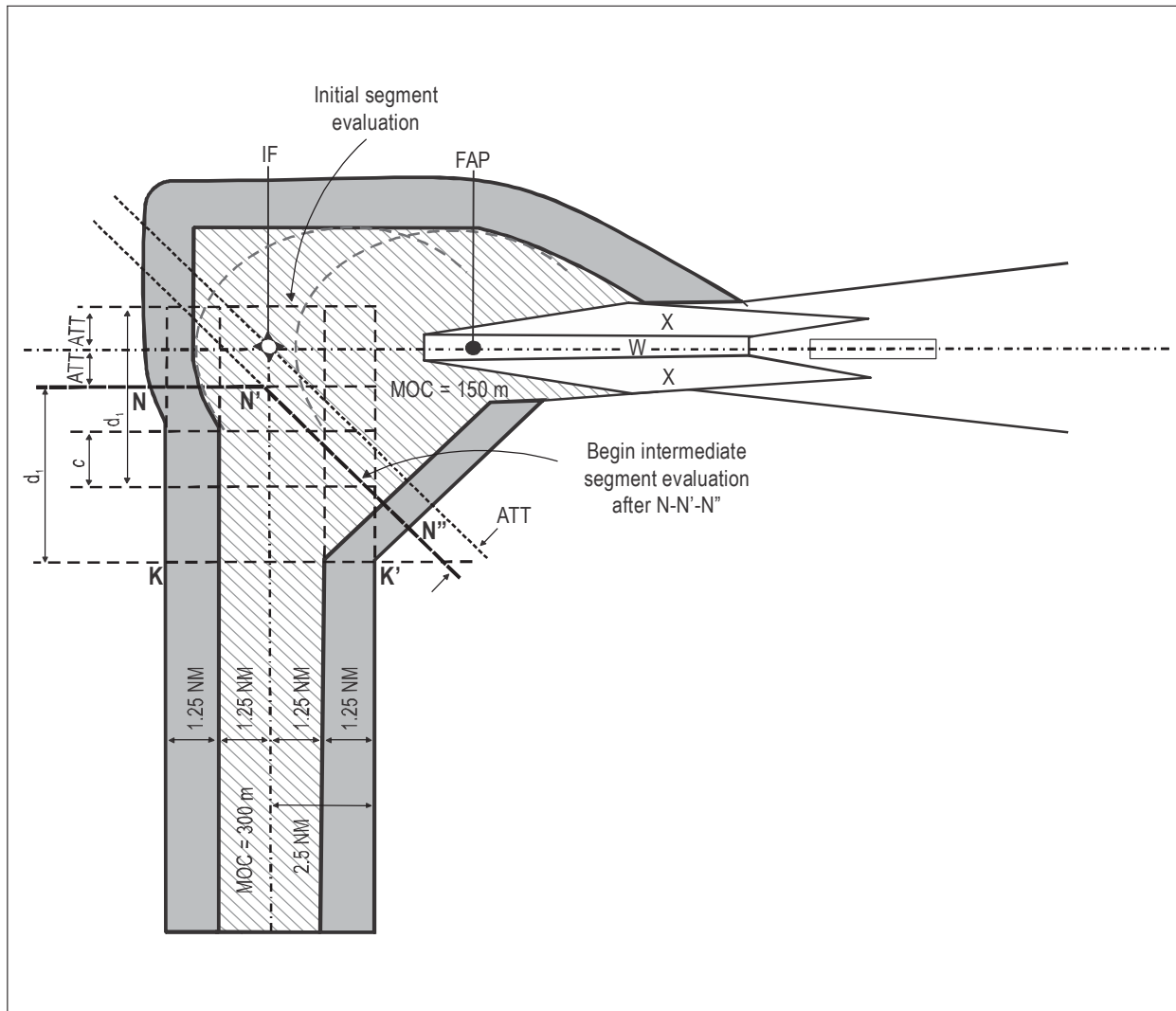


Figure III-3-6-6. Example: RNAV 1 or RNP 1 to GBAS transition, 90° intercept (max)

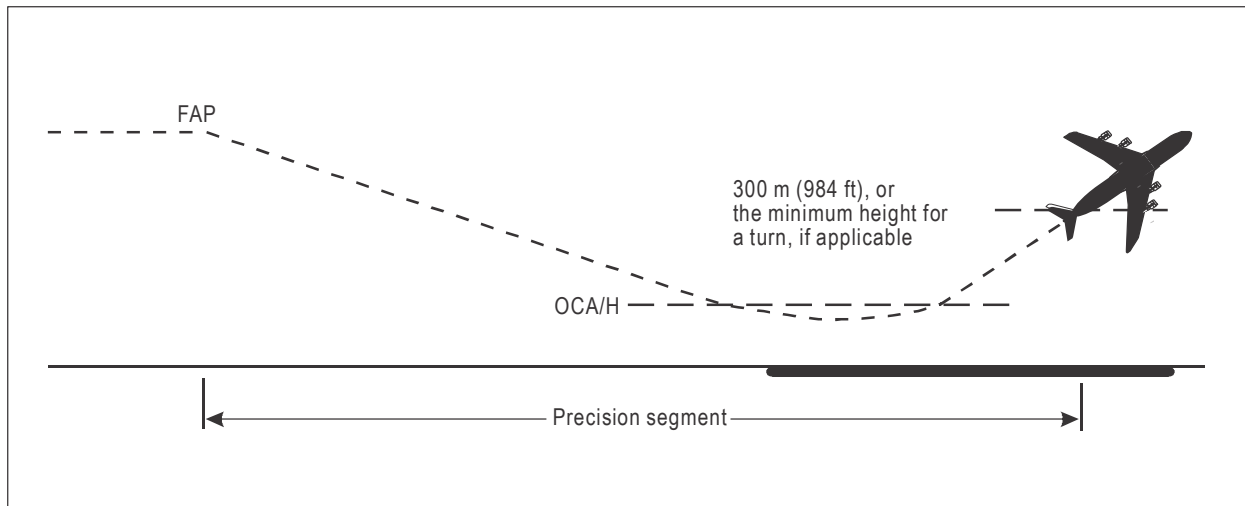


Figure III-3-6-7. Precision segment

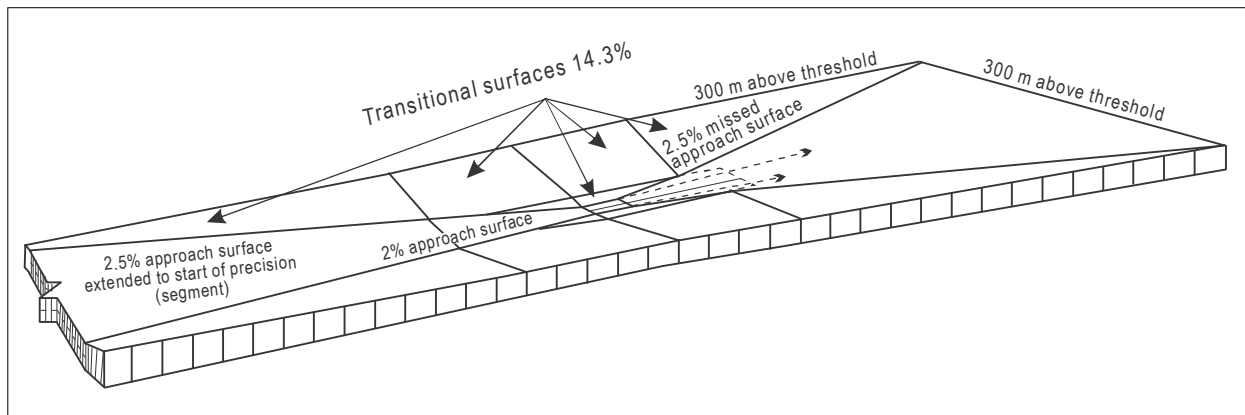


Figure III-3-6-8. Illustration of basic ILS surfaces as described in Part III, Section 3, Chapter 6, 6.4.7.2

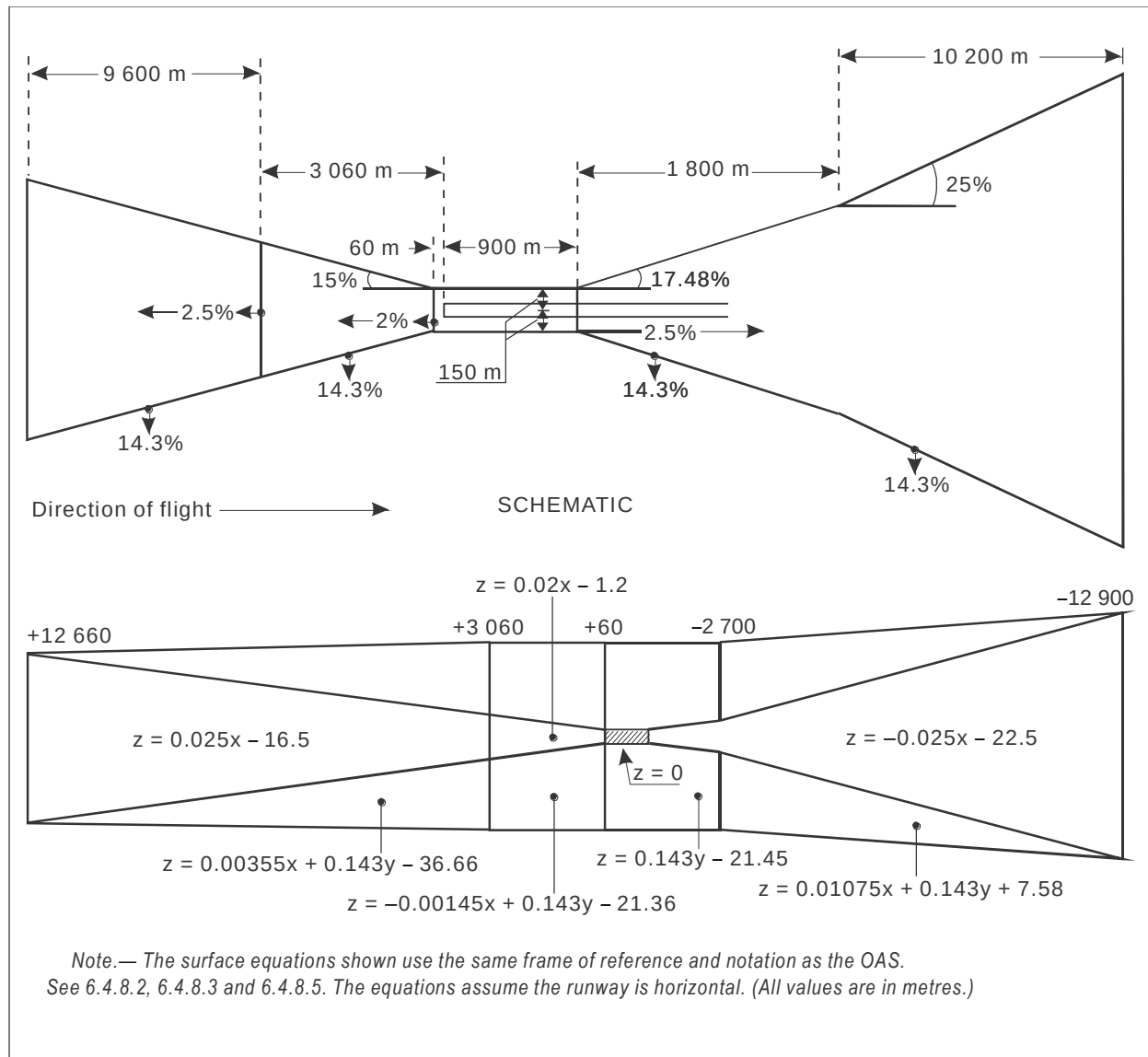


Figure III-3-6-9. Surface equations — basic ILS surfaces

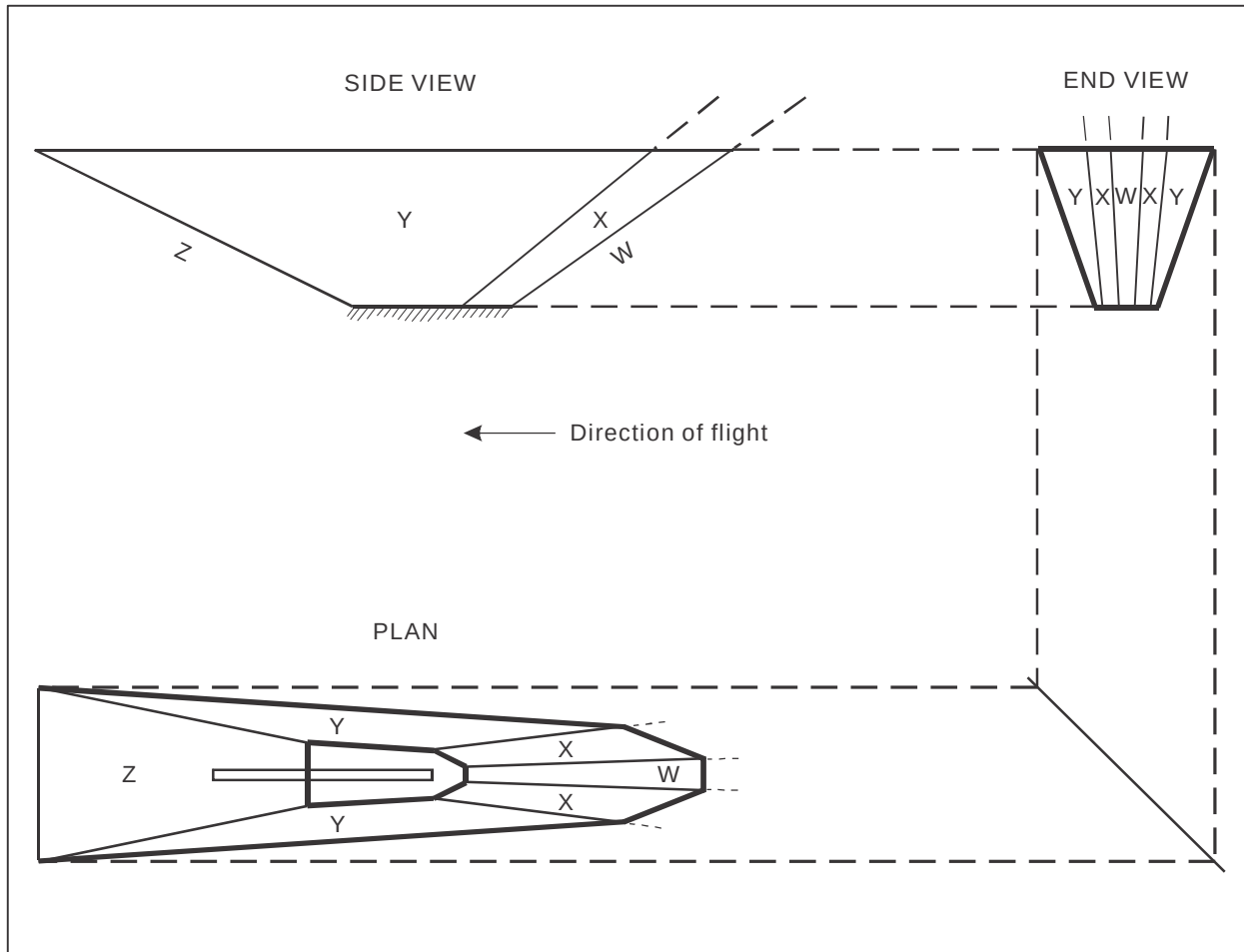


Figure III-3-6-10. Illustration of ILS obstacle assessment surfaces for GLS operations

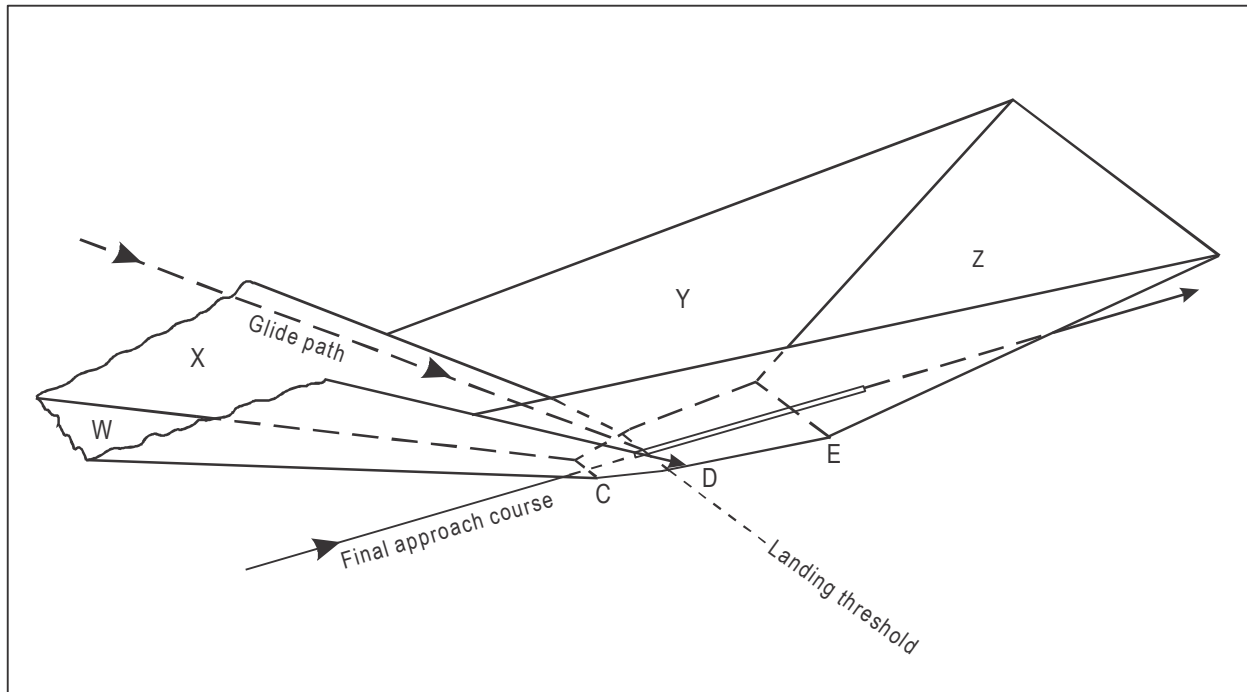


Figure III-3-6-11. Illustration of ILS obstacle assessment surfaces for GLS operations — perspective view

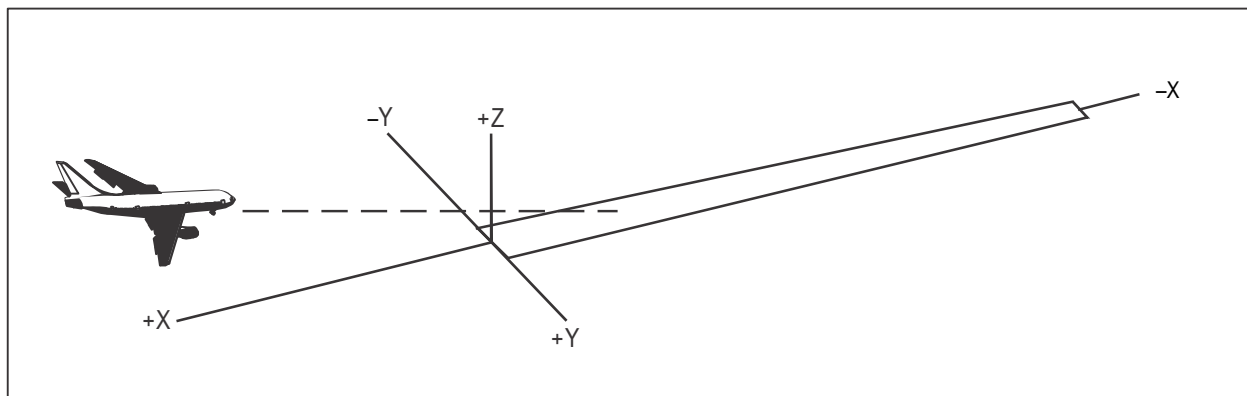
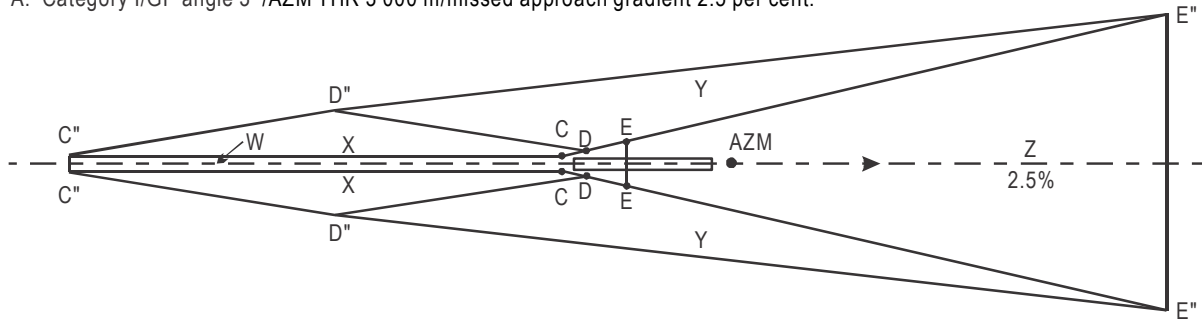


Figure III-3-6-12. System of coordinates

A. Category I/GP angle 3°/AZM THR 3 000 m/missed approach gradient 2.5 per cent.



Equations of the obstacle assessment surfaces:

$$W | z = 0.0285x - 8.01$$

$$X | z = 0.027681x + 0.1825y - 16.72$$

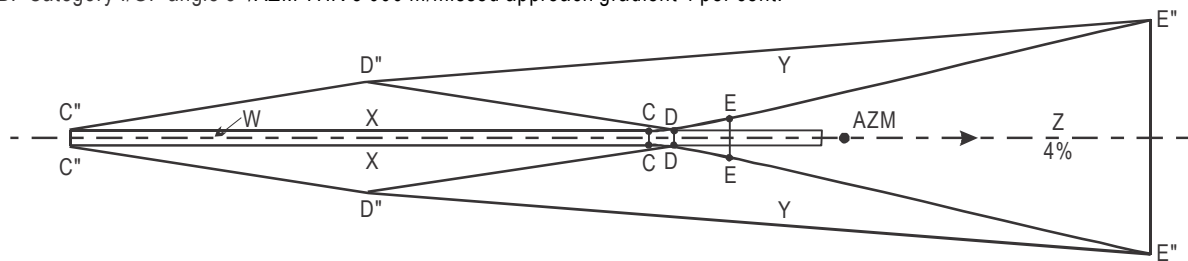
$$Y | z = 0.023948x + 0.210054y - 21.51$$

$$Z | z = -0.025x - 22.50$$

Coordinates of points C, D, E, C'', D'', E'' (m):

	C	D	E	C''	D''	E''
x	281	-286	-900	10 807	5 438	-12 900
y	49	135	205	96	910	3 001
z	0	0	0	300	300	300

B. Category I/GP angle 3°/AZM THR 3 000 m/missed approach gradient 4 per cent.



Equations of the obstacle assessment surfaces:

$$W | z = 0.0285x - 8.01$$

$$X | z = 0.027681x + 0.1825y - 16.72$$

$$Y | z = 0.020158x + 0.238021y - 26.37$$

$$Z | z = -0.04x - 36.00$$

Coordinates of points C, D, E, C'', D'', E'' (m):

	C	D	E	C''	D''	E''
x	281	-286	-900	10 807	5 438	-8 400
y	49	135	187	96	910	2 082
z	0	0	0	300	300	300

Figure III-3-6-13. Typical OAS contours for standard size aircraft

OAS

NAVIGATION AID DATA

Approach Category

Glide path (°) LOC THR Dist ⁽¹⁾ m

RDH m Course width at THR m

AIRCRAFT DATA

M/App CG (%) CAT STD ☒ Wing Semi Span m GP Wheel / Antenna height m

OAS constants

	A	B	C		A	B	C
W	0.028500	0.000000	-8.01	Y	0.023948	0.210054	-21.51
W*				Z	-0.025000	0.000000	-22.50
X	0.027681	0.182500	-16.72				

OAS Template coordinates -m (metres)

THR Elevation

	X	Y
C	281	49
D	-286	135
E	-900	205

300 m Height

	X	Y
C"	10807	96
C'''		
D"	5438	910
E"	-12900	3001

OAS height calculator

X m Y m Z m

(1) For MLS read: "Azimuth antenna to THR distance"; for GBAS read: "GARP-LTP distance".

Figure III-3-6-14. OAS output data generated by the PANS-OPS OAS software

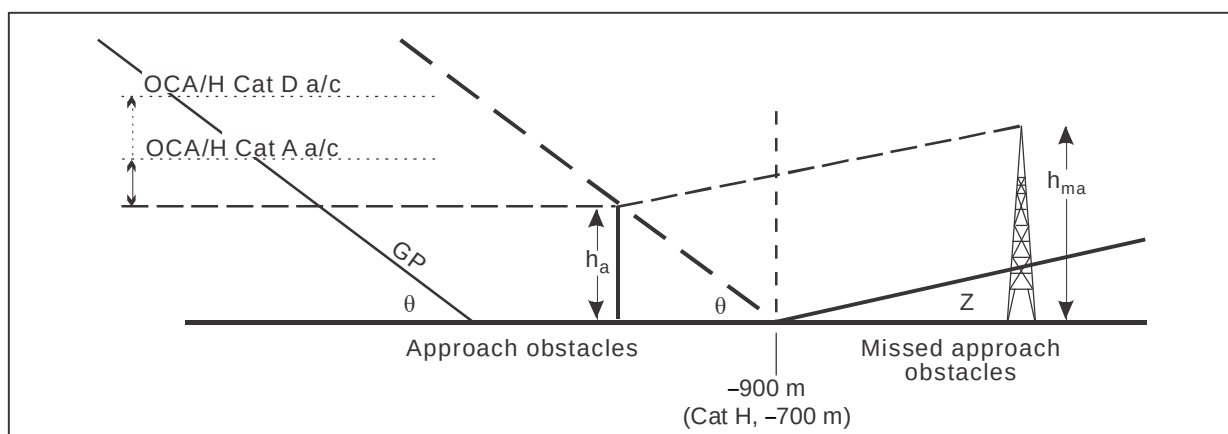


Figure III-3-6-15. Missed approach obstacle after range -900 m

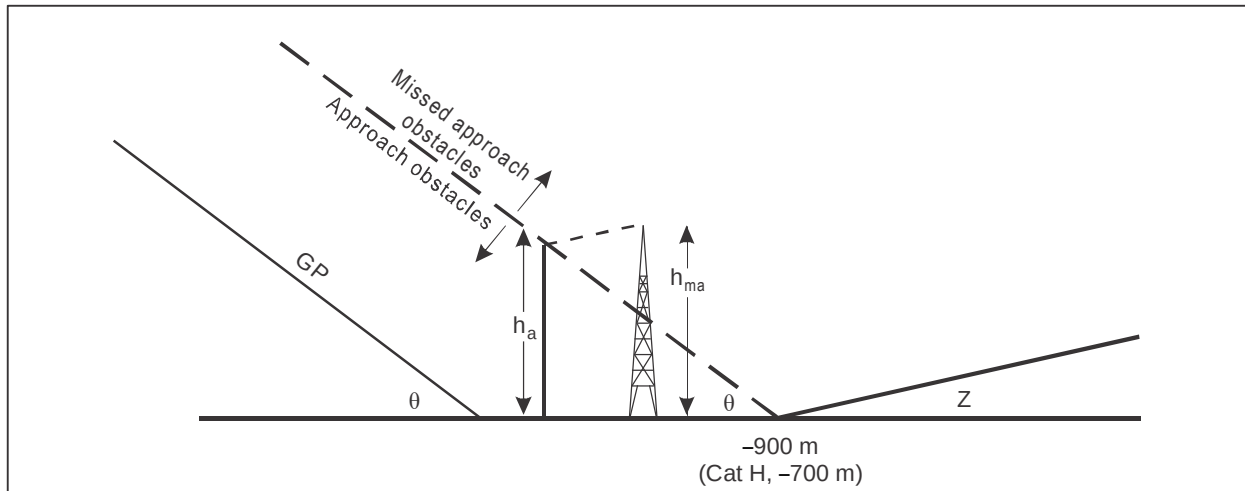


Figure III-3-6-16. Missed approach obstacle before range -900 m

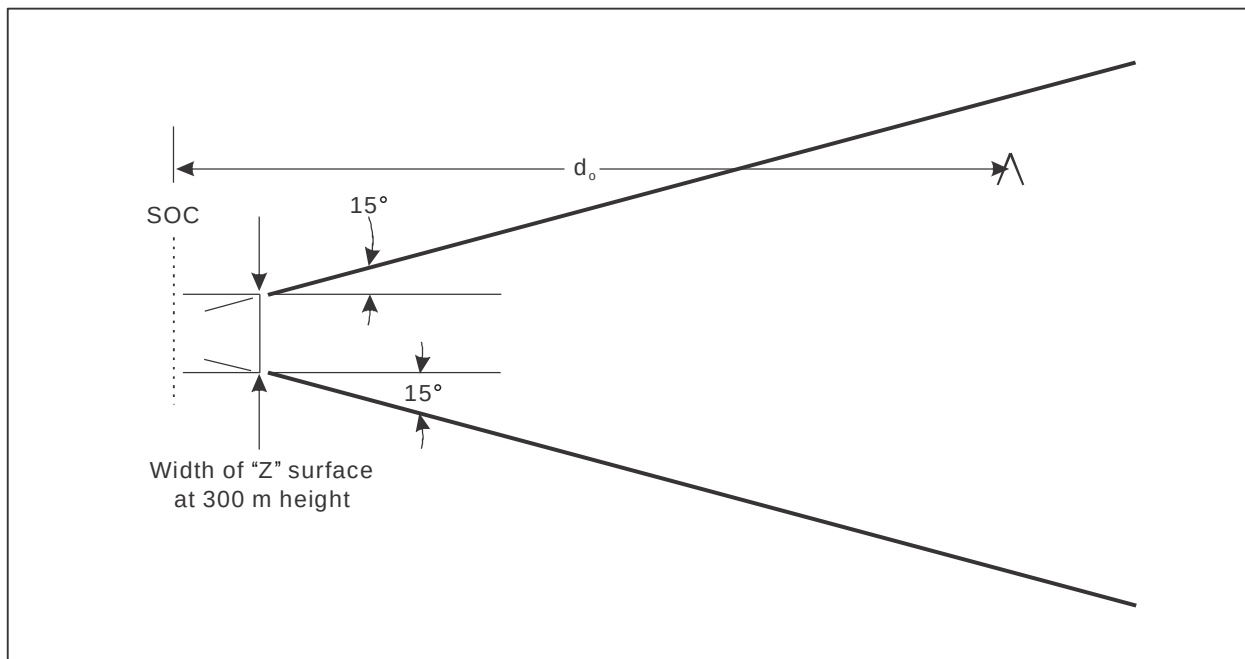


Figure III-3-6-17. Final segment of straight missed approach

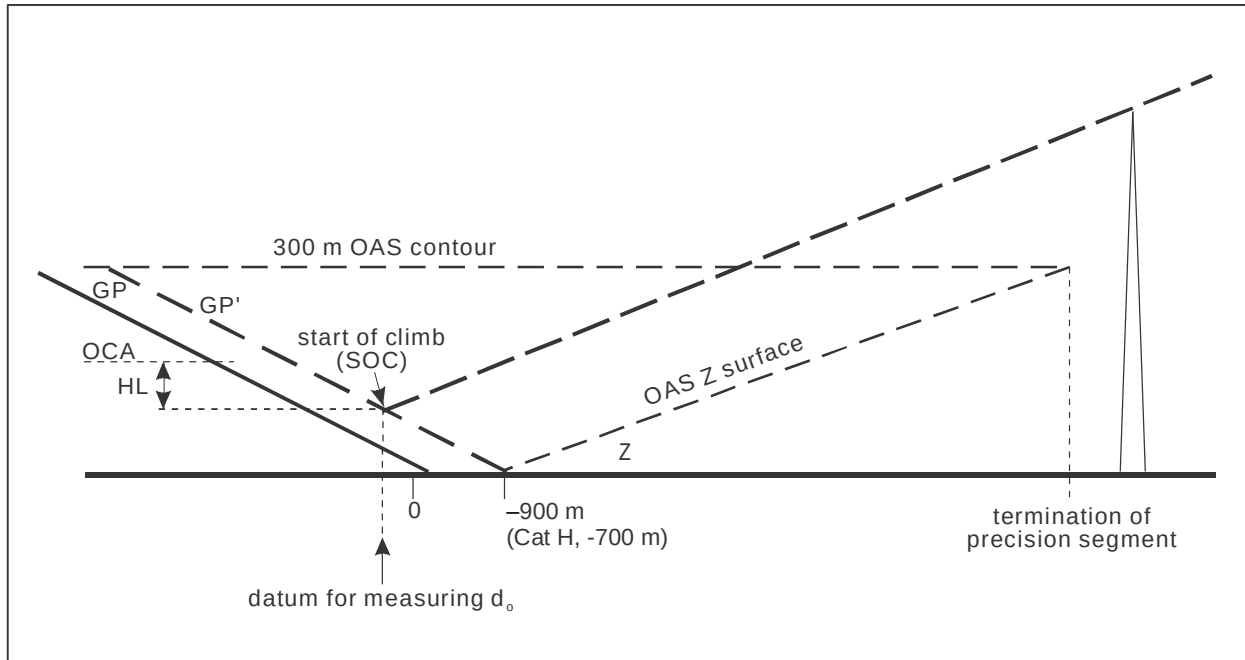


Figure III-3-6-18. Straight missed approach obstacle clearance

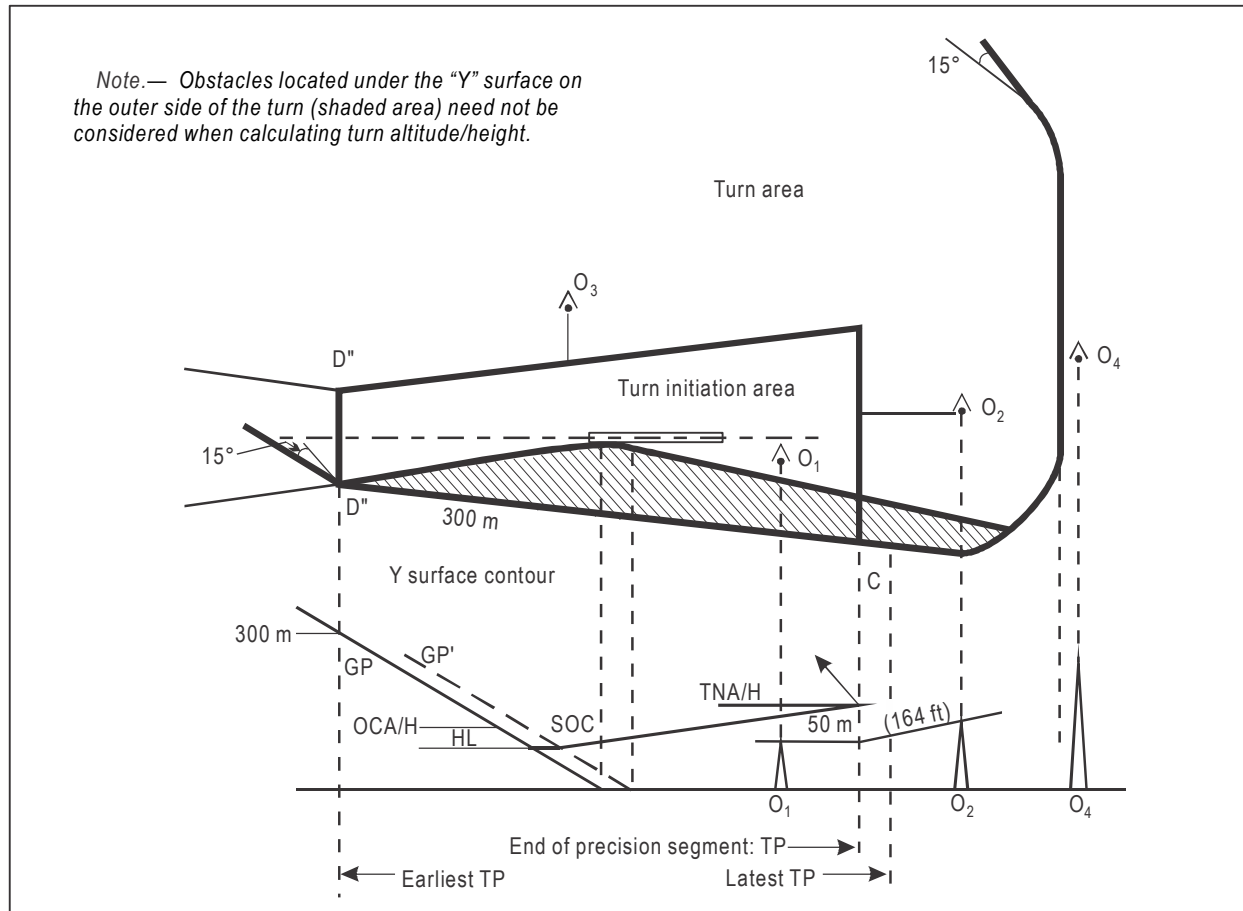


Figure III-3-6-19. Turn at a designated altitude

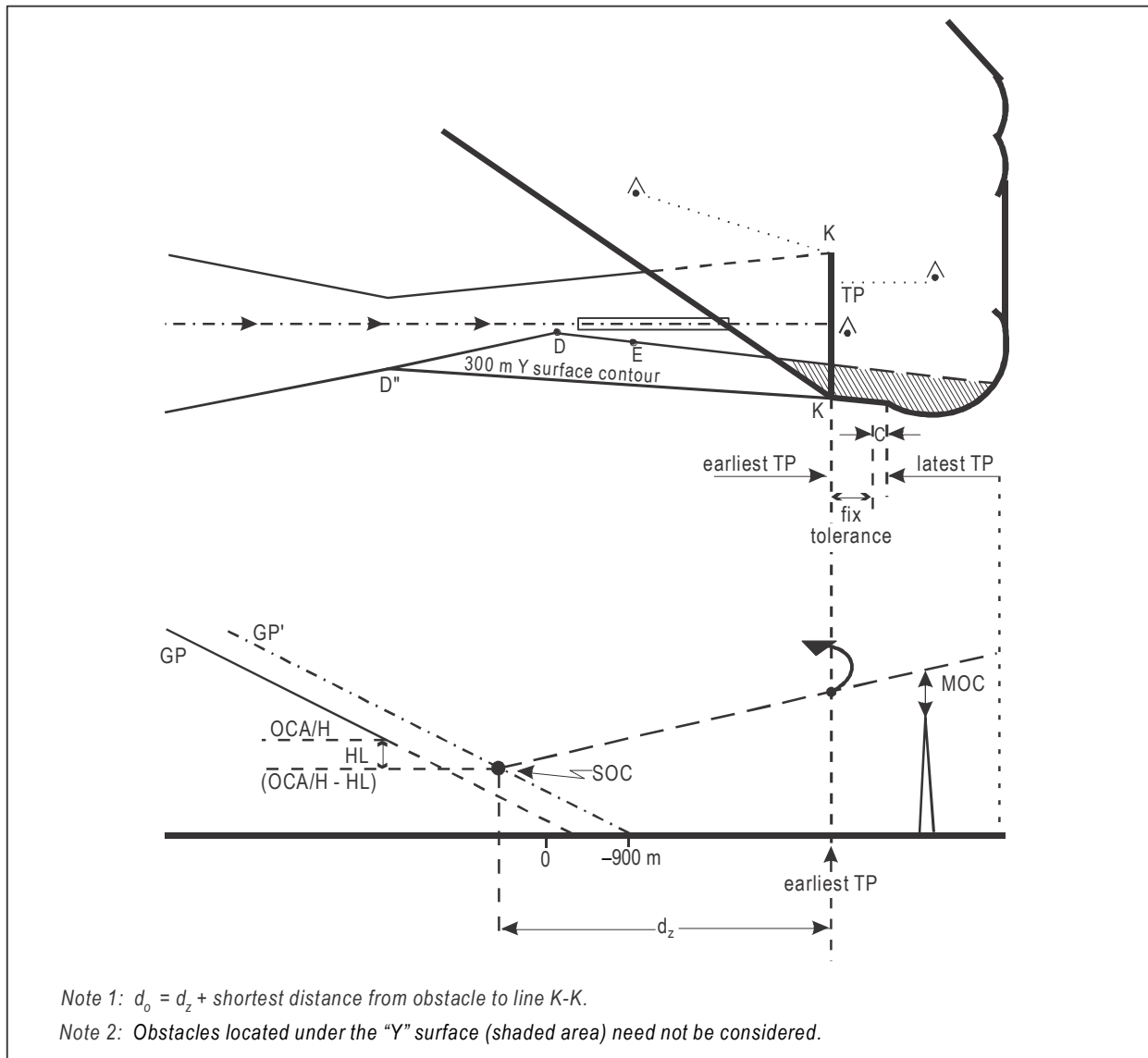


Figure III-3-6-20. Turn at designated TP (with TP fix)

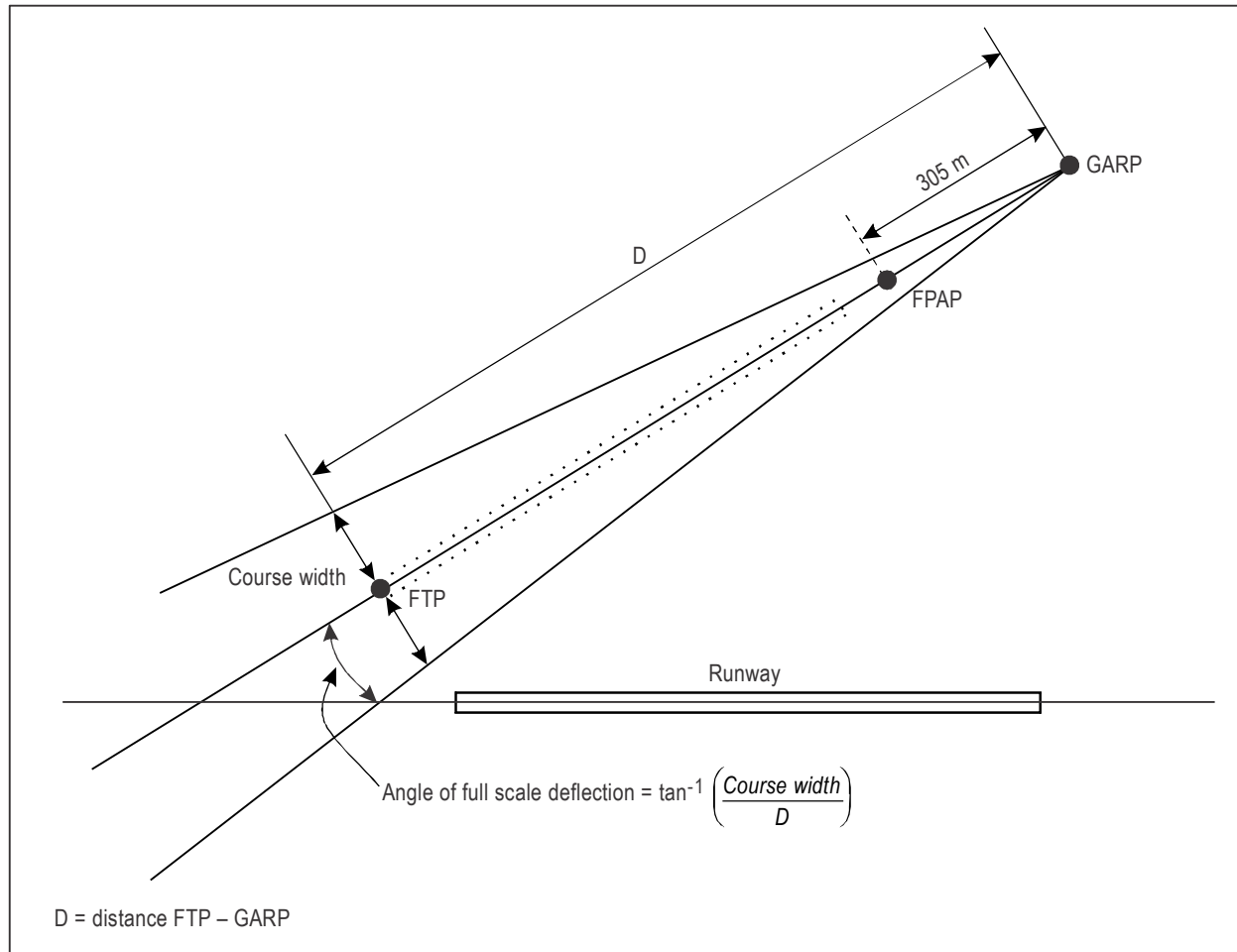


Figure III-3-6-21. GLS Cat I with offset azimuth final approach course alignment

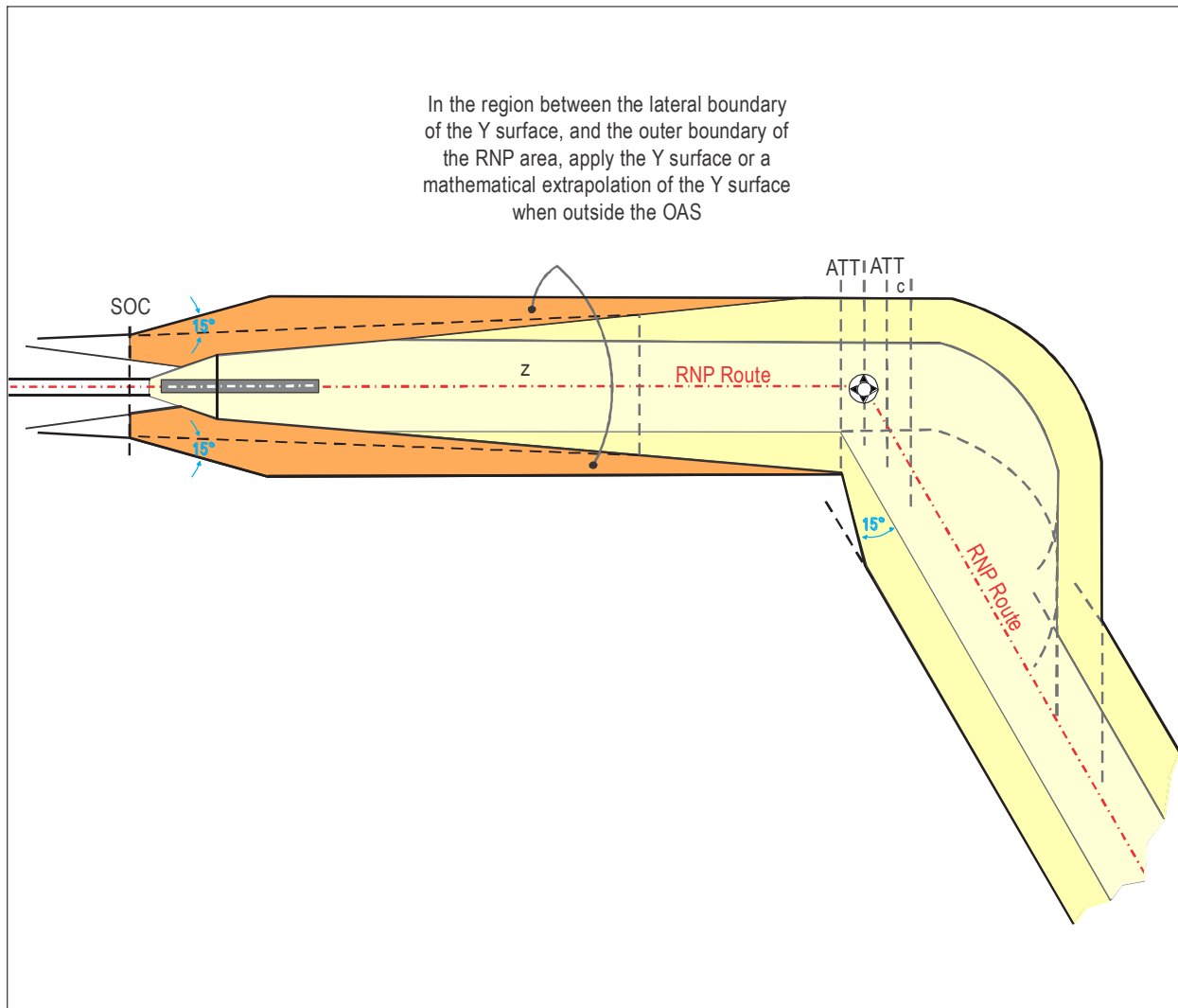


Figure III-3-6-22. Example: Transition to RNP 1, flyover turn after the precision segment

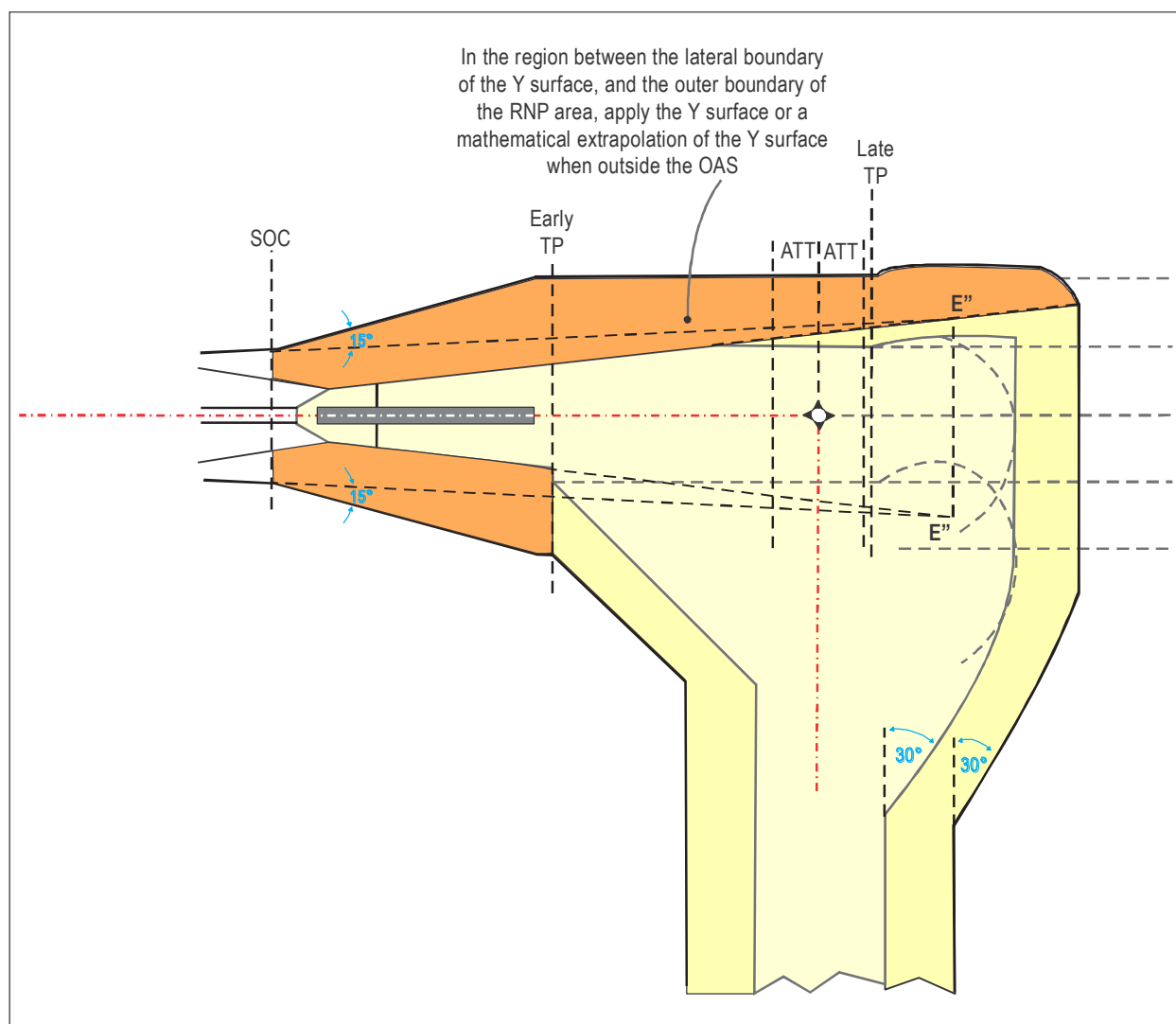


Figure III-3-6-23. Example: Transition to RNAV 1 or RNP 1, fly-by turn before the end of the precision segment

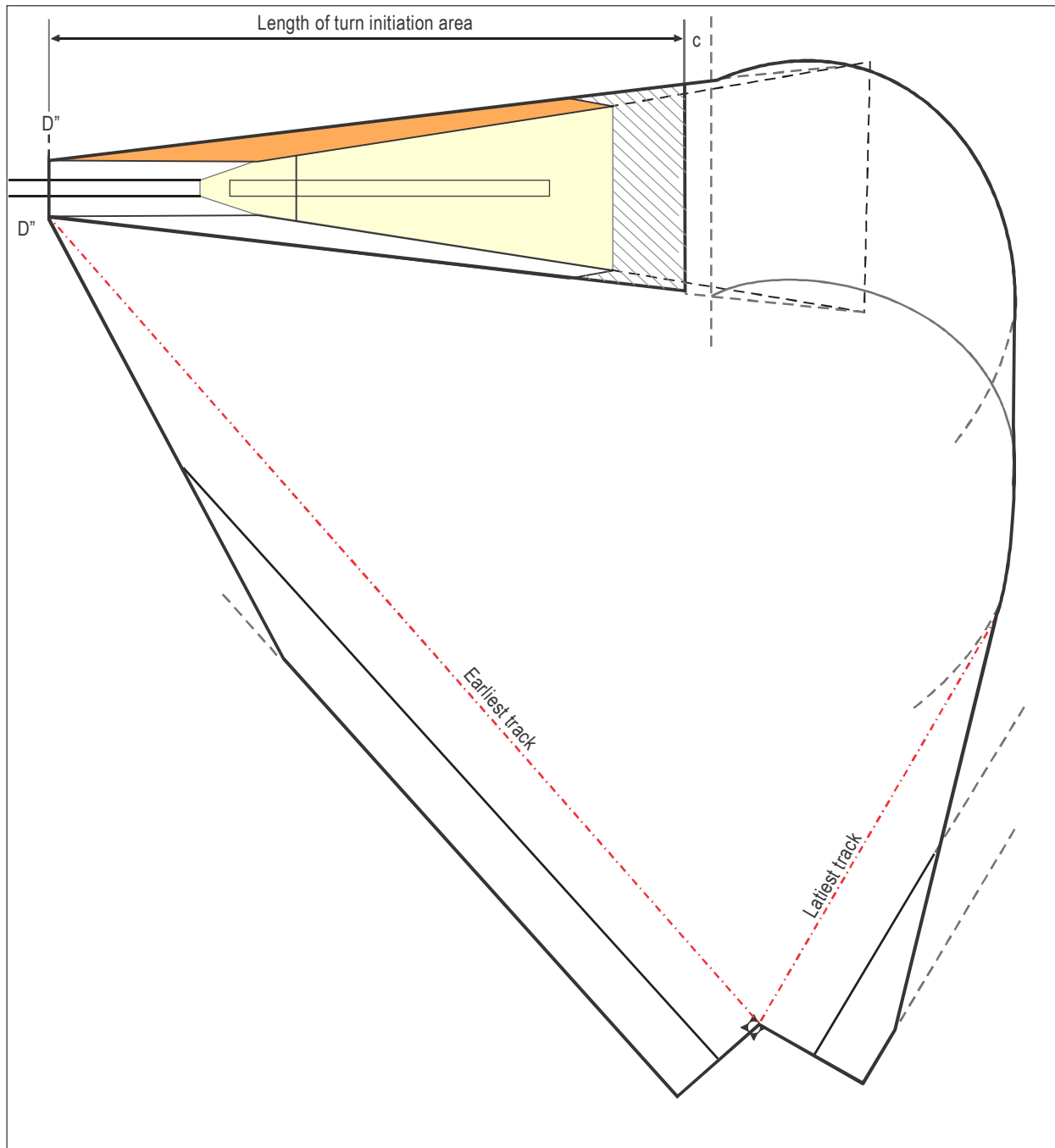


Figure III-3-6-24. Example: Turn at an altitude direct to an RNAV 1 or RNP 1 fix

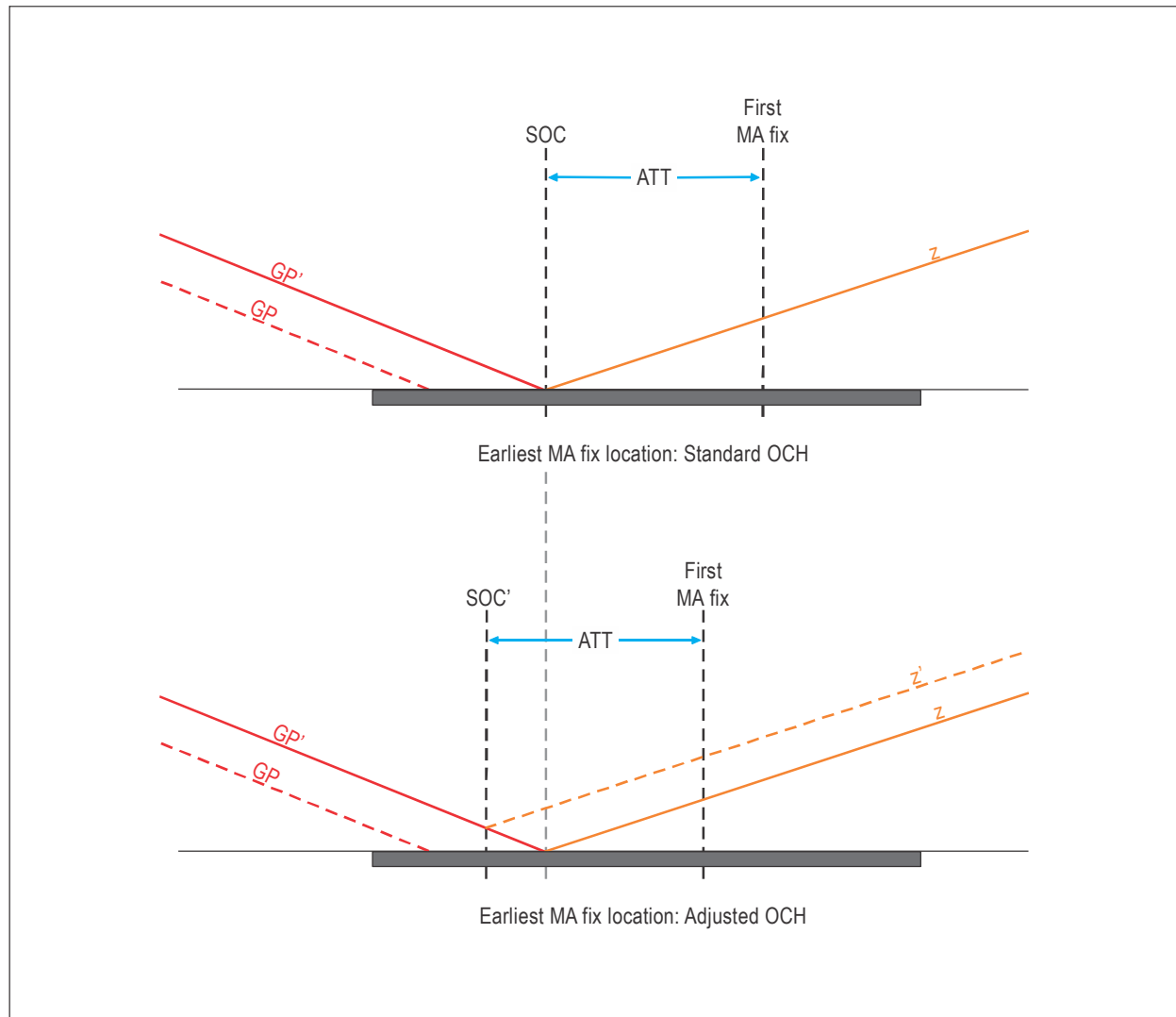


Figure III-3-6-25. Earliest fix location for RNAV or RNP missed approach

Chapter 7

HOLDING PROCEDURES

7.1 GENERAL

7.1.1 This chapter contains the criteria for PBN holding procedures.

7.1.2 Holding procedures can be designed both for operations with navigation specifications that require and do not require an aircraft to be capable of RNP holding.

7.1.2.1 *Navigation specifications requiring systems capable of RNP holding.* For HF, HA, and HM holding patterns within RNP airspace, the navigation system determines the boundaries of the holding area as defined by avionics standards. This boundary defines the maximum size of an RNP holding pattern. The navigation system has the ability to establish containment consistent with the boundaries defined in Appendix B to this Chapter. The only navigation specification requiring an area navigation system capable of RNP holding is advanced RNP (A-RNP).

7.1.2.2 *Navigation specifications not requiring systems capable of RNP holding.* For such systems, it is possible to define a holding procedure to be flown manually based on a waypoint. A conventional holding template shall be used to construct the obstacle clearance area for this type of holding.

7.1.3 *Location and number of holding patterns.* To avoid congestion only one holding pattern should be established for each procedure. The normal location would be at one of the IAFs. The holding waypoint is treated as a flyover waypoint, except for RNP holdings (according to RTCA DO-236C), where the holding waypoint will be a fly-by waypoint.

7.2 TYPES OF HOLDING PROCEDURES

7.2.1 Holding procedures may be established as waypoint holdings. The general criteria contained in Part I, Section 4, Chapter 3, Appendix C for conventional holding using an outbound leg defined by distance apply as modified by the criteria listed under each holding type.

7.2.2 *Waypoint holding.* (See Figure III-3-7-1 A.)

7.2.2.1 *Operations not requiring systems capable of RNP holding.*

- a) This type of holding will be flown manually, and PBN track guidance is provided only on the inbound track; and
- b) The end of the outbound leg of the holding is defined by timing or by a distance from the holding waypoint (WD) provided by the RNAV or RNP system.

7.3 ENTRY PROCEDURES

7.3.1 Waypoint holding

Entry procedures to a waypoint holding shall be the same as those used for conventional holding.

7.4 FIX TOLERANCE

The fix tolerance is the greater of the fix tolerances of the individual sensors allowed for in the PBN navigation specification. The waypoint tolerances are specified in Part III, Section 1, Chapters 2, 3 and 4 as applicable.

7.5 HOLDING AREA CONSTRUCTION

7.5.1 Waypoint holding area

The holding area is constructed by applying the techniques in Part II, Section 4, Chapter 1, 1.3, “Construction of holding areas” to the waypoint tolerance area.

7.5.2 Protection area construction for waypoint holdings

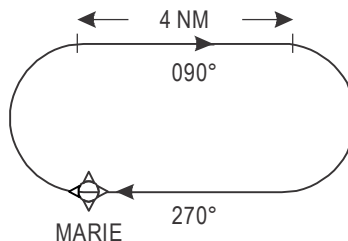
7.5.2.1 The general criteria described in 3.3, “Protection area of racetrack and holding procedures” of Part I, Section 4, Chapter 3, Appendix C, “Construction of obstacle clearance areas for reversal and holding procedures” apply as modified by the criteria in this paragraph. The criteria are broken down into three steps (see Appendix A, paragraph 1).

7.5.2.2 This additive tolerance method, the template tracing technique (TTT), is described in Appendix A to this chapter.

7.6 PROMULGATION OF PBN HOLDING PROCEDURES

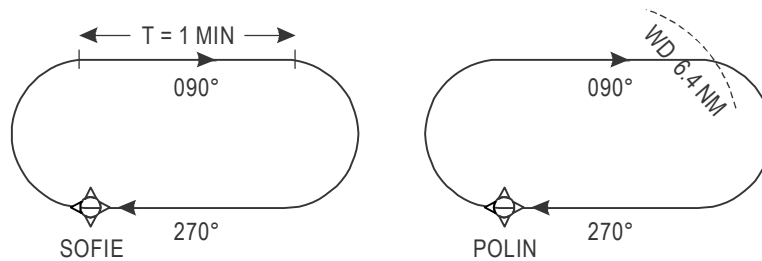
7.6.1 RNP holding

For this type of holding the outbound leg is defined by its length. The outbound length shall be published. This type of holding requires the on-board system to be capable of remaining within a certain area when flying a holding pattern. The holding pattern itself is defined by a maximum diameter and an outbound leg length. The inbound and outbound tracks of the holding pattern and the outbound leg length shall be charted.



7.6.2 PBN holding for all systems

7.6.2.1 For holding flown manually the outbound leg is either defined by a timing or a distance from the holding waypoint.



7.6.2.2 Where the outbound leg is defined by a distance from the holding waypoint, the waypoint distance (WD) shall be published on the approach chart expressed in tenths of kilometres (tenths of nautical miles).

Note.— The holding waypoint may not be charted as a flyover waypoint, but the pilot and/or aircraft navigation system is expected to treat the waypoint as a flyover waypoint while flying the holding (see Appendix A for background and rationale).

7.6.2.3 See Appendix B to this chapter for the definition of RNP holding entry sectors.

Appendix A to Chapter 7

CONSTRUCTION OF OBSTACLE CLEARANCE AREA FOR PBN HOLDING BASED ON GNSS, DME/DME OR VOR/DME

1. GENERAL

The holding area construction is broken down into the following three steps:

- a) construction of the template;
- b) basic area construction; and
- c) construction of entry area.

2. CONSTRUCTION OF THE TEMPLATE

2.1 Use of an RNAV or RNP system with conventional holding techniques

Construct a conventional holding template according to Part I, Section 4, Chapter 3, Appendix C, 3.3.2.

3. CONSTRUCTION OF THE BASIC AREA

3.1 Holding point tolerance area

Draw around holding point A the RNAV fix tolerance associated with the waypoint and locate points A1, A2, A3 and A4 on the four corners of this area.

3.2 Basic area construction

3.2.1 Use of an RNAV or RNP system with conventional holding techniques

3.2.1.1 Outbound leg of the holding defined by timing (See Figure III-3-7-App A-1.)

- a) place the conventional template point “a” successively on A1, A2, A3 and A4 to draw curves “1”, “2”, “3” and “4”; and
- b) draw the common tangents to curves “1” and “2”, “2” and “4”, “3” and “4”, “3” and “1”.

3.2.1.2 Outbound leg of the holding defined by distance (See Figure III-3-7-App A-2.)

3.2.1.2.1 Protection of the outbound turn and the outbound leg

- a) place the conventional template point “a” on A1, with axis parallel to the inbound track, and draw curve “1” (part of the outline of the template) and circle with centre “g” and radii W_G ;
- b) place the conventional template point “a” on A3, with axis parallel to the inbound track, and draw curve “2” (part of the outline of the template) and line “3” (protection of the outbound leg in the direction of the non-maneuvring side); and
- c) draw the common tangent to curves “1” and “2” and extend the straight part of curve “1” and the line “3” in the direction of the outbound end.

3.2.1.2.2 Calculation of the limiting outbound distance: WD

WD is the distance between the holding waypoint and the vertical projection of the end of the outbound track onto the WGS-84 ellipsoid.

The distance parameters are chosen and calculated in the following sequence:

- a) choice of the outbound length: ds

ds is the horizontal length of the outbound leg; ds should conform to the relationship $ds > vt$, where t is the outbound timing, as specified in Part I, Section 4, Chapter 3, 3.5.5, “Outbound time” for racetrack procedures and in Part II, Section 4, Chapter 1, 1.3.2.2, “Outbound timing” for holding procedures;

- b) calculation of the limiting outbound distance: WD

WD is the horizontal distance between the holding point and the vertical projection of the end of the outbound track.

$$WD = (ds^2 + 4r^2)^{1/2}$$

- c) minimum value for WD

Additionally, in order to guarantee that this distance does not cross the area containing the end of the outbound turn, the limiting outbound distance is such that it does not cross the area containing the end of the outbound turn. Practically, a circle with centre “holding point” and radius WD shall not interfere with the circle with centre “g” and radii W_G as drawn as described in 3.2.2.2.1 a).

If the distance calculated in point b) is such that an interference occurs, the WD shall be increased up to a value not smaller than calculated as follows:

$$WD = [(ATT+11v)^2 + (2r+XTT)^2]^{1/2} + W_g$$

- d) WD is then rounded to the next higher tenth of a km (or NM).

3.2.1.2.3 Area containing the end of the outbound leg

- a) draw from A the inbound track “RP” and two lines “RP1” and “RP2” at a distance equal to XTT on each side of it:

where XTT is the holding point cross-track tolerance;

b) with centre on A, draw arcs “D” with a radius WD, “D1” with a radius WD-ATT, “D2” with a radius WD + ATT;

where ATT is the holding point along-track tolerance;

c) locate point C1 at the intersection of the extension of curve “1” with the arc D2;

d) locate points C2 and C3 at the intersection of the extension of line “3” with arcs D1 and D2; and

e) if the aircraft intercepts the RNAV holding inbound track before reaching the limiting outbound distance WD, the pilot is assumed to follow outbound the RNAV track without drifting any further from the procedure axis, so:

where C2 and C3 are further from the procedure axis than RP2 (see Figure III-3-7-App A-3 a)), replace C2 and C3 by the intersection of RP2 with arcs D1 and D2 respectively; and

where C3 only is further from the procedure axis than RP2 (see Figure III-3-7-App A-3 b)), replace C3 by the intersection of RP2 with D2, add point C’2 at the intersection of line “3” with RP2.

3.2.1.3 Protection of the inbound turn

Rotate the template 180°, then:

a) place template point “a” on C1, with axis parallel to the inbound track, and draw curve “4” (part of the protection line of a turn of more than 180°);

b) move the template point “a” along arc D2 from C1 to C3 (with axis parallel and opposite to the inbound track) and draw curve “5”;

c) place template point “a” on C2, C3 and eventually on C’2 and draw curves “6”, “7” and eventually “8” and their common tangent; and

d) draw the tangent to curve “7” and “2”.

4. CONSTRUCTION OF THE ENTRY AREAS

4.1 Use of an RNAV or RNP system with conventional holding techniques

4.1.1 Outbound leg of the holding defined by timing

(See Figure III-3-7-App A-4.)

Draw the circle centred on “A” passing through A1 and A3; apply the same method as explained in Part I, Section 4, Chapter 3, Appendix C, 3.3.3.2.1, using the conventional holding template.

4.1.2 Outbound leg of holding defined by distance

(See Figure III-3-7-App A-5.)

Omnidirectional entry areas are defined using the same principles as for holdings at a VOR/DME fix (inbound to the station). See Part II, Section 4, Chapter 1, 1.2.1, “Entry procedures” and Part I, Section 4, Appendix C to Chapter 3, 3.3.4.1.2 for the construction of areas.

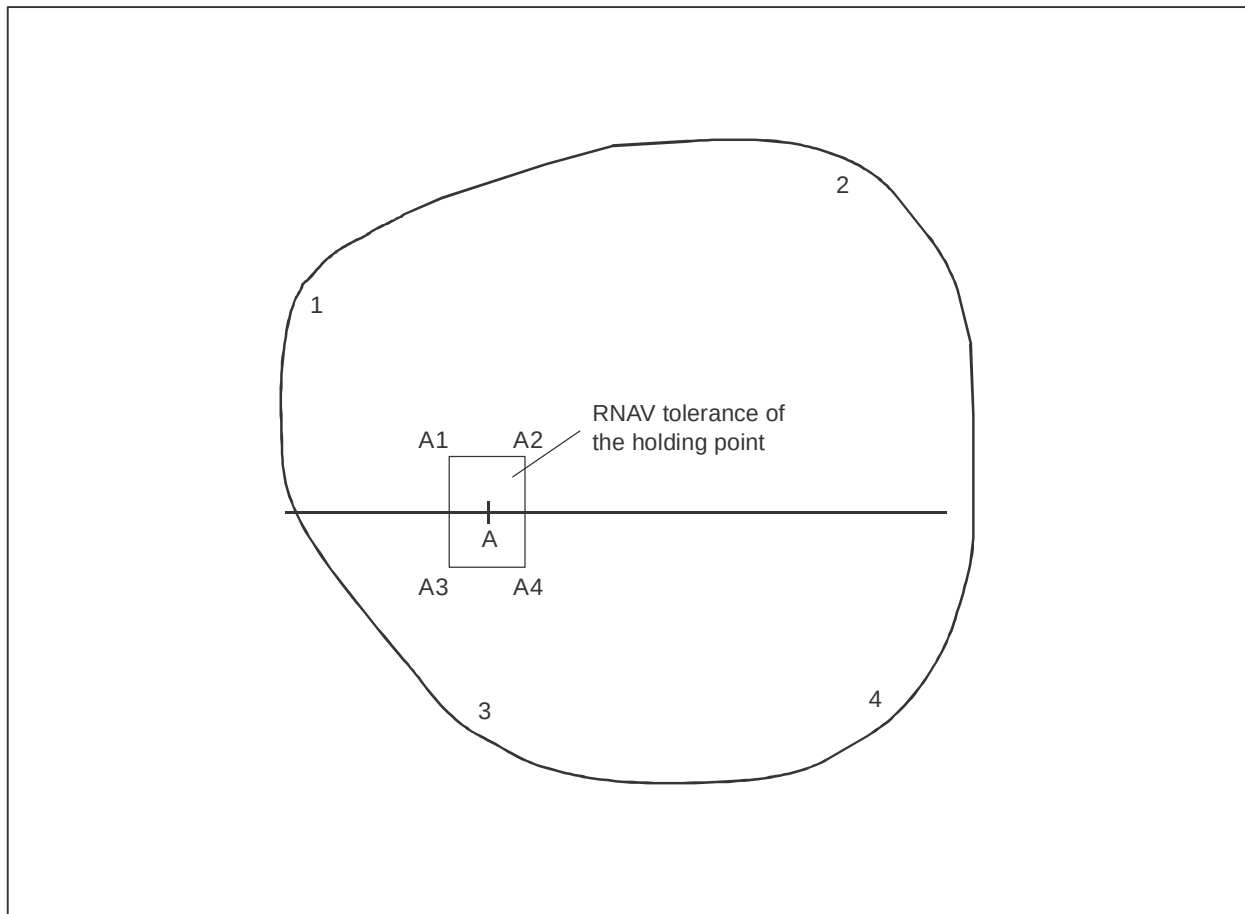


Figure III-3-7-App A-1. Holding basic area for all RNAV or RNP systems — outbound leg defined by timing

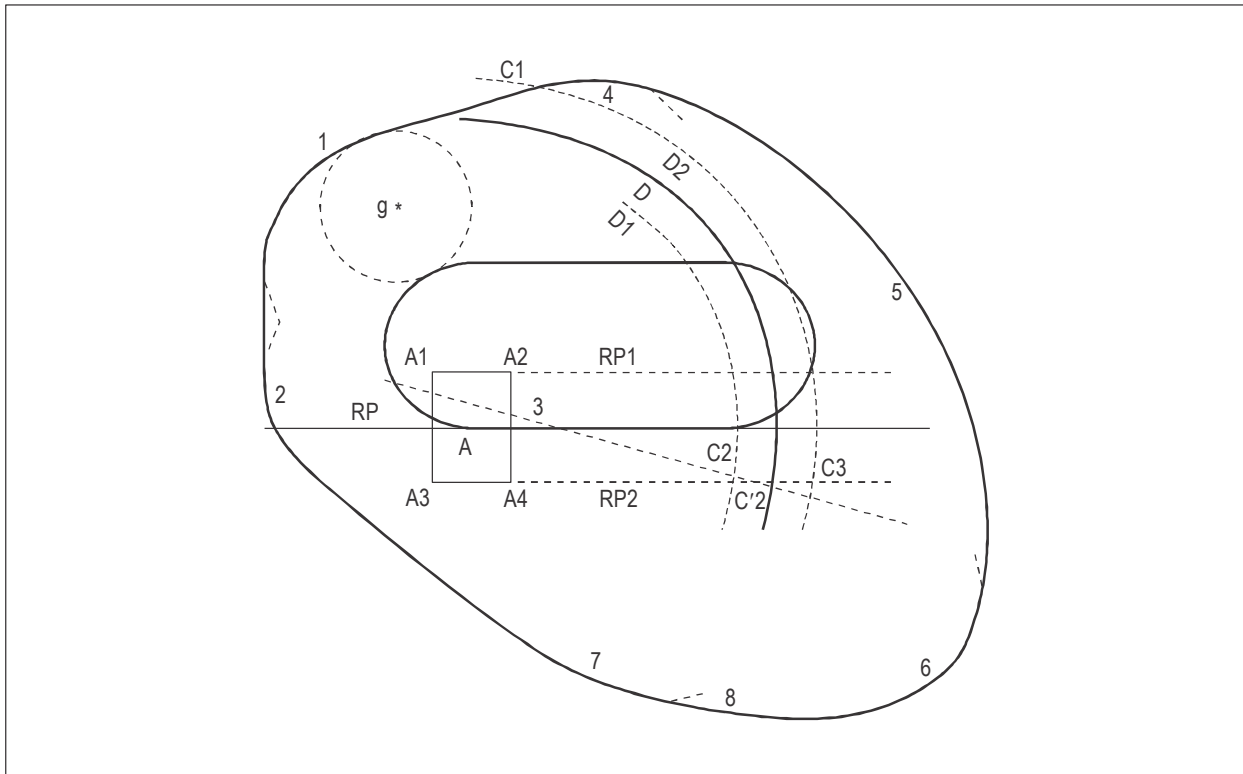


Figure III-3-7-App A-2. RNAV holding basic area for all RNAV systems — outbound leg defined by distance

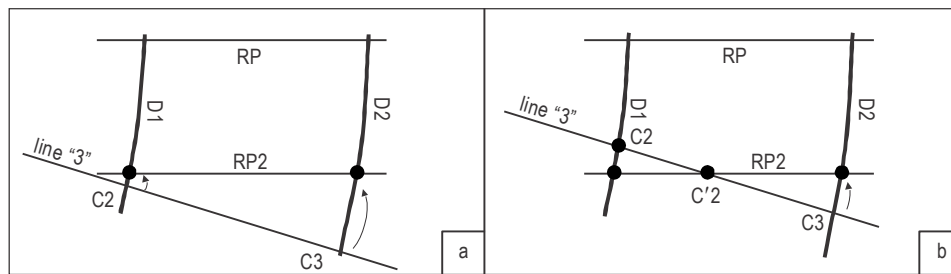


Figure III-3-7-App A-3. Area containing the end of the outbound leg (see 3.2.1.2.3 e)

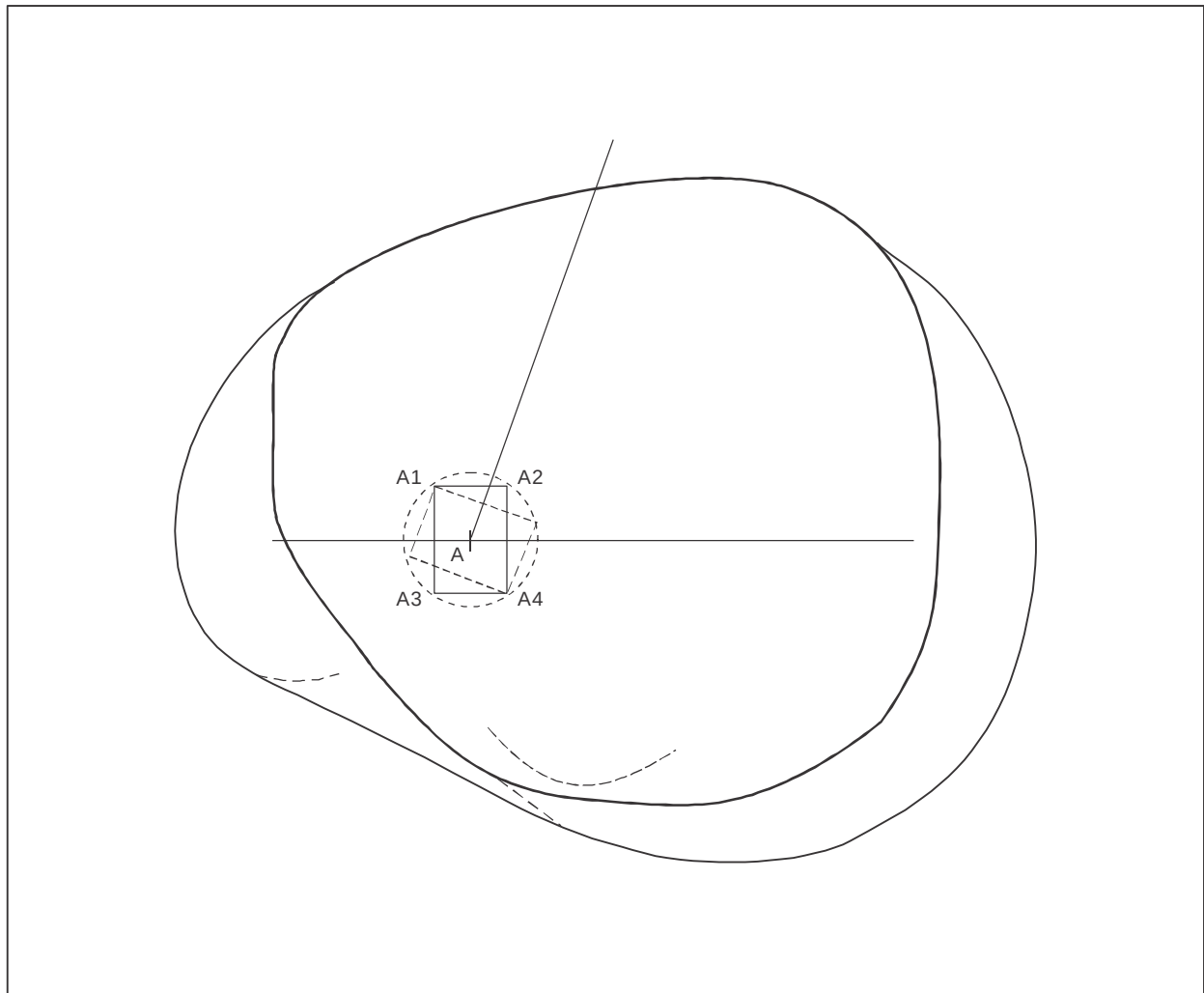


Figure III-3-7-App A-4. RNAV holding area including protection of entry procedures for all RNAV systems — outbound leg defined by timing

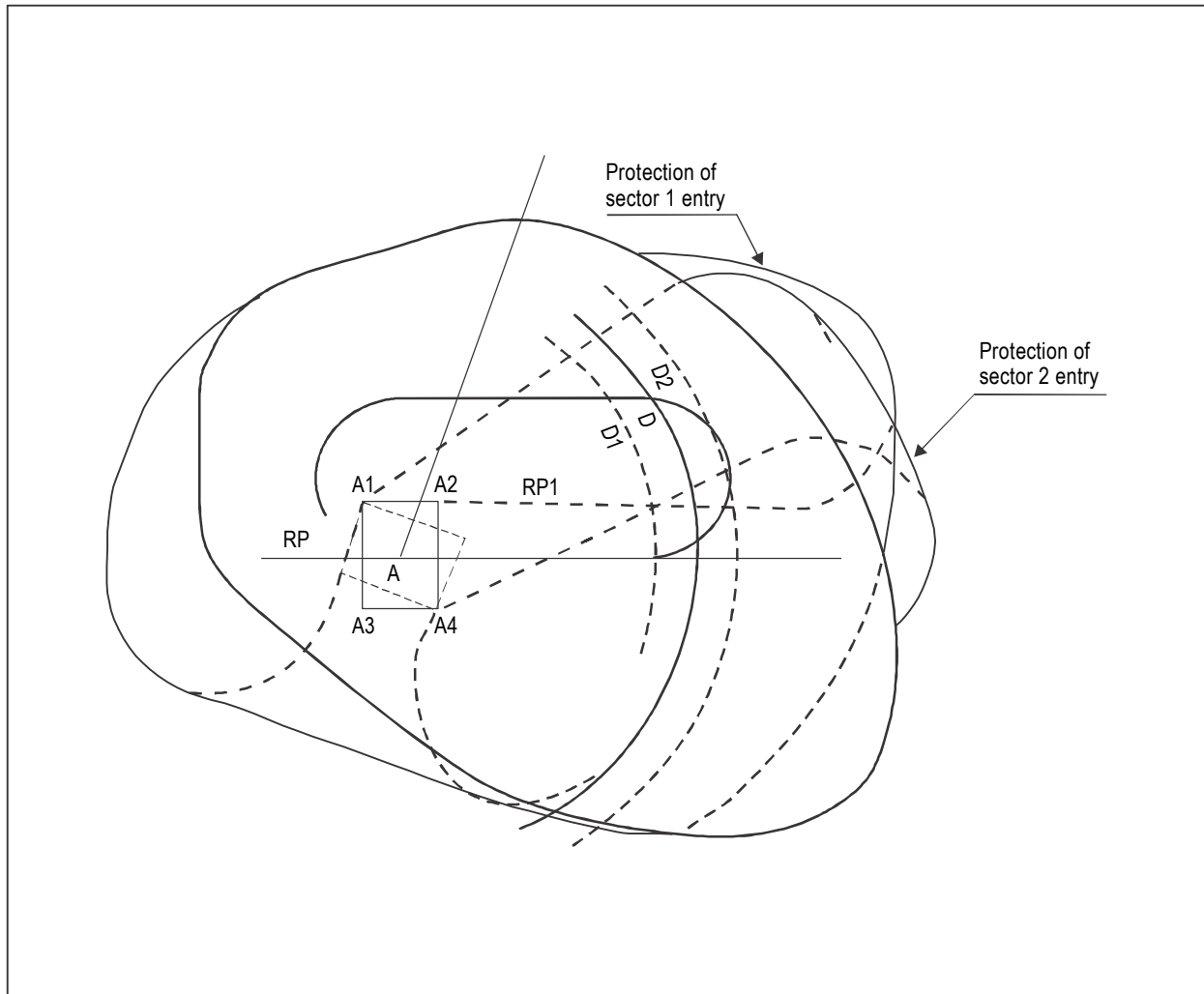


Figure III-3-7-App A-5. RNAV holding area including protection of entry procedures for all RNAV systems — outbound leg defined by distance

Appendix B to Chapter 7

RNP HOLDING

1.1 INTRODUCTION

1.1.1 RNP holdings are characterized by a maximum track geometrically defined by the length of the inbound track and the diameter of turn. Avionics compliant with RTCA DO-236C are capable of remaining (during 95 per cent of the time) within the limits of the RNP holding.

1.1.2 The maximum RNP holding pattern is defined by:

- a) holding waypoint in WGS-84 latitude and longitude;
- b) a minimum and maximum altitude;
- c) a maximum holding indicated airspeed;
- d) an inbound track to the holding fix;
- e) a length (d1) of the inbound track;
- f) a diameter of turn (d2); and
- g) the RNP value (d3).

1.2 DIAMETER Of TURN

1.2.1 The diameter of turn d2 is defined as that which can be followed throughout the turn at the defined IAS at ISA+15°, taking into account the maximum wind speed at the maximum holding altitude, assumed to be a tail wind throughout the turn and a defined bank angle (23° for FL < 245 and 15° for FL > 245).

$$d2 = 2 \times \frac{(V + W)}{20\pi R}$$

Where V is the TAS in either km/h or kt, R is the Rate of turn in degrees per second and W is wind in either km/h or kt.

1.3 RNP HOLDING PLUS SECTOR 4 ENTRY LIMITS

1.3.1 The distance d4 is used to draw the protection limit for sector 4 entries (see 1.5. 2 below) and is calculated using the formula:

$$d4 = d2 \times (1 - \sin 20) \times 2 \cos 20 \text{ (NM)}$$

1.3.2 The RNP holding plus sector 4 entries is a result of a combination of the RNP holding pattern and the sector 4 protection limit (see Figure III-3-7-App B-3).

1.4 OBSTACLE CLEARANCE

1.4.1 *Holding area.* The holding area includes the basic RNP area and the additional protection for entries from sector 4. A value (d3) equal to RNP is applied around the maximum track on straight segments of the RNP holding. A value equal to 1.414 x d3 is applied around the maximum track defined for the circular parts of the holding, until the limit reaches the limit defined for straight segments (see Figure III-3-7-App B-2).

1.4.2 *Buffer area.* The width of the buffer area is the greater of:

RNP + 3.7 km (2.0 NM)

or

Category A to E 9.3 km (5.0 NM), Category H 3.7 km (2.0 NM)

Obstacle clearance in the buffer area shall be according to the general criteria. The total buffer area width is divided into five buffers of equal size with the respective MOC.

1.5 ENTRY PROCEDURES

1.5.1 Introduction

Conventional entries described in Part I, Section 3 Chapter 5 are based on the fact that for VOR or NDB procedures it is necessary to overfly the station or holding fix at the beginning of the entry. A consequence is the need for additional protection for entry procedures. With RNP holding capability according to RTCA DO-236C, it is no longer necessary to overfly the holding way-point. This paragraph describes the entry procedures as deemed adequate by RTCA DO-236C.

1.5.2 Entry sectors

- a) draw the outline of the holding pattern; and
- b) draw a line making an angle of 70° with the axis of the inbound leg through the holding waypoint.

These two lines divide the space into four sectors: (1, 2, 3 and 4) as shown in Figure III-3-7-App B-3.

1.5.3 Entry procedures

See Figures III-3-7-App B-4 to B-7 for illustration.

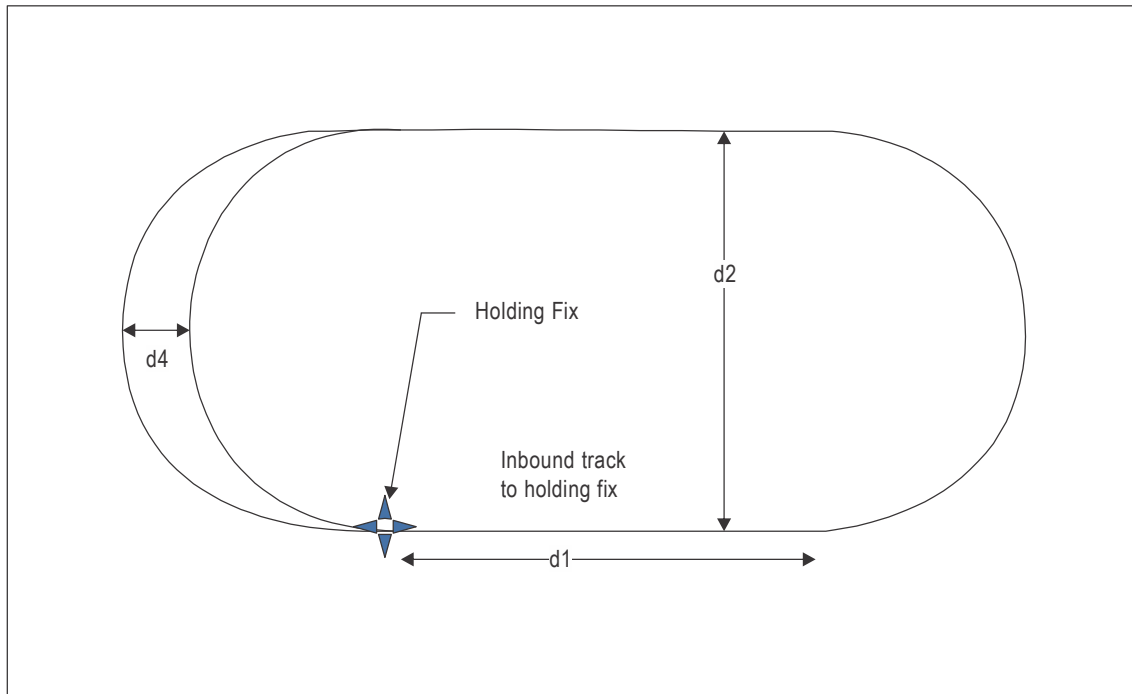


Figure III-3-7-App B-1. RNP Holding geometry

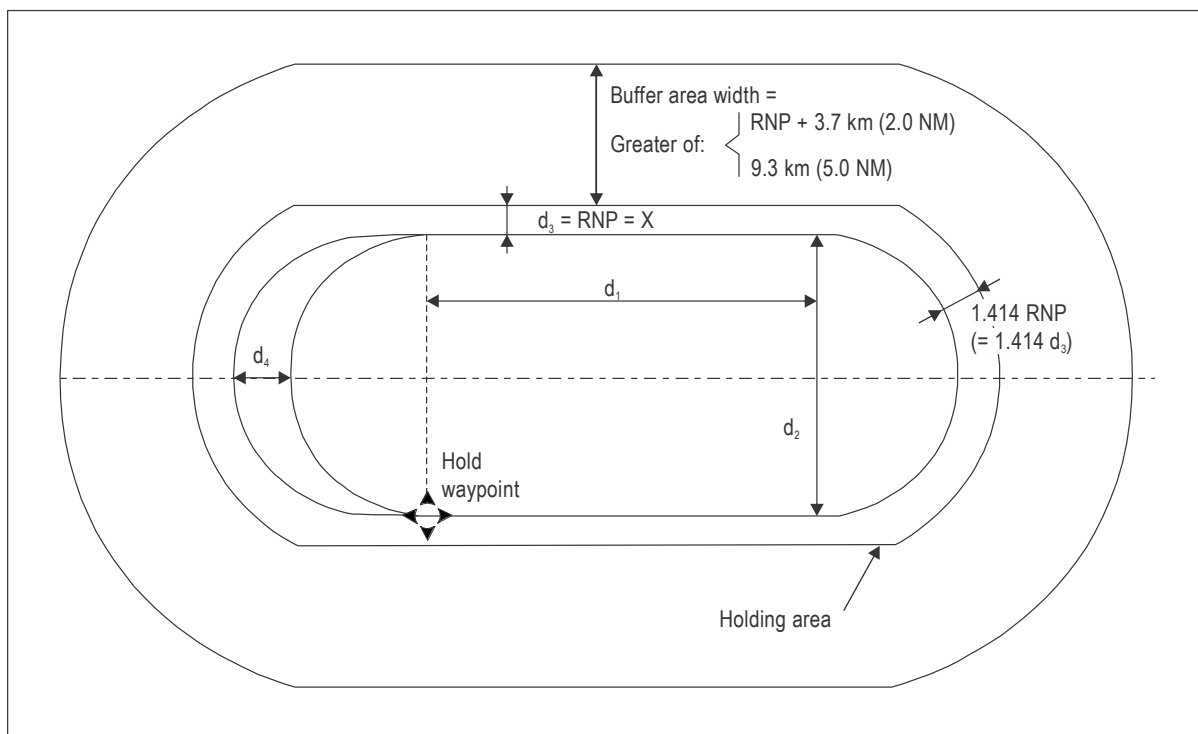


Figure III-3-7-App B-2. RNP Holding area

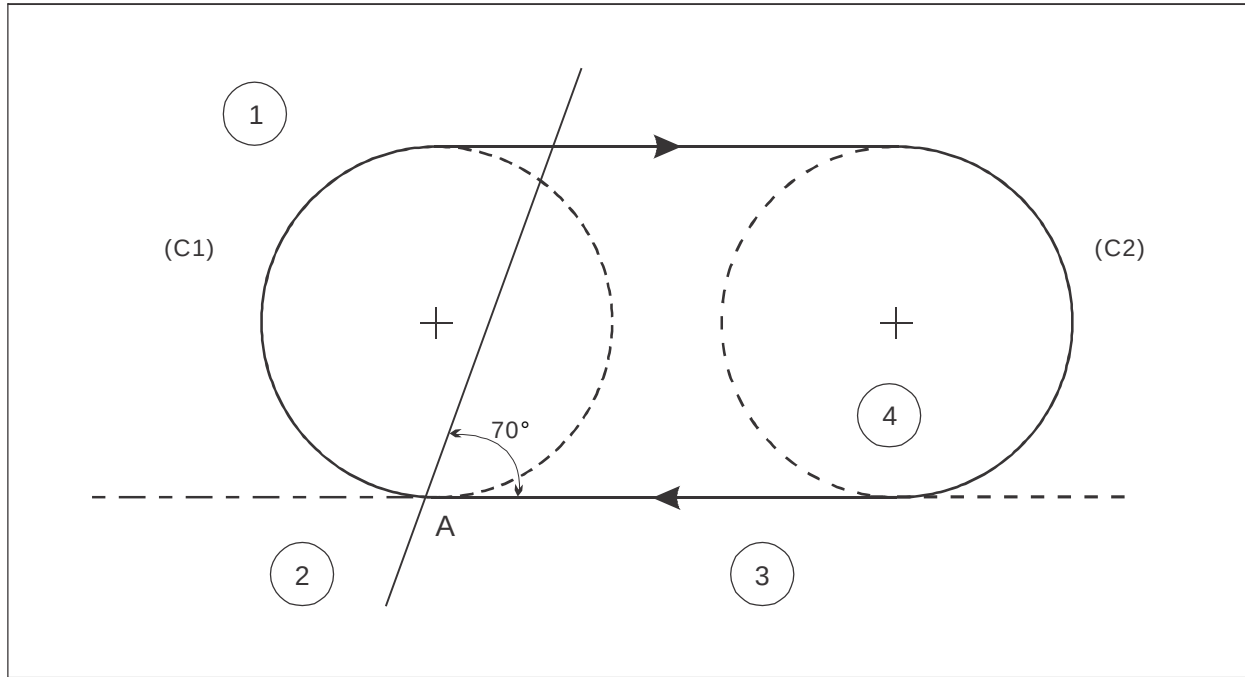


Figure III-3-7-App B-3. Entry sectors

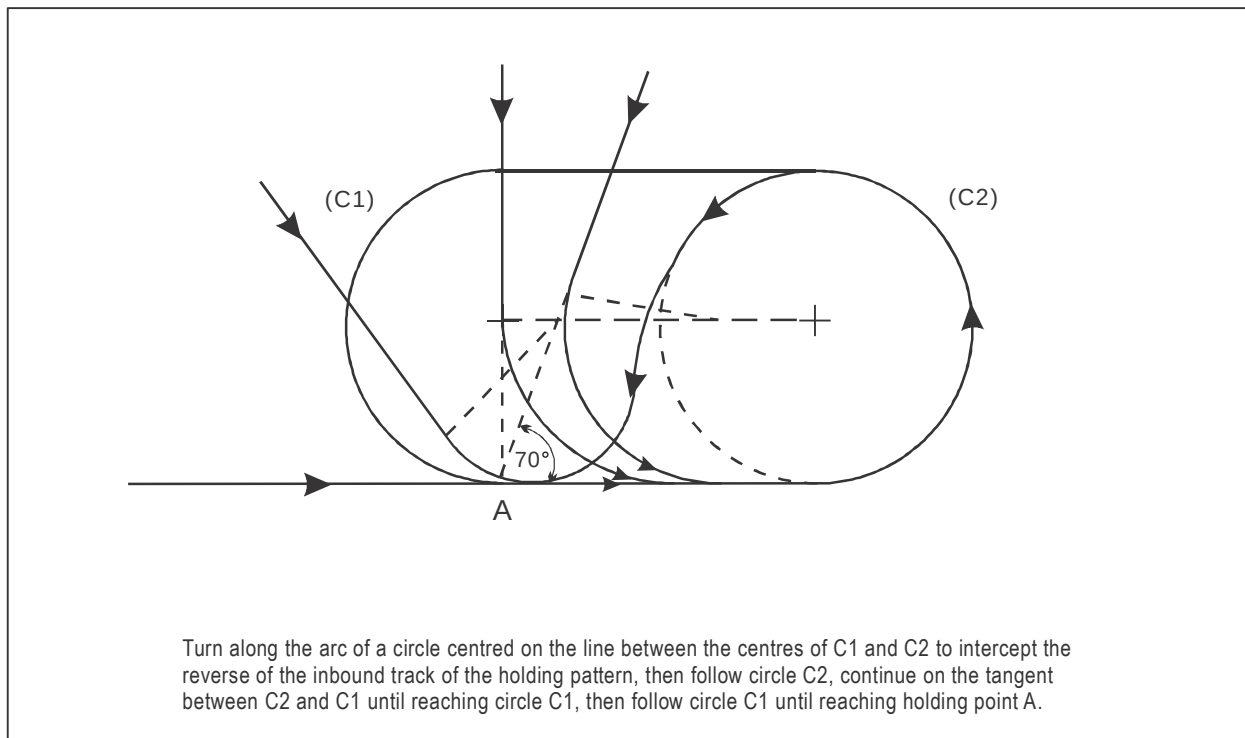
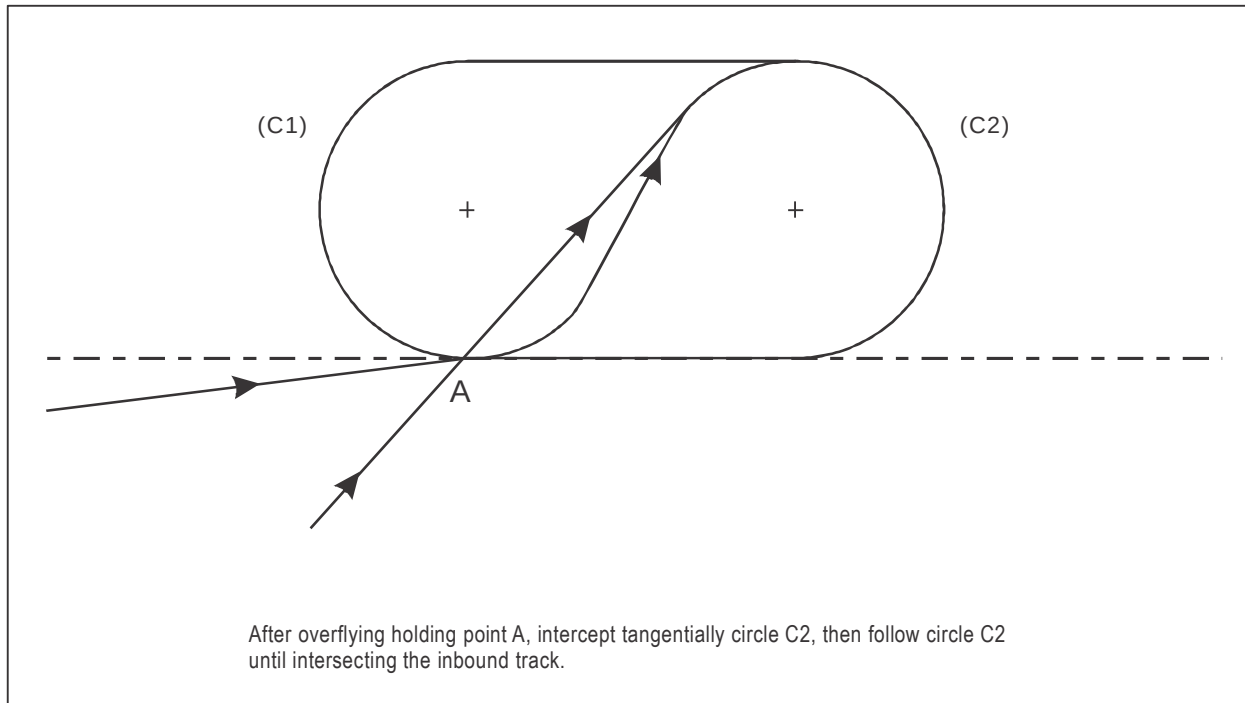
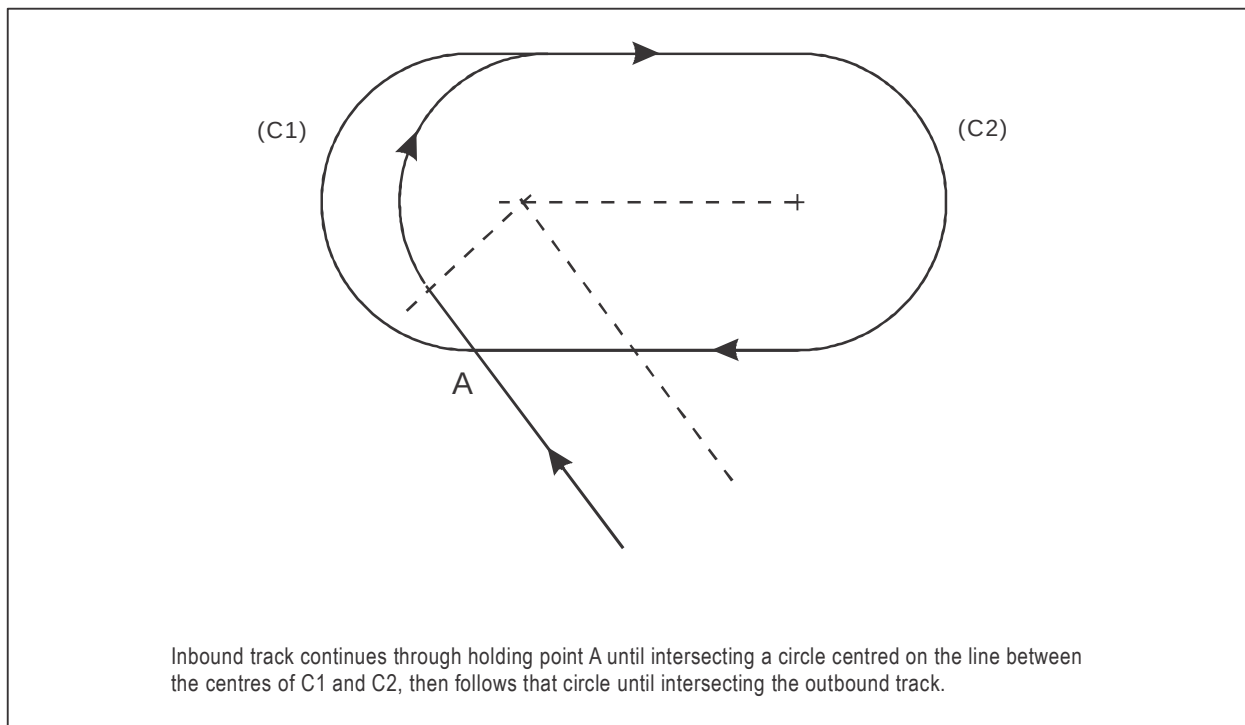
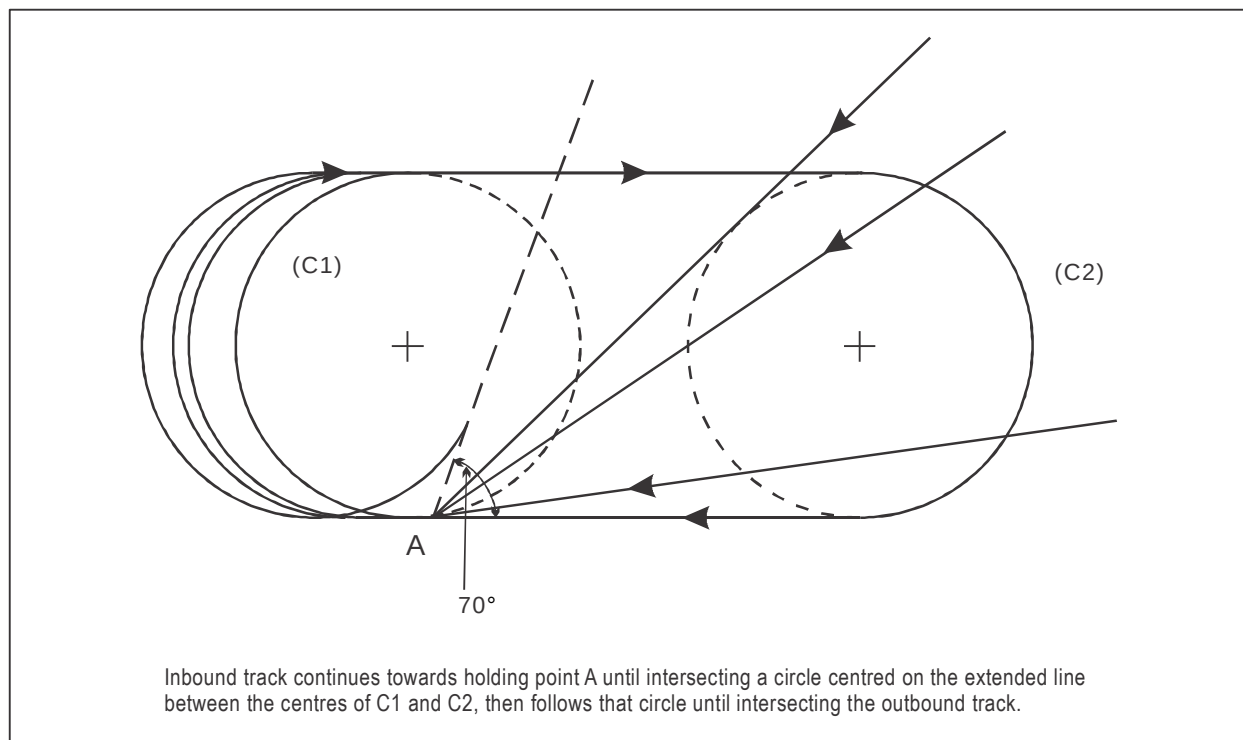


Figure III-3-7-App B-4. Sector 1 Entry

**Figure III-3-7-App B-5. Sector 2 entry****Figure III-3-7-App B-6. Sector 3 entry**

**Figure III-3-7-App B-7. Sector 4 entry**

Chapter 8

EN-ROUTE PROCEDURES

8.1 GENERAL

8.1.1 Application

8.1.1.1 This chapter describes the en-route criteria for RNAV 5, RNP 4, RNAV 2, RNP 2, Advanced RNP, RNAV 1 and RNP 0.3. The general criteria of Part II, Section 3, “En-route criteria” apply with the following modification: on the straight segments, the area has a constant width (angular limits do not apply).

8.1.1.2 The general criteria of Part I and Part III, Sections 1 and 2, as amplified or modified by the criteria in this chapter, apply.

8.1.2 Secondary areas

The general criteria for secondary areas apply (see Part I, Section 2, Chapter 1, 1.2 and 1.3).

8.1.3 Area widths

For the calculation of area widths and the underlying tolerances involved in these calculations, see the paragraph entitled “XTT, ATT and area semi-width” in Part III, Section 1 for the appropriate sensor. These are:

- a) GNSS, Section 1, Chapter 2, 2.2;
- b) DME/DME, Section 1, Chapter 3, 3.6; and
- c) VOR/DME, Section 1, Chapter 4, 4.6.

The total area width results from joining the various area widths at the relevant fixes, with the exception of the interfaces between the en-route segment and the initial approach segment, and between the en-route segment and the departure segment where the methodology detailed in Part III, Section 1, Chapter 1, 1.4 should be applied.

8.1.4 Definition of turns

The turn at a fly-by waypoint is the only turn allowed in en-route RNAV operations.

8.1.5 Turn at a fly-by waypoint

8.1.5.1 General

For turn protection and obstacle assessment see Section 2, Chapter 2, 2.3, “Wind spiral/Bounding circles method.”

8.1.6 Minimum altitudes

See Part II, Section 3, Chapter 1.

8.1.7 Charting requirements

Note.— *Guidance material for charting navigation specifications on area navigation routes is contained in the Aeronautical Chart Manual (Doc 8697).*

Chapter 1

PUBLICATION AND CHARTING — GENERAL

1.1 INTRODUCTION

The general criteria in Part I, Section 3, Chapter 5, “Published Information for departure procedures”, and Part I, Section 4, Chapter 9, “Charting/AIP” apply as modified in this chapter. See Chapter 2 for specific aeronautical database publication requirements.

1.2 NOTIFICATION OF DIFFERENCES IN AIP

The obstacle protection afforded to PBN procedures is, in most cases, predicated upon a ground track. This track is defined by a number of parameters including, inter alia, waypoint location and type, vertical path angle, maximum speed, minimum altitude, minimum bank angle and the path terminator associated with each procedure leg. If some of these parameters differ from the ICAO Standards, Recommended Practices and Procedures, they should be specified as generic to all PBN procedures with a clear statement in the GEN section of the AIP (See PANS-AIM, Appendix 2, Part 1, GEN 1.7).

1.3 DEPARTURES AND ARRIVALS

1.3.1 *Chart titles.* Charts shall be titled in accordance with Annex 4, 2.2. The required navigation specification for any published procedure shall be published in the State AIP, either on the chart or in the ENR 1.5 section.

1.3.2 Chart identification

1.3.2.1 The chart shall be identified in accordance with Annex 4, 9.5 for departures and 10.5 for arrivals and shall include the term RNAV or RNP, depending on the navigation specification.

1.3.2.2 The chart should include an identifier which is unique for that aerodrome and which may include reference to either a runway, fix or NAVAID.

1.3.3 Route designation

1.3.3.1 Each route shall be assigned a designator that is unique for that aerodrome. The designator shall be defined in accordance with Annex 11, Appendix 3. In addition, the first 4 letters of any 5LNC used in a route designator shall be unique for the aerodrome.

Note 1.— Airborne navigation databases use a maximum of 6 characters to identify a route. If the coded route designator is longer than 6 characters, the fifth character of the 5 LNC is not coded in navigation database route designation.

Note 2.— The coded route designator and the navigation specification name may be charted alongside the route in the plan view.

1.3.3.2 Separate charts should be published only if the routes differ laterally or vertically. When operationally required, separate charts may be published for each sensor or for a combination of sensors.

1.3.4 *Chart notes.* Additional procedure requirements shall be provided as chart notes. PBN items shall be separated out and published in a PBN requirements box and shall include the identification of the navigation specification used in the procedure design, as well as any additional functionality that is optional within the navigation specification, but required for the specific procedure. Examples of possible combinations are detailed in Table III-5-1-1. When a limitation to the sensors allowed within the navigation specification exists, the sensor limitation shall also be identified within the PBN requirements box (e.g. GNSS required, GNSS or DME/DME/IRU required).

Table III-5-1-1. Examples of PBN requirements box – departures and arrivals

<i>Navigation specification</i>	<i>Optional functionality, if required</i>
RNAV 5	-
RNAV 2	-
RNAV 1	-
RNP 1	RF required
Advanced RNP	-
RNP 0.3	RF required

Note 1.— Lengthy text may be shown on the verso of the chart.

Note 2.— Guidance material for charting the PBN requirements box on PBN departure and arrival procedures is contained in the Aeronautical Chart Manual (Doc 8697).

Note 3.— Core and optional functionalities for each navigation specification are provided in the Performance-based Navigation (PBN) Manual (Doc 9613).

1.3.5 Depiction

The RF leg shall be depicted as shown in Figure III-5-1-2 below. The along track distance of the RF leg shall be charted but no course value shall be shown on the RF leg. The radius and the arc centre shall not be charted but shall be included in the procedure description.

1.4 APPROACH

1.4.1 *Chart titles.* Charts shall be titled in accordance with Annex 4, 2.2.

1.4.2 Chart identification

1.4.2.1 The chart shall be identified in accordance with Annex 4, 11.6.

1.4.2.2 Approach charts depicting procedures that meet the RNP APCH navigation specification criteria shall include the term RNP in the identification (e.g. RNP RWY 23). The identification shall also include a parenthetical suffix when exceptional conditions occur as described in Table III-5-1-2.

Table III-5-1-2. Conditions when suffix shall be applied in chart designation

<i>Condition</i>	<i>Suffix</i>	<i>Example</i>
Procedure has only an LPV line of minima	LPV only	RNP RWY 23 (LPV only)
Procedure has only an LNAV/VNAV line of minima	LNAV/VNAV only	RNP RWY 23 (LNAV/VNAV only)
Procedure has both LPV and LNAV/VNAV lines of minima but no LNAV minima	LPV, LNAV/VNAV only	RNP RWY 23 (LPV, LNAV/VNAV only)
Procedure has only an LP line of minima	LP only	RNP RWY 23 (LP only)

1.4.2.3 Approach charts depicting procedures that meet the RNP AR APCH navigation specification shall include the term RNP in the identification with a parenthetical suffix (AR). (e.g. RNP RWY 23 (AR)).

1.4.2.4 The chart identification shall include the runway identification for straight-in landing, or a letter designator (a, b, c, etc.) for circling approach (see Part I, Section 4, Chapter 9).

1.4.2.5 When more than one PBN approach procedure exists for the same runway, the duplicate identification criteria defined in Part 1, Section 4, Chapter 9 apply. When a PBN approach procedure is combined with another PBN approach procedure on the same chart, the multiple procedure criteria defined in Part 1, Section 4, Chapter 9 apply.

Note.— The text in parentheses that is part of the procedure identification does not form part of the ATC clearance.

1.4.3 *Chart notes.* Additional procedure requirements shall be provided as chart notes. PBN items shall be separated out and published in a PBN requirements box and shall include the identification of the navigation specification used in the procedure design, as well as any additional functionality that is optional within the navigation specification, but required for the specific procedure. Examples of possible combinations are detailed in Table III-5-1-3. When a limitation to the sensors allowed within the navigation specification exists, the sensor limitation shall also be identified within the PBN requirements box (e.g. GNSS required, GNSS or DME/DME/IRU required).

Table III-5-1-3. Examples of PBN requirements box - approach

<i>Navigation specification</i>	<i>Optional functionality, if required</i>
RNAV 1	-
RNP 1	RF required
RNP APCH	RF required
RNP AR APCH	Missed approach RNP < 1.00
Advanced RNP	-
RNP 0.3	RF required

Note 1.— Guidance material for charting the PBN requirements box on PBN approach procedures is contained in the Aeronautical Chart Manual (Doc 8697).

Note 2.— Core and optional functionalities for each navigation specification are provided in the Performance-based Navigation (PBN) Manual (Doc 9613).

1.4.4 Depiction

1.4.4.1 Any RF requirement shall be charted in accordance with paragraph 1.3.5 above.

1.4.4.2 Procedures that have a combination of PBN and non-PBN segments, or that use multiple navigation specifications, shall have the navigation specification(s) for the PBN segments identified in a PBN requirements box in the plan view. An indication of which segments the navigation specification applies to is only required when the navigation specification is not mandatory for the execution of the procedure as a whole.

1.4.4.3 For RNP AR APCH procedures, navigation accuracy values shall be charted to the hundredth decimal place. Navigation accuracy values less than 1.00 NM within initial and intermediate segments shall be charted at the waypoint where they first take effect and continue to apply until another navigation accuracy value is charted or until the final segment is reached. Navigation accuracy values associated with the final segment shall be charted within the approach minima box along with the associated OCA/H. Navigation accuracy values less than 1.00 NM within missed approach segments shall be charted at the waypoint where they first take effect, starting with the MAPt. These values continue to apply until another navigation accuracy value is specified or until a charted accuracy value of 1.00 NM terminates the need for further charting of missed approach accuracy values. When the missed approach includes navigation accuracy values less than 1.00 NM, this shall also be stated within the PBN requirements box (e.g. Missed approach RNP < 1.00).

1.4.5 Minima

Minima for approach procedures shall be labelled on the chart as follows:

<i>Minima label</i>	<i>Associated navigation specification</i>
LNAV	RNP APCH
LNAV/VNAV	RNP APCH
LP	RNP APCH
LPV	RNP APCH
RNP 0.x	RNP AR APCH

1.5 FORMAL TEXTUAL OR TABULAR DESCRIPTION OF THE PROCEDURE

1.5.1 *General.* An accurate, complete and unambiguous RNAV procedure description is an essential publication requirement for database coding. This is accomplished by the combination of an appropriate chart, and an additional textual or tabular description of the procedure, to be used by the database coder. Where standard assumptions have been applied to the procedure design in areas such as speed and bank angle, there is no requirement to include these particular parameters in every procedure description.

1.5.2 An RNAV procedure is defined by one or a number of waypoints, each defined by a waypoint name, a path and terminator, and a set of constraints. Although “recommended” and “expected” altitudes/flight levels may be charted, they are not coded within navigation databases.

1.5.3 The textual or tabular description of the procedure, solely to support navigation database coding, shall incorporate all the data elements as specified in Chapter 2, and shall be published on the verso of the appropriate chart or as a separate properly referenced sheet (see Annex 4, 9.9.4.3, 10.9.4.3, and 11.10.9). In the following paragraphs three examples are provided (the formal, the abbreviated and the tabular description methods. The tabular description method is provided in Table III-5-1-5). The tabular description should be used when obstacle clearance is dependent upon the application of a particular path terminator.

1.5.4 *Formal textual description.* In the formal textual description, the following principles apply:

(Waypoint) (underlined) denotes flyover.

(Waypoint) (not underlined) denotes fly-by or RF waypoint as appropriate.

To (Waypoint) denotes a TF path terminator.

To (Waypoint) on course XXX ° denotes a CF path terminator.

Direct to (Waypoint) denotes a DF path terminator.

(Waypoint) {R, NN.N, Arc Centre Identifier} denotes an RF path terminator, the radius and the centre point of a fixed radius turn in terminal airspace.

Climb on course XXX°, at or above yyy feet turn right/left denotes a CA path terminator.

From (Waypoint) on track XXX°, at or above yyy feet turn left/right denotes an FA path terminator.

Climb on heading XXX°, at or above yyy feet turn left/right denotes a VA path terminator.

Continue on heading XXX° denotes a VM path terminator.

Continue on track XXX° denotes an FM path terminator.

(Waypoint) {HM, Turn Direction, Inbound Track, Leg Distance/Time} denotes an HM path terminator.

The formal description method is illustrated in Table III-5-1-4.

1.5.5 Abbreviated description

1.5.5.1 The description may be abbreviated by placing the leg constraints (speed, track and altitude) in square brackets. If these constraints are not preceded by a waypoint name, the last calculated track shall be flown until the constraint is reached.

1.5.5.2 Each constraint is coded in the format UNNNNNCD where:

U may be one of the following letters:

A for altitude in feet AMSL

F for flight level

K for indicated air speed in knots

M for degrees magnetic

T for degrees true

NNNNN is a number from 000 to 99999

C may be one of the following:

“+” for ‘at or above’

“-” for ‘at or below’

a blank space for ‘at’

D is used to indicate turn direction in conditional and flyover transitions:

L for ‘Turn left’

R for ‘Turn right’

Multiple constraints should be separated by a semi-colon (;).

1.5.5.3 Individual waypoints in a procedure, together with their associated constraints, should be separated by a hyphen (-), except when the subsequent leg requires a DF path terminator when an arrow (→) should be used.

1.5.5.4 The formal description and the abbreviated description methods are illustrated in Table III-5-1-4.

1.6 WAYPOINT NAMING

1.6.1 Waypoints used in support of RNAV SIDs, STARs and instrument approach procedures shall be designated by either a unique, five-letter, pronounceable “name-code” or a five-alphanumeric name-code. The following principles apply:

- a) waypoints shall be designated by a five-alphanumeric name-code only if they are used for waypoints unique to one aerodrome that has a properly assigned four-letter location indicator (in accordance with Doc 7910);
- b) in the following cases a unique, five-letter, pronounceable “name-code”, in accordance with Annex 11, shall be applied:
 - 1) final waypoint of a SID;
 - 2) initial waypoint of a STAR;
 - 3) waypoints common to more than one terminal control area or used in a procedure common to more than one airport which are not used for en-route; and
 - 4) waypoints for ATC purposes.

1.6.2 The following criteria apply when five-alphanumeric name-codes are used:

- a) the five-alphanumeric name-code convention that is adopted shall be applicable to all aerodromes within the State;
- b) five-alphanumeric name-codes should contain characters taken from the airport designator, and/or characters indicating the use of the significant point, with all combinations containing no more than three digits;
- c) the convention and the rules of application shall be published in the State AIP;
- d) the five-alphanumeric name-code shall be unique within the terminal area in which it is used;
- e) as global uniqueness cannot be assured, all waypoints that have a five-alphanumeric name-code identifier should be clearly listed as terminal waypoints in the AIP; and
- f) as global uniqueness cannot be assured for waypoints containing five-alphanumeric name-codes, to avoid any potential misselection by the pilot, ATC should not use waypoints designated by five-alphanumeric name-codes in any re-routing from the en-route structure into a terminal procedure.

1.7 ARC CENTRE AND SIGNIFICANT POINT COORDINATES

The WGS 84 coordinates of arc centres and significant points used in support of RNAV arrival, departure and instrument approach procedures shall be provided in tabular form as illustrated in Table III-5-1-5.

Table III-5-1-4. Illustration of the formal and the abbreviated description methods

<i>Formal description</i>	<i>Abbreviated description</i>	<i>Expected path terminator</i>	<i>Flyover required</i>
Climb on track 047°M, at or above 800 ft turn right.	[M047, A800+; R]	CA	N
Climb on heading 047°M, at or above 800 ft turn right.	[HDG M047, A800+, R]	VA	N
Direct to ARDAG at 3 000 ft	→ARDAG[A3000]	DF	N
To PF035 at or below 2 000 ft	PF035[A2000-]	TF	Y
To PF025 at or above 4 000 ft, continue on heading 265°M and await radar vectors.	PF025[A4000], [HDG, M265]	TF, VM or FM	N
To OTR on course 090°M at 210 kts	OTR[M090; K210]	CF	N
To DF006 at 2 000 ft minimum, 4 000 ft maximum, minimum speed 210 kts	DF006[A2000+; A4000-; K210+]	TF	Y
To PD750 at 250 kts, turn right with 3.7 NM radius to PD751	PD750[K250]-PD751[R, 3.7, 0543451.2N 0021234.7E]	TF, RF	N, N
To STO at or above FL 100, turn left direct to WW039 at or above FL 070, to WW038 at or above 5 000 ft	STO[F100+; L]→WW039[F070+]-WW038[A5000+]	TF, DF, TF	Y, N, N

Table III-5-1-5. Illustration of the tabular description method

<i>Serial Number</i>	<i>Path Descriptor</i>	<i>Waypoint Identifier</i>	<i>Fly-over</i>	<i>Course/Track M(°T)</i>	<i>Magnetic Variation</i>	<i>Distance (km)</i>	<i>Turn Direction</i>	<i>Altitude (m)</i>	<i>Speed (km/h)</i>	<i>VPA/TCH</i>	<i>Navigation Specification</i>
001	CA	—	—	221 (223.5)	−2.3	—	—	@150	—	—	B-RNP 1
002	DF	FOKSI	—	—	−2.3	—	R	—	—	—	B-RNP 1
003	TF	EF974	—	043 (045.7)	−2.3	12.0	—	+1 400	—	—	B-RNP 1
004	RF Centre: EF991 r =5.240 NM	EF975	—	—	−2.3	13.7	R	—	−450	—	B-RNP 1
005	TF	EF976	Y	145 (147.3)	−2.3	9.6	—	+1 550	—	—	B-RNP 1
006	TF	TARTO	—	110 (112.3)	−2.3	28.2	—	—	—	—	B-RNP 1

<i>Serial Number</i>	<i>Path Descriptor</i>	<i>Waypoint Identifier</i>	<i>Fly-over</i>	<i>Course M(°T)</i>	<i>Magnetic Variation</i>	<i>Distance (km)</i>	<i>Turn Direction</i>	<i>Altitude (m)</i>	<i>Speed (km/h)</i>	<i>VPA/TCH</i>	<i>Navigation Specification</i>
001	IF	SUSER	—	—	+2.2	—	—	+1 550	−470	—	RNP APCH
002	TF	EF974	—	048 (045.7)	+2.2	12.0	—	+1 400	—	—	RNP APCH
003	RF Centre: EF991 r =5.240 NM	EF975	—	—	+2.2	13.7	R	—	−450	—	RNP APCH
004	TF	EF976	—	348 (345.8)	+2.2	9.6	—	@900	−270	—	RNP APCH
005	TF	RW35L	Y	348 (345.8)	+2.2	9.3	—	@150	—	−3.0/50	RNP APCH
006	FA	RW35L	—	348 (345.8)	+2.2	—	—	+250	—	—	RNP APCH
007	DF	SUSER	Y	—	+2.2	—	L	+1 550	—	—	RNP APCH
008	HM	SUSER	—	048 (045.7)	+2.2	7.4	R	+1 550	−450	—	RNP APCH

Table III-5-1-6. Waypoint and arc centre coordinates examples

<i>Waypoint Identifier</i>	<i>Coordinates</i>
EF974	43°34'23.8N 116°22'54.7W
EF975	43°35'39.1N 116°20'27.9W
<i>RF Arc Centre Identifier</i>	<i>Coordinates</i>
EF991	43°32'58.2N 116°19'41.6W

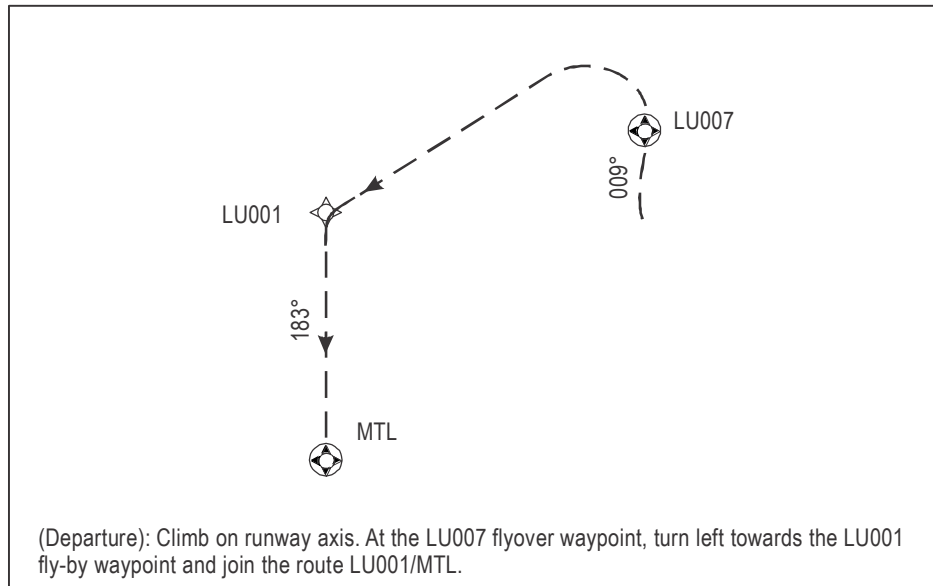


Figure III-5-1-1. Text description

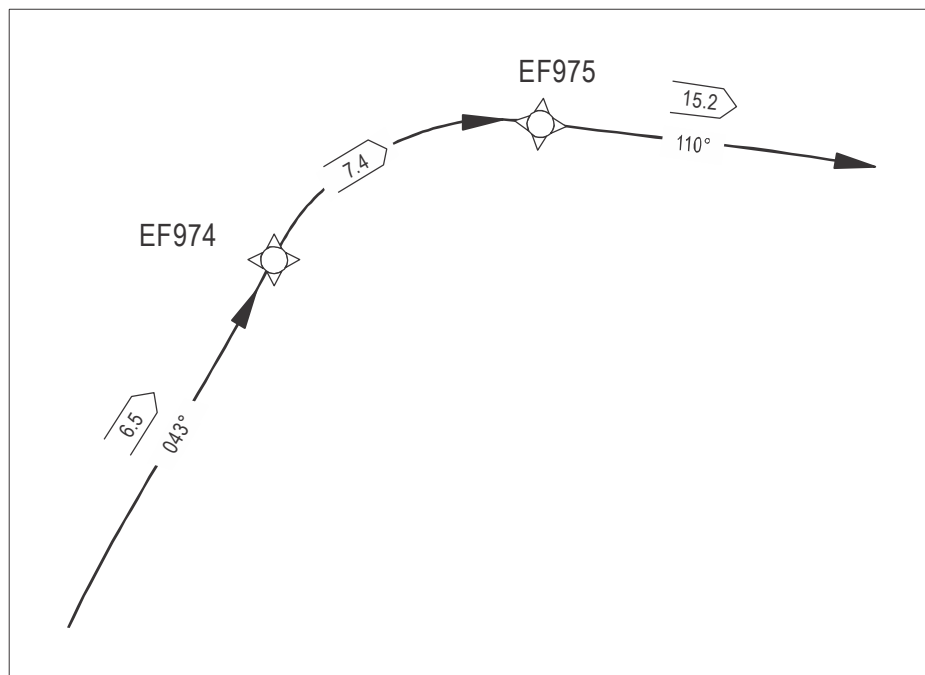


Figure III-5-1-2. RF leg depiction

Chapter 1

PBN DEPARTURE PROCEDURES FOR HELICOPTERS USING GNSS OR SBAS RECEIVERS

1.1 GENERAL

1.1.1 Application

1.1.1.1 This chapter describes the criteria for PBN (RNAV and RNP) departure procedures specific to helicopter operations and includes criteria for the design of departure procedures from both instrument heliports or runways and heliports, or landing locations. Criteria for departures from heliports and landing locations are accommodated with point-in-space (PinS) departure procedures.

1.1.1.2 The general criteria of Part I, Section 3 and Part III, Sections 1, 2 and 3 as amplified or modified by the criteria in this chapter apply to RNAV and RNP departure procedures for helicopters.

1.2 HELICOPTER DEPARTURES FROM INSTRUMENT HELIPORTS OR RUNWAYS

(To be developed)

1.3 HELICOPTER POINT-IN-SPACE (PINS) DEPARTURES FROM HELIPORTS OR LANDING LOCATIONS

1.3.1 PinS departure — General

1.3.1.1 *Description.* The PinS departure procedure consists of a visual segment followed by an instrument segment. The visual segment of the departure starts from the heliport or landing location and ends at the initial departure fix (IDF) at or above the IDF minimum crossing altitude (MCA). PinS departure procedures are constructed up to the IDF by applying the visual segment criteria, defined in this chapter, and beyond the IDF by applying the applicable PBN criteria, defined in Part III, Section I, Chapter 2. If the criteria for the visual segment can be met, the PinS departure procedure is annotated with the instruction “proceed visually”. If the criteria for the visual segment cannot be met, the procedure is annotated with the instruction “proceed VFR”. In the latter case there is no obstacle protection in the visual segment. The obstacle protection after the IDF is the same as for a PinS departure with a proceed visually instruction.

Note 1.— The term “proceed VFR”, implies that the pilot can comply with VFR in the visual segment to see and avoid obstacles and can cross the IDF at or above the MCA.

Note 2.— The term “proceed visually” implies that pilots can navigate by visual reference and see and avoid obstacles, with visibility sufficient to return to the heliport if they cannot continue visually to cross the IDF at or above the IDF MCA. Visual flight may be conducted below minima required for VFR.

1.3.1.2 The proceed visually segment can be a direct visual segment (visual segment flown directly from the heliport or landing location to the IDF) or a manoeuvring visual segment (initial take-off in a direction other than directly to the IDF).

1.3.2 PinS departure — Instrument phase — Description and protection

1.3.2.1 *Application of PBN criteria.* For the construction of the instrument segment of the departure procedure, the appropriate helicopter navigation specifications identified in Part III, Section I, Chapter 2, shall be used. The navigation information is also used during the visual segment to aid in the identification of and navigation to the IDF and, hence, only GNSS sensors may be used on these procedures.

1.3.2.1.1 *IDF fix tolerance.* The ATT, XTT and area semi-width values at the IDF are detailed in Part III, Section 1, Chapter 2.

1.3.2.1.2 Transition from visual segment to instrument segment of the PinS departure occurs at the IDF and requires the blending of the visual segment criteria with the applicable PBN protection areas at the IDF.

1.3.2.2 *Instrument segment of the PinS departure procedure.* The instrument segment consists of one or more legs and continues until the minimum en-route altitude is reached.

1.3.2.2.1 *Procedure design gradient (PDG).* The standard PDG is 5.0 per cent. It originates from the IDF MCA.

1.3.2.2.2 *Multiple PDGs.* Where multiple PDGs exist for a PinS departure, subsequent gradients in the departure shall be equal to or less than the PDG for the preceding segment.

1.3.2.2.3 *Minimum obstacle clearance (MOC).* Between the earliest IDF and the IDF, the MOC is equal to 30 m (100 ft). Between the IDF and the latest IDF, the MOC is increased by a value corresponding to the PDG, and subsequently from the latest IDF is increased by 0.8 per cent of the distance from the latest IDF until the en-route MOC (300 m (1 000 ft), 600 m (2 000 ft) mountainous area) is reached. An alternative to increasing the PDG above the 5 per cent standard value beyond the IDF, would be to increase the IDF MCA to provide the necessary clearance over an obstacle in the subsequent departure leg while maintaining a lower PDG.

1.3.2.2.4 *IFR OIS dimensions.* The OIS dimensions shall be as follows:

- a) *Origin.* The IFR OIS originates at the earliest IDF.
- b) *Area semi-widths for PinS departures (instrument segment).* See Part III, Section 1, Chapter 2.
- c) *IFR OIS vertical dimension.* The OIS is level at the IDF MCA minus MOC from the earliest IDF until the latest IDF. It then has a gradient of (PDG minus 0.8 per cent).

1.3.3 PinS departure with a “proceed visually” instruction – Direct visual segment (Direct-VS)

1.3.3.1 The Direct-VS is protected by one direct visual OCS and one visual OIS.

1.3.3.2 *Track change at the IDF.* The maximum track change at the IDF is 30°.

1.3.3.3 *Visual segment design gradient (VSDG).* The nominal VSDG shall be 13.3 per cent until reaching the IDF MCA. This is consistent with an obstacle clearance of 0.8 per cent above the direct visual OCS (see paragraph 1.3.3.7). The VSDG shall not be less than 0.8 per cent above the Annex 14 take-off climb surface.

1.3.3.4 The VSDG may be lowered if the Annex 14 take-off climb surface of the landing location corresponds to slope design category A or B (see Annex 14, Volume II, Chapter 4).

1.3.3.5 *Initial departure fix (IDF)*. The IDF shall be located:

- a) to provide sufficient visual reference from the heliport or landing location to the IDF to enable the helicopter to cross the IDF at or above the MCA; and
- b) to cater to the minimum starting height of the subsequent instrument segment.

1.3.3.6 *Visual segment length*. The length of the visual segment shall be measured from the outer edge of the heliport or landing location safety area to the IDF. The length of the visual segment should permit the helicopter to climb to the IDF MCA and to accelerate to V_{mini} by the IDF. The minimum visual segment length shall be 1.00 km (0.54 NM). The maximum length shall be 13 km (7 NM) for departures based on RNP 1/RNAV 1 and 5.6 km (3 NM) for departures based on RNP 0.3.

Note.— For visual segment lengths greater than the length of the Annex 14 take-off climb surface, the extension of the Annex 14 take-off surface should be considered (see the Heliport Manual (Doc 9261)).

1.3.3.7 *Direct visual OCS*

- a) *Alignment*. The direct visual OCS is aligned symmetrically on the centre line of the take-off climb surface.
- b) *Origin*. The direct visual OCS originates at the outer edge of the heliport or landing location safety area (SA).
- c) *Width*. The width of the direct visual OCS at its origin is equal to the width of the SA. The outer edges splay from their origins at the edge of the SA, symmetrically around the centre line of the take-off climb surface, to an overall maximum width of 120 m, at which point the outer edges parallel the centre line. For day-only operations, the splay is 10 per cent; for night operations, the splay angle is increased to 15 per cent.
- d) *Slope*. The elevation of the origin of the direct visual OCS is equal to the heliport or landing location elevation. It inclines at VSDG minus 0.8 per cent (nominally 12.5 per cent) from the heliport/landing location elevation to the point where the surface reaches the height of 30 m (100 ft) below the IDF MCA, at which it becomes level.
- e) *End*. The direct visual OCS ends at ATT after the nominal IDF.

1.3.3.8 *Visual segment obstacle identification surface (OIS)*. The purpose of the visual OIS is to identify obstacles for charting. The dimensions of the visual OIS are as follows:

- a) *Alignment*. The Visual OIS is constructed symmetrically around the direct track from the heliport/landing location to the IDF.
- b) *Origin*. The origin is perpendicular to the direct VS track at the boundary of the heliport or landing location safety area.
- c) *Width*. The area semi-width at the origin is 45 m (150 ft) and the area splays at 15° until the area connects with the instrument segment protection (see paragraph 1.3.3.12).
- d) *Slope*. The Visual OIS originates at the elevation of the heliport/landing location and rises to the IDF MCA minus 30 m (100 ft) at the nominal IDF. The visual segment OIS gradient shall be lower or equal to the visual direct OCS gradient. As a result, some combinations of IDF MCA, VSDG and VS lengths will not be feasible.

1.3.3.9 *Entering IMC on the direct visual segment.* IMC may be entered on the direct visual segment, prior to the IDF, when the following conditions are all met:

- a) the procedure description shall start with a prescribed course from the heliport or landing location to the IDF; and
- b) the direct visual segment shall have additional obstacle protection with a second OCS having dimensions described in 1.3.3.11.

Note.— Operational conditions under which IMC can be entered on the Direct-VS, prior to the IDF, are provided in PANS-OPS, Volume I, Part II, Section 7, Chapter 3.

1.3.3.10 *Early IMC entry OCS required for IMC entry prior to the IDF.* For IMC entry prior to the IDF, a level OCS, at the elevation of IDF MCA minus 30 m, shall be constructed with the lateral dimensions corresponding to the direct visual segment OIS. This OCS shall not be penetrated.

1.3.3.11 *Chart note requirement.* When a PinS departure procedure with a Direct-VS is designed with a Direct-VS early IMC entry OCS described in 1.3.3.11, such procedure shall be charted separately from PinS procedures with Manoeuvring VS and a note shall be included on the chart indicating that IMC entry at or above the MCA prior to crossing the IDF is permitted.

1.3.3.12 *Blending of visual segment with PBN criteria at the IDF.* The 15° visual OCS/OIS splay continues until it reaches the outer boundary of the fictitious IFR OIS. The fictitious IFR OIS begins at the earliest ATT of the IDF. The following figures apply:

- a) Figure IV-1-1 depicts the vertical blending of the visual OCS/OIS with an RNP 1/RNAV 1/RNP 0.3 OFR OIS at the IDF;
- b) Figure IV-1-2 a) depicts the lateral blending of the visual OCS/OIS surfaces with the IFR OIS with no track change at the IDF; (with track change at the IDF).
- c) Figure IV-1-2 b) depicts the lateral blending of the visual OCS/OIS with the IFR OIS with a track change at the IDF and a small length of the visual segment. In this case, the visual OCS/OIS may intersect the IFR OIS at the earliest IDF on the inside of the turn. From this point, the IFR OIS splays at 15° to the IFR departure track until it reaches the outer boundary of the fictitious IFR OIS;
- d) Figure IV-1-2 c) depicts the lateral blending of the visual OCS/OIS with the IFR OIS with a track change at the IDF and a large length of the visual segment. In this case, the visual OCS/OIS may reach IDF MCA minus 30 m (100 ft) prior to the IFR OIS on the outside of the turn. The visual OCS/OIS is extended to the earliest ATT of the IDF as a level surface at the IDF MCA minus 30 m (100 ft); and
- e) Figure IV-1-2 d) depicts the lateral blending of the visual OCS/OIS with the IFR OIS with a large track change and a large length of the visual segment at the IDF for RNP 0.3. In this case, the visual OCS/OIS may reach the IDF MCA minus 30 m (100 ft) prior to the IFR OIS on the outside of the turn. The visual OCS/OIS is extended up to the IFR OIS semi-width and thence parallel to the IFR OIS up to the earliest ATT of the IDF as a level surface at the IDF MCA minus 30 m (100 ft).

1.3.3.13 *Direct visual OCS penetration.* No obstacles shall penetrate the direct visual OCS. Any penetrations can be eliminated by increasing the slope of the direct visual OCS and a resulting increase of the VSDG if operationally feasible (see Figure IV-1-3). Such an increase shall be coordinated with the operators concerned.

1.3.3.14 *Visual segment OIS penetration.* Obstacles that penetrate the visual OIS shall be documented and charted. If this results in chart clutter, see Part I, Section 2, Chapter 1, 1.9 “Presentation of significant obstacles and spot elevations on charts”. The visual OIS shall be evaluated and, if recommended by an aeronautical study, any penetrating obstacles should be lit and marked.

1.3.3.15 *Mitigation of obstacle penetration in the instrument segment.* To avoid obstacle penetration of the IFR OIS, the IDF MCA should be increased such that the IFR OIS remains clear, or a turn initiated, in preference to increasing the PDG above the standard 5 per cent. The resulting VSDG is increased and is determined by the elevation change between the boundary of the heliport or landing location safety area and the revised IDF MCA (see Figure IV-1-4).

1.3.4 PinS departure with a “proceed visually” instruction — Manoeuvring visual segment

1.3.4.1 *Manoeuvring VS protection.* A manoeuvring visual segment is protected for the following manoeuvre: the pilot takes off in a direction other than directly to the IDF and then visually manoeuvres to join the initial instrument segment at the IDF.

1.3.4.2 This manoeuvring VS is protected by one sloping visual OCS and one visual OIS.

Note.— The protection provided for this VS is comparable with the one provided for PinS approaches followed by a manoeuvring VS (see Chapter 2, paragraph 2.9.3).

1.3.4.3 *VSDG for the manoeuvring VS.* The nominal VSDG shall be 13.3 per cent. This is consistent with an obstacle clearance of 0.8 per cent above the sloping initial visual OCS (see 1.3.4.5). The VSDG shall not be less than 0.8 per cent above the Annex 14 take-off climb surface.

1.3.4.4 *IDF minimum crossing height.* (MCH is the actual height of MCA above the heliport/landing location). The MCH of the IDF for a PinS departure procedure with a manoeuvring visual segment shall not be less than 90 m (295 ft) above the heliport/landing location elevation.

1.3.4.5 Sloping visual OCS

a) *Alignment.* The sloping visual OCS is aligned symmetrically on the centre line of the take-off climb surface.

Note.— If more than one take-off climb surface has to be considered, a visual OCS is designed for each.

b) *Origin.* The sloping visual OCS originates at the outer edge of the heliport or landing location safety area (SA).

c) *Width.* The width of the sloping visual OCS at its origin is equal to the width of the SA. The outer edges splay from their origins at the edge of the SA, symmetrically around the centre line of the take-off climb surface, to an overall maximum width of 120 m, at which point the outer edges parallel the centreline. For the provision of day-only operations, the splay is 10 per cent. For night operations, the splay angle is increased to 15 per cent.

d) *Slope.* The elevation of the origin of the sloping visual OCS is equal to the heliport or landing location elevation. The sloping visual OCS inclines at nominally 12.5 per cent from the heliport/landing location elevation to the point where the surface reaches the height of 152 m (500 ft) above heliport/landing location elevation.

1.3.4.6 Visual OIS

1.3.4.6.1 The visual OIS provides awareness in the area (called “manoeuvring area”) where the pilot is expected to manoeuvre visually from the initial departure track to join the initial instrument segment at the IDF.

1.3.4.6.2 Description of the “manoeuvring area”

1.3.4.6.2.1 The manoeuvring area is defined based on the following rules:

- a) the pilot initially climbs on the centre line of the take-off climb surface to reach the greater of the two following heights before manoeuvring toward the IDF:
 - 1) IDF MCH/2; and
 - 2) 90 m (295 ft) above the heliport/landing location elevation;
- b) the pilot continues to climb and accelerates so as to cross the IDF at or above the IDF MCA.

1.3.4.6.2.2 The outer boundaries of the manoeuvring area are based on the following assumed worst-case trajectories (see Figure IV-1-5):

- a) *first trajectory*: the pilot climbs in accordance with paragraph 1.3.4.5.2.1 a) above and then turns right to join the IDF;
- b) *second trajectory*: the pilot climbs in accordance with paragraph 1.3.4.5.2.1 a) above and then turns left to join the IDF.

1.3.4.6.2.3 The “manoeuvring area” is the area representing all the lines that originate at the IDF and connect with a “turn area” aligned symmetrically around the centre line of the take-off climb surface.

Note.— If more than one take-off climb surface has to be considered, the final “manoeuvring area” is the addition of all “manoeuvring areas” obtained.

1.3.4.6.2.4 Description of the “turn area” (see Table IV-1-1 and Figure IV-1-6)

1.3.4.6.2.4.1 The “turn area” is defined by an angle α each side of the centre line of the take-off climb surface (in order to consider a right turn and a left turn) and by a radius r .

Note.— The “manoeuvring area” can be reduced in size if a prominent obstacle is located near the heliport/landing location. In that case, after the initial climb, one turn direction to join the IDF is prohibited. Consequently, the “turn area” is defined only on one side of the centre line of the take-off climb surface (see Figure IV-1-9).

1.3.4.6.2.4.2 Radius (r) of the “turn area”:

- a) if the IDF MCH is equal to or less than 183 m (600 ft) above the heliport/landing location elevation, r is constant and equal to 1 482 m (0.8 NM);
- b) if the IDF MCH of the procedure is more than 183 m (600 ft) above the heliport/landing location elevation, r increases linearly (185 m (0.1 NM) for each additional 30 m (100 ft) above 183 m (600 ft)).

1.3.4.6.2.4.3 Angle (α) of the “turn area”:

- a) if the IDF MCH of the procedure is equal to or less than 183 m (600 ft) above the heliport/landing location elevation, α is constant and equal to 50°;
- b) if the IDF MCH of the procedure is more than 183 m (600 ft) above the heliport/landing location elevation and equal to or less than 304 m (1 000 ft) above the heliport/landing location elevation, α decreases linearly (5° for each additional 30 m (100 ft) above 183 m (600 ft));

- c) if the IDF MCH of the procedure is more than 304 m (1 000 ft) above the heliport/landing location elevation, α is constant and equal to 30°.

Note.— Where operationally beneficial, in order to extend the resulting “manoeuvring area”, the “turn area” can be extended by using wider angles on one side or on both sides of the centre line of the take-off climb surface.

Table IV-1-1. Definition of the “turn area” for a set of IDF MCH values

- a) Units in metres for the IDF MCH and r:

IDF MCH	93 m*	123 m*	153 m*	183 m*	213 m*	243 m*	273 m*	304 m*
r	1 482 m	1 482 m	1 482 m	1 482 m	1 667 m	1 852 m	2 037 m	2 222 m
α	50°	50°	50°	50°	45°	40°	35°	30°
* Height above the heliport/landing location elevation.								

- b) Units in ft for the IDF MCH and in NM for r:

IDF MCH	300 ft*	400 ft*	500 ft*	600 ft*	700 ft*	800 ft*	900 ft*	1 000 ft*
r	0.8 NM	0.8 NM	0.8 NM	0.8 NM	0.9 NM	1 NM	1.1 NM	1.2 NM
α	50°	50°	50°	50°	45°	40°	35°	30°
* Height above the heliport/landing location elevation.								

Note.— Taking into account the buffer of 741 m (0.4 NM) that will be applied to these values to define the OIS (see paragraph 1.3.4.5.3.2), these values are acceptable for all the operational scenarios regarding the temperature and the heliport or landing location elevation.

1.3.4.6.3 OIS definition

1.3.4.6.3.1 The OIS surface is defined by the “manoeuvring area” with an additional buffer equal to 741 m (0.4 NM) (see Figure IV-1-7).

Note.— If the initial take-off can be performed in an omni-directional fashion, the OIS is a surface aligned symmetrically on the course between the HRP and the IDF and that connects tangentially with the following: a circle centred on the IDF, having a radius equal to 741 m (0.4 NM), and a circle centred on the HRP, having a radius equal to (r + additional buffer value) (see Figure IV-1-8).

1.3.4.6.3.2 The OIS identifies obstacles in the “manoeuvring area” that are at or above 150 ft below the minimum height to which the pilot will initially climb before manoeuvring toward the IDF. Based on the above assumptions, the OIS is a surface surrounding the “manoeuvring area” and defined at the greater of the following heights:

- a) IDF MCH/2 – 46 m (150 ft); and
- b) 46 m (150 ft) above the heliport/landing location elevation.

1.3.4.6.3.3 *Obstacle clearance.* No obstacles shall penetrate the sloping visual OCS. Obstacles that penetrate the OIS shall be charted, and may be marked and/or lighted when feasible.

1.3.4.7 *Maximum speed restrictions.* There are no maximum speed restrictions when operating on a manoeuvring visual segment.

1.3.4.8 *Manoeuvring visual segment length.* The length of the visual segment should permit the helicopter to climb to the IDF MCA and to accelerate to V_{mini} by the IDF. The minimum visual segment length shall be 1.00 km (0.54 NM).

1.3.4.9 *Entering IMC on the manoeuvring visual segment.* IMC shall not be entered prior to crossing the IDF at or above the MCA.

1.3.4.10 *Track change at the IDF.* The track change at the IDF shall be less than 30° for all the possible visual trajectories. Consequently, the angle between the initial instrument segment direction after the IDF and the direction of the “extreme” visual trajectories corresponding to the limits of the “manoeuvring area” shall be less than 30° (see Figures IV-1-7 and IV-1-8).

1.3.5 PinS departure with a “proceed VFR” instruction — visual segment

1.3.5.1 Where “proceed visually” instruction is not suitable or possible, a PinS departure procedure with a “proceed VFR” instruction shall be designed.

1.3.5.2 There is no obstacle protection on the visual segment. The pilot complies with VFR to see and avoid obstacles when proceeding from the heliport or landing location to the IDF, to cross at or above the IDF MCA. Since there is no obstacle protection on the visual segment, IMC shall not be entered until crossing the IDF.

1.4 PROMULGATION

Note.— Principles governing the identification of standard departure routes are contained in Annex 11, Appendix 3. Specifications for standard instrument departure charts are contained in Annex 4.

1.4.1 The plan view shall include a note that the procedure is Cat H only.

1.4.2 The IDF shall generally be charted as a “fly-by” waypoint. If for operational reasons, the IDF needs to be a “fly-over” waypoint, it should be charted as a “fly-over” waypoint.

1.4.3 *Climb gradients in instrument segment.* PDGs greater than 5 per cent shall be charted together with the point or altitude to which they apply.

1.4.4 *Climb gradients in the visual segment.* The VSDG for the direct VS and the manoeuvring VS shall be charted.

1.4.5 *Charting of the MCA.* The IDF MCAs and all other established MCAs shall be charted on the plan view adjacent to the waypoints to which they apply.

1.4.6 *Segment tracks and lengths.* Segment tracks and lengths shall be charted.

1.4.7 *Obstacles.* Obstacles penetrating the visual OIS shall be charted.

1.4.8 Additional information for the direct and manoeuvring VS

1.4.8.1 The centre line(s) and direction(s) of the take-off climb surface(s) taken into account for the protection of the direct and/or manoeuvring visual segment shall be indicated on the chart.

1.4.8.2 The direct and/or manoeuvring visual segment shall be represented on the chart either in an inset on the plan view, or on a continuation sheet or the verso of the chart. Information depicted shall be charted to scale. If the direct and/or manoeuvring visual segment is not depicted in an inset, the plan view shall contain an annotation directing the pilot to the continuation sheet or the verso of the chart. In case of a manoeuvring visual segment, the “manoeuvring area” shall be depicted.

1.4.8.3 If the “manoeuvring area” is reduced in size in order to take into account a significant obstacle, restricted use airspace or environmentally sensitive areas located near the heliport/landing location, the following elements shall be indicated on the chart:

- a) the boundaries of the manoeuvring area;
- b) the location of the significant obstacle/restricted use airspace/environmentally sensitive area; and
- c) the boundaries of any ‘no manoeuvring’ area annotated ‘No manoeuvring’.

1.4.8.4 The departure shall be annotated “Proceed visually to the IDF” or “Proceed VFR to the IDF” as appropriate.

1.4.8.5 *Chart note requirement.* When a PinS departure procedure with a Direct VS is designed with a Direct VS early IMC entry OCS described in 1.3.3.11, such procedure shall be charted separately from PinS procedures with Manoeuvring VS and a note shall be included on the chart indicating that IMC entry at or above the MCA prior to crossing the IDF is permitted.

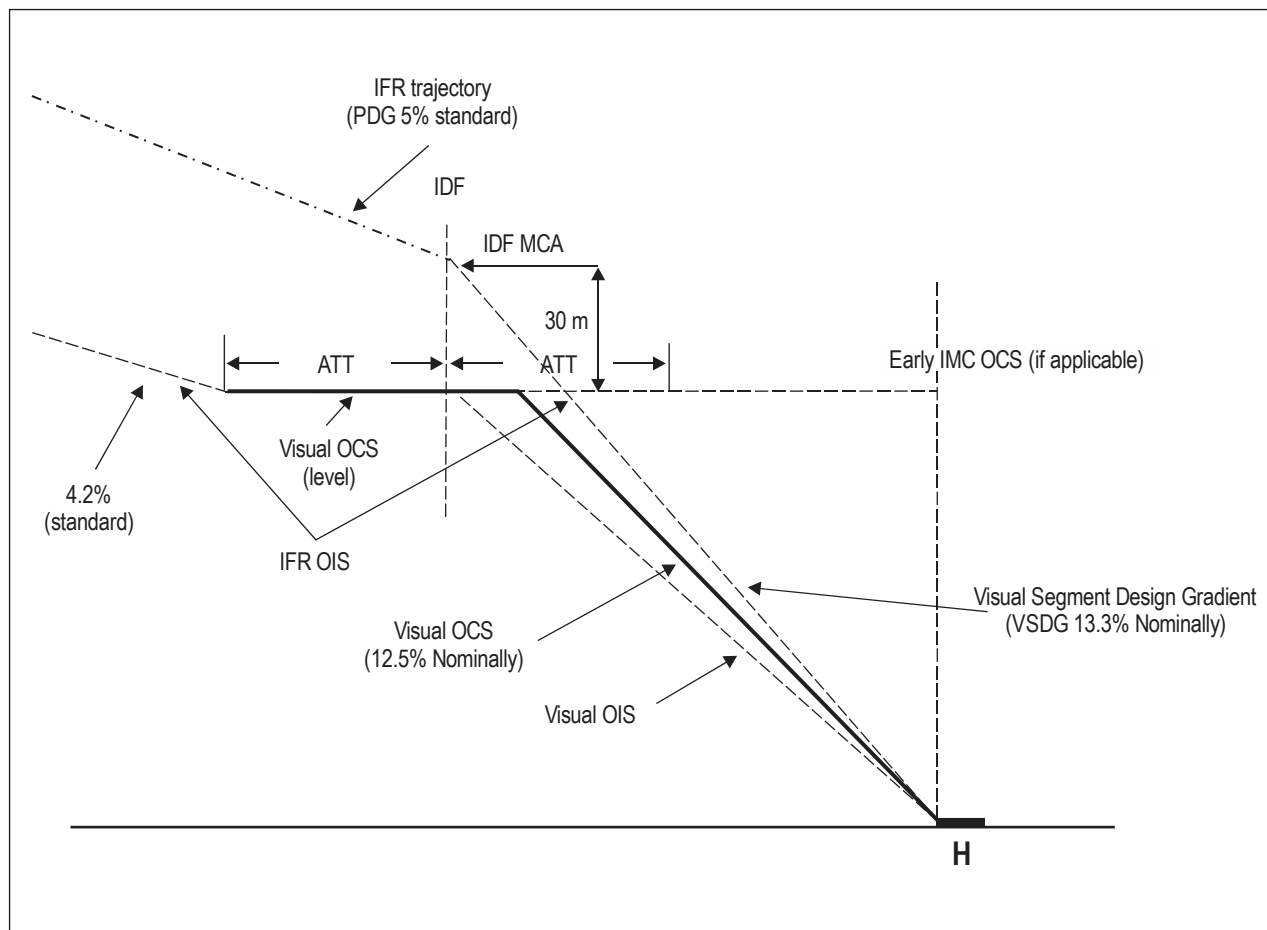


Figure IV-1-1. Vertical blending of visual segment OIS and optional early IMC entry OCS

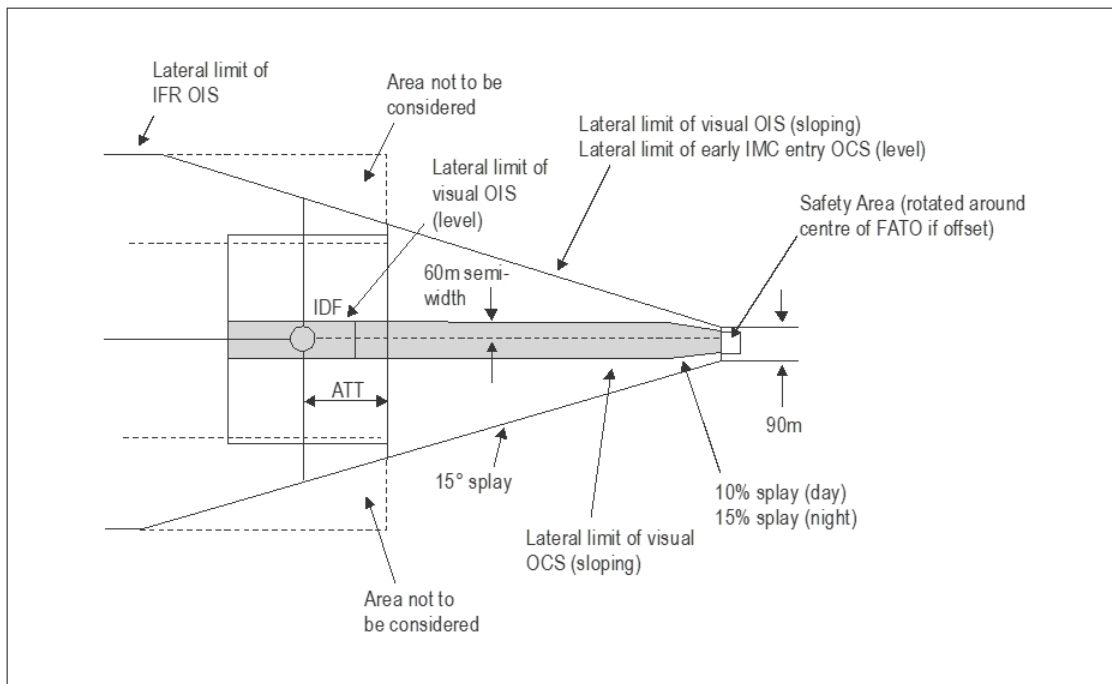


Figure IV-1-2 a). Lateral blending with no track change at the IDF

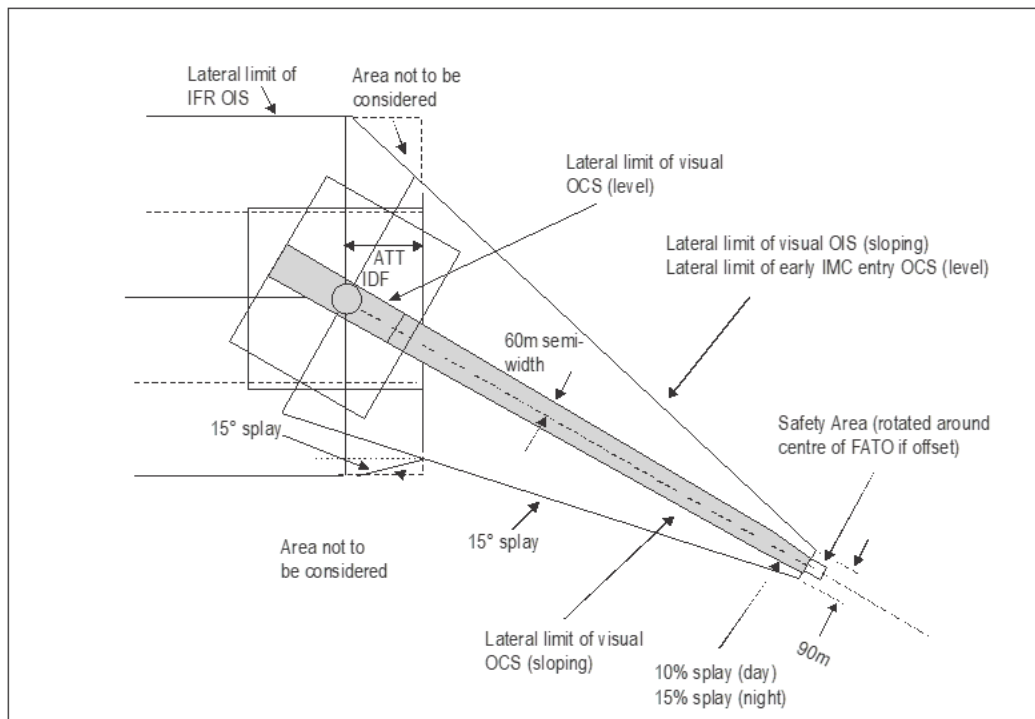


Figure IV-1-2 b). Lateral blending with track change at the IDF and small length of the visual segment

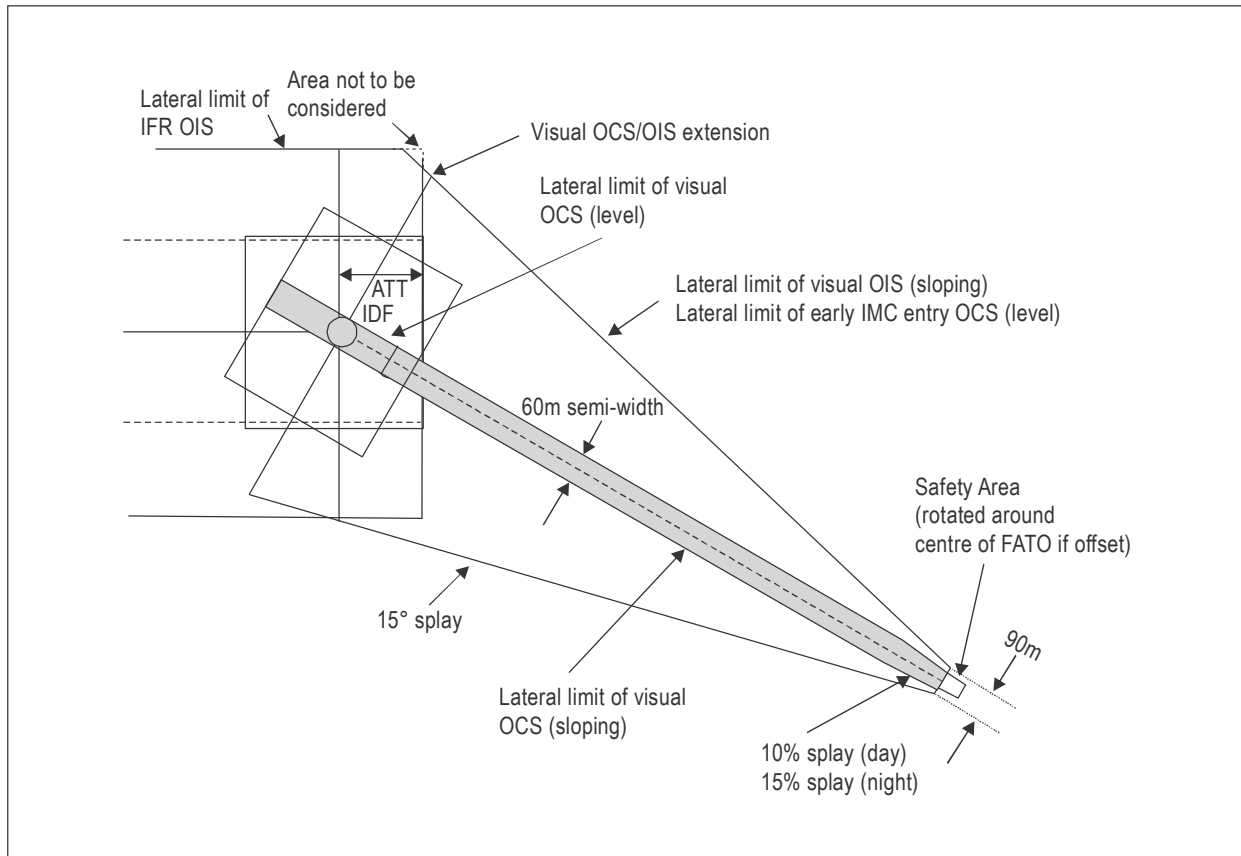


Figure IV-1-2 c). Lateral blending with track change at the IDF and large length of the visual segment

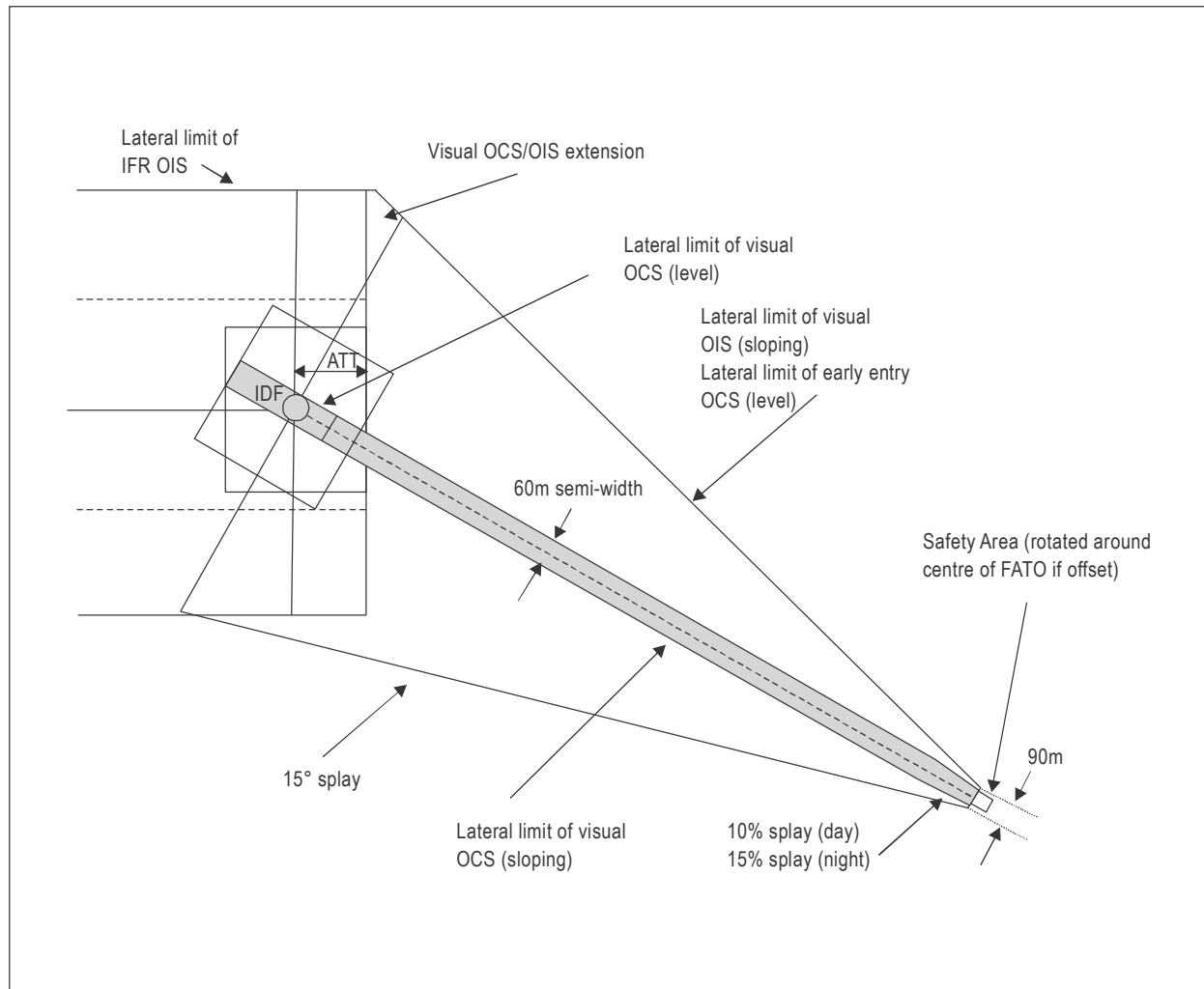


Figure IV-1-2 d). Lateral blending with large track change at the IDF and large visual segment length for RNP 0.3

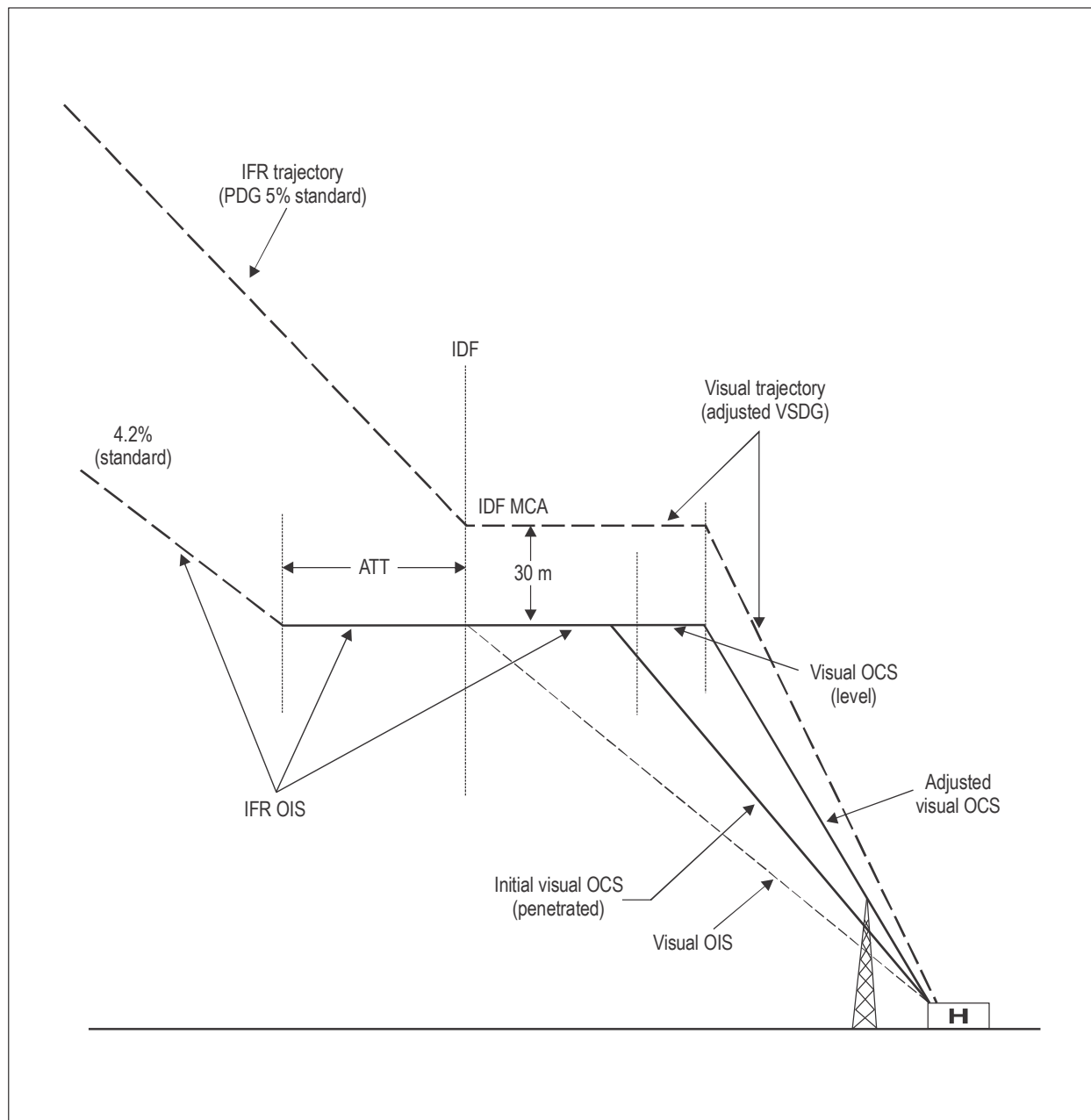


Figure IV-1-3. Mitigation of OIS penetration by raising the VSDG

Note.— Some on-board systems will not switch into the approach mode when the track change at the FAF is $>30^\circ$.

2.6.3 *Area.* See Part III, Section 3, Chapter 3, Figure III-3-3-2, Intermediate and final segments.

2.6.3.1 *Length.* The optimum length is 5.56 km (3.00 NM). It shall not be less than 3.70 km (2.00 NM), and shall not exceed 18.52 km (10.00 NM). The minimum length is governed by the magnitude of the turn required at the IF. The intermediate approach segment is designed for helicopters flying the procedure at speeds up to 220 km/h (120 KIAS). Where an operational requirement exists, the segment may be designed for an airspeed not exceeding 165 km/h (90 KIAS), in which case the approach plate will be annotated “Speed limited to 165 km/h (90 KIAS)”.

2.6.3.2 *Width.* See Part III, Section 1, Chapter 2, Tables III-1-2-9 and III-1-2-10.

2.6.4 *Obstacle clearance.* The area considered for obstacle clearance extends from the earliest IF to the nominal position of FAF. The general criteria for obstacle clearance applies, see Part I, Section 4, Chapter 4, 4.3.2, “Obstacle clearance”. The obstacle clearance in the primary area is 150 m (492 ft), tapering uniformly to zero from the edge of the primary area to the outer edge of the secondary area.

2.6.5 *Descent gradient.* Optimum descent gradient is 6.5 per cent. The maximum permissible gradient is 10 per cent. When an operational requirement exists, a gradient of as much as 13.2 per cent may be authorized, provided the speed is restricted to a maximum of 165 km/h IAS (90 kt IAS) and provided the gradient used is depicted on approach charts. The descent gradient should be calculated in accordance with Part III, Section 2, Chapter 3, 3.3.3, “Descent gradient”.

2.7 FINAL APPROACH SEGMENT

2.7.1 The final approach segment begins at the FAF (fly-by) and ends at the MAPt (flyover). All approaches will be to a point in space where the pilot should have sufficient visual reference to continue the approach and landing to the intended landing site or initiate a missed approach.

2.7.2 *Alignment.* For point-in-space approaches there are no alignment requirements in the final approach segment.

2.7.3 *Area.* See Part III, Section 3, Chapter 3, Figure III-3-3-2.

2.7.3.1 The area considered for obstacle clearance begins at the earliest FAF position and ends at the nominal position of the MAPt.

2.7.3.2 *Length.* The optimum length is 5.92 km (3.20 NM). The minimum length is governed by the magnitude of the turn required at the FAF (see Table I-4-5-1). Procedures are normally designed for helicopters flying the approach up to 130 km/h (70 KIAS). For specific cases, where the final may be designed to accommodate speeds up to 165 km/h (90 KIAS), the missed approach must also be designed to accommodate 165 km/h (90 KIAS). The maximum speed for which the final and missed approach segments are designed must be clearly annotated on the approach chart.

2.7.3.3 *Width.* See Part III, Section 1, Chapter 2, Tables III-1-2-9 and III-1-2-10.

2.7.3.4 *Merging method at FAF.* If RNP 0.3 is used on all segments, the intermediate segment width applies until the nominal FAF, where the outer edges of the protection area converge at 30 degrees until reaching the final approach segment width.

2.7.4 *Obstacle Clearance.* Primary area minimum obstacle clearance (MOC) is 75 metres (246 ft) tapering uniformly to zero from the edge of the primary area to the outer edge of the secondary area.

2.7.5 *Descent gradient.* Optimum descent gradient is 6.5 per cent. Where a higher descent gradient is necessary, the recommended maximum is 10 per cent. However, where an operational imperative exists, and the magnitude of turn at the FAF is less than or equal to 30°, a gradient of as much as 13.2 per cent may be authorized, provided the speed is restricted to a maximum of 130 km/h IAS (70 kt IAS) and provided the gradient used is depicted on approach charts. The final segment is calculated from the FAF altitude at the plotted position of the FAF to the OCA/H at the plotted position of the MAPt.

2.8 MISSED APPROACH SEGMENT

2.8.1 *General.* The missed approach segment begins at the earliest MAPt (flyover) position and ends at a holding point designated by an MAHF (flyover) or to a clearance limit. Optimum routing is straight ahead to a direct entry into holding at the MAHF.

2.8.2 *Longitudinal tolerance of the MAPt.* The longitudinal tolerance of the MAPt will be calculated as described at Part I, Section 4, Chapter 6, 6.1.6.2.1, “MAPt tolerance when MAPt is defined by a navigational facility or fix”.

2.8.3 *Calculation of start of climb (SOC).* The SOC point will be calculated as described at Part I, Section 4, Chapter 6, 6.1.6.2, “Determining SOC with an MAPt defined by a navigation facility or fix”, except that the transitional tolerance (X) is the distance a helicopter traverses during 5 seconds of flight at 130 km/h (70 KIAS) or 165 km/h (90 KIAS) converted to TAS.

2.8.4 *Missed approach area.* The missed approach area shall commence at the beginning of the MAPt longitudinal tolerance at a width equal to the final approach area at that point. At that point, the area splays at 15° on each side of the missed approach track, to account for the decrease in GNSS receiver display sensitivity from ± 0.56 km (0.30 NM) to ± 1.85 km (1.00 NM) to a total width of ± 3.43 km (1.85 NM) until 15 NM from the PinS (MAPt). If the first waypoint is reached prior to the area reaching ± 3.43 km (1.85 NM) the splay continues to 3.43 km (1.85 NM). If RNP 0.3 is chosen on all segments, the area does not splay at the early MAPt and the final approach semi-width is maintained until 15 NM from the PinS (MAPt). For missed approach procedures with GNSS receivers that do not provide continuous track guidance after the MAPt, see Figures IV-2-2 and IV-2-3. Turning missed approach with track specified to the MATF or the MAHF should be restricted to systems providing continuous track guidance after the missed approach waypoint and the approach procedure should be clearly annotated. See Figure IV-2-4.

2.8.5 *Straight missed approach.* The criteria governing straight missed approach apply (see Part I, Section 4, Chapter 6, 6.3, “Straight missed approach”). Note also that track guidance is available for the missed approach by the nomination of a GNSS fix(es).

2.8.6 *Turning missed approach.* The general criteria in Part I, Section 4, Chapter 6, 6.4.2, “General”, 6.4.3, “Turn parameters”, 6.4.6.4, “Obstacle clearance in the turn area” and 6.4.7, “Turn specified at the MAPt” apply. For turn protection and obstacle assessment, see Part III, Section 2, Chapter 2, “Turn protection and obstacle assessment.” For missed approach procedures with GNSS receivers which do not provide continuous track guidance after the MAPt, see Figures IV-2-2 and IV-2-3. Turning missed approach with track specified to MAHF should be restricted to systems providing continuous track guidance after the missed approach waypoint and the approach procedure should be clearly annotated. See Figure IV-2-4.

2.8.6.1 Turn parameters.

2.8.6.1.1 *Indicated airspeed.* The speed for the final missed approach is 165 km/h (90 KIAS). However, where operationally required to avoid obstacles, reduced speeds as slow as 130 km/h (70 KIAS) may be used, provided the procedure is annotated “**Missed approach turn limited to 130 km/h (70 KIAS) maximum**”.

2.8.6.1.2 *Alignment.* The maximum difference between the inbound track and outbound track at MATF is a maximum of 120°. This restriction does not apply in case of a flyover waypoint followed by a DF leg.

2.8.6.1.3 *Length.* Where an operational requirement exists to avoid obstacles, an MATF may be used. In this case, the applicable turn anticipation distance for the turn point must be applied after SOC. The minimum length after the turn is determined by the MSD required for the outbound segment. Refer to the method in Part III, Section 2, Chapter 1.

2.8.7 *Climb gradient.* The nominal climb gradient of the missed approach surface is 4.2 per cent (24:1). Higher gradients may be considered with operational approval when an operational requirement exists. When a gradient other than the nominal gradient is used in the construction of the missed approach procedure the gradient required must be annotated on the instrument approach chart. In addition to the OCA/H for the specified gradient, the OCA/H applicable to the nominal gradient must also be shown.

2.8.8 The MOC is 40 m (130 ft) for turns exceeding 15°. (See Part I, Section 4, Chapter 6.)

2.9 PinS APPROACH PROCEDURES WITH A “PROCEED VISUALLY” INSTRUCTION

Note.— In circumstances where a “proceed visually” instruction is not suitable or possible, a PinS approach procedure with a “proceed VFR” instruction can be designed (see section 2.10).

2.9.1 PinS approach — general

2.9.1.1 *Description.* A direct visual segment or a manoeuvring visual segment connects the PinS (the MAPt) to the heliport or the landing location. This provides the pilot flying a PinS instrument approach procedure with a visual segment to proceed visually from the MAPt to the heliport or landing location.

2.9.2 PinS approach with a "proceed visually" instruction — Direct-visual segment

2.9.2.1 *Description.* The Direct-VS connects the PinS to the landing location; this can be either direct to the landing location or via a descent point where a limited track change may occur. The Direct-VS provides the pilot flying a PinS instrument approach procedure with a visual segment to proceed visually from the MAPt to the landing location.

2.9.2.1.1 *General.* The following paragraphs provide a description of the components of the Direct-VS. See Figures IV-2-5 and IV-2-6.

2.9.2.1.1.1 The landing location shall meet the dimensions of the non-instrument heliport final approach and take-off area (FATO) and safety area (SA) as defined in Annex 14, Volume II.

2.9.2.1.1.2 The obstacle clearance surface (OCS) is continued from the MAPt to the landing location and is similar to an obstacle limitation surface (OLS) provided in Annex 14, Volume II, and a visual segment surface (VSS).

2.9.2.1.1.3 The three obstacle identification surfaces (OIS) provide the pilot obstacle awareness in the vicinity of the landing location.

2.9.2.1.1.4 A descent point (DP) defined by track and distance from the MAPt may be necessary to identify the point at which the aircraft can descend below the OCA and begin a visual descent to the landing location.

2.9.2.1.1.5 The visual segment descent angle (VSDA) describes the nominal descent path of the aircraft in the visual segment. It is the angle from the MDA at either the MAPt or DP to the landing location HRP at HCH. The nominal VSDA is 8.3°. This is consistent with an OCS of 1.12° below the VSDA (see paragraph 2.9.2.2.1). The VSDA shall be at least 1.12° above the Annex 14 take off/climb surface.

2.9.2.1.1.6 The VSDA may be lowered if the Annex 14 take-off climb surface of the landing location corresponds to slope design category A or B (see Annex 14, Volume II, Chapter 4).

2.9.2.1.1.7 A higher VSDA may be chosen in coordination with the operators concerned.

2.9.2.2 *Obstacle clearance surface (OCS) and obstacle identification surface (OIS)*

2.9.2.2.1 *OCS alignment, slope, origination base width, and lateral dimensions.* The OCS is aligned symmetrically on the course between HRP and MAPt. If a DP is established with a change of track at the DP, the OCS is aligned symmetrically on the course between the HRP and the DP. See Figures IV-2-7, IV-2-8 and IV-2-9.

2.9.2.2.1.1 The OCS originates at the outer edge of the landing location SA.

2.9.2.2.1.2 The width of the OCS at its origin is equal to the width of the SA.

2.9.2.2.1.3 The outer edges splay from their origins at the edge of the SA, symmetrically around the OCS centre line direction, to an overall maximum width of 120 m, at which point the outer edges parallel the OCS centre line. For the provision of day-only operations the splay is 10 per cent. For night operations the splay angle is increased to 15 per cent.

2.9.2.2.1.4 The elevation of the origin of the OCS is equal to the landing location elevation.

2.9.2.2.1.5 The OCS slopes upward at nominally 12.5 per cent from the heliport elevation to the point where the surface reaches the altitude of the OCA minus the primary area MOC established for the final approach segment (FAS). It then extends as a level surface to the early ATT of the MAPt.

Note 1.— The calculation of the SA size and the SA outer edges used to establish the origination base width is aircraft dependent and needs to be provided with other landing location information to develop a Direct-VS for a helicopter PinS instrument approach procedure.

Note 2.— The nominal 12.5 per cent OCS slope is consistent with a VSDA of 8.3° and an OCS of 1.12° below VSDA.

2.9.2.2.2 *Sloping OIS inner and outer lateral dimensions.* There are two sloping OIS areas; one on each side of the OCS.

2.9.2.2.2.1 The sloping OIS inner and outer edges originate at the outer edge of the OCS origin (at the outer edge of the SA).

2.9.2.2.2.2 The inner edge of the OIS extends to the limit of the OCS outer edge.

2.9.2.2.2.3 The outer edge of the sloping OIS is established by connecting the origin directly to the outer edge of the primary areas at the MAPt/DP.

2.9.2.2.2.4 The origin of the sloping OIS is established at the elevation of the landing location.

2.9.2.2.2.5 The sloping OIS rise in the vertical plane at the same gradient as the OCS, to the point where the surfaces reach the altitude of the OCA minus the primary area MOC established for the FAS. These then extend as level surfaces to the nominal MAPt.

2.9.2.2.3 *Level OIS.* A level OIS surrounds the outer lateral limits of the sloping OIS.

2.9.2.2.3.1 The inner edge of the level OIS abuts the outer edge of the sloping OIS.

2.9.2.2.3.2 The outer edge of the level OIS originates at each outer edge of the FAS secondary area and connects tangentially with a 750 m (0.40 NM) radius circle centred on the HRP.

2.9.2.2.3.3 The altitude of the level OIS is equal to the OCA for the instrument approach procedure minus 30 m.

2.9.2.3 *Visual segment descent angle (VSDA).* The VSDA describes the nominal descent path of the aircraft in the visual segment. The VSDA is derived from the slope of the OCS and is equivalent to the OCS slope plus 1.12°. The maximum VSDA is 8.3°. The VSDA originates at MAPt, or DP if established, at MDA/H, and terminates at HCH over the HRP.

2.9.2.4 *DP establishment, alignment, OCS dimensions, FAS extension.* If the VSDA reaches an altitude equal to OCA at a point that is between the latest ATT of the MAPt and the HRP, then a DP is established. The associated DP alignment course is between HRP and DP. In such a case, an additional OCS is required. This additional OCS is established as a level surface equal in dimension to the FAS primary area and at an altitude of OCA minus primary area MOC; it extends beyond the MAPt to the DP. The semi-width of this OCS extension is equal to the FAS primary area semi-width extended from the MAPt to abeam the DP. If a turn is established at the DP, the edge of the sloping OIS is constructed as follows (see Figure IV-2-9).

2.9.2.4.1 *Outer edge outside the turn.* A circular arc with the radius of the instrument segment primary area is constructed at the DP. The outer edge of the OIS is the tangential connection to the circle above and the edge of the landing location SA at the width of the SA.

2.9.2.4.2 *Outer edge inside the turn.* At the point where the OCS becomes level (OCA minus primary area MOC), a perpendicular line to the track DP-HRP is constructed. Where this perpendicular line reaches the width of the primary area of the instrument segment parallel to the instrument segment final approach track, the OIS outer edge connects to the edge of the landing location SA at the width of the SA.

2.9.2.5 *Obstacle clearance.* No obstacles shall penetrate the Direct-VS OCS. Obstacles that penetrate the sloping OIS and/or the level OIS shall be documented and should be charted.

2.9.2.6 *Direct-visual segment length.* The length of the visual segment should be selected to provide sufficient visual references from the MAPt to the landing location whilst also providing sufficient distance to decelerate, descend and land the aircraft at the landing location.

2.9.2.6.1 The maximum visual segment length shall be 3.00 km (1.62 NM).

2.9.2.6.2 The optimum visual segment length is dependent on the maximum speed in the final approach segment of the instrument procedure and is as follows:

130 km/h (70 KIAS): 1.20 km (0.65 NM)
165 km/h (90 KIAS): 2.00 km (1.08 NM).

2.9.2.6.3 The minimum visual segment length is dependent on the maximum speed in the final approach segment of the instrument procedure and shall be as follows:

130 km/h (70 KIAS): 1.00 km (0.54 NM)
165 km/h (90 KIAS): 1.60 km (0.85 NM).

2.9.2.7 *Course change.* Course changes are permitted at either the MAPt or the DP (if established) but not at both. The maximum course change is 30°.

2.9.3 PinS approach with a "proceed visually" instruction — Manoeuvring-visual segment (VS)

2.9.3.1 A Manoeuvring-VS is designed to protect a manoeuvre whereby the pilot visually acquires the heliport or landing location, or visual references associated with it, by the MAPt and visually manoeuvres around the heliport or landing location to land from a direction other than directly from the MAPt.

2.9.3.1.1 This Manoeuvring-VS is protected by a level obstacle clearance surface (OCS), the obstacle limitation surfaces (sloping OCS and OIS) and the safety area (SA) associated with the FATO (See Annex 14, Volume II, Chapter 3, section 3.1).

2.9.3.1.2 The OCH for a PinS procedure followed by a Manoeuvring-visual segment shall not be less than 90 m (295 ft) above the heliport/landing location elevation.

2.9.3.1.3 *VSDA for the Manoeuvring-VS.* The nominal VSDA is 8.3°. This is consistent with a sloping OCS of 1.12° below the VSDA (see paragraph 2.9.3.4).

2.9.3.1.4 A higher VSDA may be chosen in coordination with the operators concerned.

2.9.3.2 Level obstacle clearance surface (OCS)

2.9.3.2.1 The level OCS is a level surface at an altitude of OCA – 76 m (250 ft).

2.9.3.2.2 From the MAPt to abeam the HRP the level OCS is aligned symmetrically on the course between HRP and MAPt with a semi-width of 741 m (0.4 NM). Beyond the HRP, this surface connects tangentially with a circle centred on the HRP, this circle having a radius equal to 741 m (0.4NM).

2.9.3.3 Obstacle identification surface (OIS)

2.9.3.3.1 The OIS provides awareness in the area where the pilot is expected to manoeuvre before being aligned on the centre line of the approach surface.

2.9.3.3.2 Definition of the “manoeuvring area”

2.9.3.3.2.1 The “manoeuvring area” corresponds to the area where the pilot is expected to manoeuvre from the MAPt to the point where it is aligned on the final landing.

2.9.3.3.2.2 The “manoeuvring area” is the area enclosed by all the lines that originate at the MAPt and connect with a “base turn area” aligned symmetrically around the centre line of the approach surface (see Figures IV-2-10 and IV-2-11).

Note 1.— If more than one approach direction has to be considered, the final “manoeuvring area” is the combination of all the “manoeuvring areas” obtained.

Note 2.— Trajectories that have been considered to define the shape of this “manoeuvring area” are:

- a) the pilot flies at the OCA/H directly from the MAPt to the heliport/landing location and then performs a base turn to descend and align on the centre line of the approach surface;
- b) the pilot starts from the MAPt but diverges from the "MAPt-HRP" axis in order to manoeuvre to align on the centre line of the approach surface.

2.9.3.3.2.3 Description of “base-turn area” (see Table IV-2-1 and Figure IV-2-12).

2.9.3.3.2.3.1 The “base turn area” is defined by an angle α each side of the centre line of the approach surface (to protect both possible manoeuvres of a base turn (on the right side and on the left side) to join the centre line of the approach surface) and by a radius r .

2.9.3.3.2.3.2 *Radius (r) of the “base turn area”:*

- a) if the OCH of the procedure is equal to or less than 183 m (600 ft) above the heliport/landing location elevation, r is constant and equal to 1 482 m (0.8 NM);
- b) if the OCH of the procedure is more than 183 m (600 ft) above the heliport/landing location elevation, r increases linearly (185 m (0.1 NM) for each additional 30 m (100 ft) above 183 m (600 ft)).

2.9.3.3.2.3.3 *Angle (α) of the “base turn area”:*

- a) if the OCH of the procedure is equal to or less than 183 m (600 ft) above the heliport/landing location elevation, α is constant and equal to 50°;

- b) if the OCH of the procedure is more than 183 m (600 ft) above the heliport/landing location elevation and equal to or less than 304 m (1 000 ft) above the heliport/landing location elevation, α decreases linearly (5° for each additional 30 m (100 ft) above 183 m (600 ft));
- c) if the OCH of the procedure is more than 304 m (1 000 ft) above the heliport/landing location elevation, α is constant and equal to 30° .

Table IV-2-1. Definition of the “base turn area” for a set of OCH values

a) Units in metres for OCH and r:

OCH	93 m*	123 m*	153 m*	183 m*	213 m*	243 m*	273 m*	304 m*
r	1 482 m	1 482 m	1 482 m	1 482 m	1 667 m	1 852 m	2 037 m	2 222 m
α	50°	50°	50°	50°	45°	40°	35°	30°
* Height above the heliport/landing location elevation.								

b) Units in feet for the OCH and in NM for r:

OCH	300 ft*	400 ft*	500 ft*	600 ft*	700 ft*	800 ft*	900 ft*	1 000 ft*
r	0.8 NM	0.8 NM	0.8 NM	0.8 NM	0.9 NM	1 NM	1.1 NM	1.2 NM
α	50°	50°	50°	50°	45°	40°	35°	30°
* Height above the heliport/landing location elevation.								

Note.— Taking into account the buffer of 741 m (0.4 NM) that will be applied to these values to define the OIS (see paragraph 2.9.3.3.3.2), these values are acceptable for all the operational scenarios regarding the temperature and the heliport or landing location elevation.

2.9.3.3.2.3.4 Base turn area on one side of the MAPt-HRP axis. When turns are limited to one side of the MAPt-HRP axis, and the turn over the HRP is greater than 30° , the base turn area may not be sufficient to contain the manoeuvre. In these circumstances (when $\alpha > 30^\circ$), the angle between the edge of the base turn area and the extended MAPt-HRP axis is held at 30° and results in an increase to the base turn area (see Figure IV-2-11 b)).

2.9.3.3.2.3.5 Where operationally beneficial, in order to extend the resulting “manoeuvring area”, the “base turn area” can be extended by using wider angles on one side or on both sides of the centre line of the approach surface.

2.9.3.3.3 OIS definition

2.9.3.3.3.1 The OIS is a level surface at a height of (OCH (height above the heliport/landing location elevation)/2 – 46 m (150 ft)) or at a height of 46 m (150 ft) above the heliport/landing location elevation, whichever is greater.

2.11.2 *HAS diagram requirements.* The radius of the HAS diagram, centred on the MAPt of the PinS approach procedure with a “proceed VFR” instruction, is at least 1.5 km (0.8 NM). This minimum value may be increased depending on State-specific requirements for helicopter VFR operations. The difference in height between the OCA and the elevation of the highest terrain or water within 1.5 km (0.8 NM), or other higher value required by the State, of the MAPt shall be charted. The inbound course to the MAPt shall also be charted. An example of a HAS diagram is depicted in Figure IV-2-18. The HAS value in the example is 467 ft and the final approach course to the MAPt is 028°.

2.12 PROMULGATION

2.12.1 *Procedure identification.* PinS approaches shall be titled “RNAV(GNSS) or RNP XXX”, depending on the navigation specification, in accordance with Part III, Section 5, Chapter 1, where XXX is the final approach course. PinS approach procedures shall not be published on the same IAP chart as aeroplane (CAT A, B, C, D) and helicopter (CAT H) procedures to runways.

2.12.2 The plan view shall include:

- a) heliport/landing location name and elevation to the nearest metre or foot;
- b) bearing to the nearest degree, and distance to the nearest two-tenths of a kilometre or tenth of a nautical mile from the missed approach point (MAPt) to the heliport/landing location;
- c) textual instructions to “Proceed VFR from xxxx” (MAPt identifier) or “Proceed visually from xxxx” (MAPt identifier) or “Manoeuvre visually from xxxx” (MAPt identifier), as appropriate;
- d) obstacles, if not included in an inset; and
- e) a note that the procedure is for CAT H only.

2.12.3 For point-in-space approaches annotated “Proceed visually from (MAPt)” serving more than one heliport, the heliport name(s), heliport elevation(s), the bearing (to the nearest degree) and the distance (to the nearest two-tenths of a kilometre (tenth NM)) from MAPt to each HRP shall be included; e.g. **MCCURTAIN MEMORIAL HOSPITAL, ELEV 693’, 123/3.2.**

2.12.4 *Speed limitation.* Airspeed restrictions shall be depicted on the chart textually as “Maximum airspeed xxx km/h (xxx KIAS)” when less than 165 km/h (90 KIAS).

2.12.5 *Descent gradient.* Where an operational requirement exists, a gradient of as much as 13.2 per cent (800 ft/NM) may be authorized, provided that the gradient used is depicted on the approach chart.

2.12.6 Appropriate obstacles shall be charted (see 2.9.3.5.1).

2.12.7 An inset shall be used to show the following:

- a) obstacles that penetrate the OIS;
- b) final approach course to the MAPt;
- c) text for either “Proceed VFR from (MAPt)” or “Proceed visually from (MAPt)”, as appropriate;
- d) for “Proceed visually” PinS procedures with a direct visual segment, the descent point (DP), if established, and bearings and distances from MAPt to DP and from MAPt or DP to the heliport/landing location;

- e) for “Proceed visually” PinS procedures with a manoeuvring visual segment(s), only the ingress track(s) and the boundary of the manoeuvring area without dimensions are charted;
- f) for “Proceed visually” PinS procedures with a “no manoeuvring” area, the text “no manoeuvring” will be shown, along with the boundary of the “no manoeuvring” area. The “no manoeuvring” area shall be hachured;
- g) for “Proceed visually” PinS procedures where over-flight of the heliport or landing location is prohibited, the bearing and distance, from the MAPt to the heliport or landing location, on a line from the MAPt to the boundary of the prohibited over-flight area; and
- h) for “Proceed VFR” procedures, only a height above surface (HAS) diagram, which shall include the difference in height between the OCA and the elevation of the highest terrain or water surface and any relevant obstacles within 1.5 km (0.8NM) of the MAPt.

Note.— The inset specified above is a separate framed diagram, located on the plan view, on the verso of the chart, or on a continuation sheet, which is charted to scale and is used to show pertinent information “close-in” to the heliport or landing location.

2.12.8 The inset for either type of PinS procedure shall not be used to depict instructions, non-operational notes, descent gradients/angles, or missed approach track or instructions.

2.12.9 The profile view shall contain information relating to the instrument procedure profile and the direct visual segment profile, if it exists, with the text “Proceed VFR” or “Proceed visually”, as appropriate. The profile view of a PinS approach procedure shall include:

- a) fixes, altitudes and distances up to the MAPt;
- b) the profile and track from the MAPt to the heliport or landing location;
- c) the descent point if established;
- d) the descent angle from the MAPt or DP;
- e) the heliport crossing height (HCH);
- f) the text “Proceed visually”, which shall be located under the visual segment profile; and
- g) a descent table should be shown indicating descent angle and descent rate in metres per second (feet per minute) for appropriate speeds for applicable segments, i.e. final approach fix (FAF) to step down fix (SDF) and SDF to missed approach point (MAPt).

Note.— The descent table may be placed in the lower left or right corner of the plan view directly above the profile view.

2.12.10 For “Proceed visually” PinS procedures with a direct visual segment and/or a manoeuvring visual segment, the VSDA for the direct-VS and/or the descent gradient angle for final landing ingress shall be published.

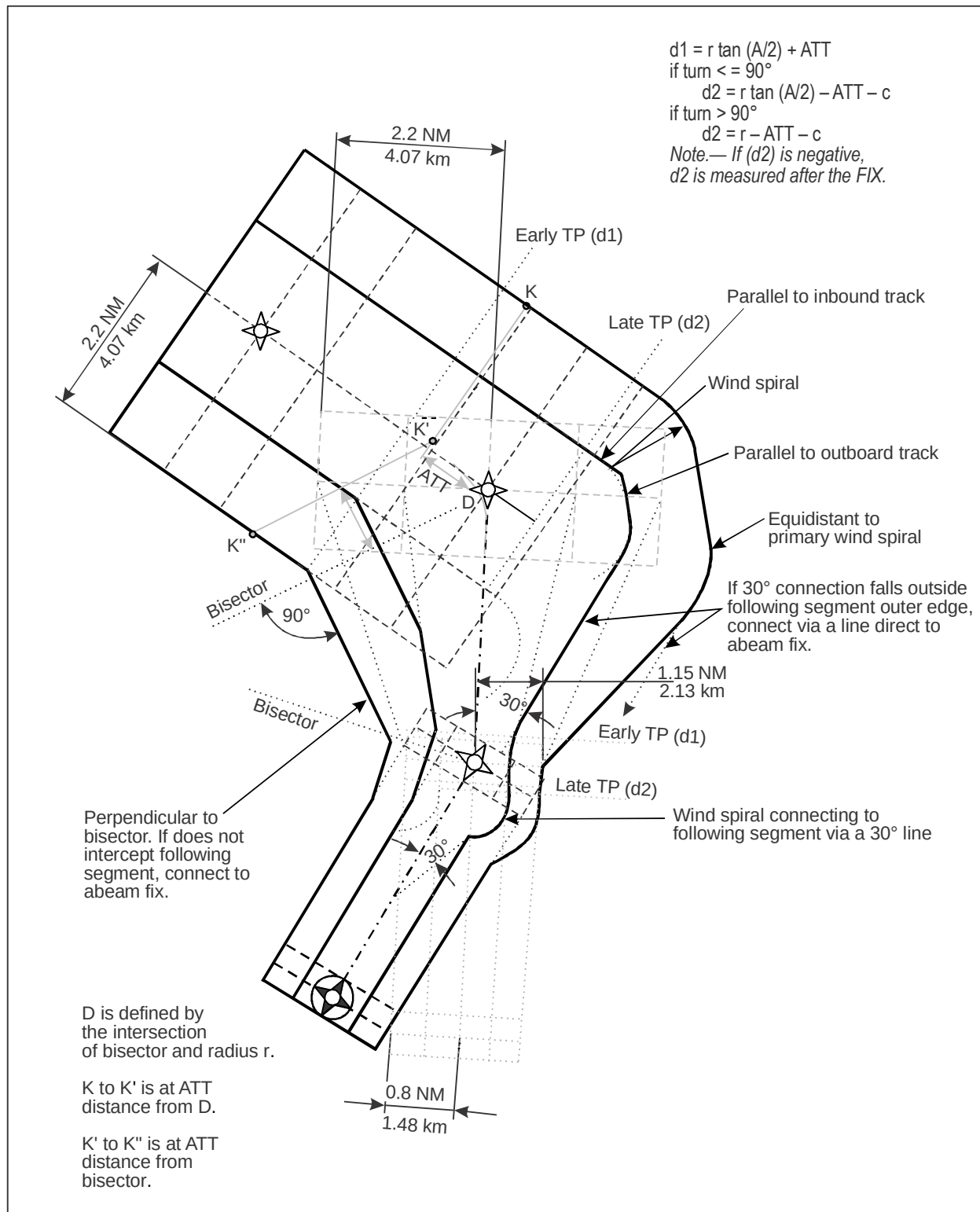


Figure IV-2-1. Initial, intermediate and final approach segments

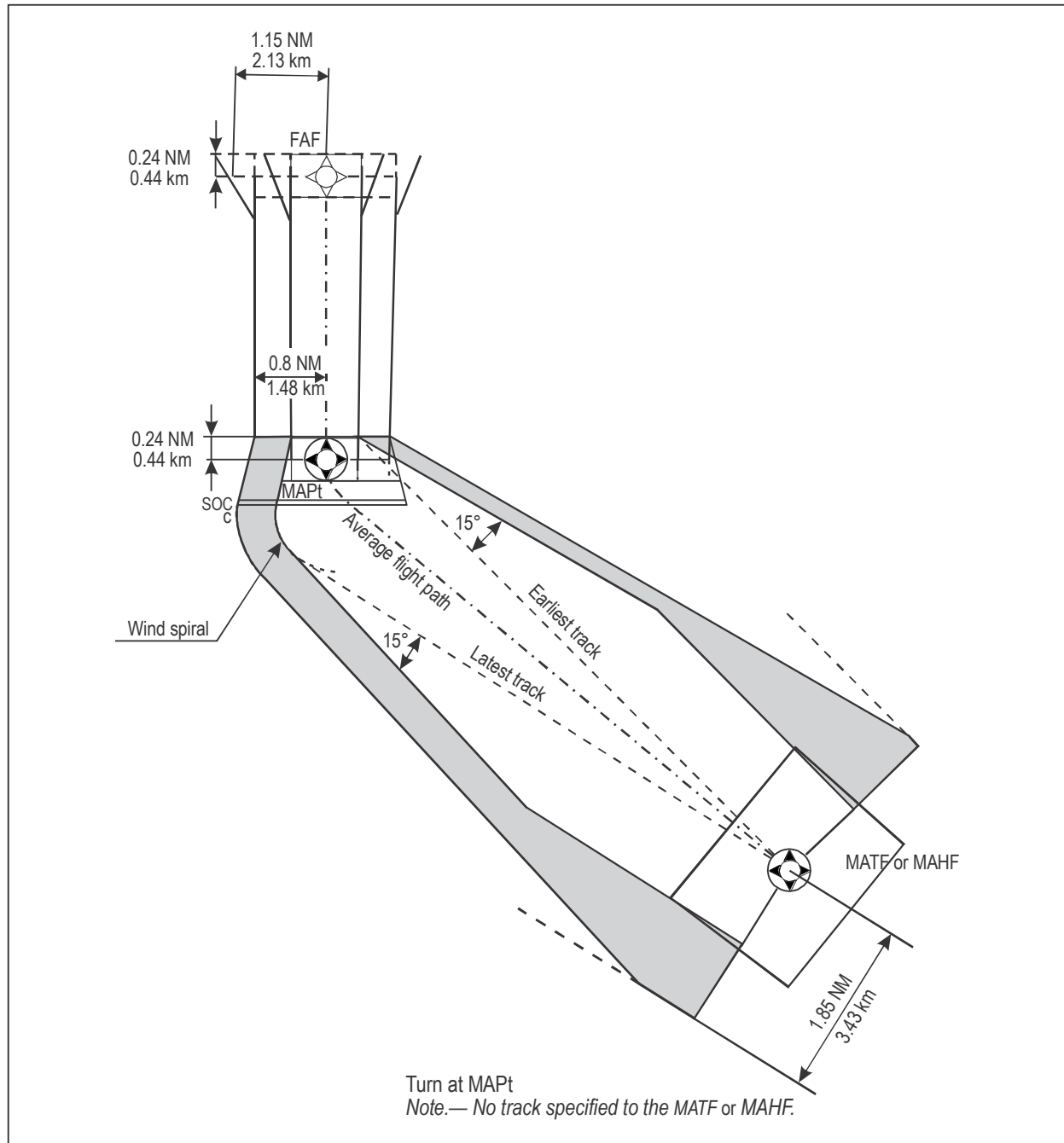


Figure IV-2-2. Turning missed approach

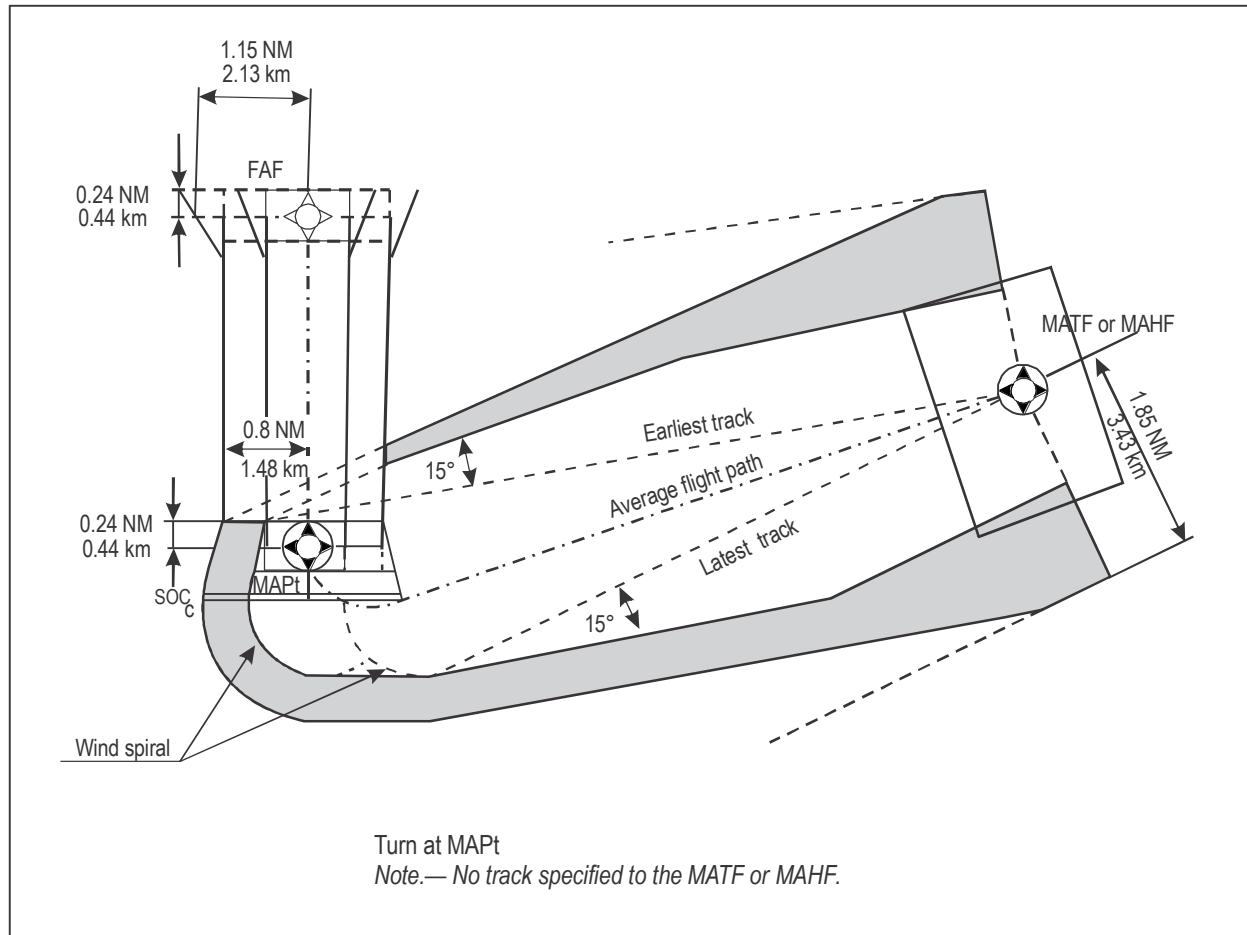


Figure IV-2-3. Turning missed approach with turn more than 90°

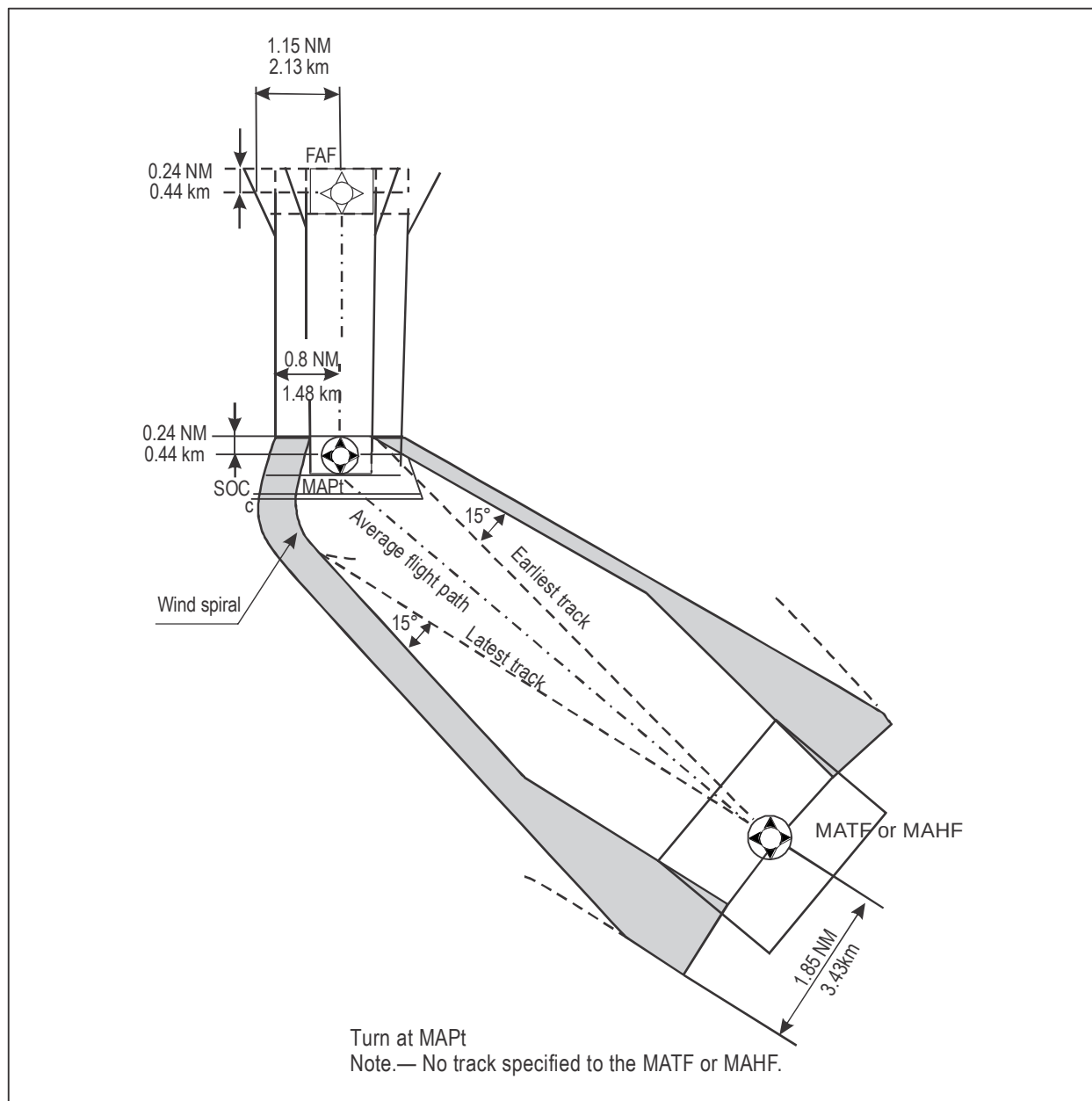


Figure IV-2-4. Turning missed approach with turn less than or equal to 90°

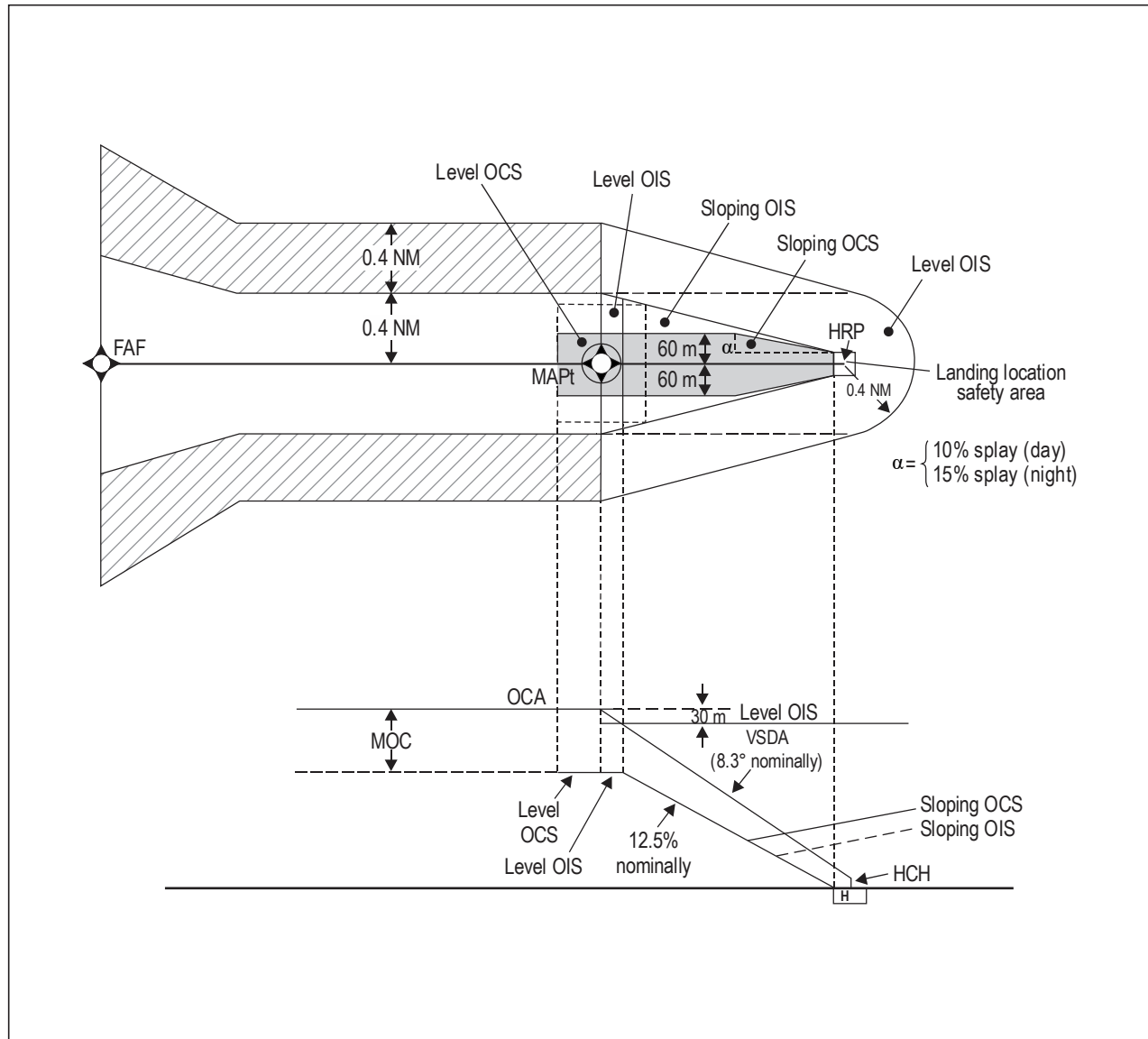


Figure IV-2-5. Direct-VS without DP and without course change

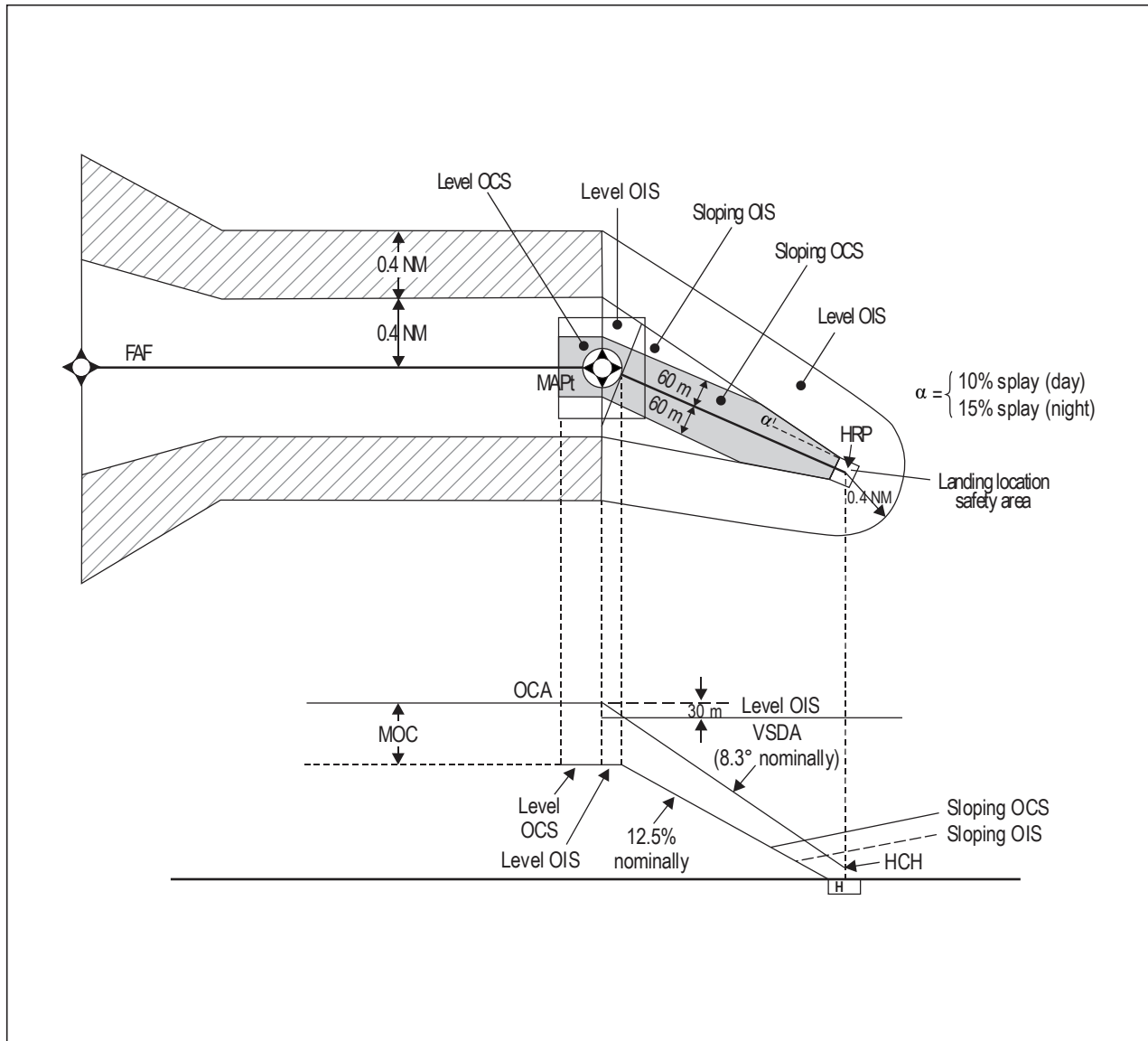


Figure IV-2-6. Direct-VS without DP and with 30° course change at MAPt

Chapter 3

POINT-IN-SPACE (PinS) RNP APCH APPROACH PROCEDURES FOR HELICOPTERS DOWN TO LPV MINIMA

3.1 GENERAL

The general criteria in Part III, Section 3, Chapter 5 and Part IV, Chapter 2, as amplified or modified by the criteria in this chapter, apply to PinS RNP APCH approaches for helicopters down to LPV minima.

3.2 FINAL APPROACH SEGMENT (FAS)

3.2.1 *FHP (fictitious heliport point) and PinS locations.* The final approach segment, ending at the PinS, is oriented on an FHP (fictitious heliport point). The distance between the PinS and the FHP is equal to 800 m (see Figure IV-3-App B-1 in Appendix B). The FHP elevation is equal to the elevation of the landing heliport where the FHPCH constraint in 3.2.3 can be complied with. This is illustrated in Figure IV-3-1. Otherwise, the FHP elevation is adjusted as per 3.2.4.

3.2.2 *Definition of the FAS data block parameters.* Possible encoding for the FAS data block fields for PinS SBAS procedures is described in Appendix B. The following values are fixed:

- a) the distance between the GARP and the FHP is equal to 3 000 m;
- b) the FHP course width is equal to +/-105 m.

3.2.3 *Definition of an operational FAS.* Once the PinS and the FHP locations have been defined, an operational FAS can be defined. The following assumption is used: the OCA/H of the precision segment (OCA/Hps) is equal to the altitude/height of the nominal glide path at the PinS location. Based on this assumption, the operational FAS is defined by fixing two of the following three values: the Glide Path Angle (GPA), the OCA/Hps at the PinS and the crossing height of the flight path angle above the FHP (FHPCH). The following constraints apply for those values:

- a) $GPA \leq 9^\circ$ (15.8%);
- b) $OCHps \geq 250$ ft;
- c) $18 \text{ m (60 ft)} \geq FHPCH \geq 0 \text{ m (0 ft)}$.

Note.— The OCHps is always referenced to the heliport or landing location elevation.

3.2.4 *Adjustment of the FHP elevation.* Where an FHP elevation equal to the elevation of the landing heliport leads to an FHPCH greater than 18 m (60 ft) or less than 0 m (0 ft), a different FHP elevation shall be determined using an FHPCH value of 15 m (50 ft). Attention should be given to ensure that obstacles and OAS are based on the appropriate reference point. Further details can be found in Appendix A.

3.2.5 *Validation of the chosen operational FAS.* Once the operational FAS has been defined, the general protection criteria described in Part III, Section 3, Chapter 5 shall be applied. The system of coordinates used to express protection

surfaces shall be based on the FHP location. General protection criteria shall be applied using Cat H parameters. However, because the OCA/Hps is geometrically fixed by the operational FAS, the following iterative process shall be performed to achieve the most efficient procedure, through application/determination of the lowest possible OCH and the lowest acceptable GPA:

- a) if there is no penetration of the protection surfaces by an obstacle leading to an OCA/H greater than the OCA/Hps, the defined operational FAS is acceptable. However, if the FAS is not the most efficient, a different FAS may be defined (for instance by decreasing the OCA/Hps and FAF elevation or by keeping the same FAF elevation and increasing the GP) and the obstacle protection surfaces shall be checked to determine whether lower minima can be achieved;
- b) if there is a penetration of the protection surfaces by an obstacle leading to an OCA/H greater than the OCA/Hps, the defined FAS is not acceptable. In this case, a new operational FAS shall be defined (for instance by increasing both the OCA/Hps and FAF elevation or by keeping the same FAF elevation and decreasing the GP) and the new protection surfaces shall be checked.

3.3 VISUAL SEGMENT: ADJUSTMENT OF THE OCA/H AND PROTECTION

3.3.1 *Adjustment of the OCA/H.* In order to ensure adequate transition between the instrument phase of flight and the visual phase of flight for “proceed visually” procedures with manoeuvring visual segment and for “proceed visually” procedures with direct visual segment with DP, the final OCA/H is calculated by including an “add-on” value to the OCA/Hps defined in paragraphs 3.2.3 and 3.2.4. This “add-on” value is directly linked to the GPA and is calculated by using the following formula:

$$\text{“add-on” value (ft)} = (1\,460/102) * \text{GPA (degree)}$$

The results of the calculation for a set of GPA values are detailed in Table IV-3-1.

Table IV-3-1. “Add-on” for a set of GPA values

<i>GPA</i>	<i>Add-on value (ft)</i>	<i>Add-on value (m)</i>
3°	43	13.1
3.5°	50	15.3
4°	57	17.5
5°	72	21.9
6°	86	26.2
7°	100	30.5
8°	115	34.9
9°	129	39.3

Note.— No add-on applies to procedures with “proceed VFR” and/or “proceed visually” with direct visual segment without DP.

3.3.2 *Protection of the visual segment.* Criteria used for the definition and the protection of the visual segment described in paragraph 2.9 apply with the following exceptions:

- a) as the SBAS OAS do not have primary and secondary areas, the OIS outer edge should be connected to a semi-width of 741 m (0.4 NM) and the level OIS should be connected to a semi-width of 1 482 m (0.8 NM) at the nominal location of the PinS (MaPt);
- b) the OCA/H shall be replaced by the OCA/Hps value defined in paragraph 3.2.4;
- c) the MOC shall be replaced by the applicable height loss margin (see Table II-1-1-2 and Part III, Section 3, Chapter 5, 5.4.5.9.4); and
- d) the MDA/H shall be replaced by the (DA/H – “add-on”) value.

3.4 SUPPORTING PinS RNP APCH WITH LNAV MINIMA

When LNAV and LPV minima for a PinS RNP APCH procedure are depicted on the same chart, the PinS and vertical profiles of the two approaches shall be the same. The LNAV descent gradient shall be equal to the LPV GPA and shall not be calculated in accordance with paragraph 2.7.5. As per the definition, the LPV OCA/Hps shall be reached at the PinS (MAPt) location, and the LNAV OCA/H shall be reached before the PinS (MAPt).

3.5 MISSED APPROACH

3.5.1 The missed approach turn shall be prescribed at a designated turning point (TP) (see Part III, Section 3, Chapter 5).

Note.— Currently criteria for turns at a designated altitude/height or “as soon as practicable” are not defined but are under development. Such criteria may be necessary in some locations, due to specific obstacle limitations, to optimize the LPV minima.

3.5.2 If a missed approach climb gradient higher than the nominal climb gradient (4.2 per cent) allows an operational benefit, it may be considered the minimum practicable gradient. In this case, the OCA/H applicable to the nominal gradient cannot be shown. If a nominal missed approach climb gradient of 4.2 per cent is operationally required, a separate procedure shall be published with its applicable OCA/H.

3.6 PROMULGATION

3.6.1 PinS approaches to LPV minima shall be promulgated in accordance with Part III, Section 5, Chapter 1 and Part IV, Chapter 2, paragraph 2.11.

3.6.2 A vertical profile inset shall be charted for these procedures. Information depicted in the vertical profile inset shall include the:

- a) the visual segment profile;
- b) heliport or landing location;
- c) location of the MAPt;

- d) final portion of the final approach segment;
- e) heliport elevation;
- f) HCH;
- g) range scale originating from the MAPt to the heliport, which is also used to identify the DP, if one exists in the visual segment;
- h) visual segment track; and
- i) necessary notes needed to highlight certain attributes of the visual segment profiles.

Appendix A to Chapter 3

ADJUSTING THE FHP ELEVATION FOR POINT-IN-SPACE (PinS) RNP APCH APPROACH PROCEDURES DOWN TO LPV MINIMA

1. INTRODUCTION

1.1 In order to ensure adequate obstacle protection, where the requirement that the FHP elevation be equal to the landing heliport elevation leads to an FHPCH greater than 18 m (60 ft) or less than 0 m (0 ft), the elevation of the FHP shall be adjusted so that the new FHPCH equals 15 m (50 ft). This also prevents any detrimental impact on VDI sensitivity as the FHPCH remains at a standard value, and increases the usability of the PinS RNP APCH approach down to the LPV minima concept as it enables a FAS geometry that is completely independent from the landing heliport.

Note.— In line with the maximum ILS reference datum height in the Annex 10 SARPs, it is considered that an FHPCH of a maximum 18 m (60 ft) provides adequate obstacle clearance for PinS RNP APCH procedures for helicopters down to LPV minima. The adjustment of the FHP elevation based on an FHPCH of 15 m (50 ft) ensures an FAS geometry consistent with SBAS APV I and Category I approaches to standard runways, where the recommended TCH equals 15 m (50 ft).

2. METHOD FOR ADJUSTING THE FHP ELEVATION

2.1 The initial FHPCH based on the requirement that the FHP elevation be equal to the landing heliport elevation can be calculated with the following formula:

$$\text{FHPCH (m)} = \text{OCHps} - 800 * \tan(\text{GPA}) \text{ (m)}$$

2.2 In case of an initial FHPCH greater than 18 m (60 ft), the adjusted FHP elevation can be determined by adding the difference between the initial FHPCH and an FHPCH of 15 m (50 ft) to the landing heliport elevation. In case of an initial FHPCH of less than 0 m (0 ft), the adjusted FHP elevation can be determined by subtracting the difference between an FHPCH of 15 m (50 ft) and the initial FHPCH from the landing heliport elevation.

3. OBSTACLE CLEARANCE OF THE SBAS APV I OR CAT I SEGMENT

3.1 The FHP, with its adjusted FHP elevation, defines the new frame of reference for the obstacle assessment, causing the OAS to rise or drop accordingly, as shown in Figure IV-3-App A-1. When reducing the heights of all missed approach obstacles to the heights of equivalent approach obstacles to determine the OCA/H, the heights of the missed approach obstacles shall refer to the adjusted FHP elevation and consequently the resulting equivalent approach obstacle heights are referenced to the adjusted FHP elevation. The OCA/H shall be determined by adding the appropriate “height loss altimeter margin” to the height of the highest approach obstacle (real or equivalent), and, in order to account for the adjustment of the FHP elevation, by adding or subtracting the difference between the adjusted FHP elevation and the landing heliport elevation as appropriate. The OCH shall comply with the OCHps constraint in Chapter 3, 3.2.3, “Definition of an operational FAS”. The OCA/H shall be subject to the validation of the chosen operational FAS as stipulated in Chapter 3, 3.2.5, “Validation of the chosen operational FAS”.

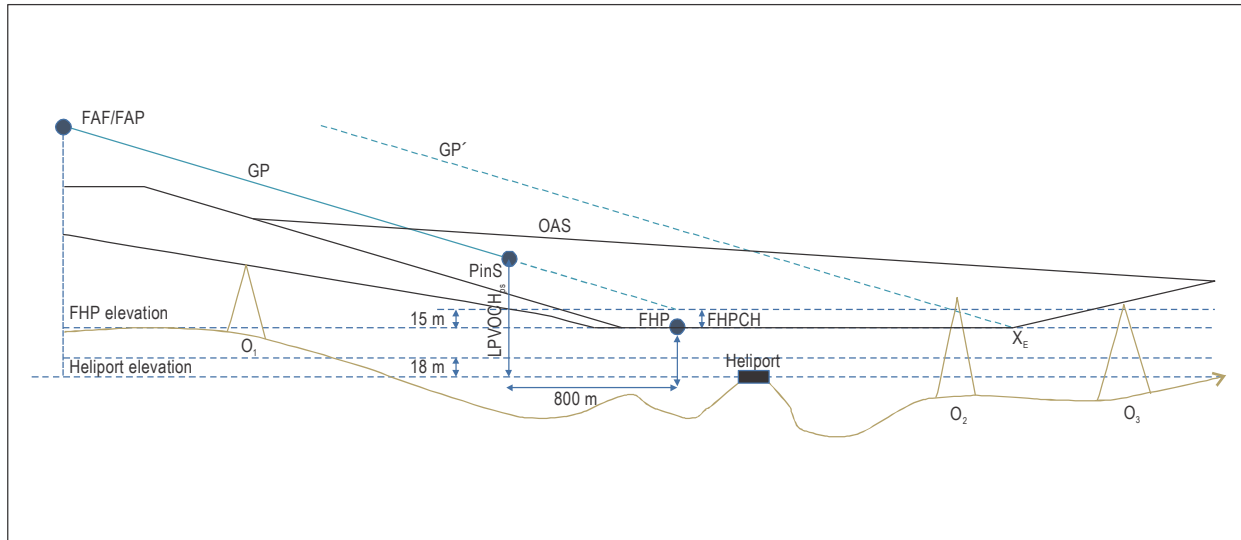


Figure IV-3-App A-1. Raised FHP and OAS for PinS approach procedures down to LPV minima

4. ENCODING OF THE FAS DATA BLOCK

4.1 The coding of the FAS data block shall ensure in any case that the nominal GP crosses the PinS/MAPt at the appropriate OCHps for which obstacle protection is guaranteed. This shall be done by coding the ellipsoidal height of the adjusted FHP along with the nominal FHPCH of 15 m (50 ft).

Appendix B to Chapter 3

ENCODING OF THE SBAS HELICOPTER PinS FAS DATA BLOCK AND DISPLAY SCALING

1. **FAS Data Block Applications to PinS Procedures.** The encoding of the FAS data block fields for PinS operations is based on Part III, Section 2, Chapter 6, Appendix A and should be encoded as described below:

- a) *Operation type*: 0 is reserved for straight-in or PinS procedures.
- b) *Service provider identifier*: 0 for WAAS, 1 for EGNOS, 2 for MSAS 3 for GAGAN, 4 for SDCM. A service provider ID of 15 indicates that any service provider may be used and a service provider ID of 14 indicates the FAS data block is not intended for SBAS use.
- c) *Airport identifier*: If the heliport has an identifier, it is encoded. If the heliport does not have an identifier, the procedure MAPt waypoint name, truncated to a maximum of four characters, should be used, since it is the closest described point in the procedure database to the heliport. For procedures serving multiple heliports, the procedure MAPt waypoint name should be used.
- d) *Runway number*: Runway number is interpreted as the final approach course rounded to the nearest 10 degrees (2 digits). For final approach courses between 355° and 004°, the runway number field should be encoded to 36.
- e) *Runway letter*: Since there is not a letter associated with the procedure, the field is encoded as 00.
- f) *Approach performance designator*: The approach performance designator field is intended for the use by GBAS equipment and not used for SBAS operations.
- g) *Route indicator*: Encode the same as in Part III, Section 2, Chapter 6, Appendix A.
- h) *Reference path data selector (RPDS)*: A numerical identifier used to select the FAS data block (desired approach). It is intended for GBAS and is not used for SBAS operations.
- i) *Reference path identifier*: Since these procedures are not flown to runways, the two-digit runway number is replaced with the FAS track rounded to the closest 10 degrees. For FAS tracks 355° to 004°, the runway number portion of the field should be encoded to 36.

Note.— This coding is consistent with a PinS procedure that supports approaches to more than one landing site.

- j) *Landing threshold point (LTP)/fictitious threshold point (FTP)-latitude*: Encode the heliport/fictitious heliport point (HP/FHP) latitude the same as the LTP/FTP is encoded in Part III, Section 2, Chapter 6, Appendix A.
- k) *Landing threshold point (LTP)/fictitious threshold point (FTP)-longitude*: Encode the HP/FHP longitude the same as the LTP/FTP is encoded in Part III, Section 2, Chapter 6, Appendix A.
- l) *LTP/FTP height above ellipsoid (HAE)*: Encode the HP/FHP height above ellipsoid the same as the LTP/FTP HAE is encoded in Part III, Section 2, Chapter 6, Appendix A.

- m) *Δ flight path alignment point (FPAP)-latitude*: This is the Δ latitude of a point located on a geodesic line beyond the HP/FHP that is aligned with the PinS FAS. Encode the same as in Part III, Section 2, Chapter 6, Appendix A.
- n) *Δ flight path alignment point (FPAP)-longitude*: This is the Δ longitude of a point located on a geodesic line beyond the HP/FHP that is aligned with the PinS FAS. Encode the same as in Part III, Section 2, Chapter 6, Appendix A.
- o) *Threshold crossing height (TCH)*: The designated crossing height of the flight path angle above the heliport/fictitious heliport point crossing height (FHPCH). Encode the same as in Part III, Section 2, Chapter 6, Appendix A.
- p) *TCH units selector*: Encode the same as depicted in Part III, Section 2, Chapter 6, Appendix A.
- q) *Glidepath angle*: Encode the same as shown in Part III, Section 2, Chapter 6, Appendix A.
- r) *Course width at threshold*: This is replaced with the course width at the heliport/fictitious heliport. For approach procedures based on SBAS, the FHP course width is equal to +/- 105 m.
- s) *Δ length offset*: Since there is not a runway associated with the procedure, the field is encoded with a 0.
- t) *Horizontal alert limit (HAL)*: Encode as shown in Part III, Section 2, Chapter 6, Appendix A. PinS procedures have HAL = 40.
- u) *Vertical alert limit (VAL)*: For PinS procedures with lateral only guidance, VAL = 0. When vertical guidance is provided, VAL ≤ 50.
- v) *Final approach segment CRC remainder*: Calculate and encode as shown in Part III, Section 2, Chapter 6, Appendix A.

2. **Display Scaling and Fictitious Heliport Orientation.** Figure IV-3-App B-1 depicts the fictitious heliport orientation and display scaling that flight testing has demonstrated to be satisfactory. At the FHP, the lateral course width is +/-105 m. With the 3 000 m distance between the FHP and GARP, the resulting angular splay is 2°. Extensive flight testing has demonstrated the best combination of procedure flyability and obstacle protection requirements results in a distance of 800 m between the PinS/MAPt and the FHP with a lateral course width of +/-133 m at the PinS/MAPt location. The extreme flexibility in the application of the FAS data block to PinS procedures is noted.

3. Figure IV-3-App B-2 depicts the vertical display scaling and the scaling relationship to the fictitious heliport. Vertical display scaling includes the option of linear scaling once angular scaling becomes greater than +/-150 m or smaller than 15 m. The location along the vertical path where the optional linearization of display begins is a function of the glidepath angle angular splay (glidepath angle/4°) where the full scale width is less than or equal to ±15 m, or greater than or equal to 150 m. The FHP and the glide path intercept point (GPIP) lay in the same plane. The horizontal distance from the FHP to the PinS/MAPt location is maintained at 800 m regardless of the glide path angle to provide for proper lateral display scaling requirements.



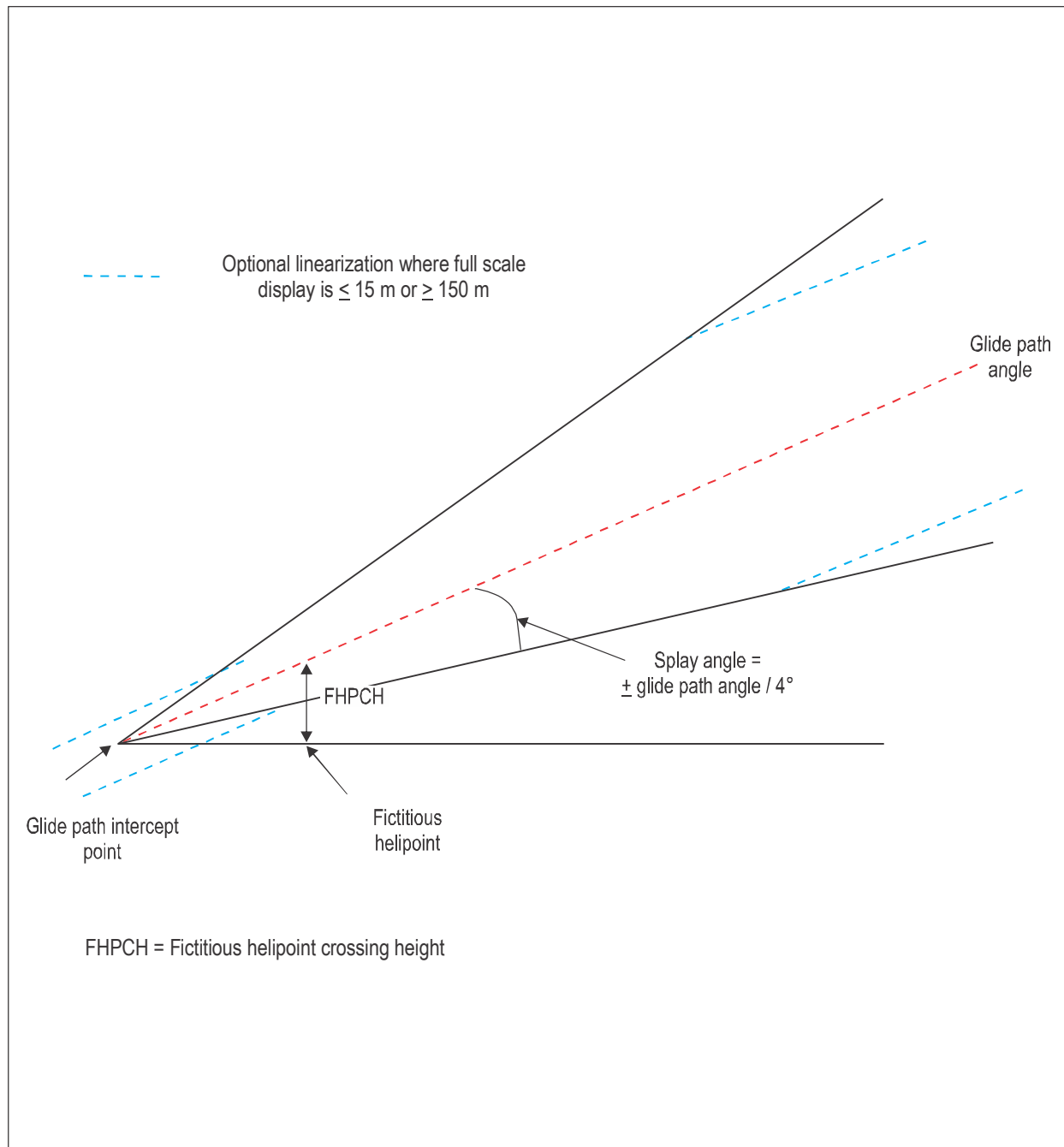


Figure IV-3-App B-2. Fictitious helipoint orientation and vertical display scaling

Appendix C to Chapter 3

STEEP GLIDE PATH ANGLE APPROACHES GREATER THAN 6.3 DEGREES (11 per cent)

1. GENERAL

1.1 For helicopter procedures, glide path angles above 5.7° (10 per cent) or 7.5° (13.2 per cent), when authorized and provided the speed is restricted to a maximum of 130 km/h IAS (70 kt IAS), are non-standard and require special approval.

1.2 The use of the SBAS APV I or Category I criteria in this appendix is limited to PinS approaches with a glide path angle greater than 6.3° (11 per cent) and smaller than or equal to 9° (15.8 per cent).

Note.— For PinS approaches with glide path angles smaller than or equal to 6.3° (11 per cent), refer to Part III, Section 3, Appendix to Chapter 5.

2. PROCEDURE DESIGN

2.1 Obstacle clearance criteria

The obstacle clearance criteria for the height loss/altimeter margin (see 3) should be adjusted for the specific glide path angle.

2.2 Determination of the SBAS APV OAS coefficients

2.2.1 *W surface:* The W surface coefficients for 6.3° glide path are used for all glide path angles greater than 6.3°.

2.2.2 *W' surface:* The W' surface coefficients for 6.3° glide path are used for all glide path angles greater than 6.3°.

2.2.3 *X and Y surfaces:* The X and Y surface coefficients for 3.5° glide path at the appropriate GARP/threshold distance are used for all glide path angles greater than 6.3°.

2.2.4 *Z surface:* The Z surface coefficients are used for all glide path angles.

2.3 Determination of the height of equivalent approach obstacle

Use the formula:

$$h_a = [h_{ma} \cot Z + (x - X_E)]/(\cot Z + \cot \theta)$$

where

h_a	=	height of equivalent approach obstacle
h_{ma}	=	height of missed approach obstacle
θ	=	glide path angle
Z	=	angle of missed approach surface
X_E	=	new coordinate of the Z surface origin
x	=	range of obstacle relative to threshold (negative after threshold).

2.4 Re-survey of obstacles

As the configuration of the SBAS APV OAS is changed, a re-survey of obstacles may be required.

2.5 Promulgation

If the glide path angle exceeds the standard values, a special note shall be included on the instrument approach chart stating that appropriate aircraft and crew authorizations are required to use such a procedure (see Annex 4, 11.10.8.8).

3. HEIGHT LOSS MARGIN

The height loss margin can be obtained by extrapolation from the formulas in Part II, Section 1, Chapter 1, 1.4.8.8.3.1 and Chapter 3, 3.4.8.8.3.1, both entitled “Height loss (HL)/altimeter margins”. However, this extrapolation may not be valid when the nominal rate of descent exceeds 5 m/s (1 000 ft/min).

Note.— Steep approach effects and operational considerations are assessed during the certification process. It is considered that the extrapolated height loss margins provide adequate obstacle protection so long as the aircraft is operated within the certified airspeed, altitude and automatic flight control system limitations.

— END —